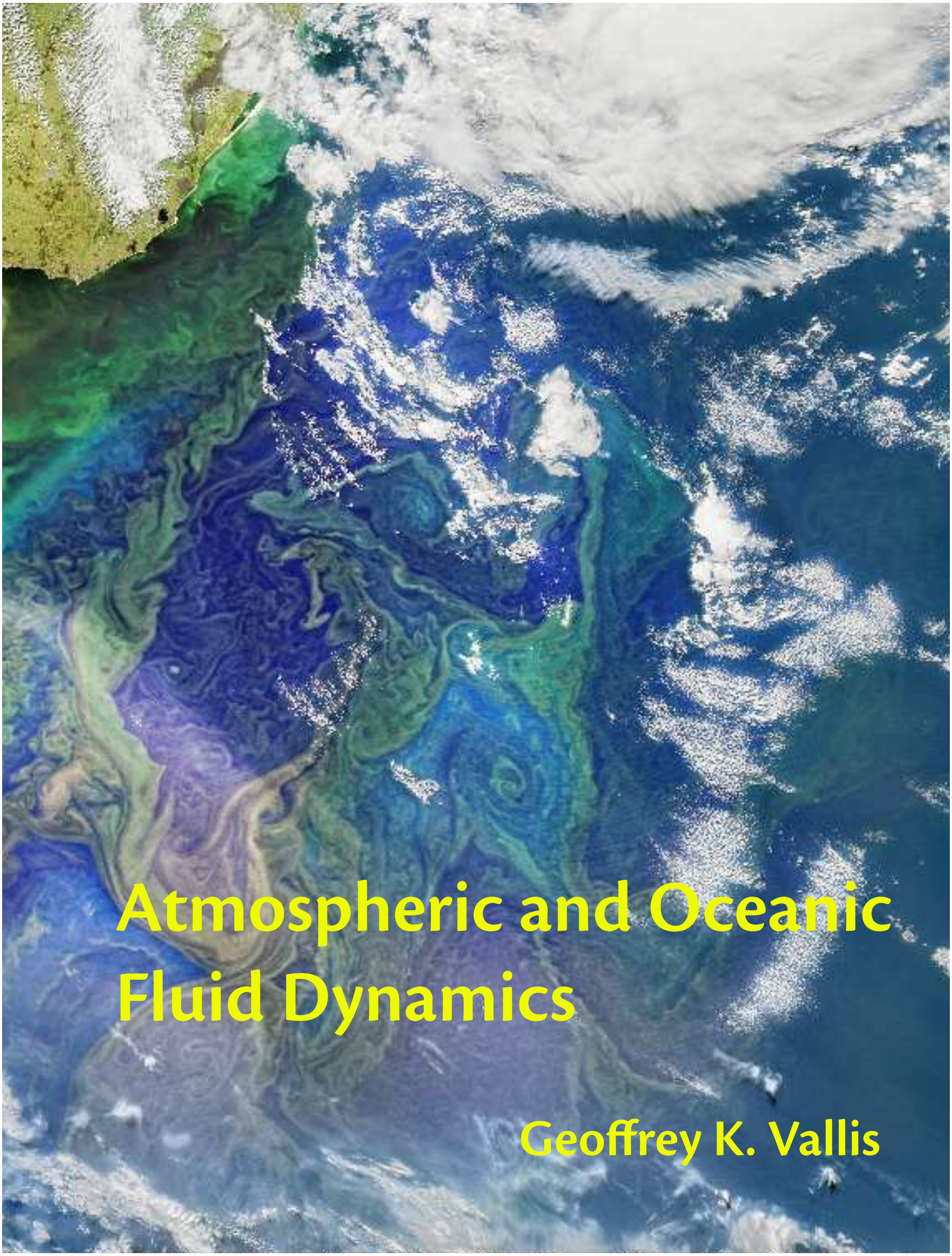

Atmospheric and Oceanic Fluid Dynamics

Geoffrey K. Vallis

Placeholder

Contents

In the main text, sections that are more advanced or that contain material that is peripheral to the main narrative are marked, like difficult ski slopes, with a black diamond, ✦. Sections that contain material that is still not settled or are at the edge of knowledge are marked with a dagger, †.

Look again at that dot. That's here. That's home. That's us... Our planet is a lonely speck in the great enveloping cosmic dark. In our obscurity, in all this vastness, there is no hint that help will come from elsewhere to save us from ourselves... Like it or not, for the moment the Earth is where we make our stand.

Carl Sagan, *Pale Blue Dot,* 1994.

Preface

September 20, 2016.
The preface is yet to be written,

Part I

FUNDAMENTALS OF GEOPHYSICAL FLUID DYNAMICS

Part II

WAVES, INSTABILITIES AND TURBULENCE

Part III

LARGE-SCALE ATMOSPHERIC CIRCULATION

Part IV

LARGE-SCALE OCEANIC CIRCULATION

Old Ocean, none knoweth thy story;
Man cannot thy secrets unfold,
Thy blue waves sing songs of thy glory
But where are thy treasures untold?
Martha Lavinia Hoffman (1865–1900), *Lines to the Ocean.*

CHAPTER 19

Wind-Driven Gyres

UNDERSTANDING THE CIRCULATION OF THE OCEAN involves a combination of observations, comprehensive numerical modelling, and more conceptual modelling, or 'theory'. All are essential, but in this chapter and the ones following our emphasis is on the last of the tripos. Its (continuing) role is not to explain every feature of the observed ocean circulation, nor to necessarily describe details best left to numerical simulations. Rather, it is to provide a conceptual and theoretical framework for understanding the circulation of the ocean, for interpreting observations and suggesting how new observations may best be made, and to aid the development and interpretation of experiments with numerical models.

The aspect of the ocean that most affects the climate is the sea-surface temperature (SST), as illustrated in Fig. 19.1, and aside from the expected latitudinal variation there is significant zonal variation too — the western tropical Pacific is particularly warm, and the western Atlantic is noticeably warmer than the corresponding latitude in the east. These variations owe their existence to ocean currents, and the main ones are sketched — in a highly schematic and non-quantitative fashion — in Fig. 19.2. Over most of the ocean, the vertically averaged currents have a similar sense to the surface currents, one exception being at the equator where the surface currents are mainly westwards but the vertical integral is dominated by the eastward undercurrent.

Two dichotomous aspects stand out: (i) the complexity of the currents as they interact with topography and the geography of the continents; (ii) the simplicity and commonality of the large-scale structures in the major ocean basins, and in particular the ubiquity of subtropical and subpolar gyres. Indeed these gyres, sweeping across the great oceans carrying vast quantities of water and heat, are perhaps the single most conspicuous feature of the circulation. The subtropical gyres are anticyclonic, extending polewards to about 45°, and the subpolar gyres are cyclonic and polewards of this, primarily in the Northern Hemisphere. The existence of the great gyres, and that they are strongest in the west, has been known for centuries; this *western intensification* leads to such well- known currents as the Gulf Stream in the Atlantic (charted by Benjamin Franklin), the Kuroshio in the Pacific, and the Brazil Current in the South Atlantic.

For much of this chapter we consider a model, and variations about it, that explains the large-scale features of ocean gyres and that lies at the core of ocean circulation theory — the steady, forced-dissipative, homogeneous model of the ocean circulation first formulated by Stommel.[1] There is perhaps no more elegant model in all of the geosciences than this one.

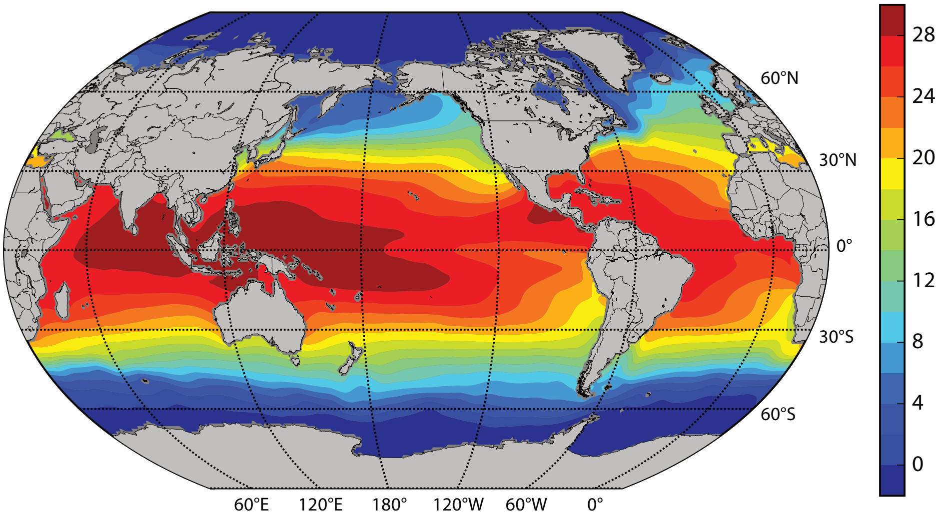

Fig. 19.1 The sea-surface temperature (SST, °C) of the worlds ocean, as detemined from a great many observations, combined in the World Ocean Circulation Experiment (WOCE).

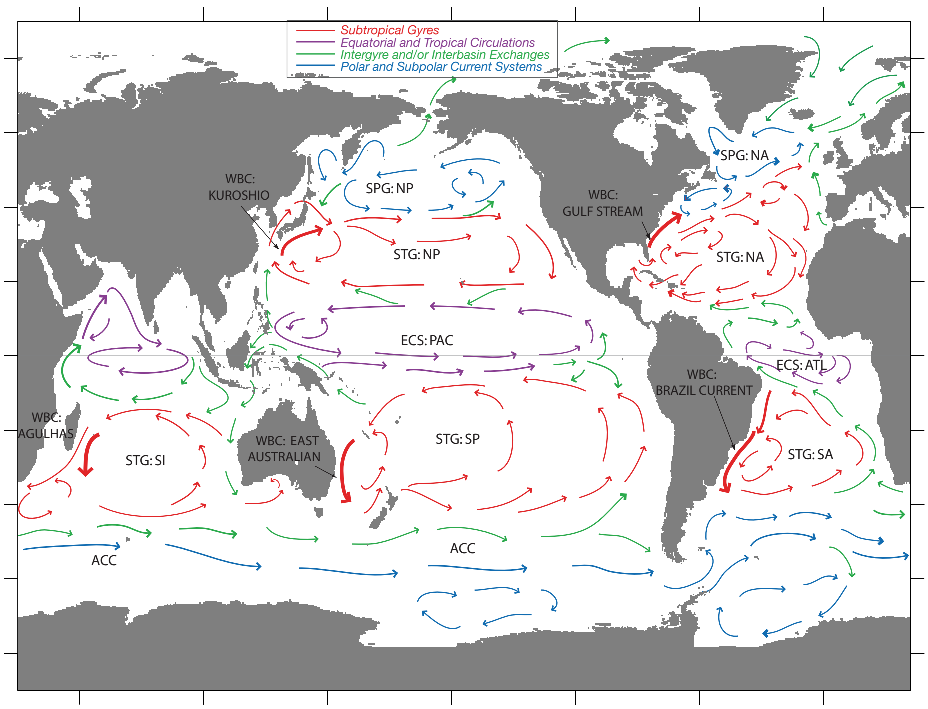

Fig. 19.2 A sketch of the main currents of the global ocean. Key: STG – Subtropical Gyre; SPG – Subpolar Gyre; WBC – Western Boundary Current; ECS – Equatorial Current System; NA – North Atlantic; SA – South Atlantic; NP – North Pacific; SP – South Pacific; SI – South Indian; ACC – Antarctic Circumpolar Current; ATL – Atlantic; PAC – Pacific. The figure is a qualitative, and not quantitative, representation of the actual flow. Most of the currents are manifested at the surface, but in the equator the figure shows the equatorial undercurrent.

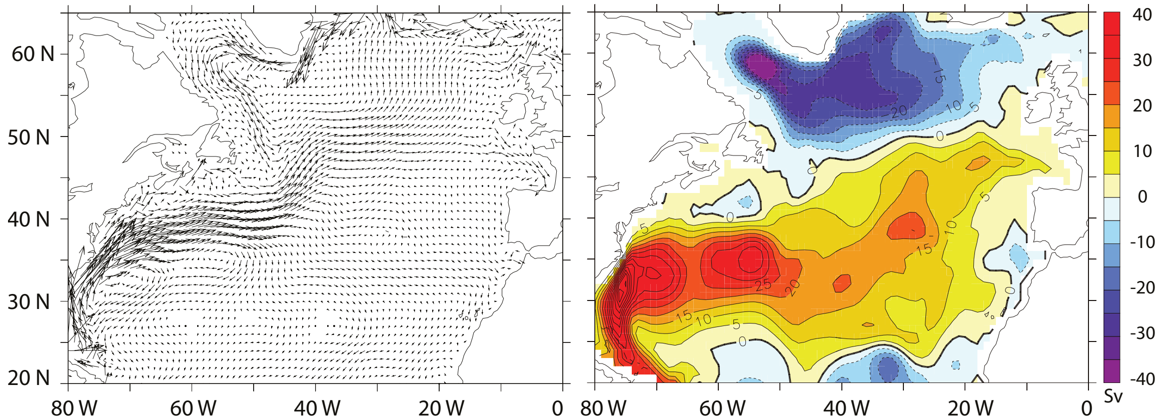

Fig. 19.3 Left: the time averaged velocity field at a depth of 75 m in the North Atlantic, obtained by constraining a numerical model to observations (so giving a 'state estimate'). Right: the streamfunction of the vertically integrated flow, in Sverdrups (1 Sv $= 10^9\ \mathrm{kg\,s^{-1}}$). Note the presence of an anticyclonic subtropical gyre (clockwise circulation, shaded red), a cyclonic subpolar gyre (anticlockwise, blue), and intense western boundary currents.[3]

19.1 THE DEPTH INTEGRATED WIND-DRIVEN CIRCULATION

Although even today we barely have sufficient observations to produce a detailed synoptic map of the ocean currents, except at the surface, the large-scale mean currents are fairly well mapped and Fig. 19.3 illustrates the average current pattern of the North Atlantic using a combination of observations and a numerical model, and the Gulf Stream is clearly visible. Similar features are seen in all the major ocean basins (Fig. 19.4) where we see subtropical and subpolar gyres, all of them intensifies in the west.[2] Our goal in this chapter is to explain the main features seen in these figures in as simple and straightofrward a manner as is possible.

The equations that govern the large-scale flow in the oceans are the planetary-geostrophic equations. Greatly simplified as these are compared to the Navier–Stokes equations, or even the hydrostatic Boussinesq equations, they are still quite daunting: a prognostic equation for buoyancy is coupled to the advecting velocity via hydrostatic and geostrophic balance, and the resulting problem is formidably nonlinear. However, it turns out that thermodynamic effects can effectively be eliminated by the simple device of vertical integration; the resulting equations are linear, and the only external forcing is that due to the wind stress. The resulting model then, at the price of some comprehensiveness, gives a useful picture of the *wind-driven* circulation of the ocean. We will consider the vertical structure of this flow in the next chapter.

19.1.1 The Stommel model

The planetary-geostrophic equations for a Boussinesq fluid are:

$$\frac{\mathrm{D}b}{\mathrm{D}t} = \dot{b}, \qquad \nabla_3 \cdot \boldsymbol{v} = 0, \tag{19.1a,b}$$

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi + \frac{1}{\rho_0}\frac{\partial \boldsymbol{\tau}}{\partial z}, \qquad \frac{\partial \phi}{\partial z} = b. \tag{19.2a,b}$$

These equations are, respectively, the thermodynamic equation (19.1a), the mass continuity equation (19.1b), the horizontal momentum equation (19.2a), (i.e., geostrophic balance, plus a stress term), and the vertical momentum equation (19.2b) — that is, hydrostatic balance. These equations are derived more fully in chapter 5, but they are essentially the Boussinesq primitive equations with the advection terms omitted in the horizontal momentum equation, on the basis of small Rossby number. In this chapter we will henceforth absorb the factor of ρ_0 into the $\boldsymbol{\tau}$, so that $\boldsymbol{\tau}$ denotes the kinematic stress, and the gradient operator will be two dimensional, in the x-y plane, unless noted.

Take the curl of (19.2a) (that is, cross differentiate its x and y components) and integrate over the depth of the ocean to give

$$\int f\nabla\cdot\boldsymbol{u}\,\mathrm{d}z + \frac{\partial f}{\partial y}\int v\,\mathrm{d}z = \mathrm{curl}_z(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B), \tag{19.3}$$

where the operator curl_z is defined by $\mathrm{curl}_z\boldsymbol{A} \equiv \partial A^y/\partial x - \partial A^x/\partial y = \mathbf{k}\cdot\nabla\times\boldsymbol{A}$, and the subscripts T and B are for top and bottom. The divergence term vanishes if the vertical velocity is zero at the top and bottom of the ocean. Strictly, at the top of the ocean the vertical velocity is given by the material derivative of height of the ocean's surface, $\mathrm{D}h/\mathrm{D}t$, but on the large-scales this has a negligible effect and we may make the rigid-lid approximation and set it to zero. At the bottom of the ocean the vertical velocity is only zero if the ocean in flat-bottomed; otherwise it is $\boldsymbol{u}\cdot\nabla\eta_B$, where η_B is the orographic height at the ocean floor. The neglect of this topographic term is probably the most restrictive single approximation in the model. Given this neglect, (19.3) becomes

$$\beta\overline{v} = \mathrm{curl}_z(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B) \tag{19.4}$$

where henceforth, in this section, quantities with an overbar are understood to be the vertical integral over the depth of the ocean. If the stresses depend only on the velocity fields then thermodynamic fields do not affect the vertically integrated flow.

At the top of the ocean, the stress is given by the wind. At the bottom, in the absence of topography we assume that the stress may be parameterized by a linear drag, or Rayleigh friction, as might be generated by an Ekman layer; it is this assumption that particularly characterizes this model as being due to Stommel. (Note that we parameterize the friction by a drag acting on the vertically integrated velocity. Using the velocity at the bottom of the ocean would be more realistic, but this wrinkle is beyond the scope of vertically integrated models.) Equation (19.4) then becomes

$$\beta\overline{v} = -r\overline{\zeta} + F_\tau(x, y), \tag{19.5}$$

where $F_\tau = \mathrm{curl}_z\boldsymbol{\tau}_T$ is the wind-stress curl at the top of the ocean and is a known function. Because the velocity is divergence-free, we can define a streamfunction ψ such that $\overline{u} = -\partial\psi/\partial y$ and $\overline{v} = \partial\psi/\partial x$. Equation (19.5) then becomes

$$r\nabla^2\psi + \beta\frac{\partial\psi}{\partial x} = F_\tau(x, y). \tag{19.6}$$

This equation is often referred to as the *Stommel problem* or the *Stommel model,* and may be posed in a variety of two dimensional domains.

19.1.2 Alternative formulations

A number of alternative formulations leading to (19.6) are possible. None are perhaps as well justified as the derivation via the planetary-geostrophic equations but the differences in the specific assumptions made give some indication of the robustness of the derivation, and show how the model might be extended to include topographic or nonlinear effects.

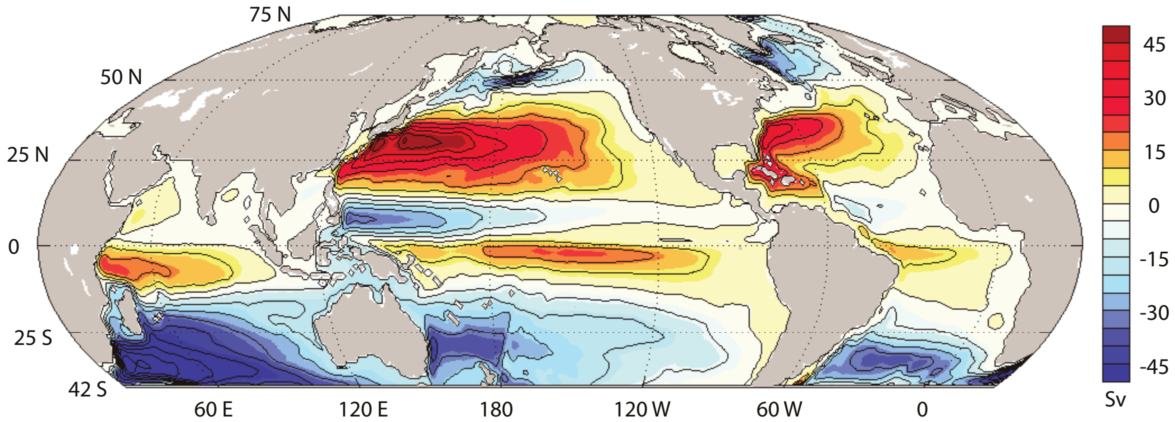

Fig. 19.4 A state estimate of the streamfunction of the vertically integrated flow for the near global ocean. Red shading indicates clockwise flow, and blue shading anticlockwise, but in both hemispheres the subtropical (subpolar) gyres are anticyclonic (cyclonic).[4]

I. A Homogeneous model

Rather than vertically integrating, we may suppose that the ocean is a homogeneous fluid obeying the shallow water equations (chapter 3). The potential vorticity equation [cf. (3.96) on page 121] is:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{\zeta+f}{h}\right)=\frac{F}{h}, \tag{19.7}$$

where F represents friction and forcing. In an ocean with a rigid-lid and flat bottom (19.7) gives the barotropic vorticity equation,

$$\frac{\mathrm{D}\zeta}{\mathrm{D}t}+\beta v=F, \tag{19.8}$$

where the term F again represents the wind-stress curl and a linear drag. Further, since the horizontal velocity is divergence-free (because of the flat-bottom and rigid-lid) we may represent it as a streamfunction, whence we obtain the closed equation

$$\frac{\mathrm{D}}{\mathrm{D}t}\nabla^2\psi+\beta\frac{\partial\psi}{\partial x}=F_\tau(x,y)-r\nabla^2\psi, \tag{19.9}$$

where F_τ again represents the wind forcing. This equation is the 'time-dependent nonlinear Stommel problem'. The steady nonlinear problem is sometimes of interest, too, and this is just

$$J(\psi,\nabla^2\psi)+\beta\frac{\partial\psi}{\partial x}=F_\tau(x,y)-r\nabla^2\psi. \tag{19.10}$$

To obtain the original Stommel model we just ignore the advective derivative, which will be valid if $|\zeta|\sim Z\ll\beta L$ where $Z=U/L$ is a representative value of vorticity. This condition is equivalent to

$$R_\beta\equiv\frac{U}{\beta L^2}\ll 1. \tag{19.11}$$

R_β is called the *beta-Rossby number*. On sufficiently large scales, $\beta\sim f/L$ and (19.11) is similar to a small Rossby number assumption.

II. Quasi-geostrophic formulation

In the planetary-geostrophic formulation, the horizontal velocity is divergence-free only because we have vertically integrated. If the scales of motion are not too large the horizontal flow at every level is divergence-free for another reason: because it is in geostrophic balance. In reality, over a single oceanic gyre (say from 15° to 40° latitude), variations in Coriolis parameter are not large, and this prompts us to formulate the model in terms of the quasi-geostrophic equations. Formally, such a model would then be restricted to length scales, L, of no more than $\mathcal{O}(Ro^{-1})$ larger than the deformation radius, and for gyre scales this criterion is marginally satisfied if $L_d = 100\,\mathrm{km}$. An advantage of the quasi-geostrophic equations is that they readily allow for the inclusion of both nonlinearity and stratification. [For an informal summary of quasi-geostrophy, see the box on page 193.] The quasi-geostrophic vorticity equation for a Boussinesq system is:

$$\frac{\mathrm{D}\zeta}{\mathrm{D}t} + \beta v = f_0 \frac{\partial w}{\partial z} + \mathrm{curl}_z \frac{\partial \boldsymbol{\tau}}{\partial z}. \tag{19.12}$$

If we neglect the advective derivative, and vertically integrate, we obtain

$$\beta \int_B^T v \,\mathrm{d}z = f_0 [w]_B^T + \mathrm{curl}_z [\boldsymbol{\tau}]_B^T. \tag{19.13}$$

where T denotes the ocean top and B the bottom. We now make one of two virtually equivalent choices.

(i) We suppose that the integration is over the entire depth of the ocean, in which case the term $[w]_B^T$ vanishes given a rigid lid and a flat bottom. If the stress at the top of the ocean is given by the wind stress, and at the bottom of the ocean it is parameterized by a linear drag, then we obtain

$$\beta \overline{v} = F_\tau(x, y) - r\overline{\zeta} \tag{19.14}$$

just as in (19.5), and where an overbar denotes a vertical integral. Writing $\overline{v} = \partial\psi/\partial x$ and $\overline{\zeta} = \nabla^2\psi$ then gives (19.6).

(ii) We suppose that the integration is between two thin Ekman layers at the top and bottom of the ocean. The stress is zero at the interior edge of these layers, but the vertical velocity is not. At the base of the upper Ekman layer, at $z = -\delta_T$, the vertical velocity is given by:

$$w(x, y, -\delta_T) = \mathrm{curl}_z(\boldsymbol{\tau}_T / f_0), \tag{19.15}$$

where the top of the ocean is at $z = 0$ and δ_T is the thickness of the top Ekman layer. Similarly, at the top of the lower Ekman layer, the vertical velocity is:

$$w(x, y, -H + \delta_B) = \delta_B \zeta, \tag{19.16}$$

where $z = -H$ at the ocean bottom and δ_B is the thickness of the bottom Ekman layer. Neglecting the advective derivative, and integrating over the ocean between the two Ekman layers, (19.12) becomes

$$\beta \overline{v} = \mathrm{curl}_z \boldsymbol{\tau}_T - f_0 \delta_B \zeta = \mathrm{curl}_z \boldsymbol{\tau}_T - f_0 \delta_B \overline{\zeta}/H, \tag{19.17}$$

where, to obtain the second equality, we assume that the bottom drag may be parameterized using the vertically integrated vorticity, $\overline{\zeta}$. Defining the drag coefficient r by $r = f_0\delta_B/H$, and introducing a streamfunction gives

$$r\nabla^2\psi + \beta \frac{\partial \psi}{\partial x} = \mathrm{curl}_z \boldsymbol{\tau}_T, \tag{19.18}$$

as before.

19.1.3 Approximate solution of Stommel model

Sverdrup balance

Equation (19.6) is linear and it is possible to obtain an exact, analytic solution. However, it is more insightful to approach the problem perturbatively, by supposing that the frictional term is small, meaning there is an approximate balance between wind stress and the β-effect.[5] Friction is small if $|r\zeta| \ll |\beta v|$ or

$$\frac{r}{L} = \frac{f\delta_B}{HL} \ll \beta \tag{19.19}$$

using $r = f\delta_B/H$, and where L is the horizontal scale of the motion, and generally speaking this inequality is well satisfied for large-scale flow. The vorticity equation becomes

$$\beta\overline{v} \approx \mathrm{curl}_z \boldsymbol{\tau}_T, \tag{19.20}$$

which is known as *Sverdrup balance*.[6] (Sometimes Sverdrup balance is taken to mean the linear geostrophic vorticity balance $\beta v = f\partial w/\partial z$, but we will restrict its use to mean a balance between the beta effect and wind stress curl.) The observational support for Sverdrup balance is rather mixed, discrepancies arising not so much from the failure of (19.19), but from the presence of small-scale eddying motion with concomitantly large nonlinear terms, and the presence of non-negligible vertical velocities induced by the interaction with bottom topography.[7] Nevertheless, Sverdrup balance provides a useful, if not impregnable, foundation on which to build.

Boundary-layer solution

For simplicity, consider a square domain of side a and rescale the variables by setting

$$x = a\widehat{x}, \qquad y = a\widehat{y}, \qquad \tau = \tau_0\widehat{\tau}, \qquad \psi = \widehat{\psi}\frac{\tau_0}{\beta}, \tag{19.21}$$

where τ_0 is the amplitude of the wind stress. The hatted variables are non-dimensional and, assuming our scaling to be sensible, these are $\mathcal{O}(1)$ quantities in the interior. Equation (19.18) becomes

$$\frac{\partial\widehat{\psi}}{\partial\widehat{x}} + \epsilon_S\nabla^2\widehat{\psi} = \mathrm{curl}_z\widehat{\boldsymbol{\tau}}_T, \tag{19.22}$$

where $\epsilon_S = (r/a\beta) \ll 1$, in accord with (19.19). For the rest of this section we will drop the hats over non-dimensional quantities. Over the interior of the domain, away from boundaries, the frictional term in (19.22) is small. We can take advantage of this by writing

$$\psi(x, y) = \psi_I(x, y) + \phi(x, y), \tag{19.23}$$

where ψ_I is the interior streamfunction and ϕ is a boundary layer correction. Away from boundaries ψ_I is presumed to dominate the flow, and this satisfies

$$\frac{\partial\psi_I}{\partial x} = \mathrm{curl}_z\boldsymbol{\tau}_T. \tag{19.24}$$

The solution of this equation (called the 'Sverdrup interior') is

$$\psi_I(x, y) = \int_0^x \mathrm{curl}_z\boldsymbol{\tau}(x', y)\,\mathrm{d}x' + g(y), \tag{19.25}$$

where $g(y)$ is an arbitrary function of integration that gives rise to an arbitrary zonal flow. The corresponding velocities are

$$v_I = \mathrm{curl}_z\boldsymbol{\tau}, \qquad u_I = -\frac{\partial}{\partial y}\int_0^x \mathrm{curl}_z\boldsymbol{\tau}(x', y)\,\mathrm{d}x' - \frac{\mathrm{d}g(y)}{\mathrm{d}y}. \tag{19.26}$$

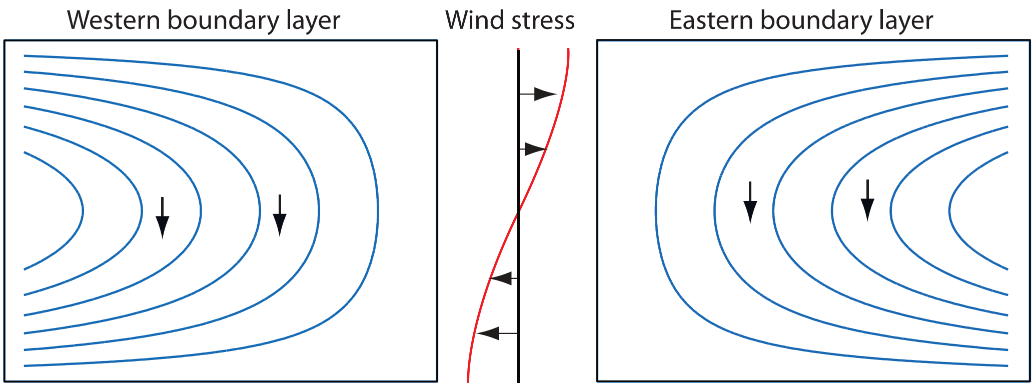

Fig. 19.5 Two possible Sverdrup flows, ψ_I, for the wind stress shown in the centre. Each solution satisfies the no-flow condition at either the eastern or western boundary, and a boundary layer is therefore required at the other boundary. Both flows have the same, equatorward, meridional flow in the interior. Only the flow with the western boundary current is physically realizable, however, because only then can friction produce a curl that opposes that of the wind stress, so allowing the flow to equilibrate.

The dynamics is most clearly illustrated if we now restrict our attention to a wind-stress curl that is zonally uniform, and that vanishes at two latitudes, $y = 0$ and $y = 1$. An example is

$$\tau_T^y = 0, \qquad \tau_T^x = -\cos(\pi y), \tag{19.27}$$

for which $\mathrm{curl}_z \boldsymbol{\tau}_T = -\pi \sin(\pi y)$. The Sverdrup (interior) flow may then be written as

$$\psi_I(x, y) = [x - C(y)]\mathrm{curl}_z \boldsymbol{\tau}_T = \pi[C(y) - x] \sin \pi y, \tag{19.28}$$

where $C(y)$ is the arbitrary function of integration $[C(y) = -g(y)/\mathrm{curl}_z \boldsymbol{\tau}]$. If we choose C to be a constant, the zonal flow associated with it is $C\,\mathrm{curl}_z \boldsymbol{\tau}_T$. We can then satisfy $\psi = 0$ at *either* $x = 0$ (if $C = 0$) *or* $x = 1$ (if $C = 1$). These solutions are illustrated in Fig. 19.5 for the particular stress (19.27).

Regardless of our choice of C we cannot satisfy $\psi = 0$ at both zonal boundaries. We must choose one, and then construct a *boundary layer* solution (i.e., we determine ϕ) to satisfy the other condition. Which choice do we make? On intuitive grounds it seems that we should choose the solution that satisfies $\psi = 0$ at $x = 1$ (the solution on the left in Fig. 19.5), for then the interior flow then goes round in the same direction as the wind: the wind is supplying a clockwise torque, and to achieve an angular momentum balance anticlockwise angular momentum must be supplied by friction. We can imagine that this would be provided by the frictional forces at the western boundary layer if the interior flow is clockwise, but not by friction at an eastern boundary layer when the interior flow is anticlockwise. Note that this argument is not dependent on the sign of the wind-stress curl: if the wind blew the other way a similar argument still implies that a western boundary layer is needed. We will now see if and how the mathematics reflects this intuitive but non-rigorous argument.

Asymptotic matching

Near the walls of the domain the boundary layer correction $\phi(x, y)$ must become important in order that the boundary conditions may be satisfied, and the flow, and in particular $\phi(x, y)$, will vary rapidly with x. To reflect this, let us *stretch* the x-coordinate near this point of failure (i.e., at either $x = 0$ or $x = 1$, but we do not know at which yet) and let

$$x = \epsilon\alpha \qquad \text{or} \qquad x - 1 = \epsilon\alpha. \tag{19.29a,b}$$

Here, α is the stretched coordinate, which has values $\mathcal{O}(1)$ in the boundary layer, and ϵ is a small parameter, as yet undetermined. We then suppose that $\phi = \phi(\alpha, y)$, and using (19.23) in (19.22), we obtain

$$\epsilon_S(\nabla^2\psi_I + \nabla^2\phi) + \frac{\partial \psi_I}{\partial x} + \frac{1}{\epsilon}\frac{\partial \phi}{\partial \alpha} = \mathrm{curl}_z \boldsymbol{\tau}_T, \tag{19.30}$$

where $\phi = \phi(\alpha, y)$ and $\nabla^2\phi = \epsilon^{-2}\partial^2\phi/\partial\alpha^2 + \partial^2\phi/\partial y^2$. Now, by choice, ψ_I exactly satisfies Sverdrup balance, and so (19.30) becomes

$$\epsilon_S\left(\nabla^2\psi_I + \frac{1}{\epsilon^2}\frac{\partial^2\phi}{\partial\alpha^2} + \frac{\partial^2\phi}{\partial y^2}\right) + \frac{1}{\epsilon}\frac{\partial\phi}{\partial\alpha} = 0. \tag{19.31}$$

We now choose ϵ to obtain a physically meaningful solution. An obvious choice is $\epsilon = \epsilon_S$, for then the leading-order balance in (19.31) is

$$\frac{\partial^2\phi}{\partial\alpha^2} + \frac{\partial\phi}{\partial\alpha} = 0, \tag{19.32}$$

the solution of which is

$$\phi = A(y) + B(y)\mathrm{e}^{-\alpha}. \tag{19.33}$$

Evidently, ϕ grows exponentially in the negative α direction. If this were allowed, it would violate our assumption that solutions are small in the interior, and we must eliminate this possibility by allowing α to take only positive values in the interior of the domain, and by setting $A(y) = 0$. We therefore choose $x = \epsilon\alpha$ so that $\alpha > 0$ for $x > 0$; the boundary layer is then at $x = 0$, that is, it is a *western boundary,* and it decays eastwards in the direction of increasing α — that is, into the ocean interior. We now choose $C = 1$ in (19.28) to make $\psi_I = 0$ at $x = 1$ in (19.28) and then, for the wind stress (19.27), the interior solution is given by

$$\psi_I = \pi(1 - x)\sin\pi y. \tag{19.34}$$

This alone satisfies the boundary condition at the eastern boundary. The function $B(y)$ is chosen to satisfy the additional condition that

$$\psi = \psi_I + \phi = 0 \qquad \text{at} \quad x = 0, \tag{19.35}$$

and using (19.34) this gives

$$\pi\sin\pi y + B(y) = 0. \tag{19.36}$$

Using this in (19.33), with $A(y) = 0$, then gives the boundary layer solution

$$\phi = -\pi\sin\pi y\,\mathrm{e}^{-x/\epsilon_S}. \tag{19.37}$$

The composite (boundary layer plus interior) solution is the sum of (19.34) and (19.37), namely

$$\boxed{\psi = (1 - x - \mathrm{e}^{-x/\epsilon_S})\pi\sin\pi y.} \tag{19.38}$$

With dimensional variables this is

$$\psi = \frac{\tau_0\pi}{\beta}\left(1 - \frac{x}{a} - \mathrm{e}^{-x/(a\epsilon_S)}\right)\sin\frac{\pi y}{a}. \tag{19.39}$$

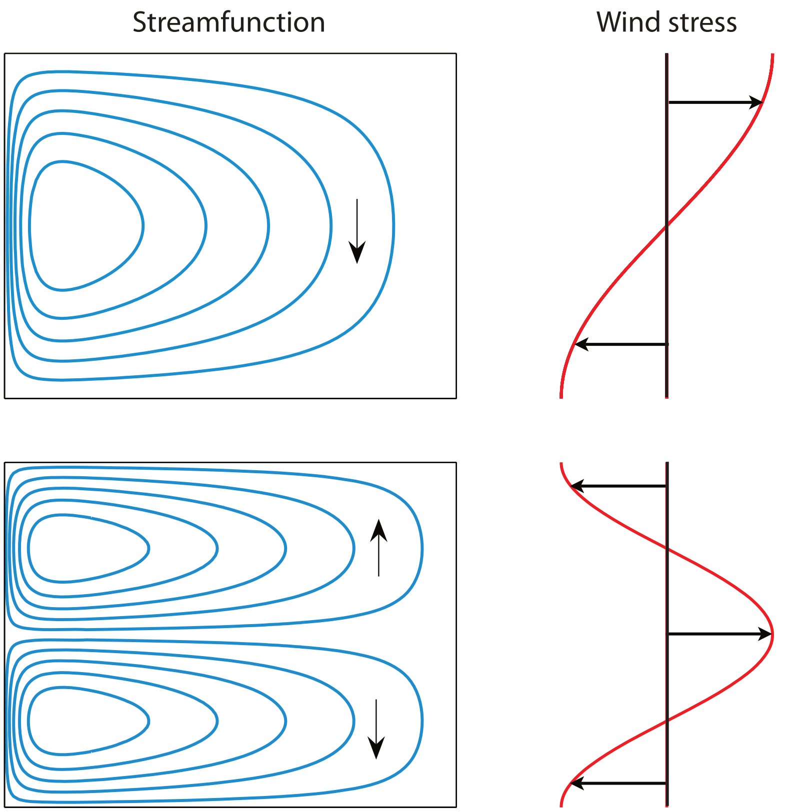

Fig. 19.6 Two solutions of the Stommel model. Upper panel shows the streamfunction of a single-gyre solution, with a wind stress proportional to $-\cos(\pi y/a)$ (in a domain of side a), and the lower panel shows a two-gyre solution, with wind stress proportional to $\cos(2\pi y/a)$. In both cases $\epsilon_S = 0.04$.

This is a 'single gyre' solution. Two or more gyres can be obtained with a different wind forcing, such as $\tau^x = -\tau_0 \cos(2\pi y)$, as in Fig. 19.6.

It is a relatively straightforward matter to generalize to other wind stresses, provided these also vanish at the two latitudes between which the solution is desired. It is left as a problem to show that in general

$$\psi_I = \int_{x_E}^{x} \mathrm{curl}_z \boldsymbol{\tau}(x', y)\,\mathrm{d}x', \tag{19.40}$$

and that the composite solution is

$$\psi = \psi_I - \psi_I(0, y)\mathrm{e}^{-x/(x_E \epsilon_S)}. \tag{19.41}$$

19.2 USING VISCOSITY INSTEAD OF DRAG

A natural variation on the Stommel problem is to use a harmonic viscosity, $\nu\nabla^2\zeta$, in place of the drag term $-r\zeta$ in the vorticity equation, the argument being that the wind-driven circulation does not reach all the way to the ocean bottom so that an Ekman drag is not appropriate. This variation is called the 'Munk problem' or 'Munk model',[8] and if both drag and viscosity are present we have the 'Stommel–Munk' model. The particular form of the lateral friction used in the Munk problem is still somewhat hard to justify because it relies on an ill-founded eddy diffusion of relative vorticity (chapter 13). Our treatment is brief, focusing on aspects that differ from the Stommel problem. The problem is to find and understand the solution to the (dimensional) equation

$$\beta\frac{\partial\psi}{\partial x} = \mathrm{curl}_z \boldsymbol{\tau}_T + \nu\nabla^2\zeta = \mathrm{curl}_z \boldsymbol{\tau}_T + \nu\nabla^4\psi \tag{19.42}$$

The Stommel and Munk models of the Wind-Driven Circulation

Formulation

- Vertically integrated planetary-geostrophic equations, or a homogeneous fluid with nonlinearity neglected.
- Friction parameterized by a linear drag (Stommel model) or a harmonic Newtonian viscosity (Munk model) or both (Stommel–Munk model).
- Flat bottomed ocean.

Variations on the theme include allowing nonlinearity in the vorticity equation, posing the problem in domains of various shapes, and allowing bottom topography (in particular sloping sidewalls). The solutions are most usefully calculated in the boundary-layer approximation, but some exact solutions exist.

Properties

- The transport in the Sverdrup interior is equatorwards for an anti-cyclonic wind-stress curl. This transport is exactly balanced by the poleward transport in the western boundary layer.
- There must be a boundary layer to satisfy mass conservation, and this must be a *western* boundary layer if the friction acts to provide a force of opposite sign to the motion itself. As there is a balance between friction and the β-effect, it is a 'frictional boundary layer'. The western location does not depend on the sign of the wind stress, nor on the sign of the Coriolis parameter, but it does depend on the sign of β, and so on the direction of rotation of the Earth.
- In the Stommel model the boundary layer width arises by noting that the terms $r\nabla^2\psi$ and $\beta\partial\psi/\partial x$ are in approximate balance in the western boundary layer, implying boundary-layer scale of $L_S = (r/\beta)$. If r, the inverse frictional time, is chosen to be $1/20\,\text{days}^{-1}$ then $L_S \approx 60\,\text{km}$, similar to the width of the Gulf Stream. Unless the wind has a special form the Sverdrup flow is non-zero on the zonal walls and there must also be boundary layers there, but they are weaker and less visible.
- In the Munk model the balance in western boundary layer is between $\nu\nabla^4\psi$ and $\beta\partial\psi/\partial x$, implying a scale of $L_M = (\nu/\beta)^{1/3}$. There are also weak boundary layers on the eastern and zonal walls, to satisfy the no-slip (or free slip) condition.

in a given domain, for example a square of side a. We need two boundary conditions at each wall to solve the problem uniquely, and as before for one of them we choose $\psi = 0$ to satisfy the no-normal-flow condition. For the other condition, two possibilities present themselves.

(i) Zero vorticity, or $\zeta = 0$. Since $\psi = 0$ along the boundary, this possibility is equivalent to $\partial^2\psi/\partial n^2 = 0$ where $\partial/\partial n$ denotes a derivative normal to the boundary. This is known as the 'free-slip' condition. At $x = 0$, for example, the condition becomes $\partial v/\partial x = 0$; that is, there is no horizontal shear at the boundary.

(ii) No flow along the boundary (the no-slip condition). That is $\psi_n = 0$ where the subscript denotes the normal derivative of the streamfunction. At $x = 0$ we have $v = 0$.

There is little a priori justification for choosing either of these. The second choice would be demanded if ν were a molecular viscosity, but then we would have to resolve a molecular boundary layer which perhaps would be a few millimetres thick. Instead, ν must be interpreted as some form of eddy viscosity, as we discuss in chapter 13. In that case, one might argue that the free-slip condition should be preferred, but in the absence of a proper theoretical basis for such eddy viscosities there is no truly rational way to make the choice. We will solve the no-slip problem.

Let the wind stress be the canonical $\tau^x = -\cos(\pi y/a)$. Then the interior (Sverdrup) flow is given by (19.34), as for the Stommel problem. This satisfies the no-slip boundary conditions at $y = 0, 1$, namely $\partial_y \psi = 0$, automatically. However, we will need boundary layers at both the western and eastern boundaries, because the interior solution cannot satisfy all four boundary conditions required by (19.42). The eastern boundary layer will be relatively weak, and needed only to satisfy the no-slip condition but, as in the Stommel problem, there will be a strong western boundary layer, needed to satisfy the no-normal flow condition. How thick will this be? Inspection of (19.42) suggests that the frictional term and the β term will balance in a boundary layer of thickness of order L_M where

$$L_M = \left(\frac{\nu}{\beta}\right)^{1/3}. \tag{19.43}$$

Non-dimensionalizing (19.42) in a similar way to the Stommel problem yields

$$-\epsilon_M \nabla^4 \widehat{\psi} + \frac{\partial \widehat{\psi}}{\partial \widehat{x}} = \mathrm{curl}_z \widehat{\boldsymbol{\tau}}_T, \tag{19.44}$$

where $\epsilon_M = (\nu/\beta a^3)$. Considering only the western boundary layer correction, we let the solution be the sum of an interior (Sverdrup) streamfunction plus a boundary layer correction:

$$\widehat{\psi} = \psi_I + \phi_W(\alpha, \widehat{y}) \tag{19.45}$$

where α is a stretched coordinate such that $\widehat{x} = \epsilon\alpha$ where ϵ is some small parameter. Substituting (19.45) into (19.44) and subtracting the Sverdrup balance gives

$$-\epsilon_M \left(\nabla^4 \psi_I + \frac{1}{\epsilon^4}\frac{\partial^4 \phi_W}{\partial \alpha^4}\right) + \frac{1}{\epsilon}\frac{\partial \phi_W}{\partial \alpha} = 0. \tag{19.46}$$

A non-trivial balance is obtained when $\epsilon = \epsilon_M^{1/3}$, implying a dimensional western boundary thickness of $\epsilon a = (\nu/\beta)^{1/3}$, consistent with (19.43). Equation (19.46) then becomes, at leading order,

$$-\frac{\partial^4 \phi_W}{\partial \alpha^4} + \frac{\partial \phi_W}{\partial \alpha} = 0. \tag{19.47}$$

The boundary conditions on (19.47) are that:

(i) $\phi_W \to 0$ as $\alpha \to \infty$: this states that the perturbation decays as it extends into the interior;

(ii) $\phi_W = -\psi_I$ at $\widehat{x} = \alpha = 0$: this is the no-normal-flow condition on the meridional boundary.

(iii) $\partial\phi_W/\partial\widehat{x} = -\partial\psi_I/\partial\widehat{x}$ at $\widehat{x} = \alpha = 0$: this is the no-slip condition.

In addition, zonal boundary layers must exist at $\widehat{y} = 0, 1$ and another meridional boundary layer must exist at $\widehat{x} = 1$, in order to satisfy the no slip condition. Solving the full problem is a straightforward albeit non-trivial algebraic exercise and, omitting the weak zonal boundary layers at $\widehat{y} = 0, 1$ but including the eastern boundary layer correction, we eventually find the solution

$$\begin{aligned}\widehat{\psi} = \psi_I - \mathrm{e}^{-\widehat{x}/(2\epsilon)} &\left\{\psi_I(0, \widehat{y})\left[\cos\left(\frac{\sqrt{3}\widehat{x}}{2\epsilon}\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}\widehat{x}}{2\epsilon}\right)\right] + \frac{2\epsilon}{\sqrt{3}}\sin\left(\frac{\sqrt{3}x}{2\epsilon}\right)\left.\frac{\partial \psi_I}{\partial \widehat{x}}\right|_{x=0}\right\} \\ &- \epsilon \mathrm{e}^{(\widehat{x}-1)/\epsilon}\left.\frac{\partial \psi_I}{\partial \widehat{x}}\right|_{x=1},\end{aligned} \tag{19.48}$$

where $\epsilon = (\nu/\beta a^3)^{1/3}$. With the canonical wind stress, (19.27), the interior solution is $\psi_I = \pi(1 -$

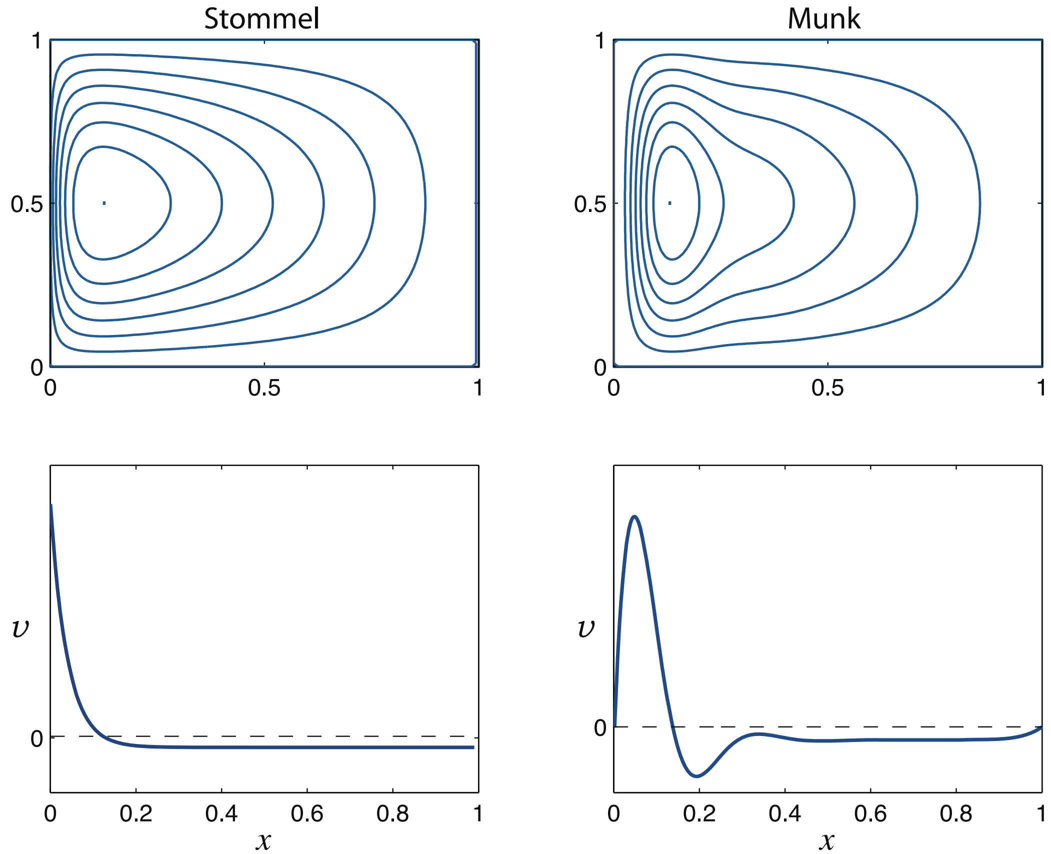

Fig. 19.7 The Stommel and Munk solutions, (19.49) with $\epsilon_S = \epsilon_M^{1/3} = 0.04$, with the wind stress $\tau = -\cos \pi y$, for $x, y \in (0, 1)$. Upper panels are contours of streamfunction in the x-y plane, and the flow is clockwise. The lower panels are plots of meridional velocity, v, as a function of x, in the centre of the domain ($y = 0.5$). The Munk solution can satisfy both no-normal flow and one other boundary condition at each wall, here chosen to be no-slip.

x) $\sin \pi y$ and the above solution becomes

$$\widehat{\psi} = \pi \sin(\pi \widehat{y}) \left\{ 1 - \widehat{x} - e^{-\widehat{x}/(2\epsilon)} \left[\cos\left(\frac{\sqrt{3}\widehat{x}}{2\epsilon} \right) + \frac{1 - 2\epsilon}{\sqrt{3}} \sin\left(\frac{\sqrt{3}\widehat{x}}{2\epsilon} \right) \right] + \epsilon e^{(\widehat{x}-1)/\epsilon} \right\}. \tag{19.49}$$

The solutions of this are plotted in Fig. 19.7. Note how the Munk layers bring the tangential as well as the normal velocity to zero. The eastern boundary layer has a similar thickness to the western boundary layer, but is not as dynamically important since its *raison d'etre* is to enable the no-slip condition to be satisfied, a relatively weak frictional constraint that manifests itself by a boundary layer in which the flow parallel to the boundary is slowed down. On the other hand the western boundary layer exists in order that the no-normal flow condition can be satisfied, which causes a qualitative change in the flow pattern. It should be emphasized that neither the Stommel nor the Munk models are accurate descriptors of the real ocean, but taken together the similarities of their solutions are a powerful argument for the relative insensitivity of the qualitative form of the solution to the detailed form of the friction. The models do in fact produce reasonably realistic patterns of large-scale flow in the major basins of the world, as illustrated in Fig. 19.8.

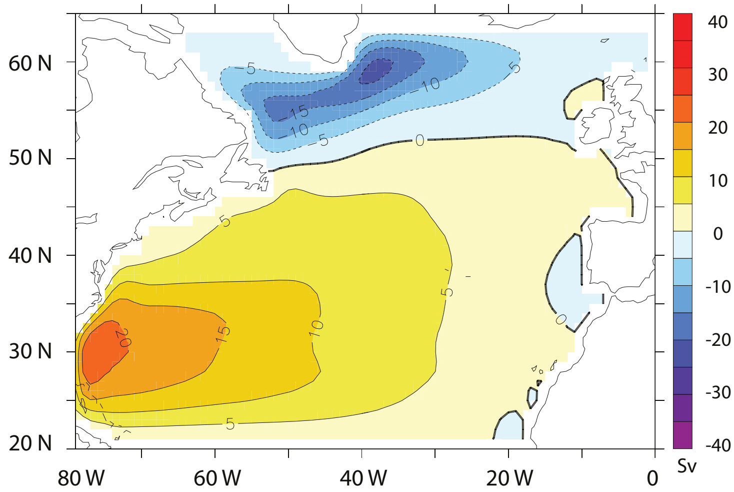

Fig. 19.8 The solution (streamfunction, in Sverdrups) to the Stommel–Munk problem numerically calculated for the North Atlantic, using the observed wind field. The model has realistic geometry, but is flat-bottomed. The calculation reproduces the observed large-scale patterns, but the Gulf Stream and its extension are too diffuse, and its separation from the coast is a little too far north. Compare with figures 19.3 and 19.4.

19.3 ZONAL BOUNDARY LAYERS

The canonical wind stress $[\tau^x = -\tau_0 \cos(\pi y/a)]$ is special because its curl vanishes at $y = 0$ and a, and so the interior solution satisfies $\psi_I = 0$ at $y = 0$ and a. We cannot expect the real wind to be so accommodating. Consider, then, the Stommel problem forced by the linear wind profile,

$$\tau^x = \frac{\tau_0}{a}\left(y - \frac{a}{2}\right). \tag{19.50}$$

Scaling the variables in the usual way leads to the non-dimensional problem

$$\epsilon_S \nabla^2 \widehat{\psi} + \frac{\partial \widehat{\psi}}{\partial \widehat{x}} = -1. \tag{19.51}$$

The interior flow obeys $\partial\widehat{\psi}/\partial\widehat{x} = -1$ everywhere, and evidently does not satisfy $\widehat{\psi} = 0$ at either $\widehat{y} = 0$ or $\widehat{y} = 1$; thus, boundary layers are needed at the meridional and zonal boundaries and we let the solution be the sum of five parts,

$$\widehat{\psi} = \psi_I + \phi_W + \phi_E + \phi_N + \phi_S, \tag{19.52}$$

with self-explanatory notation. The interior solution, ψ_I, is the solution of $\partial\psi_I/\partial\widehat{x} = -1$, so that a solution satisfying $\psi_I = 0$ at $\widehat{x} = 1$ is $\psi_I = (1 - \widehat{x})$ and there is no need for an eastern boundary layer. By the same methods we used in section 19.1.3 the western boundary layer correction is easily found to be

$$\phi_W = -\mathrm{e}^{-\widehat{x}/\epsilon_S}. \tag{19.53}$$

It remains to find ϕ_N and ϕ_S, the boundary layer corrections at the northern and southern boundaries.

Consider the boundary layer correction at $y = 1$, and introduce the stretched coordinate α where $\epsilon'\alpha = y - 1$ and where ϵ' is a small parameter. Thus, we let $\phi_N = \phi_N(x, \alpha)$, and on substituting (19.52) into (19.51) we find

$$\epsilon_S\left(\nabla^2\psi_I + \frac{\partial^2\phi_N}{\partial\widehat{x}^2} + \frac{1}{\epsilon'^2}\frac{\partial^2\phi_N}{\partial\alpha^2}\right) + \frac{\partial\psi_I}{\partial\widehat{x}} + \frac{\partial\phi_N}{\partial\widehat{x}} = -1, \tag{19.54}$$

having neglected the small contributions from the other boundary layer streamfunctions (such as ϕ_S). To obtain a non-trivial balance we choose $\epsilon'^2 = \epsilon_S$ and obtain the dominant balance

$$\frac{\partial^2\phi_N}{\partial\alpha^2} + \frac{\partial\phi_N}{\partial\widehat{x}} = 0. \tag{19.55}$$

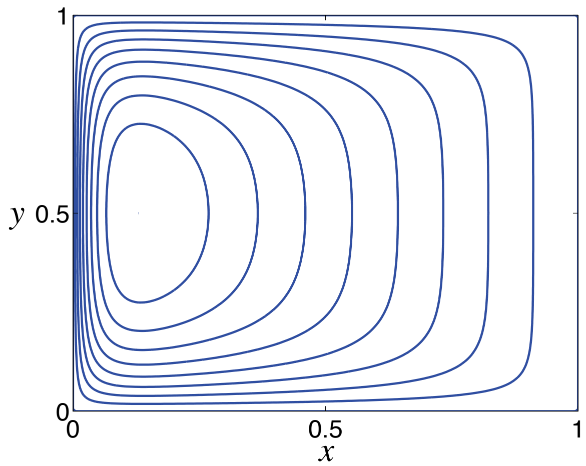

Fig. 19.9 Solutions to the Stommel problem with a wind stress that increases linearly from $y = 0$ to $y = 1$, as in (19.50). The interior solution is $\psi_I = (1 - x)$, or $v_I = -1$, necessitating zonal boundary layers at $y = 0$ and $y = 1$, as well as a western boundary layer at $x = 0$.

The boundary conditions necessary to complete the solution are:

(i) the total streamfunction (interior plus boundary correction) must vanish at the northern boundary; that is, $\phi_N(\hat{x}, \alpha = 0) = -\psi_I(\hat{x}, \hat{y} = 1) = -(1 - \hat{x})$;

(ii) the boundary solution should match to the interior streamfunction far from the boundary; that is $\phi_N(\hat{x}, \alpha \to -\infty) = 0$;

(iii) at the eastern boundary ϕ_N should also vanish, else it would provide a velocity into the eastern wall; that is $\phi_N(1, \alpha) = 0$.

The solution at the southern boundary is obtained in an analogous way, and the complete solution is illustrated in Fig. 19.9 (obtaining the solution analytically is quite algebraically tedious and is easier done numerically). The non-dimensional thickness of the zonal (northern and southern) boundary layers is $\epsilon_S^{1/2}$, which, because it scales like the half power of a small number, is much thicker than the western boundary current. The thickness arises from the dimensional equations that dominate at the boundary, namely

$$r\frac{\partial^2 \psi}{\partial y^2} + \beta\frac{\partial \psi}{\partial x} = 0. \tag{19.56}$$

This follows from (19.18) by noting that the Laplacian operator must be dominated by the derivatives in the y-direction, and the β term has an (interior) component that annihilates the wind-stress curl, plus a boundary layer correction to balance the Laplacian. Inspection of (19.56) yields a dimensional thickness $L_Z \sim \sqrt{ra/\beta} = \epsilon_S^{1/2} a$, where a is the length scale in the x-direction.

19.4 ✦ THE NONLINEAR PROBLEM

In the nonlinear problem we seek solutions to

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) + \beta\frac{\partial \psi}{\partial x} = -r\nabla^2\psi + \mathrm{curl}_z \boldsymbol{\tau}_T + \nu\nabla^2\zeta, \tag{19.57}$$

which we have written in dimensional form. In the Stommel problem we set $\nu = 0$ and in the Munk problem we set $r = 0$. In general solutions will be time-dependent and turbulent and this will create motion on small scales, so that ν cannot be neglected. The 'steady nonlinear Stommel–Munk problem' is

$$J(\psi, \zeta) + \beta\frac{\partial \psi}{\partial x} = -r\nabla^2\psi + \mathrm{curl}_z \boldsymbol{\tau}_T + \nu\nabla^2\zeta. \tag{19.58}$$

We can scale this by first supposing that the leading order balance is Sverdrupian (i.e., $\beta\partial\psi/\partial x \sim \mathrm{curl}_z\boldsymbol{\tau}_T$), from which we obtain the scales $\Psi = |\tau|/\beta$ and $U = |\tau|/(\beta L)$. Equation (19.58) may then be non-dimensionalized to yield

$$R_\beta J(\widehat{\psi},\widehat{\zeta}) + \frac{\partial\widehat{\psi}}{\partial\widehat{x}} = -\epsilon_S\nabla^2\widehat{\psi} + \mathrm{curl}_z\widehat{\boldsymbol{\tau}}_T + \epsilon_M\nabla^2\widehat{\zeta}, \tag{19.59}$$

where $R_\beta = U/\beta L^2 = |\tau|/(\beta^2 L^3)$, the β-Rossby number for this problem, is a measure of the nonlinearity. Evidently, the nonlinear term increases in importance with increasing wind stress and for a smaller domain.

19.4.1 A perturbative approach

A direct attack on the full nonlinear problem (19.58) is possible only through numerical methods, so first we shall explore the problem perturbatively, assuming the nonlinear term to be small; the analysis is straightforward, albeit messy. We begin with the Stommel problem, (19.59) with $\epsilon_M = 0$, and expand the streamfunction in terms of R_β,

$$\widehat{\psi} = \psi_0 + R_\beta\psi_1 + \cdots. \tag{19.60}$$

Now substitute this into (19.59) and equate powers of R_β. The lowest-order problem is simply

$$\epsilon_S\nabla^2\psi_0 + \frac{\partial\psi_0}{\partial\widehat{x}} = \mathrm{curl}_z,\widehat{\boldsymbol{\tau}} \tag{19.61}$$

which is the Stommel problem we have already solved. At the next order,

$$\epsilon_S\nabla^2\psi_1 + \frac{\partial\psi_1}{\partial\widehat{x}} = J(\psi_0,\zeta_0). \tag{19.62}$$

This equation has precisely the same form as the Stommel problem, with the known nonlinear term on the right-hand side playing the part of the wind stress. The algebra to obtain the solution is rather tedious, because the right-hand side varies with both $\widehat{x}$ and $\widehat{y}$, but this is much ameliorated by the use of computer algebraic manipulation languages. For the canonical wind stress $\tau^x = -\tau_0\cos(\pi\widehat{y})$ the corrected solution, in the boundary layer approximation and ignoring any corrections at the zonal boundaries, is found to be[9]

$$\widehat{\psi} \approx \sin(\pi\widehat{y})(1 - \widehat{x} - \mathrm{e}^{-\widehat{x}/\epsilon_S}) - \frac{R_\beta\pi^3}{2\epsilon_S^3}\sin(2\pi\widehat{y})\widehat{x}\mathrm{e}^{-\widehat{x}/\epsilon_S}. \tag{19.63}$$

The solution is illustrated in Fig. 19.10. The perturbation is antisymmetric about $\widehat{y} = 1/2$, being positive for $\widehat{y} > 1/2$ and negative for $\widehat{y} < 1/2$. This tends to move the centre of the gyre polewards, narrowing and intensifying the flow in the poleward half of the western boundary current, whereas the western boundary current equatorwards of $\widehat{y} = 1/2$ is broadened and weakened. The net effect is that the centre of the gyre is pushed polewards — essentially because the western boundary current is advecting the vorticity of the gyre polewards. In the perturbation solution the advection is both by and of the linear Stommel solution; thus, negative vorticity is advected polewards, intensifying the gyre in its poleward half, weakening it in its equatorward half. The solution illustrated in Fig. 19.10 has $\epsilon = 0.04$ and $R_\beta = 10^{-4}$; for larger values of R_β the perturbation itself starts to dominate.

The problem with this perturbative approach is that a boundary layer solution to the Stommel problem does not calculate derivatives accurately, so that the nonlinear term $J(\psi,\nabla^2\psi)$ is poorly approximated in the western boundary layer; however, in the interior where the errors are small the perturbative correction is negligible. A more accurate perturbative approach begins with the *exact* solution to the Stommel problem, and then proceeds in the same way. However, the analytic effort is considerable, and the intuitive sense of the way nonlinearities affect the solution is not apparent.

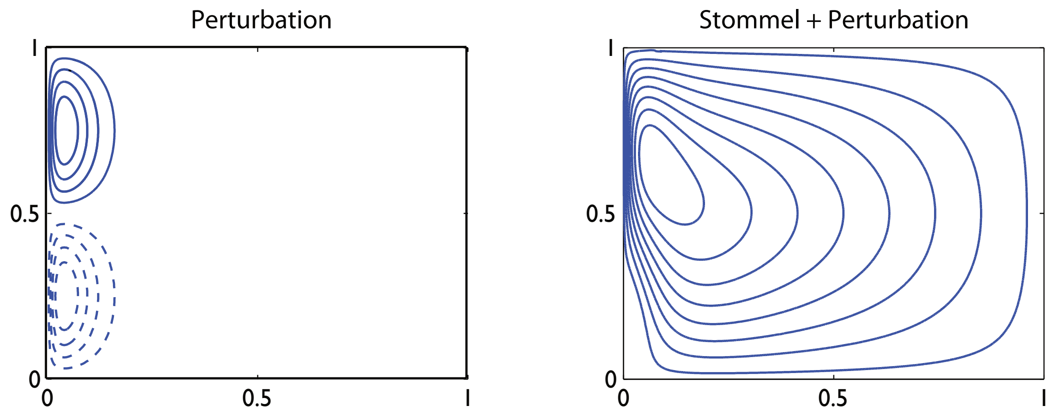

Fig. 19.10 The nonlinear perturbation solution of the Stommel problem, calculated according to (19.63). On the left is the perturbation, $-R_\beta \pi^3/(2\epsilon_S^2)\sin(2\pi y)xe^{-x/\epsilon_S}$, and on the right is the reconstituted solution, using $R_\beta = 10^{-4}$ and $\epsilon = 0.04$. Dashed contours are negative.

19.4.2 A numerical approach

Fully nonlinear solutions show qualitatively similar effects to those seen in the perturbative solutions, as we see in Fig. 19.11, where the solution to (19.59) for the Stommel and Munk problems are obtained numerically by Newton's method.[10]

Just as with the perturbative procedure, for both the Stommel and Munk problems small values of nonlinearity lead to the poleward advection of the gyre's anticyclonic vorticity in the western boundary current, strengthening and intensifying the boundary current in the northwest corner. A higher level of nonlinearity results in a strong recirculating regime in the upper westward quadrant, and ultimately much of the gyre's transport is confined to this regime. The western boundary current itself becomes less noticeable as nonlinearity increases, the more nonlinear solutions have a much greater degree of east-west symmetry than the linear ones, just as the fully nonlinear Fofonoff solutions (section 19.5.3).

The qualitative effects above do not depend on the precise formulation of the model, but the boundary conditions do play an important role in the detailed solution. For example, for a given value of R_β, nonlinearity has a stronger effect in the Munk problem with slip boundary conditions than with no-slip, because in the latter the velocity is reduced to zero at the boundary with a corresponding reduction in the advection term. However, these solutions themselves are unlikely to be relevant for larger values of nonlinearity, because then the flow becomes hydrodynamically unstable

19.5 ✦ INERTIAL SOLUTIONS

In this section we further explore inertial effects and ask: might purely inertial effects be sufficient to satisfy boundary conditions at the western boundary? Can we envision a purely inertial gyre circulation? Our question is motivated by the steady wind-driven, Rayleigh-damped quasi-geostrophic equation, namely

$$J(\psi, \nabla^2\psi) + \beta\frac{\partial\psi}{\partial x} = F_\tau - r\nabla^2\psi, \tag{19.64}$$

where F_τ is the wind forcing. Since the inertial (advective) terms are of a higher order than the linear terms, indeed they are of a higher order than the Rayleigh drag, it is natural to wonder if they

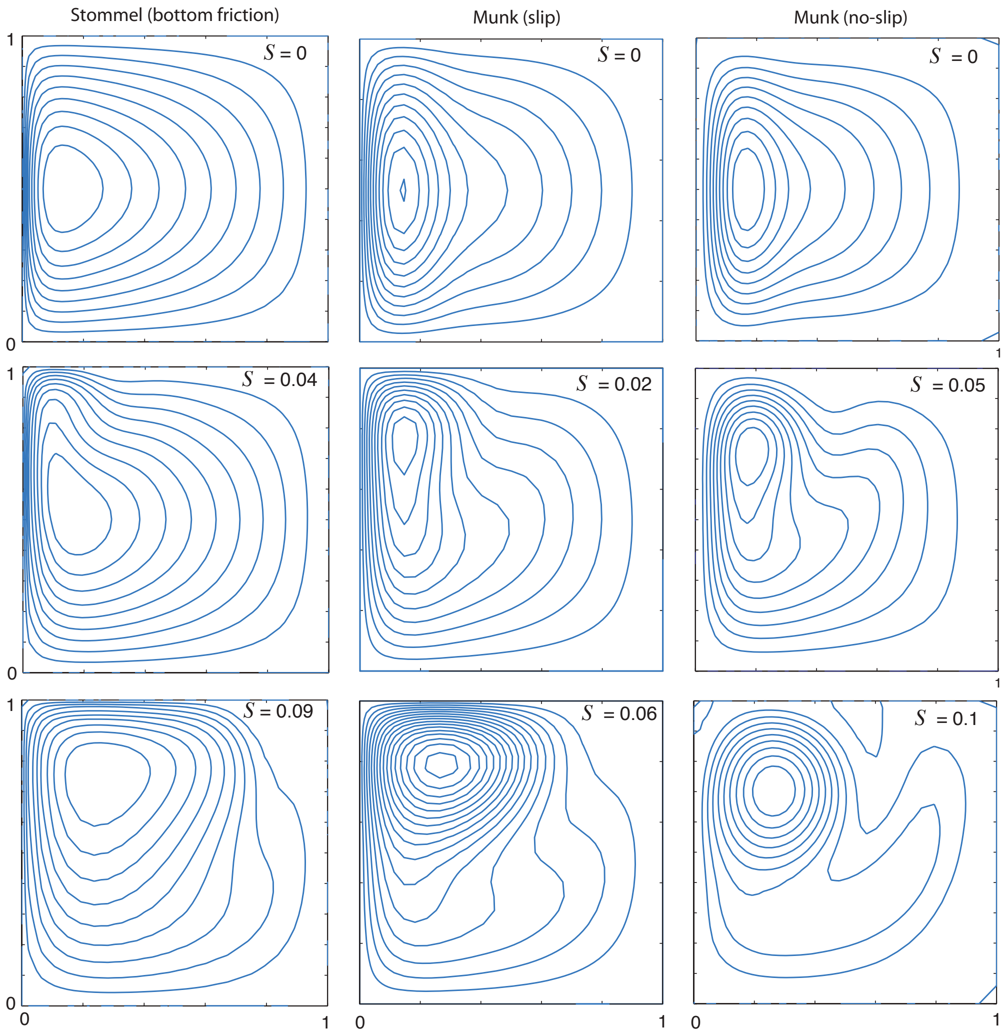

Fig. 19.11 Streamfunctions in solutions of the nonlinear Stommel and Munk problems, obtained numerically with a Newton's method, for various values of the nonlinearity parameter $S = R_\beta^{1/2}$. As in the perturbation solution, for small values of nonlinearity the centre of the gyre moves polewards, strengthening the boundary current in the north-western quadrant (for a northern-hemisphere solution). As nonlinearity increases, the recirculation of the gyre dominates, and the solutions become increasingly inertial.[11]

themselves might serve to satisfy the no-normal flow condition on the western boundary, without recourse to rather ill-defined frictional terms. The answer is no, as we see below, but nevertheless nonlinear effects may be important in the western boundary layer even if they are small in the interior.

19.5.1 The need for friction

Consider the steady barotropic flow satisfying

$$\boldsymbol{u} \cdot \nabla q = \mathrm{curl}_z \boldsymbol{\tau}_T + Fr, \tag{19.65}$$

where $q = \nabla^2\psi + \beta y$ and Fr represents frictional effects. Noting that $\boldsymbol{u}$ is divergence-free and integrating the left-hand side of (19.65) over the area, A, between two closed streamlines (ψ_1 and ψ_2, say) and using the divergence theorem we find

$$\int_A \nabla\cdot(\boldsymbol{u}q)\,\mathrm{d}A = \oint_{\psi_1} \boldsymbol{u}q\cdot\boldsymbol{n}\,\mathrm{d}l - \oint_{\psi_2} \boldsymbol{u}q\cdot\boldsymbol{n}\,\mathrm{d}l = 0. \tag{19.66}$$

Here, $\boldsymbol{n}$ is the unit vector normal to the streamline so that $\boldsymbol{u}\cdot\boldsymbol{n} = 0$. The integral of the wind-stress curl over the same area will not, in general, be zero. Now, we can take these two streamlines as close together as we wish; thus, a balance in (19.65) can only be achieved if *every* closed contour passes through a region where frictional effects are non-zero. This does not mean that nonlinear terms may not locally dominate the friction, just that friction must be somewhere important. In the Stommel and Munk problems, it means that every streamline must pass through the frictional western boundary layer.

Frictional and inertial scales

The ratio of the size of the nonlinear terms to the linear terms is given by the β-Rossby number, $R_\beta = U/(\beta L^2)$. In the western boundary layer the length scales can be expected to be much smaller than the basin scale, and if the balance in the western boundary layer were between the nonlinear and beta term, as in

$$u\frac{\partial}{\partial x}\nabla^2\psi \sim \beta\frac{\partial\psi}{\partial x}, \tag{19.67}$$

then the *inertial boundary layer thickness*, δ_I, is given by

$$\delta_I = \left(\frac{U}{\beta}\right)^{1/2}. \tag{19.68}$$

This, of course, gives $R_\beta = 1$ if $L = \delta_I$. The more energetic the flow, the wider the region where non-linearity is important, and the corresponding scale is sometimes called the Charney thickness.[12]

The linearized Stommel equation has a boundary layer of dimensional thickness of order $\delta_S = (r/\beta)$, obtained by equating $\beta\,\partial\psi/\partial x$ and $r\nabla^2\psi$. This thickness is equal to the inertial boundary layer thickness when $U = (r^2/\beta)$. If $\delta_I > \delta_S$, i.e., if $U > (r^2/\beta)$, then nonlinearity *must* be important in the western boundary layer, because the nonlinear terms are at least as important as the beta term in (19.64). On the other hand, if the Stommel boundary layer is wider than the inertial boundary layer, there is no obvious need for nonlinearity to be important, since the Stommel boundary layer generates no length scales smaller than δ_I and R_β remains small.

19.5.2 Attempting an inertial western boundary solution

Although friction must be important, it is nevertheless instructive to try to find a purely inertial solution for the western boundary layer, and to see if and how the attempt fails. Let us suppose that the interior solution has purely zonal flow, with flow either towards or away from the western boundary current, and consider the dimensional equation of motion

$$J(\psi, \nabla^2\psi + \beta y) = 0. \tag{19.69}$$

This has the general solution

$$\nabla^2\psi + \beta y = G(\psi), \tag{19.70}$$

where G is an arbitrary function of its argument. Consider first the case with flow entering the boundary layer with a local velocity $-U$ (i.e., westwards); that is, $\psi = Uy$. The potential vorticity in this region (just outside our putative boundary layer) is $Q_I = \beta y$. Thus, the relation between potential vorticity and streamfunction is $Q_I = \beta\psi_I/U$ and so

$$G(\psi) = \frac{\beta\psi}{U}. \tag{19.71}$$

Because we are assuming the flow is inviscid, the fluid will preserve this relationship between potential vorticity and streamfunction *even as it moves through the western boundary layer.* Thus, using (19.70) and (19.71), the flow in the interior *and* in the boundary layer are given by solutions of

$$\nabla^2\psi - \frac{\beta\psi}{U} = -\beta y, \tag{19.72}$$

with $\psi = 0$ on the boundary. To obtain a solution, we let $\psi = \psi_I + \phi$, where $\psi_I = Uy$ is the particular solution to (19.72) (and, of course, the interior flow). The boundary layer correction then obeys

$$\nabla^2\phi - \frac{\beta\phi}{U} = 0, \tag{19.73}$$

with $\phi = -\psi_I$ at $x = 0$. Now, in the boundary layer, length scales in the x-direction are much smaller than length scales in the y-direction, and so (19.73) becomes approximately

$$\frac{\partial^2\phi}{\partial x^2} - \frac{\beta\phi}{U} = 0. \tag{19.74}$$

(We could formalize this procedure by non-dimensionalizing and introducing a stretched coordinate, as we did for the Stommel problem.) Solutions of (19.74) are $\phi = -\psi_I \mathrm{e}^{-x/\delta_I}$, where $\delta_I = (U/\beta)^{1/2}$, and so the full solution is

$$\psi = \psi_I(1 - \mathrm{e}^{-x/\delta_I}). \tag{19.75}$$

Clearly, this solution smoothly transitions into the interior solution for large x.

What about solutions exiting the boundary layer? We might attempt a similar procedure, but now the interior boundary condition that we must match is that of eastward flow, namely $\psi_I = -Uy$. Thus, analogously (19.72), to we seek solutions to the problem,

$$\nabla^2\psi + \frac{\beta\psi}{U} = -\beta y \tag{19.76}$$

for which the homogeneous (boundary layer) equation is

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\beta\phi}{U} = 0. \tag{19.77}$$

Solutions of (19.77) are [cf. (19.75)]

$$\psi = \psi_I(1 - \mathrm{e}^{-x/\delta_I^*}), \tag{19.78}$$

where $\delta_I^* = \mathrm{i}\,\delta_I = \mathrm{i}\,(U/\beta)^{1/2}$. The solution is therefore wave-like, and does not transition smoothly to the interior flow. This suggests that friction must be important in allowing the boundary layer to smoothly connect to the interior for reasons connected to the conservation of potential vorticity, as explained heuristically in the next subsection.[13]

In addition to this transition problem, we note that (19.75) and (19.78) together do not constitute a globally acceptable inviscid solution, for the simple reason that the flow in the westwards flowing interior region has a different potential vorticity–streamfunction (q–ψ) relationship than does the eastwards flowing interior. If these two regions are connected by a boundary layer, the flow in that boundary layer must be viscous or unsteady since the value of potential vorticity on the streamlines has changed, whereas inviscid flow conserves potential vorticity.

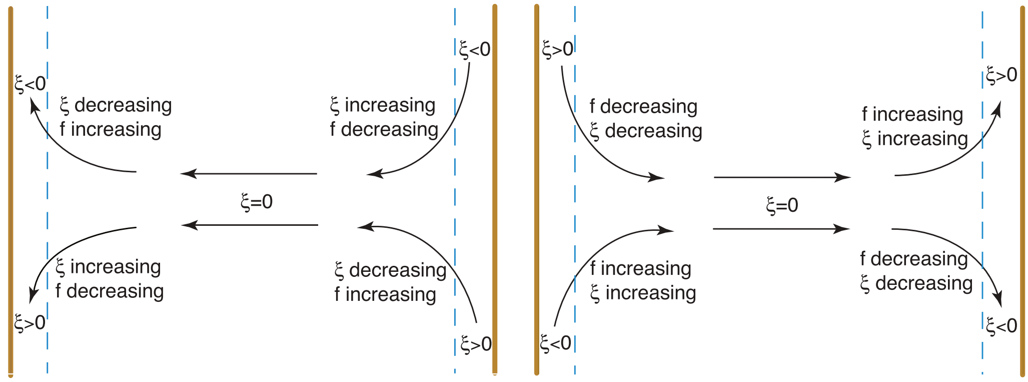

Fig. 19.12 Putative inertial boundary layers connected to a westwards flowing interior flow (left panel) or eastwards flowing interior flow (right panel), in the Northern Hemisphere. Westward flow into the western boundary layer, or flow emerging from an eastern boundary layer, is able to conserve its potential vorticity through a balance between changes in relative vorticity and Coriolis parameter. But flow cannot emerge smoothly from a western boundary layer into an eastwards flowing interior and still conserve its potential vorticity. The right panel thus has inconsistent dynamics.

The connection between the boundary layer and the interior

We have seen that solutions with a boundary layer character that blend smoothly to a flow interior exist only for westward interior flow ($u_I = -U < 0$): an inertial western boundary layer on a beta-plane evidently cannot easily release fluid into the interior. The underlying reason for this stems from the conservation of potential vorticity, as we now explain (and see Fig. 19.12).

Conservation of potential vorticity demands that, in barotropic flow, $v_x - u_y + \beta y$ is a constant on streamlines. In the interior, relative vorticity, ζ, is zero, and in the meridional boundary layers it is effectively v_x. Consider first the case with westward flow in the interior (left panel of Fig. 19.12). Fluid from the irrotational interior approaches the western boundary where it is deflected either polewards or equatorwards. In the former case its relative vorticity falls, so allowing potential vorticity to be conserved because the reduction of ζ can be balanced by an increase in the planetary vorticity, f. Similarly, flow deflected southwards produces positive relative vorticity, which is compensated for by the reduced value of f. In the eastern boundary layer, southwards (northwards) moving flow has negative (positive) relative vorticity. As it emerges into the interior its relative vorticity increases (decreases), this being balanced by a fall (rise) in the value of f. Thus, we see that the solution with a westwards flowing interior can indeed conserve potential vorticity, at both eastern and western boundaries.

On the other hand suppose the interior flow were eastwards (right panel of Fig. 19.12). Flow moving polewards in the western boundary layer has negative relative vorticity. It cannot be freely released into an irrotational interior because f and ζ would then both need to increase, violating potential vorticity conservation. Flow moving southwards with positive relative vorticity similarly is trapped within the western boundary current, unless it meets a zonal boundary which allows a eastwards moving boundary current with positive relative vorticity. Similar arguments show that an eastern boundary current cannot entrain fluid from an eastwards flowing irrotational interior.

One might ask, why cannot we simply reverse the trajectory of all the fluid parcels in an inviscid flow and thereby obtain a solution with an eastward-flowing interior? The answer is that such a flow will only be a solution if we also reverse the direction of the Earth's rotation, and so reverse the sign of β, and hence the location of the frictional boundary current.

19.5.3 A fully inertial approach: the Fofonoff model

Rather than attempt to match an inertial boundary layer with an interior Sverdrup flow, we may look for a purely inertial solution that holds basinwide, and such a construction is known as the *Fofonoff model*.[14] That is, we seek global solutions to the inviscid, unforced problem,

$$J(\psi, \nabla^2\psi + \beta y) = 0. \tag{19.79}$$

We should not regard this problem as representing even a very idealized wind-driven ocean; rather, we may hope to learn about the properties of purely inertial solutions and this might, in turn, tell us something about the ocean circulation.

The general solution to (19.79) is

$$\nabla^2\psi + \beta y = Q(\psi), \tag{19.80}$$

where $Q(\psi)$ is an arbitrary function of its argument. For simplicity we choose the linear form,

$$Q(\psi) = A\psi + B, \tag{19.81}$$

where $A = \beta/U$ and $B = \beta y_0$, where U and y_0 are arbitrary constants. Thus, (19.80) becomes

$$\left(\nabla^2 - \frac{\beta}{U}\right)\psi = \beta(y_0 - y). \tag{19.82}$$

We will further choose $\beta/U > 0$, which we anticipate will provide a westward-flowing interior flow, and which (from our experience in the previous section) is more likely to provide a meaningful solution than an eastward interior, and we will use boundary-layer methods to find a solution. A natural scaling for ψ is UL, where L is the domain size, and with this the non-dimensional problem is

$$(\epsilon_F\nabla^2 - 1)\widehat{\psi} = \widehat{y}_0 - \widehat{y}, \tag{19.83}$$

where $\epsilon_F = U/(\beta L^2)$ and $\widehat{y} = y/L$. If we take ϵ_F to be small (note that $\epsilon_F = R_\beta$) we can find a solution by boundary-layer methods similar to those used in section 19.1.3 for the Stommel problem. Thus, we write $\widehat{\psi} = \widehat{\psi}_I + \widehat{\phi}$ where $\widehat{\psi}_I = \widehat{y} - \widehat{y}_0$ [dimensionally, $\psi_I = U(y - y_0)$] and $\widehat{\phi}$ is the boundary layer correction, to be calculated separately for each boundary using

$$\epsilon_F\nabla^2\widehat{\phi} - \widehat{\phi} = 0 \tag{19.84}$$

and the boundary condition that $\widehat{\phi} + \widehat{\psi}_I = 0$. For example, at the northern boundary, $\widehat{y} = \widehat{y}_N$, the y-derivatives will dominate and (19.84) may be approximated by

$$\epsilon_F\frac{\partial^2\widehat{\phi}}{\partial\widehat{y}^2} - \widehat{\phi} = 0, \tag{19.85}$$

with solution

$$\widehat{\phi} = B\exp[-(\widehat{y}_N - \widehat{y})/\epsilon_F^{1/2}], \tag{19.86}$$

where $B = \widehat{y}_0 - \widehat{y}_N$, hence satisfying the boundary condition that $\widehat{\phi}(\widehat{x}, \widehat{y}_N) = -\widehat{\psi}_I(\widehat{x}, \widehat{y}_N)$. We follow a similar procedure at the other boundaries to obtain the full solution, and in dimensional form this is

$$\psi = U(y - y_0)\left[1 - \mathrm{e}^{-x/\delta_I} - \mathrm{e}^{-(x_E - x)/\delta_I}\right] + U(y_0 - y_N)\mathrm{e}^{-(y_N - y)/\delta_I} + U y_0\mathrm{e}^{-y/\delta_I}. \tag{19.87}$$

where $\delta_I = (U/\beta)^{1/2}$ is the boundary layer thickness. Evidently, only positive values of U corresponding to a westward interior flow give boundary-layer solutions that decay into the interior.

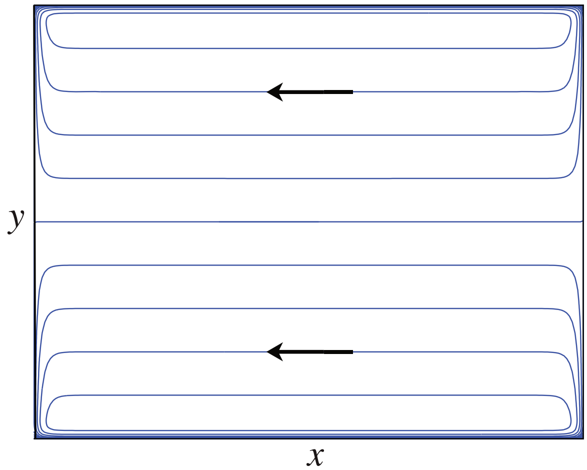

Fig. 19.13 The Fofonoff solution. Plotted are contours (streamlines) of (19.87) in the plane $0 < x < x_E$, $0 < y < y_N$ with $U = 1$, $y_N = 1$, $x_E = y_N = 1$, $y_0 = 0.5$ and $\delta_I = 0.05$. The interior flow is westwards everywhere, and $\psi = 0$ at $y = y_0$. In addition, boundary layers of thickness $\delta_I = \sqrt{U/\beta}$ bring the solution to zero at $x = (0, x_E)$ and $y = (0, y_N)$, excepting small regions at the corners.

A typical solution is illustrated in Fig. 19.13. On approaching the western boundary layer, the interior flow bifurcates at $y = y_0$. The western boundary layer, of width δ_I, accelerates away from this point, being constantly fed by the interior flow. The westward return flow occurs in zonal boundary layers at the northern and southern edges, also of width δ_I. Flow along the eastern boundary layers is constantly being decelerated, because it is feeding the interior. If one of the zonal boundaries corresponds to y_0 (e.g., if $y_N = y_0$) there would be no boundary layer along it, since ψ is already zero at $y = y_0$. Rather, there would be westward flow along it, just as in the interior. Indeed, a slippery wall placed at $y = 0.5$ would have no effect on the solution illustrated in Fig. 19.13.

19.6 ✦ TOPOGRAPHIC EFFECTS ON WESTERN BOUNDARY CURRENTS

The above sections have emphasized the role of friction in satisfying the boundary conditions in the west. However, we should certainly not think of friction as being the *cause* of the western boundary layer and in this section we shall show that if there are sloping sidewalls the role of friction is significantly different, and the western boundary current may even be largely inviscid![15] The key point is that the flow may be inviscid if it is able to follow potential vorticity contours. In a flat-bottomed western boundary layer the flow is moving to larger values of f (and not as a direct response to the wind) and so the flow *must* be frictional. However, if the sidewalls are sloping, then the flow may preserve its potential vorticity (approximately, its value of f/h) if it moves offshore as it moves polewards. In the treatment below, we will focus on homogeneous fluids, noting that the interaction of topography and stratification is a subtle and rather complex problem.

19.6.1 Homogeneous model

The potential vorticity evolution equation for a homogeneous model with topography and a rigid lid [cf. (19.7)] may be written as:

$$\frac{\mathrm{D}q}{\mathrm{D}t} = \frac{F}{h}, \tag{19.88}$$

where $q = (\zeta + f)/h$ where $h = h(x, y)$ is the time-independent depth of the fluid, and F represents vorticity forcing and frictional terms. The advecting velocity is determined by noting that the mass conservation equation is just $\nabla \cdot [\boldsymbol{u}h(x, y)] = 0$, which allows us define the mass-transport streamfunction ψ such that

$$u = -\frac{1}{h}\frac{\partial \psi}{\partial y}, \qquad v = \frac{1}{h}\frac{\partial \psi}{\partial x}. \tag{19.89}$$

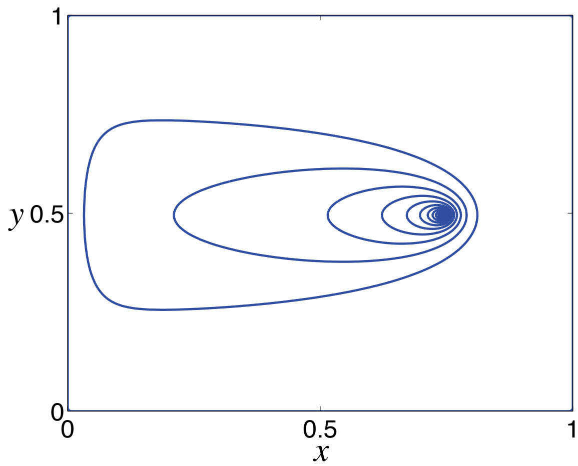

Fig. 19.14 The β-plume, namely the Green function for the Stommel problem. Specifically we plot the solution of (19.93) with $\psi = 0$ at the walls, and a delta-function source at $x = 0.75,\ y = 0.5$.

The streamfunction trails westwards from the source, as if it were a tracer being diffused while being advected westwards along lines of constant f.

The streamfunction itself is obtained from the vorticity by solving the elliptic equation,

$$\nabla \cdot \left(\frac{1}{h} \nabla \psi \right) = \zeta = qh - f. \tag{19.90}$$

Equations (19.88), (19.90) and (19.89) form a closed system. Unlike the quasi-geostrophic case, neither the Rossby number nor the topography need be small for the model to be valid. Including finite size topography but not stratification is not especially realistic *vis-à-vis* the real ocean, but nevertheless this model is physically realizable and a useful tool.

The 'topographic Stommel problem' is obtained by neglecting relative vorticity in the potential vorticity in (19.88) (i.e., let $q = f/h$) and using a linear drag acting on the vertically integrated fields for friction, and a wind-stress curl forcing. Multiplying (19.88) by h, expanding the advective term and omitting time dependence gives

$$J\left(\psi, \frac{f}{h} \right) = -r\nabla^2 \psi + \mathrm{curl}_z(\boldsymbol{\tau}_T/h), \tag{19.91}$$

where the boundary conditions are $\psi = 0$ at the domain edges and $\boldsymbol{\tau}_T$ represents the wind stress at the top.

19.6.2 Advective dynamics

An illuminating way to begin to study the problem is to write (19.91) in the form[16]

$$J(\Psi, \psi) = +r\nabla^2 \psi - \mathrm{curl}_z(\boldsymbol{\tau}_T/h), \tag{19.92}$$

where $\Psi \equiv f/h$. Thus, noting that $J(\Psi, \psi) = \boldsymbol{U} \cdot \nabla \psi$ where $\boldsymbol{U} = \mathbf{k} \times \nabla(f/h)$, we regard ψ as being advected by the pseudovelocity $\boldsymbol{U}$. This advection is along f/h contours and is quasi-westwards, meaning that high values of potential vorticity lie to the right. Equation (19.92) is then an advection-diffusion equation for the tracer ψ, with the 'source' of ψ being the wind stress curl with ψ being diffused by the first term on the right-hand side of (19.92), and advected by $\boldsymbol{U}$. This same interpretation applies to the original Stommel problem, of course, where the pseudovelocity, $-\beta\mathbf{i}$, is purely westwards, and it is useful to first revisit this problem.

Consider, then, a flat-bottomed ocean, where the wind-stress curl is just a point source at $\boldsymbol{x}_0$. With $\Psi = f$, (19.92) becomes

$$r\nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = \delta(\boldsymbol{x} - \boldsymbol{x}_0). \tag{19.93}$$

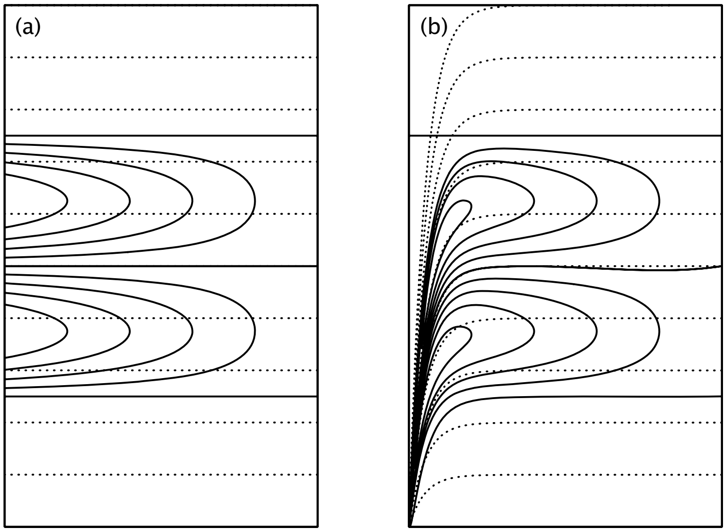

Fig. 19.15 The two-gyre Sverdrup flow (solid contours) for (a) a flat-bottomed domain and (b) a domain with sloping sidewalls. The f/h contours are dotted.[17]

This can be transformed to a Helmholtz equation by writing $\phi = \psi \exp(\beta x/2r)$, giving $\nabla^2\phi - [\beta/(2r)]^2\phi = \delta(\boldsymbol{x} - \boldsymbol{x}_0)$. This may then be solved exactly, and the solution (for ψ) is illustrated in Fig. 19.14 — this is the Green function for the Stommel problem. The tracer ψ is 'advected' westwards along f/h contours — lines of latitude in this case — spreading diffusively as it goes; the resulting structure is called a *β-plume*. The western boundary layer results as a consequence of the f/h contours colliding with the western boundary, along with the need to satisfy the boundary condition $\psi = 0$. If there were no diffusion at all, ψ would just propagate westwards from the source, and if in addition the source were spatially distributed, for example as $\sin \pi y$, the solution streamfunction would represent Sverdrup interior flow. This case is illustrated in Fig. 19.15(a), which shows the solution to (19.93) with $r = 0$ and with the right-hand side replaced by a conventional 'two-gyre' wind stress.

Now consider the case with a sloping sidewall on the western boundary. The f/h contours (the dotted lines in Fig. 19.15b) tend to converge at the southwest corner of the domain, and only where f/h contours intersect the boundary is a diffusive boundary layer required. In terms of the interpretation above, wind stress provides a source for the streamfunction ψ and the latter is advected *pseudowestwards* — i.e., along potential vorticity contours, with higher values to the right. The source in this case is distributed over the entire domain, but the contours all converge in the southwest corner. The (numerically obtained) solution to the associated Stommel problem is illustrated in Fig. 19.16, and the western boundary current in this case is no longer a frictional boundary layer. Friction is necessarily important where the flow crosses f/h contours; linear theory suggests that this will occur at the southwest corner. It also occurs on the western boundary where the f/h contours are densely packed and the vorticity in the topographic Sverdrup flow is large, and the friction enables the flow to move across the f/h contours. (In the flat bottomed case the f/h contours are zonal and friction allows the flow to move meridionally.)

19.6.3 Bottom pressure stress and form drag

In the homogeneous problem f and h appear only in the combination f/h, and we may solve the problem entirely without considering pressure effects. It is nevertheless informative to think about how the pressure interacts with the topography to produce meridional flow, and as way toward addressing the effects of stratification. The geostrophic momentum equation is

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi + \boldsymbol{F}, \tag{19.94}$$

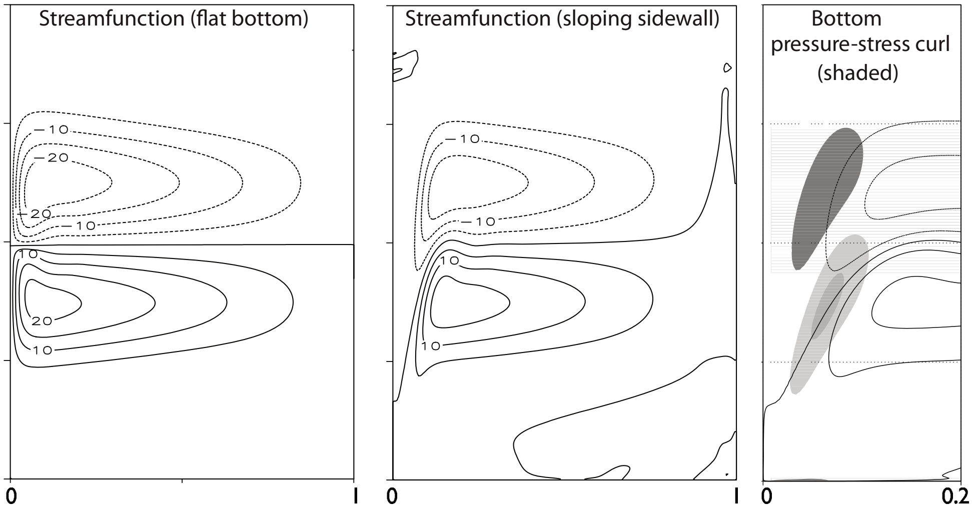

Fig. 19.16 The numerically obtained steady solution to the homogeneous problem with a two-gyre forcing and friction, for a flat-bottomed domain and a domain with sloping western sidewall. The shaded regions in the right panel show the regions where bottom pressure-stress curl is important in the meridional flow of the western boundary currents.[18]

where $\boldsymbol{F}$ represents both wind forcing and frictional terms. Integrating this over the depth of the ocean (with $z = 0$ at the top), and using the Leibnitz rule ($\nabla \int_{\eta_B}^0 \phi \,\mathrm{d}z = \int_{\eta_B}^0 \nabla\phi \,\mathrm{d}z - \phi_B \nabla \eta_B$) to evaluate the pressure term, gives

$$\boldsymbol{f} \times \overline{\boldsymbol{u}} = -\nabla\overline{\phi} - \phi_B \nabla \eta_B + \overline{\boldsymbol{F}}. \tag{19.95}$$

Here, the overbar denotes a vertical integral (e.g., $\overline{\boldsymbol{u}} = \int_{\eta_B}^0 \boldsymbol{u} \,\mathrm{d}z$), ϕ_B is the pressure at $z = \eta_B$ and η_B is the z-coordinate of the bottom topography. We take the top of the ocean at $z = 0$, and note that $\nabla\eta_B = -\nabla h$ where h is the fluid thickness. The second term on the right-hand side of (19.95) is the stress on the fluid due to a correlation between the pressure gradient and the topography at the ocean bottom — it is the bottom *form drag*, first encountered in section 3.6.

Taking the curl of (19.95) gives

$$\beta\overline{v} = -\mathrm{curl}_z(\phi_B \nabla \eta_B) + \mathrm{curl}_z \overline{\boldsymbol{F}} = -J(\phi_B, \eta_B) + \mathrm{curl}_z \overline{\boldsymbol{F}}. \tag{19.96}$$

This equation holds for both a stratified and homogeneous fluid. The first term on the right-hand side is the bottom pressure-stress curl, or the form-drag curl. (It is also sometimes informally referred to as the bottom pressure torque.)

Equation (19.96) is similar to (19.91), in that both arise from (19.95). To derive an equation with the same form as (19.91) but valid for a stratified fluid, we write the vertical integral of the pressure as

$$\int_{-h}^0 \phi \,\mathrm{d}z = \int_{-h}^0 \left[\mathrm{d}(\phi z) - z(\partial\phi/\partial z)\,\mathrm{d}z\right] = \int_{-h}^0 \left[\mathrm{d}(\phi z) - zb \,\mathrm{d}z\right] = h\phi_B + \Gamma, \tag{19.97}$$

using hydrostatic balance, and where $\Gamma \equiv -\int_{-h}^0 zb \,\mathrm{d}z$. Using (19.97) in (19.95) we obtain:

$$\boldsymbol{f} \times \overline{\boldsymbol{u}} = -h\nabla\phi_B + \nabla\Gamma + \overline{\boldsymbol{F}}. \tag{19.98}$$

The curl of this equation just gives back (19.96), but if we divide by h before taking the curl we obtain

$$J(\psi, f/h) + J(h^{-1}, \Gamma) = \mathrm{curl}_z(\overline{\boldsymbol{F}}/h). \tag{19.99}$$

The second term in (19.99) is known as the JEBAR TERM — joint effect of baroclinicity and relief — and it couples the depth integrated flow with the baroclinic flow.[19] Thus, if the bottom is not flat, the Stommel–Munk models are not solutions for the vertically integrated flow. If the stratification vanishes (i.e., b is constant) then Γ is function of h alone and $J(h^{-1}, \Gamma) = 0$, and (19.99) reprises (19.91), given an appropriate choice of $\overline{\boldsymbol{F}}$.

Bottom pressure stress in a homogeneous gyre

For the remainder of this section we restrict attention to a homogeneous (i.e., unstratified) gyre. If there is no forcing or friction, then from (19.96) and (19.98) we see that the flow simultaneously satisfies

$$\beta v = -\nabla\phi_B \times \nabla\eta_B \equiv -J(\phi_b, \eta_B), \qquad \text{and} \qquad J(\psi, f/h) = 0. \tag{19.100a,b}$$

The right-hand side of (19.100a), the form-drag curl, is non-zero whenever the pressure gradient has a component parallel to the topographic contour (i.e., when the isobars are not aligned with the topographic contours). From (19.100b) we may conclude that if there *is* meridional flow in an unforced, inviscid fluid it must be along f/h contours, and this meridional flow may be thought of as being driven by the curl of the form drag. If the domain is flat-bottomed then the form drag is zero, and in that case all meridional flow is forced or viscous.

Real flows are both forced and viscous. Bottom pressure stresses may — and likely do — locally dominate viscous stresses. However, the bottom pressure-stress curl cannot balance the wind-stress curl when integrated over the whole domain, or indeed when integrated over an area bounded by a line of constant ϕ_B or constant η_B, because its integral over such an area vanishes and (19.91) cannot be balanced if $r = 0$. In the numerical simulations shown in Fig. 19.16, it is the bottom pressure-stress curl term that largely balances the poleward flow term (βv) in the vorticity equation in parts of western boundary current, with friction acting in the opposite sense. That is, over some regions where the flow is crossing f/h contours we have the balance

$$[\beta v] \approx [\text{bottom pressure-stress curl}] - [\text{friction}], \tag{19.101}$$

where the terms in square brackets are positive, and friction is small. In contrast, in the flat-bottomed case in the western boundary layer we have the classical balance $[\beta v] \approx +[\text{friction}]$, with both terms positive.

Now consider the balance of momentum, integrated zonally across the domain. We write the vorticity equation (19.96) in the form

$$\nabla \cdot (f\overline{\boldsymbol{u}}) = -\mathrm{curl}_z(\phi_B \nabla\eta_B) + \mathrm{curl}_z\boldsymbol{\tau}_T - \mathrm{curl}_z\boldsymbol{\tau}_B, \tag{19.102}$$

where $\boldsymbol{u} = (u, v)$, $\boldsymbol{\tau}_T$ is the wind stress at the top and $\boldsymbol{\tau}_B$ the frictional stress at the bottom. Integrate (19.102) over the area of a zonal strip bounded by two nearby lines of latitude, y_1 and y_2, and the coastlines at either end. The term on the left-hand side vanishes by mass conservation and using Stokes' theorem we obtain:

$$\int_{y_1} \phi_B \frac{\partial\eta_B}{\partial x}\,\mathrm{d}x - \int_{y_2} \phi_B \frac{\partial\eta_B}{\partial x}\,\mathrm{d}x = \int_{y_1} (\tau_T^x - \tau_F^x)\,\mathrm{d}x - \int_{y_2} (\tau_T^x - \tau_F^x)\,\mathrm{d}x. \tag{19.103}$$

If the topography is non-zero, there is nothing in this equation to prevent the wind stress being balanced by the form stress terms, with the friction being a negligible contribution. If, for example, friction were to be confined to the southwest corner, then bottom pressure stress is the proximate driver of fluid polewards in the western boundary current. This may hold only if the scale of the sloping sidewall is greater than the thickness of the Stommel layer; if the converse holds then the sidewalls appear to be essentially vertical to the flow. If the sidewalls are truly vertical, then the form stress is confined to delta-functions at the walls. Friction must then be important even in the zonal balance, because if we restrict the integral in (19.103) to a strip that does not quite reach the sidewalls, the left-hand side vanishes identically and the wind stress can only be balanced by friction.

To conclude this discussion, we note that the effects of topography are likely greater in homogeneous fluids than in stratified fluids, because the stratification will partially shield the wind-driven upper ocean from feeling the topography, but we leave the exploration of that topic for another day.

Notes

1 Henry Stommel (1920–1992) was one of the most creative and influential physical oceanographers of the twentieth century. Spending most of his career at Woods Hole Institute of Oceanography, his enduring contributions include the first essentially correct theory of western intensification (and so of the Gulf Stream), some of the first models of abyssal flow and the thermohaline circulation (chapter 21), and his foundational work on the thermocline. His *forté* was in constructing elegantly simple models of complex phenomena — often models that were physically realizable in the laboratory — while at the same time testing and encouraging others to test the models against observations. This chapter might have been entitled 'Variations on a theme of Stommel'.

2 Both Fig. 19.3 and Fig. 19.4 are 'state estimates' — outputs of a model constrained by or combined with observations in such a way as to produce an approximation of the ocean state, hopefully more accurate than either models or data can separately produce. An atmospheric 'reanalysis' is also a state estimate, but for historical reasons meteorologists use a non-standard terminology. Because of the dearth of data in the ocean compared to the atmosphere, oceanic state estimates are less accurate than their atmospheric counterparts.

3 Courtesy of R. Zhang. See also Zhang & Vallis (2007).

4 I'm grateful to P. Heimbach for this figure, obtained using the ECCO state estimation system.

5 The asymptotic solution to this boundary value problem was obtained by Wasow (1944), a few years prior to Stommel's work, and further investigated by Levinson (1950). However, it seems unlikely these two investigators were motivated by the oceanographic problem.

6 Harald Sverdrup (1888–1957) was a Norwegian meteorologist/oceanographer who is most famous for the balance that now bears his name, but he also played a leadership role in scientific policy and was the director of Scripps Institution of Oceanography from 1936–1948. The Sverdrup unit is also named for him. Originally defined as a measure of volume transport, with $1\,\mathrm{Sv} \equiv 10^6\,\mathrm{m^3\,s^{-1}}$, it is more generally thought of as a mass transport with $1\,\mathrm{Sv} \equiv 10^9\,\mathrm{kg\,s^{-1}}$, in which case it can also be used as a measure of transport in the atmosphere. The Hadley Cell, for example, has an average transport of about 100 Sv (Figs. 14.3 and 15.21).

7 Leetmaa *et al.* (1977) and Wunsch & Roemmich (1985) offer complementary views on the matter.

8 After Munk (1950). Many thanks to Manuel Lopez Mariscal of CICESE for some useful comments and corrections.

9 Hendershott (1987), Veronis (1966a).

10 Veronis (1966a,b) was one of the first to investigate nonlinear effects in wind-driven gyres. See also Fox-Kemper & Pedlosky (2004) and references therein.

11 Solutions kindly provided by B. Fox-Kemper.

12 After Charney (1955).

13 See also Greenspan (1962). Il'in & Kamenkovich (1964) and Ierley & Ruehr (1986) show numerically that the friction must be sufficiently strong for steady boundary-layer solutions to exist.

14 After Fofonoff (1954).

15 Hughes & de Cuevas (2001).

16 Welander (1968).

17 Figure kindly provided by Laura Jackson.

18 Adapted from Jackson *et al.* (2006).

19 Sarkisyan & Ivanov (1971).

Further reading

Mainly theory and modelling

Abarbanel H. D. I. & Young, W. R., Eds., 1987. *General Circulation of the Ocean.*
Contains several useful review articles on the oceanic general circulation as it was then understood.

In increasing order of size, the following books cover a variety of topics in ocean circulation, all at about the graduate student level.

Samelson, R. M., 2011. *The Theory of Large-Scale Ocean Circulation.*

Dijkstra, H. A. 2008. *Dynamical Oceanography.*

Pedlosky, J., 1996. *Ocean Circulation Theory.*

Huang, R. X., 2010. *Oceean Circulation.*

Olbers, D., Willebrand, J. & Eden, C., 2012. *Ocean Dynamics.*
These last two books are both hefty treatises on the topic.

Treatments of El Niño are to be found in

Clarke, A. J., 2008. *An Introduction to the Dynamics of El Niño and the Southern Oscillation.*

Sarachik, E. and Cane, M., 2010. *The El Niño–Southern Oscillation Phenomenon.*

Description and Observation

Talley, L. D., Pickard, G.L., W. J. Emery, W. J. & Swift, J. H. 2011. *Descriptive Physical Oceanography: An Introduction.*
This text gives a sense of the big picture, as well as being full of useful maps of ocean properties.

Wunsch, C. 2015. *Modern Observational Physical Oceanography.*
Shows how modern observing systems (floats, satellites, etc.) can be used alongside numerical models to provide a more complete view of the ocean.

There is a tide in the affairs of men,
which, taken at the flood, leads on to fortune.
Omitted, all the voyage of their life is bound in shallows and in miseries.
On such a full sea are we now afloat,
and we must take the current when it serves, or lose our ventures.
William Shakespeare, *Julius Caesar*, c. 1599.

CHAPTER 20

Structure of the Upper Ocean

IN THE PREVIOUS CHAPTER we developed an understanding of the vertically integrated flow of the worlds oceans. If we are to proceed further we must develop an understanding of the *vertical structure* of the oceans, and that is the subject of this chapter. Our main focus will be on the upper ocean and we will proceed as follows.

1. We first explore the vertical structure of the wind-driven circulation, essentially as a continuation of the investigation of the previous chapter. We use the quasi-geostrophic equations to understand why the subsurface ocean moves at all, and we introduce the notion of potential vorticity homogenization.

2. A limitation of the quasi-geostrophic approach is that these equations take the stratification, $N(z)$, as a given and therefore cannot provide a answer to the question as to what produces the density structure itself. Thus, beginning in section 20.3, we relax the quasi-geostrophic restriction and, using the *planetary geostrophic* equations, we try to understand the dynamics that give rise to the vertical structure of density itself. We focus on *main thermocline,* the region of the upper ocean in which temperature and density vary most rapidly, in all seasons, and we discuss the structure of both the internal thermocline and the ventilated thermocline, the meaning of which will become apparent later.

As with many fluid problems the dynamics becomes intertwined with the thermodynamics, and the mean flow becomes intertwined with the smaller, turbulent, baroclinic eddies, in rather subtle ways that, to this day, are not fully understood and that large numerical models are only beginning to properly simulate. We begin by looking at the vertical structure of the wind-driven gyres, and if and how the influence of the wind can be communicated to the subsurface ocean.

20.1 VERTICAL STRUCTURE OF THE WIND-DRIVEN CIRCULATION

20.1.1 A two-layer quasi-geostrophic model

We pose the problem using the quasi-geostrophic equations, taking the background stratification of the ocean as a given.[1] The simplest system that has vertical structure is a two-layer model and that is where we start. We don't yet wish to consider the effects of mesoscale eddies, so we'll limit ourselves to motion larger than the deformation scale, although not so large that the quasi-geostrophic system itself does not hold.

Scales of motion

On scales that are sufficiently larger than the deformation radius we can ignore the relative vorticity compared to planetary vortex stretching and the β-effect. Since quasi- geostrophic scaling itself applies only to scales that are not significantly larger than the deformation scale, our analysis will be formally valid under the following set of inequalities.

$$
\begin{aligned}
\beta L &\ll f_0 && \text{(small variations in Coriolis parameter)}\\
\beta L &> U/L && \text{(to ignore relative vorticity compared to planetary vorticity)}\\
L^2 &> L_d^2 && \text{(to ignore relative vorticity compared to vortex stretching)}\\
Ro\ L^2 &\ll L_d^2 && \text{(to keep the variations in stratification small),}
\end{aligned}
$$

where L_d is the deformation radius and L the scale of the motion. The first and last of the above inequalities are standard quasi-geostrophic requirements, with the '$\gg$' symbol denoting the asymptotic ordering. The middle two inequalities are taken within the quasi-geostrophic dynamics, and are needed in order to ignore relative vorticity and give a balance between the β-effect and vortex stretching. The simultaneous satisfaction of all these conditions may seem restrictive, but the plangent dynamics contained within the quasi-geostrophic equations and the generality of the method employed below will suggest that the principle results obtained may transcend the limitations of the equations used. In the mid-latitude ocean $L_d \approx 10^5$ m and the above inequalities are in fact reasonably well satisfied for $L \approx 10^6$ m and $U \approx 0.1\ \mathrm{m\,s^{-1}}$ with $\beta = 10^{-11}\ \mathrm{m^{-1}\,s^{-1}}$ and $f_0 = 10^{-4}\ \mathrm{s^{-1}}$.

Constructing the model

We now make the following simplifications for our model ocean.

(i) We use the two-layer quasi-geostrophic equations, with layers of equal thickness.

(ii) We seek only statistically-steady solutions.

(iii) We include a frictional term coming from a downgradient flux of potential vorticity. Given the neglect of relative vorticity, this is equivalent to an interfacial drag.

(iv) We will neglect the western boundary layer.

Because of the equal-layer-thickness assumption, which makes the algebra simpler, it is best considered as a model for the upper ocean above a level where the vertical velocity is approximately zero. The equations of motion are then

$$J(\psi_1, q_1) = \frac{1}{H_0}\mathrm{curl}_z \boldsymbol{\tau}_T - \nabla\cdot\boldsymbol{T}_1, \qquad J(\psi_2, q_2) = -\nabla\cdot\boldsymbol{T}_2 \tag{20.1a,b}$$

where

$$q_1 = \beta y + F(\psi_2 - \psi_1), \qquad q_2 = \beta y + F(\psi_1 - \psi_2). \tag{20.2a,b}$$

Here, $F = f_0^2/(g' H_0) = 1/L_d^2$ is a measure of the stratification, where H_0 is the thickness of either layer, and the $\nabla\cdot\boldsymbol{T}$ terms are represent interfacial eddy stresses, which, if needed, we will parameterize by a downgradient flux of potential vorticity,

$$\boldsymbol{T}_1 = -\kappa\nabla q_1 = -\kappa(F\nabla(\psi_2 - \psi_1) + \beta\mathbf{j}), \qquad \boldsymbol{T}_2 = -\kappa\nabla q_2 = -\kappa(F\nabla(\psi_1 - \psi_2) + \beta\mathbf{j}), \tag{20.3}$$

where κ is a constant. We will mostly be interested in the limit of small κ, or more specifically $UL/\kappa \gg 1$, which is a large Péclet number condition. (The Péclet number is similar to a Reynolds number, but with the diffusivity replacing the kinematic viscosity.) So first consider the case when κ is identically zero. An *exact* solution to (20.1) has $\psi_2 = 0$, so that (20.1a) becomes $\beta\partial\psi_1/\partial x = H_0^{-1}\mathrm{curl}_z\boldsymbol{\tau}_T$, with solution

$$\psi_1 = -\frac{1}{H_0\beta}\int_x^{x_E} \mathrm{curl}_z\boldsymbol{\tau}_T\,\mathrm{d}x. \tag{20.4}$$

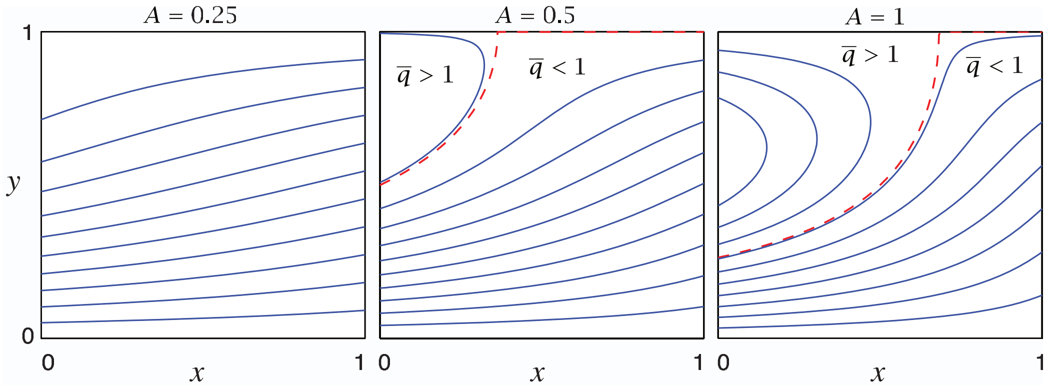

Fig. 20.1 Contours of $\overline{q} = \beta y + A \sin \pi y(1 - x)$, with $\beta = 1$, for three values of A. The red dashed line is $\overline{q} = 1$, which separates the blocked region to the east ($\overline{q} < 1$) from the closed region to the west ($\overline{q} > 1$). See Fig. 20.2 for plots of the other fields.

That is, *there is no flow in the lower layer*, and the upper layer solution is given by Sverdrup balance. The solution satisfies $\psi_1 = 0$ at $x = x_E$ and, because $\psi_2 = 0$, the nonlinear term on the left-hand side of (20.1a) vanishes identically. This is both counter-intuitive and counter-observations, for we know the subsurface ocean is not quiescent. Is there another solution?

A general solution

We now construct the solution without assuming $\psi_2 = 0$. Although the equations are nonlinear, we can obtain a linear equation for a streamfunction by adding (20.1a) and (20.1b), giving

$$J(\psi_1, \beta y + F(\psi_2 - \psi_1)) + J(\psi_2, \beta y + F(\psi_1 - \psi_2)) = \frac{1}{H_0}\mathrm{curl}_z \boldsymbol{\tau}_T. \tag{20.5}$$

The nonlinear terms cancel leaving

$$J(\overline{\psi}, \beta y) = \frac{1}{H_0}\mathrm{curl}_z \boldsymbol{\tau}_T, \qquad \text{where} \quad \overline{\psi} = \psi_1 + \psi_2, \tag{20.6a,b}$$

with solution [cf. (20.4)]

$$\overline{\psi} = -\frac{1}{H_0 \beta}\int_x^{x_E} \mathrm{curl}_z \boldsymbol{\tau}_T \, \mathrm{d}x'. \tag{20.7}$$

This simply says that the vertically integrated flow obeys Sverdrup balance. For the canonical wind stress

$$\boldsymbol{\tau}_T = -\tau_0 \cos \pi y \, \mathbf{i} \tag{20.8}$$

and we obtain $\overline{\psi} = (\pi\tau_0/\beta H_0)(x_E - x)\sin \pi y$. It is useful to define

$$\overline{q} \equiv (\beta y + F\overline{\psi}), \tag{20.9}$$

and then $\overline{q} = \beta[y + A(1 - x)\sin \pi y]$, where $A = \pi\tau_0/(\beta H_0)$ parameterizes the wind strength, and this is plotted in Fig. 20.1. For $\overline{q} < 1$ (below and to the right of the dashed line) all the geostrophic contours intersect the eastern boundary and the flow is 'blocked'. For $\overline{q} > 1$ the flow is 'closed'.

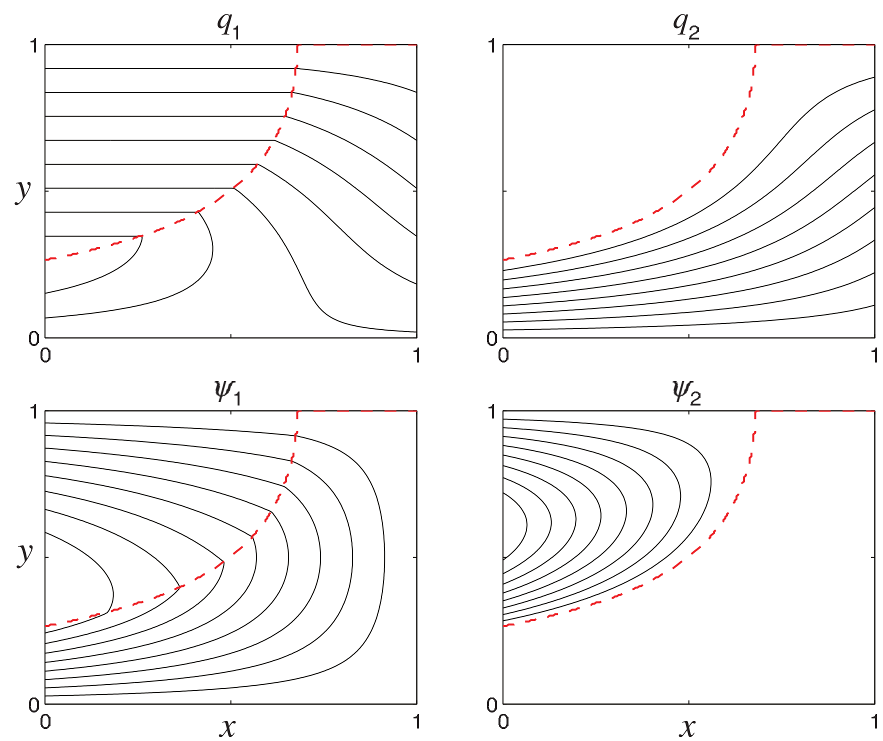

Fig. 20.2 Upper- and lower-level potential vorticity and streamfunction for the canonical wind stress (20.8). The field of $\overline{q}$ is that of Fig. 20.1 with $A = 1$. The dashed line divides the blocked region from the closed region. The lower layer streamfunction ψ_2 is non-zero only in the closed region, and here $q_2 = \beta L$ and $q_1 = 2\beta y - \beta L$. In the blocked region the upper layer carries all of the Sverdrup transport. Both the streamfunction and potential vorticity are continuous at the divide: $\psi_2 = 0$ and $q_2 = \overline{q} = \beta L$.

Lower layer

Although the full equations are nonlinear, using (20.9) we can obtain a linear equation for the lower layer. Because the Jacobian of a field with itself vanishes, (20.1b) and (20.2b) imply that

$$J(\psi_2, \overline{q}) = -\nabla \cdot \boldsymbol{T}_2, \tag{20.10}$$

and this is useful because $\overline{q}$ is a function of the wind, using (20.9). If $\nabla \cdot \boldsymbol{T}_2 = 0$ then

$$J(\psi_2, \overline{q}) = 0. \tag{20.11}$$

As well as the possibility that $\psi_2 = 0$ we now have the more general solution

$$\psi_2 = G(\overline{q}), \tag{20.12}$$

where G is an *arbitrary* function of its argument. Isolines of ψ_2 and $\overline{q}$ are then coincident. (Contours that are isolines of both streamfunction and potential vorticity are known as geostrophic contours.)

Consider a blocked isoline of $\overline{q}$; that is, one that intersects the eastern boundary (see Fig. 20.1). The ψ_2 contour coincident with this has a value of zero at the eastern boundary (by the no-normal flow condition). Thus $\psi_2 = 0$ *everywhere* in the blocked region, and $q_2 = \overline{q}$. In this region the Sverdrup transport is carried everywhere by the upper layer, and the lower layer is at rest. This

Summary of Wind-Driven, Two-layer Solution

The vertically integrated flow in a wind-driven two-layer quasi-geostrophic model is determined by Sverdrup balance. The effects of eddies may be crudely parameterized by a downgradient diffusion of potential vorticity. If this is identically zero, then the lower-layer flow is identically zero and the upper-layer flow carries all the transport. If the diffusion is small but non-zero, the lower-layer streamfunction approximately satisfies $J(\psi_2, \overline{q})$, where $\overline{q}$ is given by (20.9), and therefore ψ_2 is a function of $\overline{q}$ — that is, $\psi_2 \approx G(\overline{q})$. For a typical subtropical wind, contours of $\overline{q}$, and therefore contours of ψ_2, are naturally divided into two regions (Fig. 20.1).

(i) A blocked region (the shadow zone), in which contours of $\overline{q}$ intersect the eastern boundary, the lower layer flow is zero and the upper layer carries all the Sverdrup transport.

(ii) A closed region in which (if we envision a nearly inviscid western boundary current) the flow recirculates. In this region we posit that the lower layer potential vorticity becomes homogeneous, with a value determined by the value at the region's boundary, and this in turn is determined by tracing $\overline{q}$ back to the domain boundary.

To satisfy a circulation constraint the function $G(\overline{q})$ must be a linear function, and given this, the entire solution may be determined. If, for example, the wind is zonal and a function of y only, and $\operatorname{curl}_z \boldsymbol{\tau}_T = g(y)$ then, in both regions:

$$\overline{\psi} \equiv \psi_1 + \psi_2 = -\frac{1}{\beta H_0} g(y)(x_E - x), \qquad \overline{q} \equiv \beta y + F\overline{\psi}. \tag{OC.1a,b}$$

In the blocked region:

$$\psi_2 = 0, \qquad \psi_1 = -\frac{1}{\beta H_0}(x_E - x)g(y), \tag{OC.2a}$$

$$q_1 = \beta y + F(\psi_1 - \psi_2), \qquad q_2 = \beta y + F(\psi_2 - \psi_1). \tag{OC.2b}$$

In the closed region:

$$q_2 = \beta L \quad \text{(by homogenization)}, \tag{OC.3a}$$

$$\psi_2 = \frac{1}{2F}(\overline{q} - \beta L), \qquad \psi_1 = \overline{\psi} - \psi_2, \tag{OC.3b}$$

$$q_1 = \beta y + F(\psi_2 - \psi_1) = 2\beta y - \beta L. \tag{OC.3c}$$

For $g(y) = -\sin \pi y$ these solutions are illustrated in Figs. 20.1 and 20.2.

This approach provides a solution to the conundrum of what drives the subsurface (quasi-geostrophic) ocean, for if there are no eddy effects at all [i.e., $\boldsymbol{T}_1 = \boldsymbol{T}_2 = 0$ in (20.3)], then the lower layer flow is stationary. This solution is unrealistic, for the upper layer flow could be made quite shallow. Another solution to this issue is provided in section 20.6, wherein it is assumed that the lower layers may outcrop and so feel the wind directly.

region is called a 'shadow zone', for the fluid is in the shadow of the eastern boundary, and it will re-appear in a model of the ventilated thermocline later on in this chapter. In the region of closed contours, ψ_2 cannot be given by this argument. But if κ is sufficiently small, we can expect (20.11) to approximately hold, and that the presence of a small amount of dissipation will determine the functional relationship between ψ_2 and $\overline{q}$. Thus, in summary, there are two regions of flow:

(i) the blocked region in which $\psi_1 \approx \overline{\psi} \gg \psi_2$ and ψ_1 is approximately given by (20.4);

(ii) a closed region in which $\psi_2 = G(\overline{q}) + \mathcal{O}(\kappa)$.

20.1.2 Relation between streamfunction and potential vorticity

A general argument

In chapter 13 we showed that, within a region of closed contours, the values of a tracer that is materially conserved except for the effects of a small diffusion would become *homogeneous*. In the case at hand, potential vorticity is that tracer, so that within potential vorticity contours or closed streamlines potential vorticity will become homogenized. If we can determine the value of q_2 within the region of closed contours, then from (20.2) ψ_2 is given by

$$\psi_2 = (1/2F)(\overline{q} - q_2) \tag{20.13}$$

and the solution would be complete. Now, outside the closed region $\psi_2 \ll \psi_1$, so that the outermost contour of the closed region must be characterized by $q_2 \approx \overline{q}$, for this makes ψ_2 continuous between closed and blocked regions. Thus, the value of q_2 within the closed homogeneous region is that of $\overline{q}$ (i.e., $\beta y + F\overline{\psi}$) on its boundary. Since this contour intersects the poleward edge of the domain, where $\overline{\psi}$ is zero, the value of this contour is just βy at $y = L$; that is, βL. Thus, within the closed region,

$$q_2 = \beta L. \tag{20.14}$$

A specific calculation

Now consider the steady, lower-layer potential vorticity equation (20.1b); noting that $J(\psi_2, F(\psi_1 - \psi_2)) = J(\psi_2, F(\psi_1 + \psi_2))$, (20.1b) may be written as

$$J(\psi_2, \overline{q}) = -\nabla \cdot \boldsymbol{T}_2. \tag{20.15}$$

Integrating around a closed contour of $\overline{q}$ the left-hand side vanishes and

$$R \int (\nabla\psi_1 - \nabla\psi_2) \cdot \boldsymbol{n} \, \mathrm{d}l = 0 \tag{20.16}$$

or

$$\oint \boldsymbol{u}_1 \cdot \mathrm{d}\boldsymbol{l} = \oint \boldsymbol{u}_2 \cdot \mathrm{d}\boldsymbol{l} \tag{20.17}$$

thus, the deep circulation around a mean geostrophic contour (i.e., isoline of $\overline{q}$) is equal to the upper-level circulation.

Previously we argued that

$$\psi_2 = G(\overline{q}) = G(\beta y + F(\psi_1 + \psi_2)). \tag{20.18}$$

where G is an arbitrary function of its argument. In order to satisfy (20.17) (a linear relation between $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$) G must be a linear function, and so we write

$$\psi_2 = C\left[\frac{\beta y}{F} + (\psi_1 + \psi_2)\right] + B, \tag{20.19}$$

where C and B are constants. This may be rearranged to give

$$\psi_1 = -C\frac{\beta y}{F} + (\psi_1 + \psi_2)(1 - C) - B. \tag{20.20}$$

The above two equations are consistent with (20.17) if $C = 1/2$. With this, (20.19) gives

$$\overline{q} = 2F(\psi_2 - B) \tag{20.21}$$

and the potential vorticity in the closed contour region of the lower layer is

$$q_2 = \beta y + F\overline{\psi} - 2F\psi_2 = -2FB. \tag{20.22}$$

That is, it is constant. Outside the closed contours $\psi_2 \ll \psi_1$ so that $q_2 \approx \overline{q} = \beta y + F\psi_1$. If we trace this contour to the edge of the domain where $\psi_1 = 0$ and $y = L$ then we see that the value of $\overline{q}$ on the contour, and hence q_2 in the closed region, is βL, as in (20.14), and $B = -\beta L/(2F)$. Using (20.21) then gives

$$\psi_2 = (2F)^{-1}(\overline{q} - \beta L). \tag{20.23}$$

Given ψ_2 and q_2 [from (20.14)], we obtain q_1 and ψ_1 using (20.2) and (20.6b), giving

$$q_1 = 2\beta y - \beta L, \qquad \psi_1 = \overline{\psi} - \psi_2. \tag{20.24}$$

All these fields are illustrated in Fig. 20.2, and see the shaded box on page 765 for a summary.

20.2 ✦ A MODEL WITH CONTINUOUS STRATIFICATION

We now look at the dynamics of the continuously stratified circulation, largely by way of an extension of our two-layer procedure. Let us first consider how deep the wind's influence is.

20.2.1 Depth of the wind's influence

The thermal wind-relationship in the form $f\,\partial u/\partial z = \partial b/\partial y$ implies a vertical scale H given by

$$H = \frac{fUL}{\Delta b}, \tag{20.25}$$

where Δb is a typical magnitude of the horizontal variation of the buoyancy. We can relate this to the Ekman pumping velocity W_E using the linear geostrophic vorticity equation, $\beta v = f\partial w/\partial z$, which, (assuming that the horizontal components of velocity are roughly similar, i.e., $V = U$), implies that

$$U = \frac{fW_E}{\beta H}. \tag{20.26}$$

Equations (20.25) and (20.26) may be combined to give an estimate of the depth of the wind-driven circulation, namely

$$H = \left(\frac{f^2 W_E L}{\beta\,\Delta b}\right)^{1/2}, \tag{20.27}$$

where L may be interpreted as the gyre scale. We now use quasi-geostrophic scaling to relate the horizontal temperature gradient to the stratification using the thermodynamic equation,

$$\frac{\mathrm{D}b}{\mathrm{D}t} + wN^2 = 0, \tag{20.28}$$

with implied scaling

$$\Delta b = \frac{W_E N^2 L}{U} = \frac{N^2 \beta H L}{f_0}, \tag{20.29}$$

where the second equality uses (20.26). Using (20.27) and (20.29) gives

$$\boxed{H = \left(\frac{W_E f^3}{\beta^2 N^2} \right)^{1/3}.} \tag{20.30}$$

Potential vorticity interpretation

The estimate (20.30) can be obtained and interpreted more directly: *the wind-driven circulation penetrates as far as it can alter the potential vorticity q from its planetary value βy.* Recall that, ignoring relative vorticity,

$$q = \beta y + \frac{\partial}{\partial z}\left(\frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right). \tag{20.31}$$

The two terms are comparable if

$$\frac{f_0^2}{N^2 H^2} U L \approx \beta L \qquad \text{or} \qquad H^2 \approx \frac{f_0^2 U}{N^2 \beta}. \tag{20.32}$$

Using (20.26) to eliminate U in favour W_E recovers (20.30). Thus, for a given stratification, we have an estimate of the depth of the wind-driven circulation, or at least a scaling for depth of the vertical influence of the wind.

20.2.2 The complete solution

Armed with an estimate for the depth of the wind's influence, we can obtain a solution for the continuously stratified case analogous to that found in the two-layer case in section 20.1. Our assumptions are as follows.

(i) In the limit of small dissipation, streamfunction and potential vorticity have a functional relationship with each other.

(ii) Potential vorticity is homogenized within closed isolines of q or ψ. The value of q within the homogenized pool is that of the outermost contour, which here is the value of q at the poleward edge of the barotropic gyre.

(iii) Outside of the pool region, (i.e., below the depth of the wind's influence) the streamfunction is zero, and the potential vorticity is given by the planetary value, i.e., βy.

Given these, finding a solution is not difficult. If N^2 is constant and neglecting relative vorticity, the expression for potential vorticity is

$$q = \frac{\partial^2}{\partial z^2}\left(\frac{f_0^2}{N^2} \psi \right) + \beta y. \tag{20.33}$$

We non-dimensionalize by writing

$$z = \left(\frac{f_0^2 U}{N^2 \beta} \right)^{1/2} \widehat{z}, \quad q = \beta L \widehat{q}, \quad \psi = \widehat{\psi} U L, \quad y = L \widehat{y}, \quad w = \frac{U^2 f_0}{N^2 H} \widehat{w}, \tag{20.34}$$

where the hatted variables are non-dimensional, and the scaling for w arises from the thermodynamic equation $N^2 w \sim J(\psi, f_0 \psi_z)$. With this, (20.33) becomes

$$\widehat{q} = \frac{\partial^2 \widehat{\psi}}{\partial \widehat{z}^2} + \widehat{y}, \tag{20.35}$$

The flow is then given by solving the following equations:

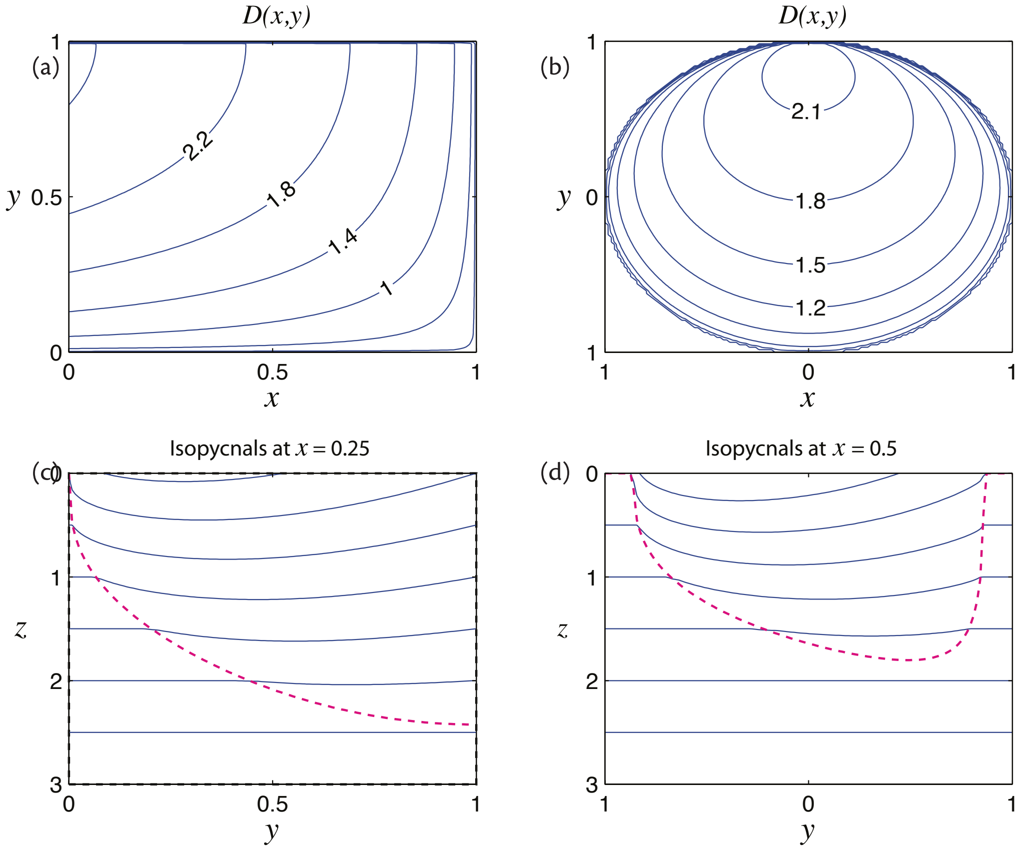

Fig. 20.3 Solutions of (20.9) for two different barotropic streamfunctions. On the left $\psi_B = (1-x)\sin \pi y$ and on the right $\psi_B = 1-(x^2+y^2)$ for $x^2+y^2 < 1$, zero elsewhere. The upper panels show contours of the depth of the wind-influenced region [solutions of (20.41)]. The depth increases to the northwest in the left panel, and to the north in the right panel, so that in both cases the area of the bowl shrinks with depth. The lower panels are contours of $z + (\beta L/f_0)\psi_z/2$, with $\beta L/f_0 = 1/2$, obtained from (20.37) or (20.42), at $x = 0.25$ and $x = -0.5$ in the two cases. These are isopycnal surfaces, with a rather large value of $\beta L/f_0$ to exaggerate the displacement in the bowl region. The dashed lines indicate the boundary of the bowl region, outside of which the isopycnals are flat.

$$\psi_{zz} + y = y_0, \qquad -D(x,y) < z < 0, \tag{20.36a}$$

$$\psi = 0, \qquad z \le -D(x,y), \tag{20.36b}$$

where D is the (to be determined) depth of the bowl, y_0 is a constant, and we have dropped the hats over the non-dimensional variables. The solution in $D(x,y) < z < 0$ corresponds to the closed region of the two-layer model, and the solution $z \le -D(x,y)$ corresponds to the blocked region of zero lower-layer flow. The constant y_0 is the non-dimensional value of potential vorticity within the pool region, and following our reasoning in the two-layer case this is the value of the potential vorticity at the northern boundary. Dimensionally this is βL, so that in non-dimensional units $y_0 = 1$.

The lower boundary condition on (20.36a) is that $\psi = \psi_z = 0$ at $z = -D$, because in the abyss $\psi = \partial\psi/\partial z = 0$ and we require that both ψ and $\partial\psi/\partial z$ be continuous (note that the buoyancy perturbation is proportional to $\partial\psi/\partial z$). The solution that satisfies this is

$$\psi = \frac{1}{2}(z + D)^2(y_0 - y), \tag{20.37}$$

and $\psi = 0$ for $z < D$.

To obtain an expression for D we first note that the non-dimensional vertical velocity at $z = 0$ is given by

$$w = -J(\psi, \psi_z), \tag{20.38}$$

which, using (20.37), gives

$$w = \frac{1}{2}(z + D)^2(y_0 - y)\frac{\partial D}{\partial x}. \tag{20.39}$$

At $z = 0$ the vertical velocity is the Ekman pumping velocity and (20.39) becomes

$$D^2\frac{\partial D}{\partial x} = \frac{2w_E}{(y_0 - y)} \, . \tag{20.40}$$

But the Ekman pumping velocity is related to the barotropic streamfunction, ψ_B, by the Sverdrup relationship, so that integrating (20.40) gives

$$\boxed{D^3 = \frac{6\psi_B}{(y_0 - y)} = -\frac{6(x_E - x)w_E}{(y_0 - y)} \, ,} \tag{20.41}$$

where the second equality holds if w_E is not a function of x. This is a solution for the depth of moving region, the bowl in which potential vorticity is homogenized. An expression for the streamfunction is then obtained by using (20.41) in (20.37), and is found to be

$$\psi = \begin{cases} \frac{1}{2}\left[z(y_0 - y)^{1/2} + (6\psi_B)^{1/3}(y_0 - y)^{1/6}\right]^2 & -D < z < 0, \\ 0 & z < -D. \end{cases} \tag{20.42}$$

The potential vorticity corresponding to this solution is

$$q = \begin{cases} y_0 & -D < z < 0, \\ y & z < -D. \end{cases} \tag{20.43}$$

Solutions are illustrated in Figs. 20.3 for cases with two different barotropic streamfunctions.

It is possible to heuristically extend models such as the one described above by appending a western boundary layer, and indeed the homogenization of potential vorticity depends upon the presence of such a region to allowing the flow to recirculate. However, as we saw in section 19.5.3, it is difficult for flow to leave a western boundary layer without the help of friction, and a neutrally stable, damped, stationary Rossby wave typically forms. The critical issue then is whether the presence of dissipation in the western boundary layer affects the homogenization of potential vorticity in the gyre itself. This problem is the province of observation and numerical simulation, and solutions with both quasi-geostrophic and primitive equation models do in fact show that potential vorticity is able to homogenize under many circumstances.[3] Let us now take a brief look at some observations.

20.2.3 Observations of potential vorticity

Homogenization of potential vorticity in the real ocean has been observed in both Pacific and Atlantic Oceans, in both hemispheres, and to a lessor degree in Indian Ocean and various maps

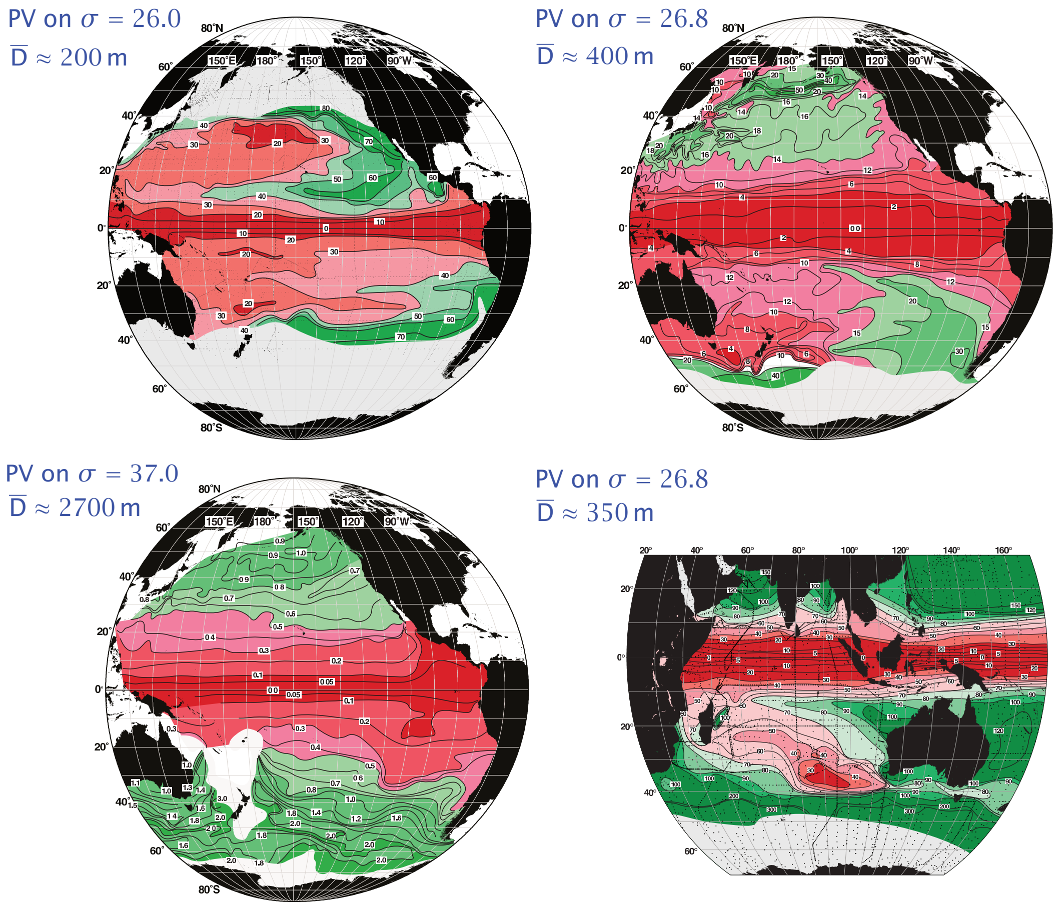

Fig. 20.4 Potential vorticity (i.e., $f(\partial\rho/\partial z)$) in the Pacific and Indian Oceans on the potential density surfaces labelled, which each have approximate average depths $\overline{\text{D}}$. Some potential vorticity homogenization can be seen in the subtropical gyre in the Pacific (the upper two plots) but less so at depth in the Pacific (lower left) and less so at all depths in the Indian Ocean (just the upper ocean is shown, lower right), which has a less pronounced gyre structure.[2]

are shown in Fig. 20.4 through Fig. 20.7.[5] In all of the plots we see that the near-equatorial variation of potential vorticity is dominated by the beta effect, more so at depth where the potential vorticity isolines are more-or-less along latitude lines until almost 20°, but in the subtropical gyres there are large regions of homogeneous potential vorticity.

Looking first at the Pacific, the upper two plots in Fig. 20.4 show the potential vorticity (here defined as $f\partial\rho/\partial z$) on potential density surfaces in the main thermocline. These surfaces slope up toward the pole, and the $\sigma = 26$ surface (i.e., a surface with density of approximately 1026 kg m^{-3}) outcrops at about 40°, a little equatorward of the boundary between the subpolar and subtropic gyre. On these surfaces there are large swaths of near-uniform potential vorticity in the subtropical gyre, perhaps a little more obviously so in the Northern Hemisphere, with strong gradients quite noticeable at the gyre edge at about 50° N. Tongues of high potential vorticity are advected by the gyre itself, sweeping equatorward and westward along the $\sigma = 26$ surface in the Northern Hemisphere.

Moving into the deep Pacific there is less homogenization, with isolines of potential vorticity generally crossing the entire Pacific at all latitudes, with a just the odd pool of closed contours.

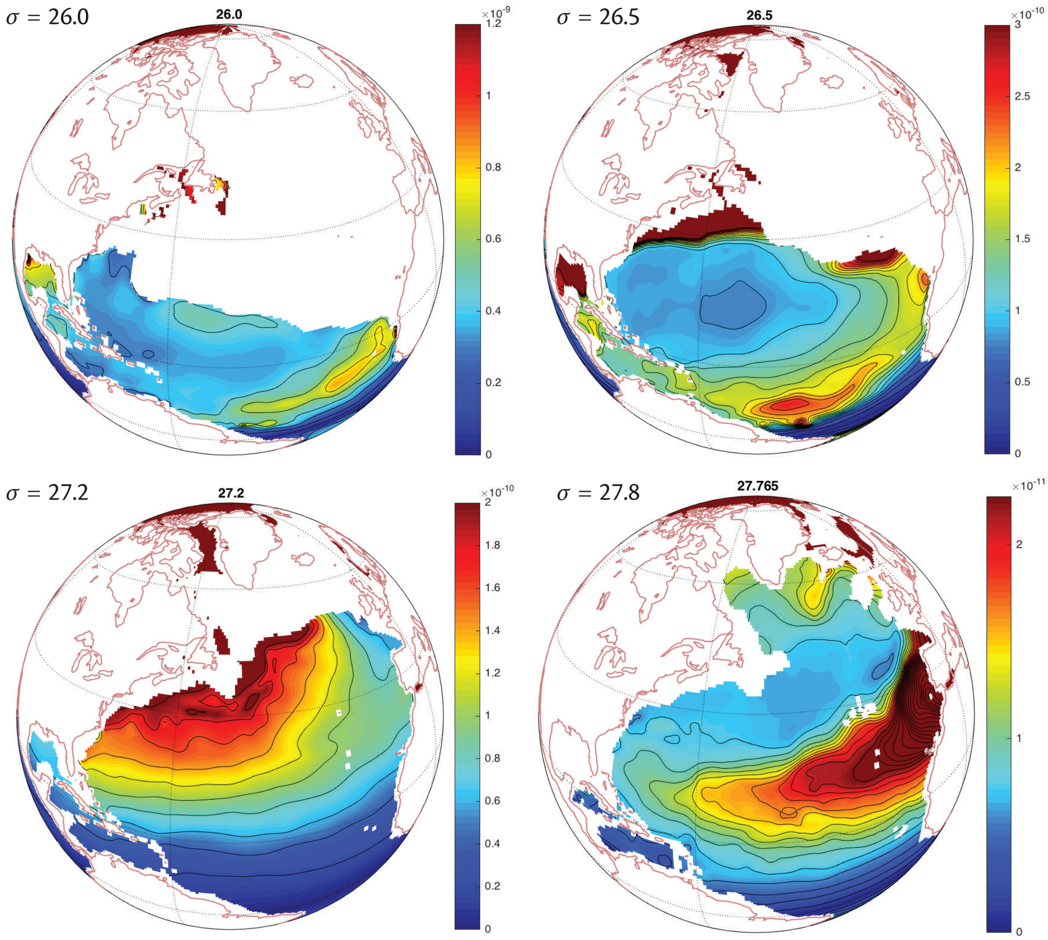

Fig. 20.5 Potential vorticity in the Atlantic Ocean on the potential density surfaces labelled, averaged over several Januaries using the MIMOC climatology. Potential vorticity is a normalized version of f/h. Specifically it is $|f|\delta\sigma/(\rho h)$ where $h(x, y)$ is the isopycnal layer thickness and $\delta\sigma$ is a fixed difference between layer interface potential densities, so here PV has units of $m^{-1}s^{-1}$. The PV is homogenized over much of the subtropical gyre around $\sigma = 26.5$. At deeper levels the potential vorticity is more dominated by the beta effect and an influx of Mediterranean water.[4]

The Indian Ocean has less potential vorticity homogenization at all depths, most likely because the subtropical gyre itself is less pronounced in the Indian Ocean, and the subpolar gyre is largely replaced by the eastward flow of the Antarctic Circumpolar Current system.

The Atlantic also shows large regions on homogenization in the upper ocean as seen in Fig. 20.6 and Fig. 20.6. These maps were constructed from a different set of observations, and using a different method, than those of Fig. 20.4, but show similar features — homogenization in the upper gyre but planetary values (and now a Mediterranean influence) dominating at depth. Consider Fig. 20.6 layer-by-layer, from the top down, where a close inspection of Fig. 20.8 will reveal the depths of each layer. The shallow, $\sigma = 26.0$m layer (typically tens of meters deep) outcrops in the middle of the subtropical gyre, receiving most of its fluid directly by Ekman-pumping from the mixed layer, and has a relatively small pool of homogenized potential vorticity. The deeper, $\sigma = 26.5$ level (with typical depths of a few hundred metres over much of the gyre) outcrops much further poleward and consequently has an extensive recirculating regime that homogenizes the potential vorticity.

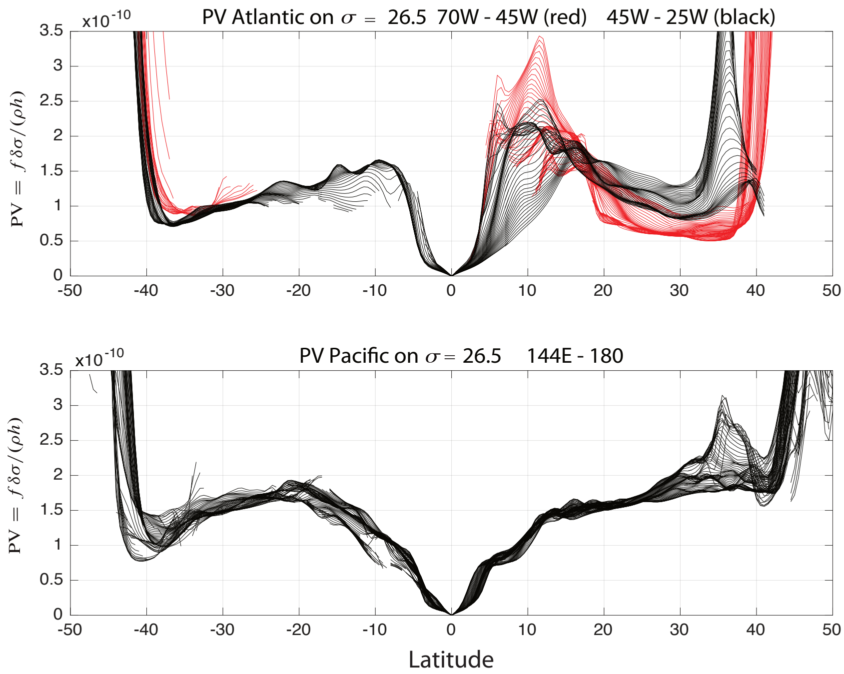

Fig. 20.6 Potential vorticity in the Atlantic and Pacific at the $\sigma_0 = 26.5$ level, using the same data as in Fig. 20.5.[4] The Coriolis parameter is latitudinally varying but taken as positive in both hemispheres for graphical convenience. See text for more discussion.

This level is plotted in Fig. 20.6 and we see a region between about 10° N and 25° N where potential vorticity increases moving southward — that is, $\partial Q/\partial y < 0$ — so enabling baroclinic instability. Going deeper, at $\sigma = 27.24$ (with typical depth of several hundred to a thousand or so metres) the planetary influence begins to dominate, with the subtropical gyre shrinking and a smaller region of potential vorticity homogenization further north. Finally, at $\sigma = 27.76$, or about 1500 m depth, the circulation is dominated by low potential vorticity Labrador Sea Water to the north and high potential vorticity from Mediterranean Salt Tongue to the south — again with a potentially baroclinically unstable flow where potential vorticity increases equatorward. The homogenization in both Pacific and Atlantic is quite strikingly displayed in Fig. 20.6, with broad plateaus of near-constant potential vorticity until the edge of the poleward edge of the gyres, where there is a sudden leap that almost certainly acts as a mixing barrier.

The story of potential vorticity is a rich one. The mesoscale eddies provide the stirring that

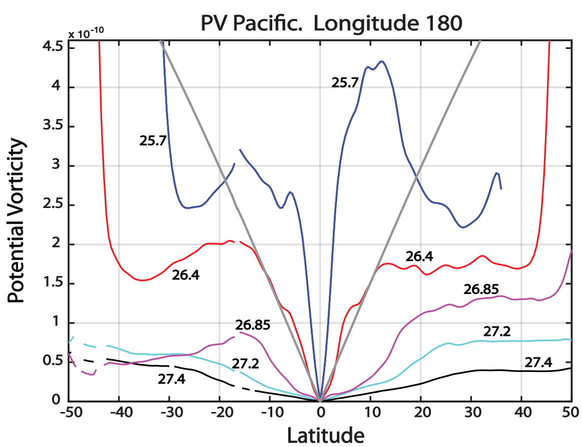

Fig. 20.7 As for Fig. 20.6, but now the potential vorticity at various sigma levels in the Pacific, on the date line at longitude 180. The v-shaped grey lines show the reference variation due to βy for the $\sigma = 26.4$ level. For the other levels, the reference variation is similar to the actual variation between about 0° and 10° latitude.

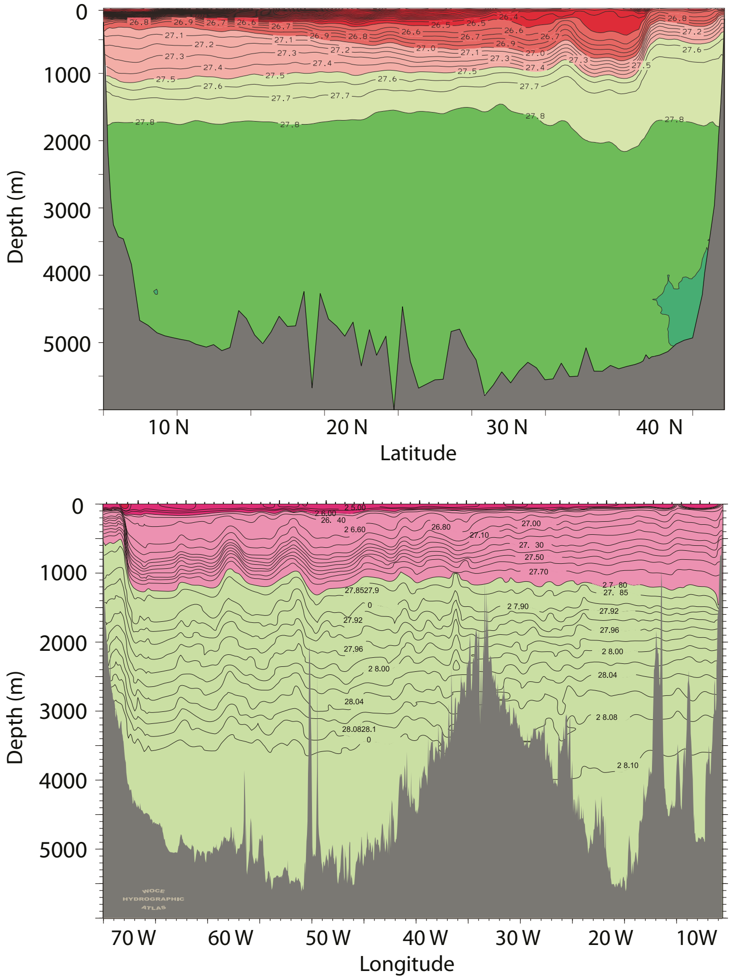

Fig. 20.8 Sections of density in the North Atlantic.

Upper panel: meridional section of potential density at 53° W, from 5° N to 45° N, with a uniform contour interval. In the upper northwestern region of the subtropical thermocline there is a region of low stratification known as MODE water: isopycnals above this outcrop in the subtropical gyre and are 'ventilated'; isopycnals below the MODE water outcrop in the subpolar gyre, north of about 45°, and/or in ACC.

Lower panel: zonal section of neutral density at 36° N, from about 75° W to 10° W. Contour interval changes where the colour changes. Note the front associated with the western boundary current at about 70° W.[6]

leads to the homogenization, but competing processes complicate matters. The eddies weaken with depth and planetary effects begin to dominate, and in the upper ocean the varied influences of the western boundary current, topographic effects and the stripping of potential vorticity sheets from solid boundaries turn the ocean into a colourful tapestry.

20.3 THE MAIN THERMOCLINE

We now approach the dynamics of the upper ocean from another angle and address the mechanisms that give rise to the actual density structure of the upper ocean, and in particular to the *main thermocline,* which is the region of the upper ocean, about 1 km deep, in which the density and temperature change most rapidly, as illustrated in Fig. 20.8 and Fig. 20.9.

We will consider the circulation in a closed, single hemispheric basin, and suppose that there is a net surface heating at low latitudes and a net cooling at high latitudes that maintains a meridional temperature gradient at the surface. Let us presume, ab initio, that there is a single overturning cell, with water rising at low latitudes before returning to polar regions, as illustrated schematically in Fig. 20.10. We will investigate the dynamics of this meridional overturning circulation (MOC) in much more detail in the next chapter, 21, but here our interest is mainly in why and how it produces a density structure similar to that observed. (Readers for whom the MOC is of primary

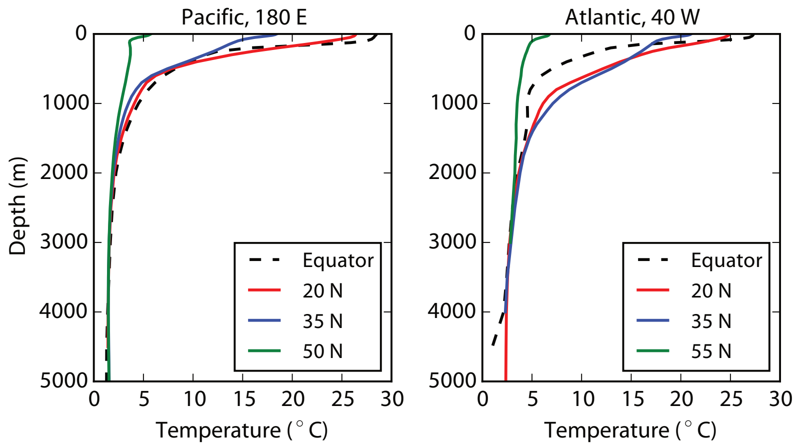

Fig. 20.9 Profiles of mean temperature in the North Pacific and Atlantic, from WOCE, at the longitudes and latitudes indicated. The profiles are considerably smoother than instantaneous ones.

Note the shallowness of the equatorial thermoclines (especially in the Atlantic), and weakness of the subpolar thermoclines.

interest may wish to read chapter 21 before preceeding with the rest of this chapter.) We will also, by and large, omit considerations of saline effects and assume a linear equation of state, so that the thermocline is synonymous with the *pycnocline,* the region where density changes rapidly.

The physical picture we have is the following. Cold, dense water at high latitudes sinks, so that dense water extends all the way to the ocean floor. By hydrostasy the pressure in the deep ocean is then higher at high latitudes than at low, where the water is warmer. Thus the water moves equatorward filling the abyss. This water is also slowly *warmed* by heat diffusion down from above, and it is this diffusion that enables the circulation to persist: if diffusion were zero, the entire ocean would eventually fill with the densest available water and the circulation would cease. Given a circulation like that envisioned above, we can see that there must be a vertical density gradient, except at the highest latitudes where the water is sinking. The water that fills the interior from the cold pole is, obviously, colder and denser than the surface waters at lower latitude so there must be vertical temperature gradient, and we indeed see in Fig. 20.9 how the temperature profile varies with latitude. However, without considering the dynamics it is hard to see what form the temperature profile will take; for example, it is conceivable that the polar waters might fill up the abyss nearly all the way to the surface, with a thermocline then only be a few metres thick. Or there might be a uniform temperature gradient from the surface to the ocean floor.

Complicating matters, the thermocline is also the region in where the gyre circulation is most prominent, so that the potential vorticity dynamics of the previous few sections must play a role. Putting that complication aside for now, let us first look at a simple kinematic model.

20.3.1 A simple kinematic model

The fact that cold water with polar origins upwells into a region of warmer water suggests that we consider the simple one-dimensional advective–diffusive balance,

$$w\frac{\partial T}{\partial z} = \kappa\frac{\partial^2 T}{\partial z^2}, \tag{20.44}$$

where w is the vertical velocity, κ is a diffusivity and T is temperature. In mid-latitudes, where this might hold, w is positive and the equation represents a balance between the upwelling of cold water and the downward diffusion of heat. If w and κ are given constants, and if T is specified at the top ($T = T_T$ at $z = 0$) and if $\partial T/\partial z = 0$ at great depth ($z = -\infty$) then the temperature falls exponentially away from the surface according to

$$T = (T_T - T_B)e^{wz/\kappa} + T_B, \tag{20.45}$$

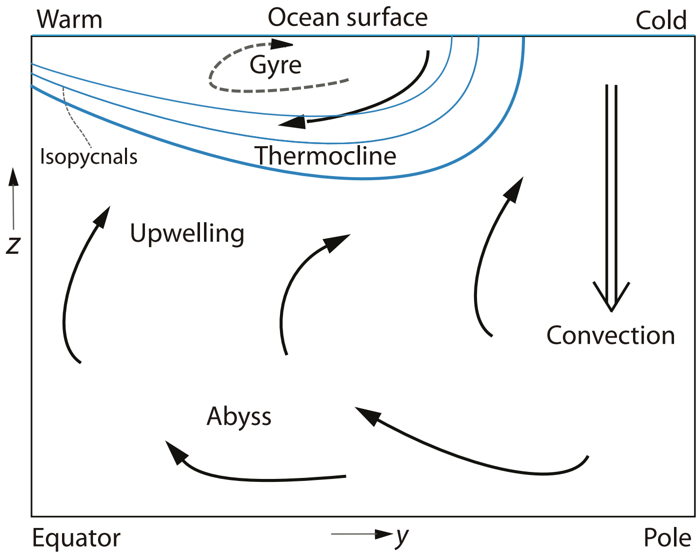

Fig. 20.10 Cartoon of a single-celled meridional overturning circulation, with a wall at the equator.

Sinking is concentrated at high latitudes and upwelling spread out over lower latitudes. The thermocline is the boundary between the cold abyssal waters, with polar origins, and the warmer near-surface subtropical water. Wind forcing in the subtropics pushes the warm surface water into the fluid interior, deepening the thermocline as well as circulating as a gyre.

where T_B is the temperature at depth. Temperature decays exponentially away from its surface value with the scale

$$\delta = \frac{\kappa}{w}, \tag{20.46}$$

and this is an estimate of the thermocline thickness. It is not particularly useful, because the magnitude of w depends on κ, as we will see. However, it is reasonable to see if the observed ocean is broadly consistent with this expression. The diffusivity κ can be measured; it is an eddy diffusivity, maintained by small-scale turbulence, and measurements produce values that range between $10^{-5}\,\mathrm{m^2\,s^{-1}}$ in the main thermocline and $10^{-4}\,\mathrm{m^2\,s^{-1}}$ in abyssal regions over rough topography and in and near continental margins, with still higher values locally.[7] The vertical velocity is too small to be measured directly, but various estimates based on deep water production suggest a value of about $10^{-7}\,\mathrm{m\,s^{-1}}$. Using this and the smaller value of κ in (20.45) gives an e-folding vertical scale, κ/w, of just 100 m, beneath which the stratification is predicted to be very small (i.e., nearly uniform potential density). Using the larger value of κ increases the vertical scale to 1000 m, which is probably closer to the observed value for the total thickness of the thermocline (Fig. 20.9), but using such a large value of κ in the main thermocline is not supported by the observations. Similarly, the deep stratification of the ocean is rather larger than that given by (20.44), except with values of diffusivity on the large side of those observed, a topic we return to in chapter 21.[8]

Aside from diffusion, mechanical forcing, and in particular the wind, will deepen the thermocline, as 20.10 suggests. The wind-stress curl forces water to converge in the subtropical Ekman layer, thereby forcing relatively warm water to downwell and meet the upwelling colder abyssal water at some finite depth, thus deepening the thermocline from its purely diffusive value. Indeed, in so far as we can separate the two effects of wind and diffusion, we can say that the strength of the wind influences the *depth* at which the thermocline occurs, whereas the strength of the diffusivity influences the *thickness* of the thermocline.

20.4 SCALING AND SIMPLE DYNAMICS OF THE MAIN THERMOCLINE

We now begin to consider the dynamics that produce an overturning circulation and a thermocline. The Rossby number of the large-scale circulation is small and the scale of the motion large, and the flow obeys the planetary-geostrophic equations,

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi, \qquad \frac{\partial \phi}{\partial z} = b, \qquad \nabla \cdot \boldsymbol{v} = 0, \qquad \frac{\mathrm{D}b}{\mathrm{D}t} = \kappa \frac{\partial^2 b}{\partial z^2}, \tag{20.47a,b,c,d}$$

in our standard notation, in which $\boldsymbol{u}$ and $\boldsymbol{v}$ refer to the two- and three-dimensional velocities. We suppose that these equations hold below an Ekman layer, so that the effects of a wind stress may be included by specifying a vertical velocity, w_E, at the top of the domain. The diapycnal diffusivity, κ, is some kind of eddy diffusivity, but since its precise form and magnitude are uncertain we proceed with due caution, and a useful practical philosophy is to try to ignore dissipation and viscosity where possible, and to invoke them only if there is no other way out. Let us therefore scale the equations in two ways, with and without diffusion; these scalings will be central to our theory.

20.4.1 A diffusive scale

Suppose that the circulation is steady and resembles that of Fig. 20.10, but with no wind forcing. Can we estimate how deep the diffusive layer will be in the subtropical gyre? We will suppose that, as in the kinematic model, the thermodynamic equation reduces to the advective-diffusive balance of (20.44), but we will use the other equations in (20.47) to give an estimate of the vertical velocity. If we take the curl of (i.e., cross differentiate) the momentum equation (20.47a) and use mass continuity we obtain the linear vorticity equation, $\beta v = f\partial w/\partial z$, and if we take the vertical derivative of the momentum equation and use hydrostasy we obtain thermal wind, $\partial \boldsymbol{u}/\partial z = \mathbf{k}\times\nabla b$. Collecting these equations together we have

$$w\frac{\partial b}{\partial z} = \kappa \frac{\partial^2 b}{\partial z^2}, \qquad \beta v = f\frac{\partial w}{\partial z}, \qquad f\frac{\partial \boldsymbol{u}}{\partial z} = \mathbf{k} \times \nabla b, \tag{20.48a,b,c}$$

with corresponding scales

$$\frac{W}{\delta} = \frac{\kappa}{\delta^2}, \qquad \beta V = \frac{fW}{\delta}, \qquad \frac{U}{\delta} = \frac{\Delta b}{fL}, \tag{20.49a,b,c}$$

where δ is the vertical scale and other scaling values are denoted with capital letters. We suppose that $V \sim U$, where U is the zonal velocity scale, and henceforth we will denote both by U, and L is the horizontal scale of the motion, which we take as the gyre or basin scale. Typical values for the subtropical gyre might bs $\Delta b = g\Delta\rho/\rho_0 = g\beta_T \Delta T \sim 10^{-2}\,\mathrm{m\,s^{-2}}$, $L = 5000\,\mathrm{km}$, $f = 10^{-4}\,\mathrm{s^{-1}}$ and $\kappa = 10^{-5}\,\mathrm{m^2\,s^{-2}}$.

Equation (20.49a) is the same as (20.46), as expected, but we can now use (20.49b,c) to obtain an estimate for the vertical velocity, namely

$$W = \frac{\beta\delta^2 \Delta b}{f^2 L}. \tag{20.50}$$

Using this and (20.49a) gives the diffusive vertical scale, and the estimates

$$\boxed{\delta = \left(\frac{\kappa f^2 L}{\beta\,\Delta b}\right)^{1/3} \qquad W = \left(\frac{\kappa^2 \beta\,\Delta b}{f^2 L}\right)^{1/3}} \tag{20.51}$$

With values of the parameters as above, (20.51) gives $\delta \approx 150\,\mathrm{m}$ and $W \approx 10^{-7}\,\mathrm{m\,s^{-1}}$.

20.4.2 An advective scale

The value of the vertical velocity obtained above is very small, much smaller that the Ekman pumping velocity at the top of the ocean, which is of order 10^{-6}–$10^{-5}\,\mathrm{m\,s^{-1}}$. This difference suggests that we might ignore the diffusive term in (20.48a) — indeed, ignore the thermodynamic term completely — and construct an adiabatic scaling estimate for the depth of the wind's influence. Further, in subtropical gyres the Ekman pumping is downward, whereas the diffusive velocity is upward, meaning that at some level we expect the vertical velocity to be zero. This depth, D_a, will approximately be the depth at which the Ekman pumping velocity fades away.

The equations of motion are just the thermal wind balance and the linear geostrophic vorticity equation, namely

$$\beta v = f\frac{\partial w}{\partial z}, \qquad \boldsymbol{f}\times\frac{\partial \boldsymbol{u}}{\partial z} = -\nabla b \tag{20.52}$$

with correspoding scales

$$\beta U = f\frac{W}{D_a}, \qquad \frac{U}{D_a} = \frac{1}{f}\frac{\Delta b}{L}. \tag{20.53}$$

We recall that $V \sim U$ and D_a is the unknown depth scale of the motion.

The thermodynamic equation does not enter, but we take the vertical velocity to be that due to Ekman pumping, W_E. From (20.53) we immediately obtain

$$\boxed{D_a = W_E^{1/2}\left(\frac{f^2 L}{\beta\,\Delta b}\right)^{1/2}.} \tag{20.54}$$

[This may be compared with the estimate of (20.27).] If we relate U and W_E using mass conservation, $U/L = W_E/D_a$, instead of using (20.52a), then we write L in place of f/β and (20.54) becomes $D_a = (W_E f L^2/\Delta b)^{1/2}$, which is not qualitatively different than (20.54) for large scales.

The important aspect of the above estimate is that the depth of the wind-influenced region increases with the magnitude of the wind stress (because $W_E \propto \mathrm{curl}_z\tau$) and decreases with the meridional temperature gradient. The former dependence is reasonably intuitive, and the latter arises because as the temperature gradient increases the associated thermal wind-shear U/D_a correspondingly increases. But the horizontal transport (the product UD_a) is fixed by mass conservation; the only way that these two can remain consistent is for the vertical scale to decrease. Taking $W_E = 10^{-6}\,\mathrm{m\,s^{-1}}$, and other values as before, gives $D_a = 500\,\mathrm{m}$, and $W_E = 10^{-5}\,\mathrm{m\,s^{-1}}$ gives $D_a = 500\,\mathrm{m}$. Such a scaling argument cannot be expected to give more than an estimate of the depth of the wind-influenced region; nevertheless, because D_a is much less than the ocean depth, the estimate does suggest that the wind-driven circulation is predominantly an upper-ocean phenomenon.

✦ *A wind-influenced diffusive scaling*

The scalings above assumes that the length scale over which thermal wind balance holds is the gyre scale itself. In fact, there is another length scale that is more appropriate, and this leads to a slightly different diffusive scaling for the thickness of the thermocline. To obtain this scaling, we first note that the depth of the subtropical thermocline is not constant: it shoals up to the east because of Sverdrup balance, and it may shoal up polewards as the curl of the wind stress falls (and is zero at the poleward edge of the gyre). Thus, referring to Fig. 20.11, the appropriate horizontal length scale $\widetilde{L}$ is given by

$$\widetilde{L} = \delta\frac{L}{D_a}. \tag{20.55}$$

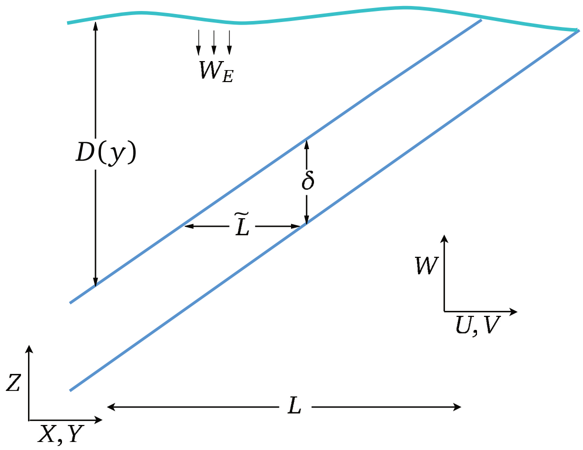

Fig. 20.11 Scaling the thermocline. The diagonal lines mark the diffusive thermocline of thickness δ and depth $D(y)$. The advective scaling for $D(y)$, i.e., D_a, is given by (20.54), and the diffusive scaling for δ is given by (20.56).

This is no longer an externally imposed parameter, but must be determined as part of the solution. Using $\widetilde{L}$ instead of L as the length scale in the thermal wind equation (20.49c) gives, using (20.54), the modified diffusive scale

$$\delta = \kappa^{1/2}\left(\frac{f^2 L}{\Delta b\, \beta D_a}\right)^{1/2} = \kappa^{1/2}\left(\frac{f^2 L}{\Delta b\, \beta W_E}\right)^{1/4}. \tag{20.56}$$

Substituting values of the various parameters results in a thickness of about 100–200 m. The thermocline thickness now scales as $\kappa^{1/2}$. The interpretation of this scale and that of (20.51) is that the thickness of the thermocline scales as $\kappa^{1/3}$ in the absence of a wind stress, but scales as $\kappa^{1/2}$ if a wind stress is present that can provide a finite slope to the base of the thermocline that is independent of κ, and this is confirmed by numerical simulations.[9] From (20.48a) the vertical velocity, and hence the meridional overturning circulation, no longer scales as $\kappa^{2/3}$ but as

$$W = \frac{\kappa}{\delta} \propto \kappa^{1/2}. \tag{20.57}$$

20.4.3 Summary of the physical picture

What do the vertical scales derived above represent? The wind-influenced scaling, D_a, is the depth to which the directly wind-driven circulation can be expected to penetrate. Thus, over this depth we can expect to see wind-driven gyres and associated phenomena. At greater depths lies the abyssal circulation, and this is not wind-driven in the same sense. Now, in general, the water at the base of the wind-driven layer will not have the same thermodynamic properties as the upwelling abyssal water — this being cold and dense, whereas the water in the wind-driven layer is warm and subtropical (look again at Fig. 20.10). The thickness δ characterizes the diffusive transition region between these two water masses and in the limit of very small diffusivity this becomes a *front*. One might say that D_a is the *depth* of the thermocline, while δ is the *thickness* of the thermocline. In the diffusive region, no matter how small the diffusivity κ is in the thermodynamic equation, the diffusive term is important. Of course if the diffusion is sufficiently large, the thickness will be as large or larger than the depth, and the two regions will blur into each other, and this may indeed be the case in the real ocean. Nevertheless, these scales are a useful foundation on which to build.[10]

20.5 THE INTERNAL THERMOCLINE

We now try to go beyond simple scaling arguments and investigate in more detail the dynamics of the thermocline. In this section we consider the diffusive, or internal, thermocline and in section 20.6 we consider the advective, or ventilated, thermocline. The advective term in the thermodynamic equation makes such an investigation difficult, and prevents us from constructing exact analytic models, but not from constructing informative models. We begin by expressing the planetary-geostrophic equations as an equation in a single unknown.

20.5.1 The *M* equation

The planetary-geostrophic equations can be written as a single partial differential equation in a single variable, although the resulting equation is of quite high order and is nonlinear. We write the equations of motion as

$$-fv = -\frac{\partial \phi}{\partial x}, \qquad fu = -\frac{\partial \phi}{\partial y}, \qquad b = \frac{\partial \phi}{\partial z}, \tag{20.58a,b,c}$$

$$\nabla \cdot \boldsymbol{v} = 0, \qquad \frac{\partial b}{\partial t} + \boldsymbol{v} \cdot \nabla b = \kappa \nabla^2 b, \tag{20.59a,b}$$

where we take $f = \beta y$. Cross-differentiating the horizontal momentum equations and using (20.59a) gives the linear geostrophic vorticity relation $\beta v = f \partial w / \partial z$ which, using (20.58a) again, may be written as

$$\frac{\partial \phi}{\partial x} + \frac{\partial}{\partial z}\left(-\frac{f^2}{\beta} w\right) = 0. \tag{20.60}$$

This equation is the divergence in (x, z) of $(\phi, -f^2 w/\beta)$ and is automatically satisfied if

$$\phi = M_z \qquad \text{and} \qquad \frac{f^2 w}{\beta} = M_x. \tag{20.61a,b}$$

where the subscripts on M denote derivatives. Then straightforwardly

$$u = -\frac{\partial_y \phi}{f} = -\frac{M_{zy}}{f}, \qquad v = \frac{\partial_x \phi}{f} = \frac{M_{zx}}{f}, \qquad b = \partial_z \phi = M_{zz}. \tag{20.62a,b,c}$$

The thermodynamic equation, (20.59b) becomes

$$\frac{\partial M_{zz}}{\partial t} + \left(\frac{-M_{zy}}{f} M_{zzx} + \frac{M_{zx}}{f} M_{zzy}\right) + \frac{\beta}{f^2} M_x M_{zzz} = \kappa M_{zzzz} \tag{20.63}$$

or

$$\frac{\partial M_{zz}}{\partial t} + \frac{1}{f} J(M_z, M_{zz}) + \frac{\beta}{f^2} M_x M_{zzz} = \kappa M_{zzzz}. \tag{20.64}$$

where J is the usual horizontal Jacobian. This is the *M equation*,[11] somewhat analogous to the potential vorticity equation in quasi-geostrophic theory in that it expresses the entire dynamics of the system in a single, nonlinear, advective–diffusive partial differential equation, although M_{zz} is materially conserved (in the absence of diabatic effects) by the three-dimensional flow. Because of the high differential order and nonlinearity of the system analytic solutions of (20.64) are hard to find, and from a numerical perspective it is easier to integrate the equations in the form (20.58) and (20.59) than in the form (20.64). Nevertheless, it is possible to move forward by approximating (20.64) to one or two dimensions, or by a priori assuming a boundary-layer structure.

A one-dimensional model

Let us consider an illustrative one-dimensional model (in z) of the thermocline.[12] Merely setting all horizontal derivatives in (20.64) to zero is not very useful, for then all the advective terms on the left-hand side vanish. Rather, we look for steady solutions of the form $M = M(x, z)$, and the M equation then becomes

$$\frac{\beta}{f^2} M_x M_{zzz} = \kappa M_{zzzz}, \tag{20.65}$$

which represents the advective–diffusive balance

$$w\frac{\partial b}{\partial z} = \kappa \frac{\partial^2 b}{\partial z^2}. \tag{20.66}$$

In proceededing this way we have assumed that the value of κ varies meridionally in the same manner as does β/f^2; without this technicality M would be a function of y, violating our premise.

If the ocean surface is warm and the abyss is cold, then (20.65) represents a balance between the upward advection of cold water and the downward diffusion of warm water. The horizontal advection terms vanish because the zonal velocity, u, and the meridional buoyancy gradient, b_y, are each zero. Let us further consider the special case

$$M = (x - x_e)W(z), \tag{20.67}$$

where the domain extends from $0 \le x \le x_e$, so satisfying $M = 0$ on the eastern boundary. Equation (20.65) becomes the ordinary differential equation

$$\boxed{\frac{\beta}{f^2} W W_{zzz} = \kappa W_{zzzz},} \tag{20.68}$$

where W has the dimensions of velocity squared. We non-dimensionalize this by setting

$$z = H\hat{z}, \qquad \kappa = \hat{\kappa}(HW_S), \qquad W = \left(\frac{f^2 W_S}{\beta}\right)\widehat{W}, \tag{20.69a,b,c}$$

where the hatted variables are non-dimensional and W_S is a scaling value of the dimensional vertical velocity, w (e.g., the magnitude of the Ekman pumping velocity W_E). Equation (20.68) becomes

$$\widehat{W}\widehat{W}_{\hat{z}\hat{z}\hat{z}} = \hat{\kappa}\widehat{W}_{\hat{z}\hat{z}\hat{z}\hat{z}}, \tag{20.70}$$

The parameter $\hat{\kappa}$ is a non-dimensional measure of the strength of diffusion in the interior, and the interesting case occurs when $\hat{\kappa} \ll 1$; in the ocean, typical values are $H = 1\,\mathrm{km}$, $\kappa = 10^{-5}\,\mathrm{m\,s^{-2}}$ and $W_S = W_E = 10^{-6}\,\mathrm{m\,s^{-1}}$ so that $\hat{\kappa} \approx 10^{-2}$, which is indeed small. (It might appear that we could completely scale away the value of κ in (20.68) by scaling W appropriately, and if so there would be no meaningful way that one could say that κ was small. However, this is a chimera, because the value of κ would still appear in the boundary conditions.

The time-dependent form of (20.70), namely $\widehat{W}_{\hat{z}\hat{z}t} + \widehat{W}\widehat{W}_{\hat{z}\hat{z}\hat{z}} = \hat{\kappa}\widehat{W}_{\hat{z}\hat{z}\hat{z}\hat{z}}$, is similar to Burger's equation, $V_t + VV_z = \nu V_{zz}$, which is known to develop fronts. (In the inviscid Burger's equation, $\mathrm{D}V/\mathrm{D}t = 0$, where the advective derivative is one-dimensional, and therefore the velocity of a given fluid parcel is preserved on the line. Suppose that the velocity of the fluid is positive but diminishes in the positive z-direction, so that a fluid parcel will catch-up with the fluid parcel in front of it. But since the velocity of a fluid parcel is fixed, there are two values of velocity at the same point, so a singularity must form. In the presence of viscosity, the singularity is tamed to a front.) Thus, we might similarly expect (20.70) to produce a front, but because of the extra derivatives the argument is not as straightforward and it is simplest to obtain solutions numerically.

Equation (20.70) is fourth order, so four boundary conditions are needed, two at each boundary. Appropriate ones are a prescribed buoyancy and a prescribed vertical velocity at each boundary, for example

$$\begin{aligned} \widehat{W} &= \widehat{W}_E, \quad -\widehat{W}_{\hat{z}\hat{z}} = B_0, && \text{at top} \\ \widehat{W} &= 0, \quad -\widehat{W}_{\hat{z}\hat{z}} = 0, && \text{at bottom,} \end{aligned} \tag{20.71}$$

where $\widehat{W}_E$ is the (non-dimensional) vertical velocity at the base of the top Ekman layer, which is negative for Ekman pumping in the subtropical gyre, and B_0 is a constant, proportional to the buoyancy difference across the domain. We obtain solutions numerically by Newton's method,[13] and these are shown in Figs. 20.12 and 20.13. The solutions do indeed display fronts, or boundary layers, for small diffusivity. If the wind forcing is zero (Fig. 20.13), the boundary layer is at the top of the fluid. If the wind forcing is non-zero, an internal boundary layer — a front — forms in the fluid interior with an adiabatic layer above and below. In the real ocean, where wind forcing is of course non-zero, the frontal region is known as the *internal thermocline.*

20.5.2 ✦ Boundary-layer analysis

The reasoning and the numerical solutions of the above sections suggest that the internal thermocline has a boundary-layer structure whose thickness decreases with κ. If the Ekman pumping at the top of the ocean is non-zero, the boundary layer is internal to the fluid. To learn more, let us perform a boundary layer analysis, much as we did when investigating western boundary currents in section 19.1.3. The nonlinearity precludes a complete solution of the equation, but we can nevertheless obtain some useful information.

One-dimensional model

Let us now *assume* a steady two-layer structure of the form illustrated in Fig. 20.14, and that the dynamics are governed by (20.70) in a domain that extends from 0 to −1. The buoyancy thus varies rapidly only in an internal boundary layer of non-dimensional thickness $\hat{\delta}$ located at $\hat{z} = -h$; above and below this the buoyancy is assumed to be only very slowly varying. Following standard boundary layer procedure we introduce a stretched boundary layer coordinate ζ where

$$\hat{\delta}\zeta = \hat{z} + h. \tag{20.72}$$

That is, ζ is the distance from $\hat{z} = -h$, scaled by the boundary layer thickness $\hat{\delta}$, and within the boundary layer ζ is an order-one quantity. We also let

$$\widehat{W}(\hat{z}) = \widehat{W}_I(\hat{z}) + \widetilde{W}(\zeta), \tag{20.73}$$

where $\widehat{W}_I$ is the solution away from the boundary layer and $\widetilde{W}$ is the boundary layer correction. Because the boundary layer is presumptively thin, $\widehat{W}_I$ is effectively constant through it and, furthermore, for $\hat{z} < -h$, $\widehat{W}$ vanishes in the limit as $\kappa = 0$. We thus take $\widehat{W}_I = 0$ throughout the boundary layer. (The small diffusively-driven upwelling below the boundary layer is part of the boundary layer solution, not the interior solution.) Now, buoyancy varies rapidly in the boundary layer but it remains an order-one quantity throughout. To satisfy this we explicitly scale $\widetilde{W}$ in the boundary layer by writing

$$\widetilde{W}(\zeta) = \hat{\delta}^2 B_0 A(\zeta), \tag{20.74}$$

where B_0 is defined by (20.71) and A is an order-one field. The derivatives of W are

$$\frac{\partial \widehat{W}}{\partial \hat{z}} = \frac{1}{\hat{\delta}}\frac{\partial \widetilde{W}}{\partial \zeta} = \hat{\delta} B_0 \frac{\partial A}{\partial \zeta}, \qquad \frac{\partial^2 \widehat{W}}{\partial \hat{z}^2} = B_0 \frac{\partial^2 A}{\partial \zeta^2}, \tag{20.75}$$

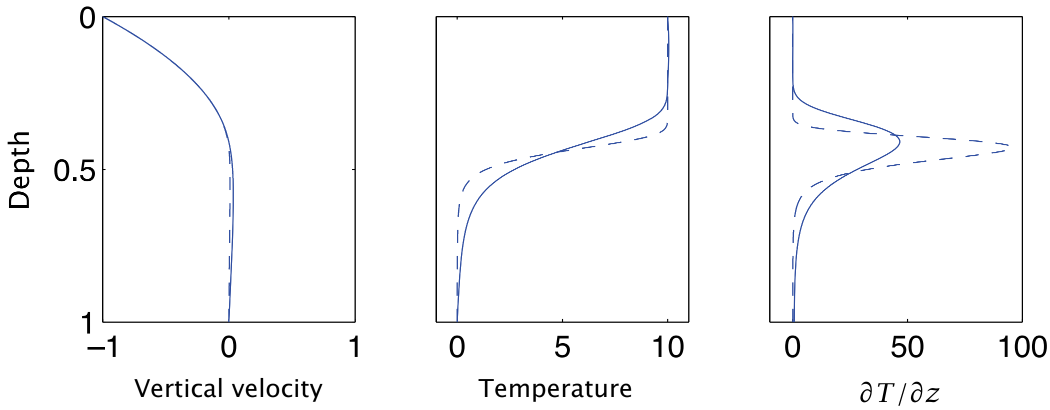

Fig. 20.12 Solution of the one-dimensional thermocline equation, (20.70), with boundary conditions (20.71), for two different values of the diffusivity: $\widehat{\kappa} = 3.2 \times 10^{-3}$ (solid line) and $\widehat{\kappa} = 0.4 \times 10^{-3}$ (dashed line), in the domain $0 \le \widehat{z} \le -1$. 'Vertical velocity' is W, 'temperature' is $-W_{\widehat{z}\widehat{z}}$, and all units are the non-dimensional ones of the equation itself. A negative vertical velocity, $\widehat{W}_E = -1$, is imposed at the surface (representing Ekman pumping) and $B_0 = 10$. The internal boundary layer thickness increases as $\widehat{\kappa}^{1/3}$, so doubling in thickness for an eight-fold increase in $\widehat{\kappa}$. The upwelling velocity also increases with $\widehat{\kappa}$ (as $\widehat{\kappa}^{2/3}$), but this is barely noticeable on the graph because the downwelling velocity, above the internal boundary layer, is much larger and almost independent of $\widehat{\kappa}$. The depth of the boundary layer increases as $\widehat{W}_E^{1/2}$, so if $\widehat{W}_E = 0$ the boundary layer is at the surface, as in Fig. 20.13.

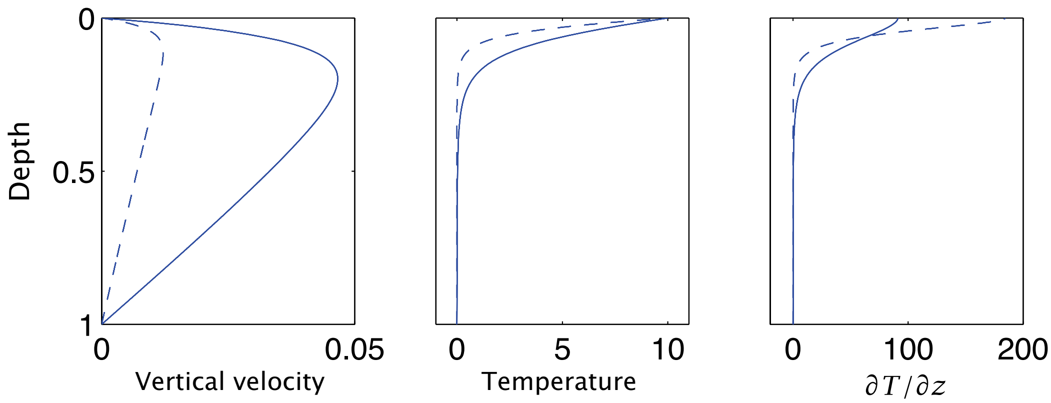

Fig. 20.13 As for Fig. 20.12, but with no imposed Ekman pumping velocity at the upper boundary ($\widehat{W}_E = 0$), again for two different values of the diffusivity: $\widehat{\kappa} = 3.2 \times 10^{-3}$ (solid line) and $\widehat{\kappa} = 0.4 \times 10^{-3}$ (dashed line). The boundary layer now forms at the upper surface. The boundary thickness again increases with diffusivity and, even more noticeably, so does the upwelling velocity — this scales as $\widehat{\kappa}^{2/3}$, and so increases fourfold for an eightfold increase in $\widehat{\kappa}$.

so that $\widehat{W}_{\widehat{z}\widehat{z}}$ is an order-one quantity. Far from the boundary layer the solution must be able to match the external conditions on temperature and velocity, (20.71); the buoyancy condition on $W_{\widehat{z}\widehat{z}}$ is satisfied if

$$A_{\zeta\zeta} \to \begin{cases} 1 & \text{as } \zeta \to +\infty \\ 0 & \text{as } \zeta \to -\infty. \end{cases} \tag{20.76}$$

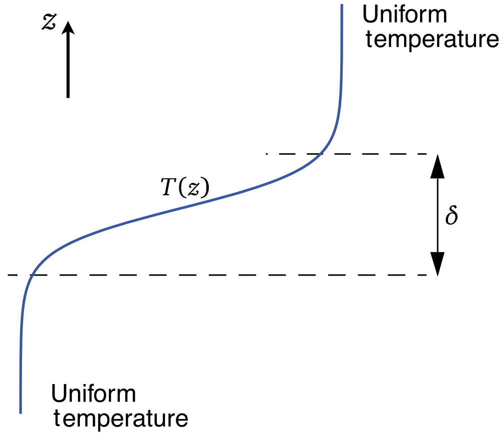

Fig. 20.14 The simplified boundary-layer structure of the internal thermocline. In the limit of small diffusivity the internal thermocline forms a boundary layer, of thickness δ in the figure, in which the temperature and buoyancy change rapidly.

On vertical velocity we require that $W \to (\hat{z}/h+1)W_E$ as $\zeta \to +\infty$, and $W \to$ constant as $\zeta \to -\infty$. The first matches the Ekman pumping velocity above the boundary layer, and the second condition produces the abyssal upwelling velocity, which as noted vanishes for $\kappa \to 0$.

Substituting (20.73) and (20.74) into (20.70) we obtain

$$B_0 A A_{\zeta\zeta\zeta} = \frac{\hat{\kappa}}{\hat{\delta}^3} A_{\zeta\zeta\zeta\zeta}. \tag{20.77}$$

Because all quantities are presumptively $\mathcal{O}(1)$, (20.77) implies that $\hat{\delta} \sim (\hat{\kappa}/B_0)^{1/3}$. We restore the dimensions of δ by using $\kappa = \hat{\kappa}(HW_S)$ and $\Delta b = B_0 L f^2 W_S/(\beta H^2)$, where Δb is the dimensional buoyancy difference across the boundary layer [note that $b = M_{zz} = (x-1)W_{zz} \sim LW_{zz} \sim LB_0 f^2 W_S/(\beta H^2)$ using (20.69)]. The dimensional boundary layer thickness, δ, is then given by

$$\delta \sim \left(\frac{\kappa f^2 L}{\Delta b\, \beta}\right)^{1/3}, \tag{20.78}$$

which is the same as the heuristic estimate (20.51). The dimensional vertical velocity scales as

$$W \sim \frac{\kappa}{\delta} \sim \kappa^{2/3} \left(\frac{\Delta b\, \beta}{f^2 L}\right)^{1/3}, \tag{20.79}$$

this being an estimate of strength of the upwelling velocity at the base of the thermocline and, more generally, the strength of the diffusively-driven component of meridional overturning circulation of the ocean.

Although the detailed properties of such one-dimensional thermocline models depend on the details of model construction, two significant features are robust.

- The thickness of the internal thermocline increases with increasing diffusivity, and decreases with increasing buoyancy difference across it, and as the diffusivity tends to zero the thickness of the internal thermocline tends to zero.
- The strength of the upwelling velocity, and hence the strength of the meridional overturning circulation, increases with increasing diffusivity and increasing buoyancy difference.

✦ *The three-dimensional equations*

We now apply boundary layer techniques to the three-dimensional M equation.[14] The main difference is that the depth of the boundary layer is now a function of x and y, so that the stretched coordinate ζ is given by

$$\hat{\delta}\zeta = z + h(x, y). \tag{20.80}$$

[The coordinates (x, y, z) in this subsection are non-dimensional, but we omit their hats to avoid too cluttered a notation.] Just as in the one-dimensional case we rescale M in the boundary layer and write

$$M = B_0\widehat{\delta}^2\widehat{A}(x, y, \zeta), \tag{20.81}$$

where the scaling factor $\widehat{\delta}^2$ again ensures that the temperature remains an order-one quantity. In the boundary layer the derivatives of M become

$$\frac{\partial M}{\partial z} = \frac{1}{\widehat{\delta}}\frac{\partial A}{\partial \zeta}, \tag{20.82}$$

and

$$\frac{\partial M}{\partial x} = \widehat{\delta}^2 B_0\left(\frac{\partial A}{\partial \zeta}\frac{\partial \zeta}{\partial x} + \frac{\partial A}{\partial x}\right) = \widehat{\delta}^2 B_0\left(\frac{\partial A}{\partial \zeta}\frac{1}{\widehat{\delta}}\frac{\partial h}{\partial x} + \frac{\partial A}{\partial x}\right). \tag{20.83}$$

Substituting these into (20.63) we obtain, omitting the time-derivative,

$$\begin{aligned}\widehat{\delta}\left[\frac{1}{f}\left(A_{\zeta x}A_{\zeta\zeta y} - A_{\zeta y}A_{\zeta\zeta x}\right) + \frac{\beta}{f^2}A_x A_{\zeta\zeta\zeta}\right] + \frac{\beta}{f^2}h_x A_\zeta A_{\zeta\zeta\zeta} \\ + \frac{1}{f}\left[h_x\left(A_{\zeta\zeta}A_{\zeta\zeta y} - A_{\zeta y}A_{\zeta\zeta\zeta}\right) + h_y\left(A_{\zeta x}A_{\zeta\zeta\zeta} - A_{\zeta\zeta}A_{\zeta\zeta x}\right)\right] &= \frac{\kappa}{B_0\widehat{\delta}^2}A_{\zeta\zeta\zeta\zeta},\end{aligned} \tag{20.84}$$

where the subscripts on A and h denote derivatives. If $h_x = h_y = 0$, that is if the base of the thermocline is flat, then (20.84) becomes

$$\frac{1}{f}\left[A_{\zeta x}A_{\zeta\zeta y} - A_{\zeta y}A_{\zeta\zeta x}\right] + \frac{\beta}{f^2}A_x A_{\zeta\zeta\zeta} = \frac{\kappa}{B_0\widehat{\delta}^3}A_{\zeta\zeta\zeta\zeta}. \tag{20.85}$$

Since all the terms in this equation are, by construction, order one, we immediately see that the non-dimensional boundary layer thickness $\widehat{\delta}$ scales as

$$\widehat{\delta} \sim \left(\frac{\kappa}{B_0}\right)^{1/3}, \tag{20.86}$$

just as in the one-dimensional model. On the other hand, if h_x and h_y are order-one quantities then the dominant balance in (20.84) is

$$\frac{1}{f}\left[h_x(A_{\zeta\zeta}A_{\zeta\zeta y} - A_{\zeta y}A_{\zeta\zeta\zeta}) + h_y(A_{\zeta x}A_{\zeta\zeta\zeta} - A_{\zeta\zeta}A_{\zeta\zeta x})\right] = \frac{\kappa}{B_0\widehat{\delta}^2}A_{\zeta\zeta\zeta\zeta} \tag{20.87}$$

and

$$\widehat{\delta} \sim \left(\frac{\kappa}{B_0}\right)^{1/2}, \tag{20.88}$$

confirming the heuristic scaling arguments. Thus, if the isotherm slopes are fixed independently of κ (for example, by the wind stress), then as $\kappa \to 0$ an internal boundary layer will form whose thickness is proportional to $\kappa^{1/2}$. We expect this to occur at the base of the main thermocline, with purely advective dynamics being dominant in the upper part of the thermocline, and determining the slope of the isotherms (i.e., the form of h_x and h_y), as in Fig. 20.11. Interestingly, the balance in the three-dimensional boundary layer equation does not in general correspond locally to $wT_z \approx \kappa T_{zz}$. Both at $\mathcal{O}(1)$ and $\mathcal{O}(\delta)$ the horizontal advective terms in (20.84) are of the same asymptotic size as the vertical advection terms. In the boundary layer the thermodynamic balance is thus $\boldsymbol{u}\cdot\nabla_z T + wT_z \approx \kappa T_{zz}$, whether the isotherms are sloping or flat. We might have anticipated this, because the vertical velocity passes through zero within the boundary layer.

What are the dynamics above the diffusive layer, presuming that it does not extend all the way to the surface? Answering this leads us into our next topic, the 'ventilated thermocline'.

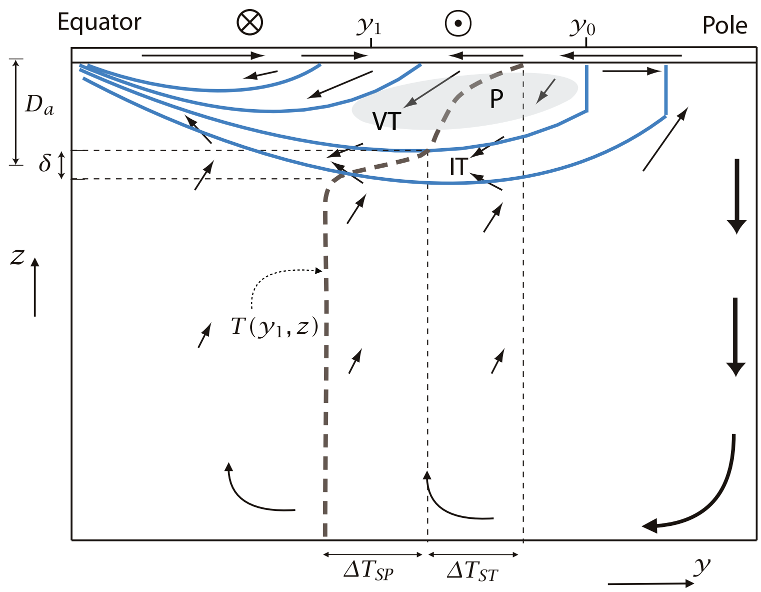

Fig. 20.15 Sketch of the large-scale circulation and structure of the main thermocline, in a single-hemisphere ocean driven by wind stress (⊙ and ⊗) and a surface temperature that decreases monotonically from equator to pole.

The thin arrows indicate the meridional overturning circulation and the flow in the Ekman layer near the ocean surface. The thick dashed line is a temperature profile, $T(y_1, z)$ at the subtropical latitude y_1, where the horizontal axis is temperature. The solid blue lines are isotherms and the homegenized western pool region is shaded grey.

Key: VT – ventilated thermocline. IT – internal thermocline. P – pool region in west. δ – thickness of the internal thermocline. Da – depth of the ventilated thermocline. ΔT_{ST} – temperature drop across subtropical gyre and across ventilated thermocline. ΔT_{SP} – temperature drop across the subpolar gyre and internal thermocline. y_0 – subtropical-subpolar gyre boundary. y_1 – a latitude in the subtropical gyre.

20.6 THE VENTILATED THERMOCLINE

We now consider the nature of the dynamics *above* the diffusive layer, presuming that the diffusivity is sufficiently small that there is a meaningful separation of the internal boundary layer and the advective dynamics above. In the advective region there is no general reason that the temperature profile should be uniform, and we envision an essentially adiabatic region that is both wind-driven and stratified. This region of the thermocline has become known, for reasons that will become apparent, as the *ventilated thermocline.* The main thermocline is composed of the internal thermocline plus a ventilated region, and to set our bearings it may be useful to refer now to the overall picture sketched in Fig. 20.15 and the shaded box on page 790.

To elucidate the structure of the ventilated thermocline we will assume the following.[15]

(i) The motion satisfies the ideal, steady, planetary-geostrophic equations.

(ii) The surface temperature, and the vertical velocity due to Ekman pumping, are given. (These surface conditions are, in reality, influenced by the ocean's dynamics, but we assume that we can calculate a solution with specified surface conditions.) At the base of the wind-influenced region we will impose $w = 0$.

(iii) Rather than use the continuously stratified equations, we will assume that the solution can be adequately represented by a small number of layers, each of constant density. The abyss is represented by a single stationary layer.

(iv) We will not take into account the possible effects of a western boundary current. In that sense the model is an extension of the Sverdrup interior of homogeneous models.

The model is thus not a complete one, yet we may hope that it is revealing about the structure of the real ocean.

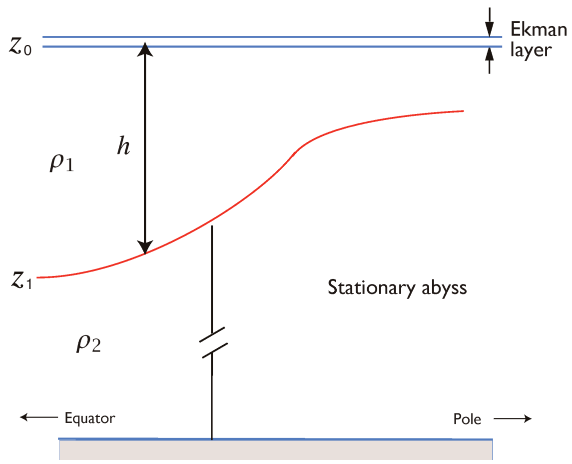

Fig. 20.16 A reduced gravity, single-layer model. A single moving layer lies above a deep, stationary layer of higher density. The upper surface is rigid. A thin Ekman layer may be envisioned to lie on top of the moving layer, providing a vertical velocity boundary condition.

20.6.1 A reduced gravity, single-layer model

The simplest possible model along these lines is to suppose the ocean is composed of just two layers, and only one moving layer, as illustrated in Fig. 20.16. The upper layer of density ρ_1 is wind-driven, whereas the lower layer of density ρ_2 is assumed to be stationary; this is called a 'one-and-a-half-layer' model or a 'reduced gravity single-layer' model. Pertinent questions are, how deep is the upper layer? What is the velocity field in it?

In the planetary-geostrophic approximation, the momentum and mass conservation equations of the reduced gravity shallow water model may be written as:

$$\boldsymbol{f} \times \boldsymbol{u} = -g' \nabla h, \qquad \nabla \cdot \boldsymbol{u} = -\frac{\partial w}{\partial z}, \tag{20.89a,b}$$

where ∇ is a two-dimensional operator (as it will be for the rest of this section) and $g' = g(\rho_2 - \rho_1)/\rho_0$ is the *reduced gravity*. Taking the curl of (20.89a) gives the geostrophic vorticity equation, $\beta v + f \nabla \cdot \boldsymbol{u} = 0$, and integrating this over the depth of the layer and using mass conservation gives

$$h\beta v = f(w_E - w_b), \tag{20.90}$$

where w_E is the velocity at the top of the layer, due mainly to Ekman pumping, and w_b is the vertical velocity at the layer base. If the flow is steady, w_b is zero for then

$$w_b = \boldsymbol{u} \cdot \nabla h = -\frac{g'}{f}\frac{\partial h}{\partial y}\frac{\partial h}{\partial x} + \frac{g'}{f}\frac{\partial h}{\partial x}\frac{\partial h}{\partial y} = 0. \tag{20.91}$$

Using this result and geostrophic balance, (20.90) becomes

$$\frac{g'}{f}\beta h \frac{\partial h}{\partial x} = f w_E \tag{20.92}$$

which integrates to

$$\boxed{h^2 = -2\frac{f^2}{g'\beta}\int_x^{x_e} w_E \, \mathrm{d}x' + H_e^2,} \tag{20.93}$$

where H_e is the (unknown) value of h at the eastern boundary x_e, and it is a constant to satisfy the no-normal flow condition. This apart, the equation contains complete information about the solution. We note that:

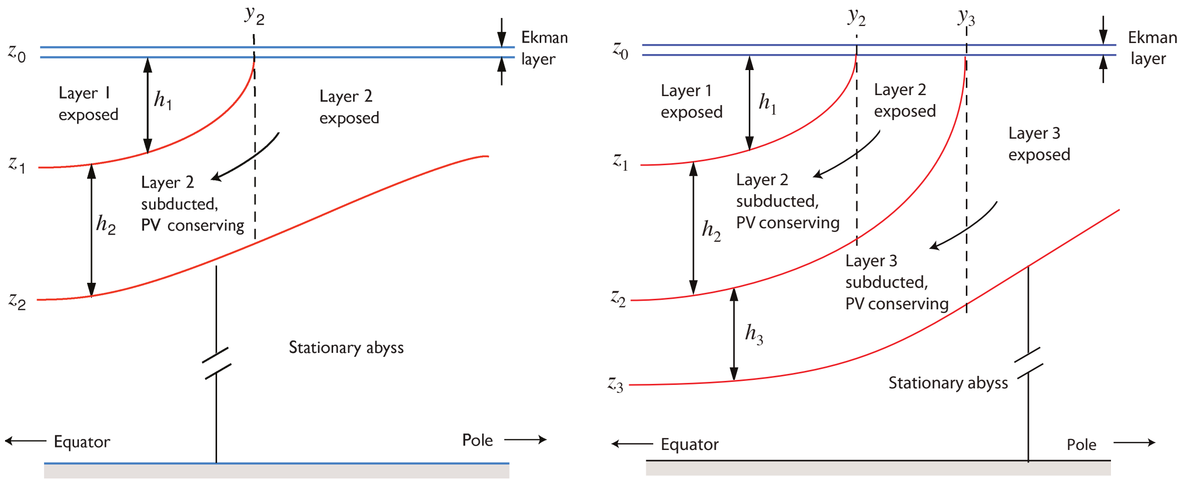

Fig. 20.17 Two-layer (left) and three-layer (right) schematics of the ventilated thermocline, each with a stationary abyss. Models with still more moving layers may be constructed, at least in principle, by extension.

- the depth of the moving layer scales as the magnitude of the wind stress (or Ekman pumping velocity) to the one-half power;
- the horizontal solution is similar to the simpler Sverdrup interior solution previously obtained in section 19.1.3;
- there is no solution if w_E is positive; that is, if there is Ekman upwelling;
- the solution depends on the unknown parameter H_e, the layer depth at the eastern boundary. (That the eastern boundary depth is undetermined is perhaps the main incomplete aspect of the theory as presented here.[16])

20.6.2 Two-layer and three-layer models

Imagine now there are two moving layers above a stationary abyss, as in Fig. 20.17. If there is a meridional buoyancy gradient at the surface then isopycnals *outcrop*, or intersect the surface. Thus, at some latitude (say $y = y_2$, which for simplicity we assume not to be a function of longitude) layer 2 passes underneath layer 1, which is of lower density, as sketched in Fig. 20.17.

1 Thus, polewards of y_2 the dynamics are just those of a single layer discussed above, whereas equatorwards of y_2 layer 2 does not feel the wind directly, and its dynamics are governed by two principles.

(i) Sverdrup balance. This still applies to the vertically integrated motion, and thus to the sum of layer 1 and layer 2.

(ii) Conservation of potential vorticity. The motion in layer 2 is shielded from the wind forcing, and the effects of dissipation are assumed to be negligible. Thus, the fluid parcels in the layer will conserve their potential vorticity.

We first use potential vorticity conservation to obtain an expression for the depths of each layer in terms of the total depth of the moving fluid, h, and then use Sverdrup balance to obtain h.

We can apply exactly the same procedure if there are three (or more) moving layers, as in the right-hand panel of Fig. 20.17, and in principle go to the limit of cotinuous stratification. However, the algrebra becomes considerably more complicated and most of the essential dynamics are contained in two layers, so that is our focus.

Potential vorticity conservation

Conservation of potential vorticity in the region equatorward of y_2 is, for steady flow,

$$\boldsymbol{u}_2 \cdot \nabla q_2 = 0 \qquad \text{for} \quad y < y_2, \tag{20.94}$$

where $q_2 = f/h_2$. Now, the velocity field in layer 2 is given by $\boldsymbol{u}_2 = (g_2'/f)\mathbf{k}\times\nabla h$, where $h = h_1 + h_2$ is the total depth of the moving fluid (see the appendix to this chapter). Thus, (20.94) becomes

$$-\frac{g_2'}{f}\frac{\partial h}{\partial y}\frac{\partial}{\partial x}\left(\frac{f}{h_2}\right) + \frac{g_2'}{f}\frac{\partial h}{\partial x}\frac{\partial}{\partial y}\left(\frac{f}{h_2}\right) = \frac{g_2'}{f}J\left(\frac{f}{h_2}, h\right) = 0. \tag{20.95}$$

This is an equation relating h and h_2 and it has the general solution

$$q_2 \equiv \frac{f}{h_2} = G_2(h), \tag{20.96}$$

where G_2 is an *arbitrary* function of its argument. However, we *know* what the potential vorticity of layer 2 is at the moment it is subducted; it is just

$$q_2(y_2) \equiv \frac{f(y_2)}{h_2} = \frac{f_2}{h}, \tag{20.97}$$

where $f_2 \equiv f(y_2)$, and $h_2 = h$ because $h_1 = 0$. This relationship must therefore hold everywhere in layer 2, equatorwards of y_2; that is,

$$G_2(h) = \frac{f_2}{h} \tag{20.98}$$

Thus, in the subducted region, and taking $z_0 = 0$,

$$\frac{f}{h_2} = \frac{f_2}{h}, \qquad \text{or} \qquad \frac{f}{z_1 - z_2} = -\frac{f_2}{z_2}. \tag{20.99}$$

From this we easily obtain expressions for the depths of each layer as functions of the total depth, h, namely

$$h_2 = z_1 - z_2 = \frac{f}{f_2}h \qquad \text{and} \qquad h_1 = -z_1 = \left(1 - \frac{f}{f_2}\right)h. \tag{20.100}$$

It remains only to find an expression for the total depth of the moving fluid, h, and this we do using Sverdrup balance. Note that because potential vorticity, f/h_2, is conserved, as the subducted fluid column moves equatorwards its thickness must decrease.

Using Sverdrup balance to find the total depth

Equations (20.100) contain the unknown total depth h, and we now use Sverdrup balance to find this and close the problem. The linear vorticity equation is $\beta v = f\partial w/\partial z$, where the velocity at the top of layer one is that due to Ekman layer and the velocity at the base of layer two is zero. Given this, we may write the Sverdrup balance as

$$\beta(h_1 v_1 + h_2 v_2) = f w_E \tag{20.101}$$

where, using (20.129), the velocities in each layer are given by

$$f v_1 = \frac{\partial}{\partial x}(g_2' h + g_1' h_1) \qquad \text{and} \qquad f v_2 = \frac{\partial}{\partial x}(g_2' h). \tag{20.102}$$

Thermocline Dynamics — an Overview

The model of the main thermocline that we have constructed in sections 20.3–20.6 is illustrated schematically in Fig. 20.15. Some of the features, and limitations, of this model are listed below.

- The main subtropical thermocline consists of an advective upper region overlying a diffusive base.
  - The diffusive base forms the *internal thermocline,* and in the limit of small diffusivity this is an internal boundary layer. The advective region forms the *ventilated thermocline.* The separation of the two regions may, in reality, not be sharp.
  - The relative thicknesses of these layers is a function of various parameters, notably the strength of the wind and the magnitude of the diffusivity.

- Above the ventilated thermocline there may be a mixed layer with a seasonally varying depth. In certain regions, for example at the poleward edge of the subtropical gyre, convection may deepen the mixed layer as far as the base of the thermocline.

- In the thermocline theories we have presented there is no explicit western boundary layer. Such a boundary layer is needed to close the circulation and the heat budget.

- The single-hemisphere model assumes that the water that sinks at high latitude either upwells through the main thermocline or returns to the subpolar gyre beneath the main thermocline. In reality some of this water may cross into the other hemisphere before upwelling — more so in the Atlantic than Pacific.
  - In this case, the diffusion-dependent overturning circulation represents only part of the overall meridional overturning circulation.
  - Nevertheless, there would remain a diffusive internal thermocline (and a ventilated thermocline above it) because there is still a boundary between the warm subtropical water and cold abyssal water.

- Within the ventilated thermocline there are two regions — the shadow zone and the western pool — whose dynamics are not determined without additional assumptions. Plausible assumption for the western pool are:
  - All the water within it is ventilated, leading to a model of mode water.
  - The potential vorticity within the pool is homogenized through the action of mesoscale eddies.

  Some combination of these might also apply. The size of the pool region increases as the poleward boundary of the pool region approaches the latitude of the outcrop (Fig. 20.21) and can extend almost across the entire gyre.

- Topics of research include questions of how potential vorticity homogenization is affected by a western boundary current, if and how the non-passive nature of potential vorticity affects the pool region, and if and how all these concepts apply in the real ocean.

Using these, Sverdrup balance becomes

$$\beta h_1 \frac{\partial}{\partial x}(g_2' h + g_1' h_1) + \beta (h - h_1) g_2' \frac{\partial h}{\partial x} = f^2 w_E, \tag{20.103}$$

or

$$\frac{\partial}{\partial x}\left(g_2' h^2 + g_1' h_1^2\right) = \frac{2f^2}{\beta} w_E. \tag{20.104}$$

On integrating, the above equation becomes

$$\left(h^2 + \frac{g_1'}{g_2'} h_1^2\right) = D_0^2 + C, \qquad \text{where} \qquad D_0^2(x,y) = -\frac{2f^2}{\beta g_2'} \int_x^{x_e} w_E(x', y)\, \mathrm{d}x', \tag{20.105a,b}$$

which by construction vanishes at the eastern wall ($x = x_e$). The constant of integration C may be interpreted as follows. Let us write $C = H_e^2 + (g_1'/g_2')H_1^2$ where H_e is the (unknown) total depth of layers 1 and 2 at the eastern boundary, and H_1 is the depth of layer 1. These must both be constants in order to satisfy the no-normal flow condition. However, H_1 must be zero, because at the outcrop line $h_1 = 0$. Thus, H_1 is zero at $y = y_2$, and therefore zero everywhere, and $C = H_e^2$.

Using (20.100) and (20.105) we obtain a closed expression for h, namely

$$h = -z_2 = \frac{(D_0^2 + H_e^2)^{1/2}}{\left[1 + (g_1'/g_2')(1 - f/f_2)^2\right]^{1/2}}. \tag{20.106}$$

Using (20.100) the depths in each layer, and the corresponding geostrophic velocities, can readily be obtained.

A typical solution is shown in Fig. 20.18. The upper layer exists only equatorwards of the outcrop latitude, $y_2 = 0.8$, and isolines of total thickness correspond to streamlines of the lower layer. We see, as expected, the overall shape of a subtropical gyre, with the circulation being closed by an implicit western boundary current that is not part of the calculation. Two regions are shaded in the figure, the 'pool' region in the west and the 'shadow zone' in the south-east. The solutions above do not apply to these, and they require some special attention.

20.6.3 The shadow zone

In the fluid interior the potential vorticity of a parcel in layer 2 is determined by tracing its trajectory back to its outcrop latitude where the potential vorticity is given. That trajectory is determined by its velocity, and this turn is determined by inverting the potential vorticity. Now, parcels subducted at y_2 sweep equatorward and westward, so that a parcel, labelled '$\boldsymbol{a}$' say, subducted at the eastern boundary will in general leave the eastern boundary tracing a southwestern trajectory. Consider another parcel, '$\boldsymbol{b}$' say, in the interior that lies eastwards of the subducted position of $\boldsymbol{a}$, in the shaded region of Fig. 20.19. It is impossible to trace $\boldsymbol{b}$ back to the outcrop line without trajectories crossing, and this is forbidden in steady flow. Rather, it seems as if the trajectory of $\boldsymbol{b}$ would emanate from the eastern wall. What is the potential vorticity there?

At the eastern boundary the condition of no normal flow at the boundary demands that h be constant (so that $u_2 = 0$), and h_1 be constant (so that $u_1 = 0$). But if a parcel in layer 2 moves *along* the boundary potential vorticity conservation demands that f/h_2 is constant, and therefore h_2 must change, contradicting the no-normal flow requirement. Thus, the velocity at the boundary can have neither a normal component nor a tangential component, and so we cannot trace parcels in the shaded region back to the wall. Rather, in the absence of closed trajectories (for example, eddying motion), we may assume that the shaded region is stagnant, and h is constant. Of course, potential vorticity is everywhere given by f/h_2, which varies spatially, but since there is no motion

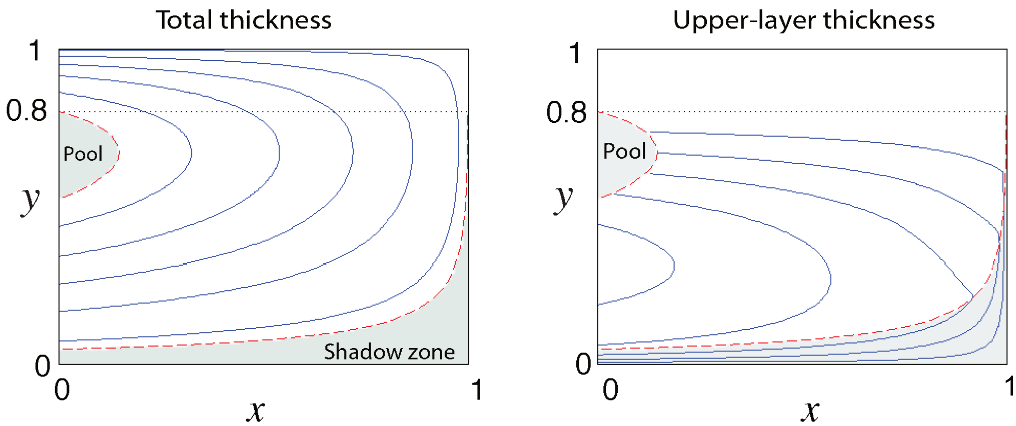

Fig. 20.18 Contour plots of total thickness and upper layer thickness in a two-layer model of the ventilated thermocline. The thickness generally increases westwards, and the flow is clockwise. The shadow zone and the western pool are shaded, and no contours are drawn in the latter. The outcrop latitude, $y_2 = 0.8$, is marked with a dotted line. [Parameters are $g_1' = g_2' = 1$, $\beta = 1$, $f_0 = 0.5$, $H_e = 0.5$, and $w_E = -\sin(\pi y)$.]

potential vorticity is still, rather trivially, conserved along trajectories. This region is aptly called the *shadow zone,* since the region falls under the shadow of the eastern boundary; an analogous region arose in the quasi-geostrophic discussion of section 20.1.

To obtain an expression for the fields within the shadow zone, first note that because h is constant, its value is equal to that on the eastern wall; that is, $h = H_e$. The wind forcing must then all be taken up by the upper layer, and Sverdrup balance then implies

$$\beta v_1 h_1 = f w_E, \tag{20.107}$$

and using (20.102) we obtain an expression for h_1, to wit

$$h_1^2 = -\frac{2f^2}{\beta g_1'} \int_x^{x_e} w_E(x', y)\,\mathrm{d}x' = \frac{g_2'}{g_1'} D_0^2, \tag{20.108}$$

which is zero at the eastern wall. In the lower layer the thickness is just $h_2 = H_e - h_1$. The boundary of the shadow zone is given by the trajectory of a fluid parcel in layer 2 that emanates from the eastern boundary at the outcrop line where $h_1 = 0$ and $h = h_2 = H_e$. Since the flow is steady, the trajectory is an isoline of h. Thus, from (20.106) we have

$$h^2 = \frac{(D_0^2(x_s, y_s) + H_e^2)}{[1 + g_1'/g_2'(1 - f/f_2)^2]} = H_e^2, \tag{20.109}$$

where (x_s, y_s) denotes the boundary of the shadow zone. (Note that $x_s = x_e$ at $y = y_2$.) The above equation yields

$$D_0^2(x_s, y_s) = H_e^2 \left[\frac{g_1'}{g_2'} \left(1 - \frac{f}{f_2} \right)^2 \right], \tag{20.110}$$

which, given the wind stress, determines the shadow zone boundary x_s as a function of y.

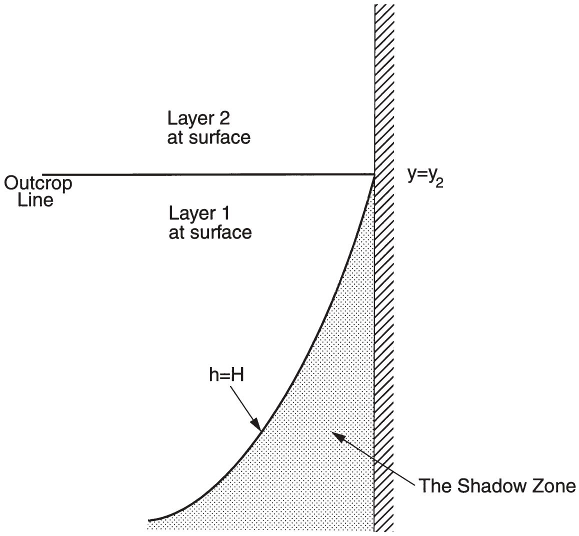

Fig. 20.19 The shadow zone in the ventilated thermocline. Layer 2 outcrops at $y = y_2$. A column moving equatorward along the eastern boundary in layer 2 is subducted at y_2.

The column cannot remain against the eastern wall and both preserve its potential vorticity, which implies the column shrinks, at the same time that the no-normal flow condition is satisfied, as by geostrophy this implies the layer depth is constant. Thus, the column must move westward, along the boundary of a 'shadow zone' within which there is no motion. The streamline it follows is the of constant total thickness of the two moving layers — see (20.129) or (20.131c).

20.6.4 † The western pool

Polewards of the outcrop latitude the fluid of layer 2 feels the wind directly and the layer thickness is determined by Sverdrup balance. Equatorwards of the outcrop latitude the properties of this layer are determined by potential vorticity conservation, with the potential vorticity being determined by the layer thickness at the outcrop. However, just as there is a region in the east where trajectories cannot be traced back to the outcrop, there is a 'pool' region in the west that is bounded by the trajectory that emerges from the western boundary at the outcrop latitude. Within the pool, trajectories cannot be traced back to the outcrop (Fig. 20.18), and one might suppose that they emerge from the western boundary current. There are two plausible hypotheses for determining the layer depths within this region.

(i) Within layer 2, potential vorticity is homogenized.

(ii) Because there is no source for layer-2 water, layer-2 water does not exist and the pool consists solely of ventilated, layer-1 water.

We may remark that neither of the above can be derived from the governing equations of motion without making additional physical assumptions that are neither a priori true nor obvious. We discuss both hypotheses briefly below, followed by a more general discussion of how the pool region fits together with the earlier discussions about potential vorticity homogenization in the quasi-geostrophic equations.

(i) Potential vorticity homogenization

The pool region is a region of recirculation, receiving water from and depositing water into the western boundary current. Thus, following the ideas described in chapter 13 and employed in section 20.1, we hypothesize that the potential vorticity within this region becomes homogenized. The value of potential vorticity within the pool is just the value of potential vorticity at its boundary, and this is given by $f_2/h_2(w)$, where $h_2(w)$ is the thickness of layer 2 at the western boundary at the outcrop latitude. This is given using (20.93) with $f = f_2$ and $g' = g'_2$, and thus the potential vorticity in the pool is given by

$$q_{pool} = \frac{f_2}{D_w^2 + H_e^2}, \qquad (20.111)$$

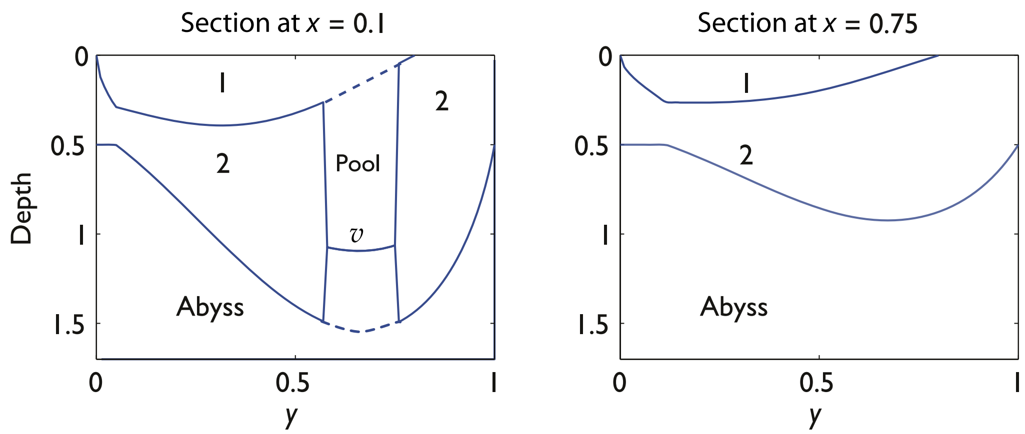

Fig. 20.20 Two north-south section of layer thickness, at different longitudes, from the same solution as Fig. 20.18. The numbers refer to the fluid layer. The section on the left passes through the western pool region. In the homogenized PV model of the pool, the dashed lines should be solid and the line labelled v should be removed. In the ventilated pool model, the dashed lines should be removed. The region near $y = 0$ in both plots where the total depth of the thermocline is constant is the shadow zone.

where $D_w^2 = -2(f_2^2/g_2'\beta)\int_{x_w}^{x_e} w_E(x', y_2)\,\mathrm{d}x'$. The thickness of layer 2 in the pool must be consistent with this, and so is given by

$$h_2 = \frac{f}{q_{pool}}. \tag{20.112}$$

The thickness of layer 1 is determined by using Sverdrup balance, (20.101), which, given h_2 and geostrophy, reduces to an equation for h_1.

The extent of the pool region is dependent upon the outcrop latitude of the moving layer, since the boundary of the pool is a thickness contour. As the outcrop latitude moves poleward toward the gyre boundary then the pool region expands, as seen in Fig. 20.21. By the same token, if we have more moving layers then, since the deeper layers outcrop further poleward (Fig. 20.17) those deeper layers will have a more extensive pool region. That is, the pool expands with depth and the layer that outcrops just equatorward of the gyre boundary may have a pool region reaching across the gyre, as illustrated in Fig. 20.22, which shows the pool boundaries in a calculation (not shown) with a three-layer model.

(ii) The ventilated pool

'I came for the waters.' 'What waters?' 'I was misinformed.'
From *Casablanca* (1942).

The homogenization hypothesis, although entirely plausible, depends on the assumption of down-gradient diffusion of potential vorticity by eddies. Also, because there is no source of layer-2 water in the pool, we must suppose that it is ventilated by eddy pathways that meander down from the surface. An alternative hypothesis, and one that does not rely on the properties of mesoscale eddies, is to suppose that the western pool is filled with water that is directly ventilated from the surface. That is, if there is no surface source for a water mass, we simply suppose that that water mass does not exist. In the two-layer model, this means that the western pool is filled entirely with layer-1 fluid. If a non-ventilated (e.g., layer-2) fluid is present initially, then we hypothesize that it is slowly expunged by the continuous downwards Ekman pumping of layer-1 water into the pool.[17]

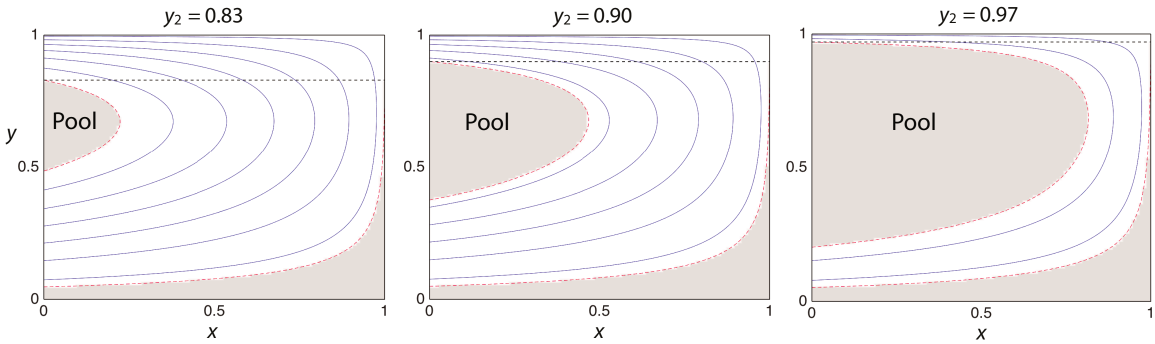

Fig. 20.21 Contour plots of total thickness in a two-layer model of the ventilated thermocline, with parameters as in Fig. 20.18, except for the outcrop latitude, y_2, that takes values as indicated. The pool and the shadow zone are shaded and contours within are not shown. The pool expands considerably as the outcrop latitude approaches the gyre boundary at $y = 1$.

Because the layer-2 fluid is absent, the layer-1 fluid extends all the way down to the stagnant abyss; it takes up all the Sverdrup transport, and this determines the depth of the ventilated pool. Thus, rounding up the usual equations, we set $h = h_1$ in (20.105) to give

$$h_1^2 = D_1^2 + g_2' H_e^2, \tag{20.113}$$

where

$$D_1^2(x, y) = -\frac{2f^2}{\beta g_{1a}'} \int_x^{x_e} w_E(x', y)\,\mathrm{d}x', \tag{20.114}$$

with $g_{1a}' = g_1' + g_2'$ being the reduced gravity between layer 1 and the abyss, and H_e, as before, being the thickness of layer 2 at the eastern boundary. Because $g_{1a}' > g_2'$ this pool will generally be shallower than the total depth of moving fluid $(h_1 + h_2)$ just outside, but the depth of layer-1 fluid alone will be much greater; that is, there will be discontinuities in layer depths at the pool boundary. A section through the pool region is shown in Fig. 20.20. The figure also shows the configuration of the pool if the homogeized potential vorticity hypothesis is used.

Although we may be shocked by the appearance of discontinuities in layer depths in a fluid model, the model does provide a simple mechanism for the appearance of *mode water*. This is a distinct mass of weakly stratified, low potential vorticity water appearing in the north-west corner of the North Atlantic subtropical gyre (where it is sometimes called '18 degree water'), with analogues in the other gyres of the world's oceans; it is so-called because it appears as a distinct mode in a census of water properties. The proximate mechanism for mode water formation is convection in winter, but for such convection to occur the large-scale ocean circulation must maintain a weakly stratified region, and it is the ventilated pool that enables this, and sets the formation in the context of thermocline structure. In reality, the vertical isopycnals predicted by the simple model will be highly baroclinically unstable, and the ensuing mesoscale eddies will erode the pool interface and cause the isopycnals to slump, so that the discontinuities in layer depths will be manifest only as rapid changes or fronts.

Observations show that mode water exists only over a small region in the northestern corner of the subtropical gyre, as the pool region in Fig. 20.18 might suggest. Regions of potential vorticty homogenization can extend much further, as in Fig. 20.4, with a similar degree of homogenization in the Atlantic. It may be that the effects of the wind are only able to omogenize the water masses in the upper regions of the pool. At greater depths the effects of baroclinic instabiilty and the consequent mesoscale eddies dominate the wind effects and produced regions of homogenized potential vorticity of greater extent.

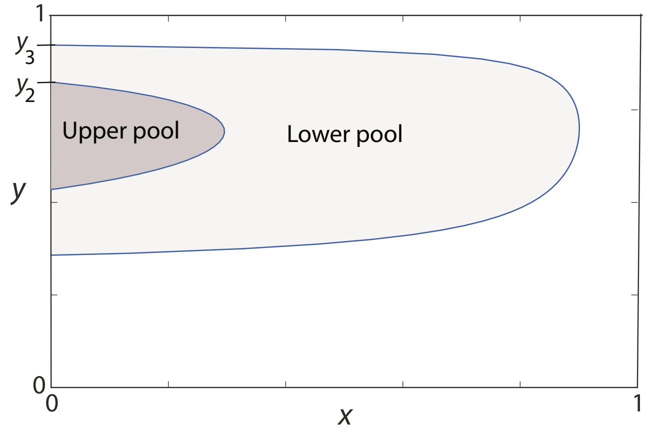

Fig. 20.22 The two pool regions in a calculation of the ventilated thermocline with three moving layers. The upper pool refers to the western pool in the layer 2, which is the upper subducted layer that outcrops at y_2, and the lower pool is that in layer 3, which outcrops at y_3 (see right-hand panel of Fig. 20.17).

20.6.5 † Remarks on thermocline structure

In the first two sections of this chapter we discussed a quasi-geostrophic, wind-driven view of the upper thermocline and found potentially large regions in which the flow re-circulated and potential vorticity was homogenized. One might call this an *elliptic* view of the upper ocean circulation, with the value of potential vorticity being determined by an elliptic boundary value problem with some of the characteristics of a diffusion problem. Following this we discussed the thermocline in a rather different way: in the ventilated thermocline the potential vorticity is subducted in from the surface, more along the lines of a *hyperbolic* problem, or a problem in characteristics, with an unavoidably *diffusive* internal boundary layer below connecting the thermocline to the abyss.

These are not competing theories, rather, these three regions (elliptic, hyperbolic, and diffusive, representing the regions of potential vorticity homogenization, the ventilated thermocline, and the internal thermocline, respectively) can co-exist in the upper ocean, as sketched in Fig. 20.15. The theory presented above suggests how they fit together, but cannot provide a truly quantitative comparison with observations. Depending on the value of the diapycnal diffusivity, the diffusive and internal thermoclines may be almost indistinguishable, observationally if not dynamically, and, depending on the stratification and wind structure the regions of potential vorticity homogenization may be larger or smaller.

Finally we remark that potential vorticity is not a passive scalar, which means that the boundary of the pool region will be affected by the homogenization process. The homogenization process may entrain additional water into the pool, which then grows and may prevents subducted water from entering into it. However, the picture of if and how the homogenization of potential vorticity precisely affects the structure of the ventilated thermocline is a little murky and transparency will also require high resolution numerical simulations and more complete observations to guide us to the truth.

20.A APPENDIX: MISCELLANEOUS RELATIONSHIPS IN A LAYERED MODEL

Here we collect various expressions relating pressure, density and velocity in a geostrophic and Boussinesq layered model. The layers and the interfaces are numbered, increasing downwards, as in Fig. 20.23, and the bottom layer is stationary.

20.A.1 Hydrostatic balance

Hydrostatic balance is $\partial p/\partial z = -\rho g$. We can integrate this to give, in layers n and $n-1$,

$$p_n = -\rho_n g z + p'_n(x, y), \qquad p_{n-1} = -\rho_{n-1} g z + p'_{n-1}(x, y). \tag{20.115}$$

Since pressure is continuous, at $z = z_{n-1}$ these two expressions are equal so that

$$-\rho_n g z_{n-1} + p'_n = p_{n-1} = -\rho_{n-1} g z_{n-1} + p'_{n-1} \tag{20.116}$$

whence

$$g'_{n-1} z_{n-1} = \frac{1}{\rho_0}(p'_n - p'_{n-1}) \tag{20.117}$$

where $g'_n = g(\rho_{n+1} - \rho_n)/\rho_0$ is the *reduced gravity*, and ρ_0 is the constant, reference, value of the density used in the Boussinesq approximation, taken to be equal to ρ_1.

20.A.2 Geostrophic and thermal wind balance

In the Boussinesq approximation, geostrophic balance is:

$$\rho_0 f \boldsymbol{u}_n = \mathbf{k} \times \nabla p_n. \tag{20.118}$$

Using (20.118) with (20.117) gives

$$\boldsymbol{u}_{n+1} - \boldsymbol{u}_n = \frac{g'_n}{f} \mathbf{k} \times \nabla z_n, \tag{20.119}$$

which is the appropriate form of thermal wind balance for this system. Let us suppose that at sufficient depth there is no motion, and in particular that layer $N+1$ is stationary and contains no pressure gradients. That is, $p'_{N+1} = 0$ and so, from (20.116) and (20.118)

$$p'_N = -(\rho_{N+1} - \rho_N) g z_N = -g'_N \rho_0 z_N. \tag{20.120}$$

Integrating upwards we obtain the pressure in each layer,

$$p'_n = -\rho_0 \sum_{i=n}^{i=N} g'_i z_i \tag{20.121}$$

where $n \leq N$. Thus, the geostrophic velocities in each layer are given by

$$f \boldsymbol{u}_n = -\mathbf{k} \times \nabla \left(\sum_{i=n}^{i=N} g'_i z_i \right). \tag{20.122}$$

The quantity in brackets on the right-hand side is not a velocity streamfunction because it is $f\boldsymbol{u}_n$, and not $\boldsymbol{u}_n$ that is given by its curl. Nevertheless, the velocity is normal to its gradient, and therefore its isolines define streamlines.

The upper surface of the ocean is assumed to be fixed; this is the 'rigid-lid' approximation. Thus, $z_0 = 0$ and $h_1 = -z_1$. More generally, the layer thicknesses and the interfaces between the layers are related by

$$z_n = -\sum_{i=M}^{i=n} h_i \tag{20.123}$$

where M is the index of the uppermost layer, and $n \geq M$. If there is no outcropping, then $M = 1$.

The geostrophic velocity in the lowest moving layer is given by

$$f \boldsymbol{u}_N = -g'_N \mathbf{k} \times \nabla z_N = g'_N \mathbf{k} \times \nabla h. \tag{20.124}$$

This means that lines of constant depth of the lowest layer are also streamlines; the velocity moves parallel to the depth contours. The vertical velocity at the base of the lowest layer is given by, for steady flow,

$$w(z = -h) = \boldsymbol{u}_N \cdot \nabla h = \frac{1}{f} g'_N (\mathbf{k} \times \nabla h) \cdot \nabla h = 0. \tag{20.125}$$

That is, there is no vertical motion at the base of the moving layers.

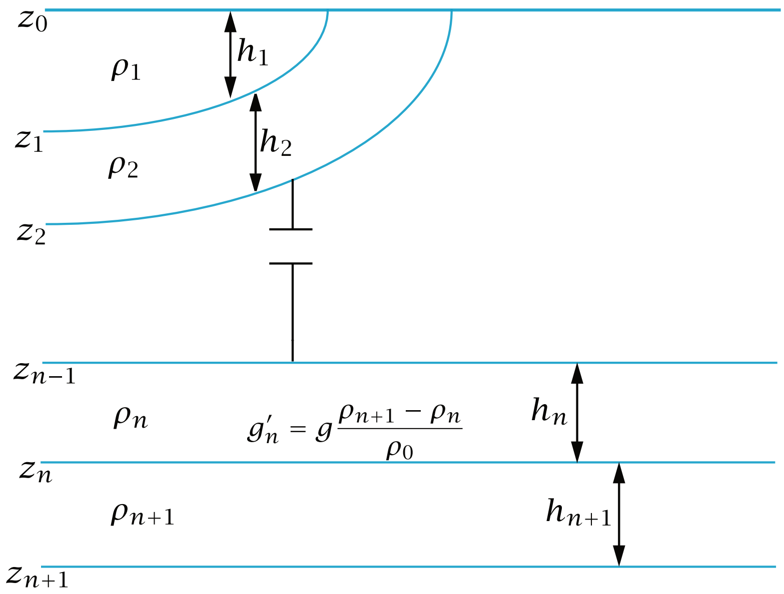

Fig. 20.23 Structure and notational conventions used for a multi-layered model.

20.A.3 Explicit cases

A one-layer reduced-gravity model

The perturbation pressure in the moving layer (layer 1) is

$$p'_1 = -\rho_0 g'_1 z_1 = \rho_0 g'_1 h_1. \tag{20.126}$$

The geostrophic velocity is given by

$$f\boldsymbol{u}_1 = \frac{1}{\rho_0}\mathbf{k}\times\nabla p'_1 = g'_1\mathbf{k}\times\nabla h_1. \tag{20.127}$$

(In a single-layer model, the subscripts are often omitted.)

A two-layer model

The perturbation pressure in the upper and lower moving layers are given by

$$p'_1 = -\rho_0(g'_2 z_2 + g'_1 z_1) = \rho_0(g'_2 h + g'_1 h_1), \tag{20.128a}$$
$$p'_2 = -\rho_0 g'_2 z_2 = \rho_0 g'_2(h_1 + h_2) = \rho_0 g'_2 h, \tag{20.128b}$$

[cf. (20.120)], where $h = h_1 + h_2 = -z_2$.

The corresponding geostrophic velocities are

$$f\boldsymbol{u}_1 = \mathbf{k}\times\nabla(g'_2 h + g'_1 h_1), \tag{20.129a}$$
$$f\boldsymbol{u}_2 = \mathbf{k}\times\nabla(g'_2 h). \tag{20.129b}$$

A three-layer model

The perturbation pressures in the three moving layers are

$$p_1 = -\rho_0[g'_3 z_3 + g'_2 z_2 + g'_1 z_1] = \rho_0[g'_3 h + g'_2(h_1 + h_2) + g'_1 h_1], \tag{20.130a}$$
$$p_2 = -\rho_0[g'_2 z_2 + g'_3 z_3] = \rho_0[g'_2(h_1 + h_2) + g'_3 h], \tag{20.130b}$$
$$p_3 = -\rho_0 g'_3 z_3 = \rho_0 g'_3 h, \tag{20.130c}$$

where $h = h_1 + h_2 + h_3 = -z_3$. The corresponding geostrophic velocities are:

$$f\boldsymbol{u}_1 = \mathbf{k} \times \nabla[g'_3 h + g'_2(h_1 + h_2) + g'_1 h_1] \tag{20.131a}$$
$$f\boldsymbol{u}_2 = \mathbf{k} \times \nabla[g'_2(h_1 + h_2) + g'_3 h] \tag{20.131b}$$
$$f\boldsymbol{u}_3 = \mathbf{k} \times \nabla[g'_3 h]. \tag{20.131c}$$

Notes

1 Drawing from Rhines & Young (1982b). Young & Rhines (1982) also considered the problem of a western boundary layer.

2 Indian Ocean results are from McCarthy & Talley (1999). Pacific plots are courtesy of L. Talley. The data is mainly from CTD profiles and bottle casts. The potential density is σ_0 in the upper ocean plots, and σ_2 for the deep ocean.

3 See Rhines & Young (1982a) and Holland *et al.* (1984) for a numerical example of PV homogenization.

4 I am very grateful to Peter Rhines for constructing figures 20.5, 20.6 and 20.7 from MIMOC data, and for his insightful interpretations. The MIMOC climatology is decribed at http://www.pmel.noaa.gov/mimoc/ and by Schmidtko *et al.* (2013). The date comes mainly from Argo CTDs, supplemented by shipboard and ice-tethered profiler CTDs, and is put onto a 0.5° grid.

5 See Keffer (1985), Talley (1988), Lozier *et al.* (1996) and McCarthy & Talley (1999) for earlier observations of potential vorticity.

6 Sections courtesy of L. Talley and the WOCE hydrographic atlas. Approximate-neutral density, or more commonly just neutral density, is a quantity such that the buoyancy force is locally perpendicular to its iso-surfaces, so that a parcel that is displaced adiabatically along a neutral density iso-surface will remain neutrally buoyant (Jackett & McDougall 1997, Eden & Willebrand 2008, de Szoeke 2000, Nycander 2011). Continuous, global neutral-density surfaces cannot be constructed exactly because of the thermobaric term in the equation of state for seawater and a parcel will not necessarily return to its level of departure when displaced to the same (x, y) position. The concept is nonetheless useful and for the purposes of this figure neutral density is potential density.

7 See, among others, Toole *et al.* (1994), Polzin *et al.* (1997), Gregg (1998) and Ledwell *et al.* (1998).

8 For estimates of the strength of the overturning circulation in the ocean, and its relation to diapycnal diffusivity and the observed stratification, see Munk (1966), revisited by Munk & Wunsch (1998). See also Huang (1999) and, for a review, Wunsch & Ferrari (2004). If the abyssal flow is along rather than across isopycnals, smaller values of diffusivity suffice to maintain deep stratification. See section 21.6.

9 Vallis (2000).

10 The modern development of the theory of the thermocline began with two back-to-back papers in 1959 in the journal *Tellus*. Welander (1959) suggested an adiabatic model, based on the ideal-fluid thermocline equations (i.e., the planetary-geostrophic equations, with no diffusion terms in the buoyancy equation), whereas Robinson & Stommel (1959) proposed a model that is intrinsically diffusive. In this model [developed further by Stommel & Webster (1963), Salmon (1990), and others] the thermocline is an internal boundary layer or front that forms at the convergence of two different homogeneous water types, warm surface fluid above and cold abyssal fluid below. Meanwhile, the adiabatic model continued its own development (see Veronis 1969), culminating in the ventilated thermocline model of Luyten *et al.* (1983) and its continuous extensions (Killworth 1987, Huang 1986). Signs that the two classes of theory might not be wholly incompatible came from Welander (1971b) and Colin de Verdière (1989) who noted that the diffusion might become important below an adiabatic near-surface flow, and Samelson & Vallis (1997) eventually suggested a model in which the upper thermocline is adiabatic, as in the ventilated thermocline model, but

has a diffusive base, constituting an internal boundary layer. Mesoscale eddies play a role in homogenizing potential vorticity in the pool regions (e.g. Henning & Vallis 2004, Cessi & Fantini 2004, Maze & Marshall 2011). Some extensions of the ventilated-style of model to the subpolar gyre are provided by Bell (2015b,a).

11 Welander (1971a).

12 Drawing from Salmon (1990).

13 Newton's method is an iterative way to numerically solve certain types of differential equations. The solutions here are obtained using about 1000 uniformly spaced grid points to span the domain, taking just a few seconds of computer time. Because of the boundary layer structure of the solutions employing a non-uniform grid would be even more efficient for this problem, but there is little point in designing a streamlined hat to reduce the effort of walking.

14 Following Samelson (1999b).

15 Following Luyten *et al.* (1983).

16 It may be that the eastern boundary depth is determined by global thermodynamic and/or mass constraints; see, for example, Boccaletti *et al.* (2004). There must also be a poleward transport across the boundary of the subtropical–subpolar gyre to balance the equatorward transport in the Ekman layer and that of the meridional overturning circulation, and this balance requirement may influence the boundary depth.

17 Dewar *et al.* (2005). This paper also discusses the nature of discontinuities at the pool boundaries, and their treatment via shock conditions.

In the shadow zone, layer-2 fluid also has no direct surface source, so one may wonder why this is also not expunged. However, the shadow zone is not a recirculating regime and the Ekman induced displacement will be much less efficient. More directly, the eastern boundary condition $h_1(x = x_e) = 0$ precludes the vanishing of layer-2 water there, and this boundary condition is propagated westwards into the interior.

The ocean has its silent caves,
Deep, quiet and alone;
Though there be fury on the waves,
Beneath them there is none.
The awful spirits of the deep
Hold their communion there.

Nathaniel Hawthorne, *The Ocean*, c. 1833.

CHAPTER 21

The Meridional Overturning Circulation and the Antarctic Circumpolar Current

THE *MERIDIONAL OVERTURNING CIRCULATION*, or the MOC, of the ocean is the circulation associated with sinking mostly at high latitudes and upwelling elsewhere, with the meridional transport mostly taking place below the main thermocline. Understanding thes circulation is one of the main goals of this chapter. The theory explaining the MOC is not nearly as settled as that of the quasi-horizontal wind-driven circulation discussed in chapter 19, but considerable progress has been made, in particular with a significant re-thinking of the fundamentals occurring in the late 20th and early 21st century, as we will discover. Our other main goal is to glean an understanding of the Antarctic Circumpolar Current, or ACC, the theory of which has also undergone a transformation over that same period. The ACC is important not only in its own right, but because it mediates the MOCs of the individual ocean basins, connecting them into a global circulation.

That there *is* a deep circulation has been known for a long time, largely from observations of tracers such as temperature, salinity, and constituents such as dissolved oxygen and silica.[3] We can also take advantage of numerical models that are able to assimilate observations (mainly from hydrographic measurements, floats and satellites) and produce a state estimate of the overturning circulation that is consistent with both the observations and the equations of motion, and one such estimate is illustrated in Fig. 21.1. We see that the water does not all upwell in the subtropics as we tacitly assumed in the previous chapter. In fact, much of the mid-depth circulation more-or-less follows the isopycnals that span the two hemispheres (Fig. 21.2), sinking in the North Atlantic and upwelling in the Southern Ocean, with the transport in between being, at least in part, adiabatic.

The MOC used to be known as the 'thermohaline' circulation, reflecting the belief that it was primarily driven[4] by buoyancy forcing arising from gradients in temperature and salinity. Such a circulation requires that the diapycnal mixing must be sufficiently large, but many measurements have suggested this is not the case and that has led to a more recent view that the MOC is at least partially, and perhaps primarily, mechanically driven, mostly by winds, and so *along* isopycnals instead of across them. However, the situation is not wholly settled, and it is almost certain that both buoyancy and wind forcing, as well as diapycnal diffusion, play a role.

In the first half of the chapter we mainly discuss somewhat classical topics associated with the buoyancy forcing. Then, beginning in section 21.6, we discuss the role of wind forcing in producing a MOC. This forces us to take an extended diversion into the dynamics of the ACC in section 21.7

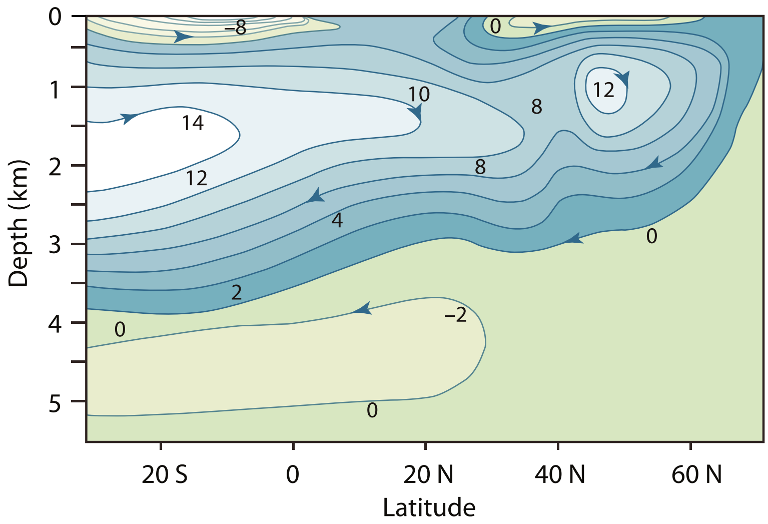

Fig. 21.1 An estimate of the mean meridional overturning circulation of the Atlantic (i.e., the streamfunction of the zonally averaged meridional flow) in Sverdrups.[1]

and then, in the last two sections, we present a theory of the MOC that incorporates both wind and buoyancy effects. We start by considering a simple but revealing fluid model of buoyancy forcing at the surface in a very idealised setting.

21.1 SIDEWAYS CONVECTION

Perhaps the simplest and most obvious fluid dynamical model of the overturning circulation is that of *sideways convection*. The physical situation is sketched in Fig. 21.3. A fluid (two- or three-dimensional) is held in a container that is insulated on all of its sides and its bottom, but its upper surface is non-uniformly heated and cooled. In the purest fluid dynamical problem the heat enters the fluid solely by conduction at the upper surface, and one may suppose that here the temperature is imposed. Thus, for a simple Boussinesq fluid the equations of motion are

$$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + \boldsymbol{f} \times \boldsymbol{v} = -\nabla\phi + b\mathbf{k} + \nu\nabla^2\boldsymbol{v}, \qquad \frac{\mathrm{D}b}{\mathrm{D}t} = \kappa\nabla^2 b, \qquad \nabla\cdot\boldsymbol{v} = 0, \tag{21.1a,b,c}$$

with boundary conditions

$$b(x, y, 0, t) = g(x, y), \tag{21.2}$$

where $g(x, y)$ is a specified field, and $\partial_n b = 0$ on the other boundaries, meaning that the derivative normal to the boundary, and so the buoyancy flux, is zero. The oceanographic relevance of (21.1) and (21.2) should be clear: the ocean is heated *and* cooled from above, and although the thermal forcing in the real ocean may differ in detail (being in part a radiative flux, and in part a sensible and latent heat transfer from the atmosphere), (21.2) is a useful idealization. An alternative upper boundary condition would be to impose a flux condition whereby

$$\text{flux} = \kappa\frac{\partial}{\partial z}b(x, y, 0, t) = h(x, y), \tag{21.3a}$$

where $h(x, y)$ is given. In some numerical models of the ocean, the heat input at the top is parameterized by way of a relaxation to some specified temperature. This is a form of flux condition in which

$$\kappa\frac{\partial b}{\partial z} = C(b^*(x, y) - b), \tag{21.3b}$$

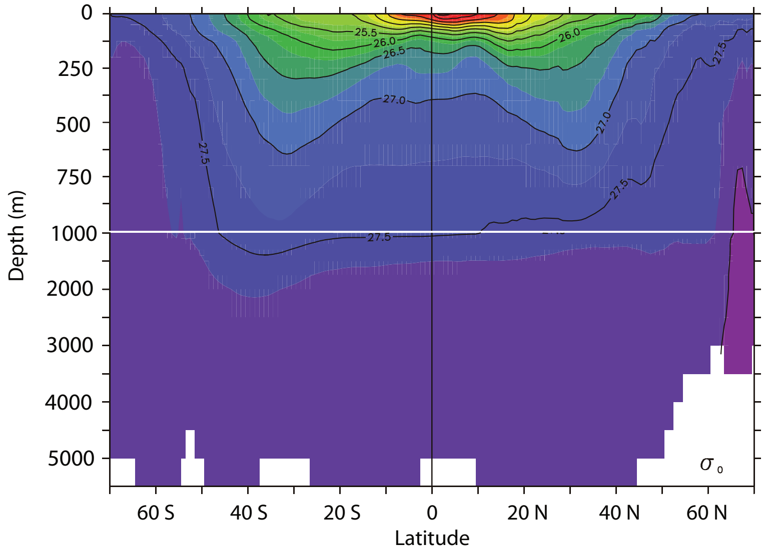

Fig. 21.2 The climatological zonally-averaged potential density (σ_θ) in the Atlantic ocean. Note the break in the vertical scale at 1000 m.

The region of rapid change of density (and temperature) is concentrated in the upper kilometre, in the main thermocline, below which the density is more uniform. The flow of the MOC is largely, but not exactly, parallel to the isopycnals. [2]

and C is an empirical constant and $b^*(x, y)$ is given.[5] Although this may be a little more relevant than (21.2) for the real ocean, which of the three boundary conditions is chosen will not affect the arguments below, and we use (21.2).

21.1.1 Two-dimensional convection

We may usefully restrict attention to the two-dimensional problem, in latitude and height. The two-dimensional flow may then loosely be thought of as representing the large-scale, statistically steady, zonally averaged flow of the ocean. The zonally averaged zonal flow is then small, and concomitantly so is the Coriolis force. The incompressibility of the flow then allows one to define a streamfunction such that

$$v = -\frac{\partial \psi}{\partial z}, \qquad w = \frac{\partial \psi}{\partial y}, \qquad \zeta = \nabla_x^2 \psi = \left(\frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right), \tag{21.4}$$

where ζ is the vorticity in the meridional plane. We will omit the subscript x on the Laplacian operator where there is no ambiguity.

Taking the curl of Boussinesq equations of motion (21.1) then gives

$$\frac{\partial \nabla^2 \psi}{\partial t} + J(\psi, \nabla^2 \psi) = \frac{\partial b}{\partial y} + \nu \nabla^4 \psi \tag{21.5a}$$

$$\frac{\partial b}{\partial t} + J(\psi, b) = \kappa \nabla^2 b \tag{21.5b}$$

where $J(a, b) \equiv (\partial_y a)(\partial_z b) - (\partial_z a)(\partial_y b)$.

Non-dimensionalization and scaling

We non-dimensionalize (21.5) by formally setting

$$b = \Delta b\, \hat{b}, \qquad \psi = \Psi \widehat{\psi}, \qquad y = L\widehat{y}, \qquad z = H\widehat{z}, \qquad t = \frac{LH}{\Psi}\hat{t}, \tag{21.6}$$

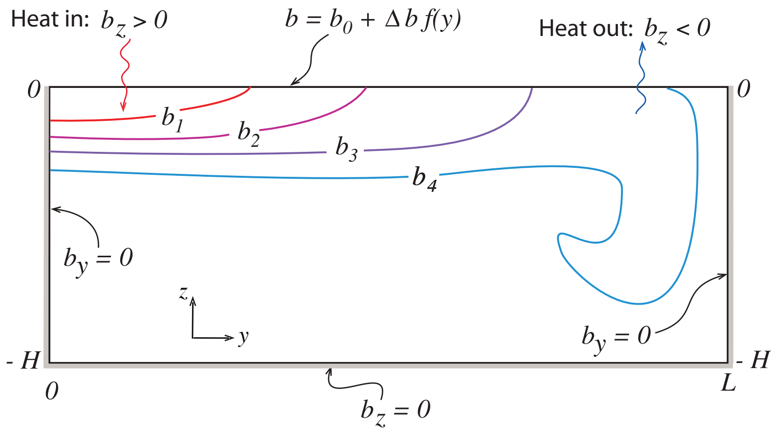

Fig. 21.3 Sketch of sideways convection. The fluid is differentially heated and cooled along its top surface, whereas all the other walls are insulating.

The result is, typically, a small region of convective instability and sinking near the coldest boundary, with generally upwards motion elsewhere.[6]

where the hatted variables are non-dimensional, Δb is the temperature difference across the surface, L is the horizontal size of the domain, and Ψ, and ultimately the vertical scale H, are to be determined. Substituting (21.6) into (21.5) gives

$$\frac{\partial \widehat{\nabla}^2 \widehat{\psi}}{\partial \hat{t}} + \hat{J}(\widehat{\psi}, \nabla^2 \widehat{\psi}) = \frac{H^3 \Delta b}{\Psi^2} \frac{\partial \hat{b}}{\partial \hat{y}} + \frac{\nu L}{\Psi H} \widehat{\nabla}^4 \widehat{\psi}, \tag{21.7a}$$

$$\frac{\partial \hat{b}}{\partial \hat{t}} + \hat{J}(\widehat{\psi}, \hat{b}) = \frac{\kappa L}{\Psi H} \widehat{\nabla}^2 \hat{b}, \tag{21.7b}$$

where $\widehat{\nabla}^2 = (H/L)^2 \partial^2/\partial \hat{y}^2 + \partial^2/\partial \hat{z}^2$ and the Jacobian operator is similarly non-dimensional. If we now use (21.7b) to choose Ψ as

$$\Psi = \frac{\kappa L}{H}, \tag{21.8}$$

so that $t = H^2 \hat{t}/\kappa$, then (21.7) becomes

$$\frac{\partial \widehat{\nabla}^2 \widehat{\psi}}{\partial \hat{t}} + \hat{J}(\widehat{\psi}, \widehat{\nabla}^2 \widehat{\psi}) = Ra \sigma \alpha^5 \frac{\partial \hat{b}}{\partial \hat{y}} + \sigma \widehat{\nabla}^4 \widehat{\psi}, \tag{21.9}$$

$$\frac{\partial \hat{b}}{\partial \hat{t}} + \hat{J}(\widehat{\psi}, \hat{b}) = \widehat{\nabla}^2 \hat{b}, \tag{21.10}$$

and the three non-dimensional parameters that govern the behaviour of the system are

$$Ra = \left(\frac{\Delta b L^3}{\nu \kappa} \right), \qquad \text{(the Rayleigh number)}, \tag{21.11a}$$

$$\sigma = \frac{\nu}{\kappa}, \qquad \text{(the Prandtl number)}, \tag{21.11b}$$

$$\alpha = \frac{H}{L}, \qquad \text{(the aspect ratio)}. \tag{21.11c}$$

Sometimes H is used instead of L in the Rayleigh number definition; we use L here because it is an external parameter. The Rayleigh number is a measure of the strength of the buoyancy forcing relative to the viscous term, and in the ocean it will be very large indeed, perhaps $\sim 10^{24}$ if molecular values are used.

For steady non-turbulent flows, and also perhaps for statistically steady flows, then we can demand that the buoyancy term in (21.9) is $\mathcal{O}(1)$. If it is smaller then the flow is not buoyancy

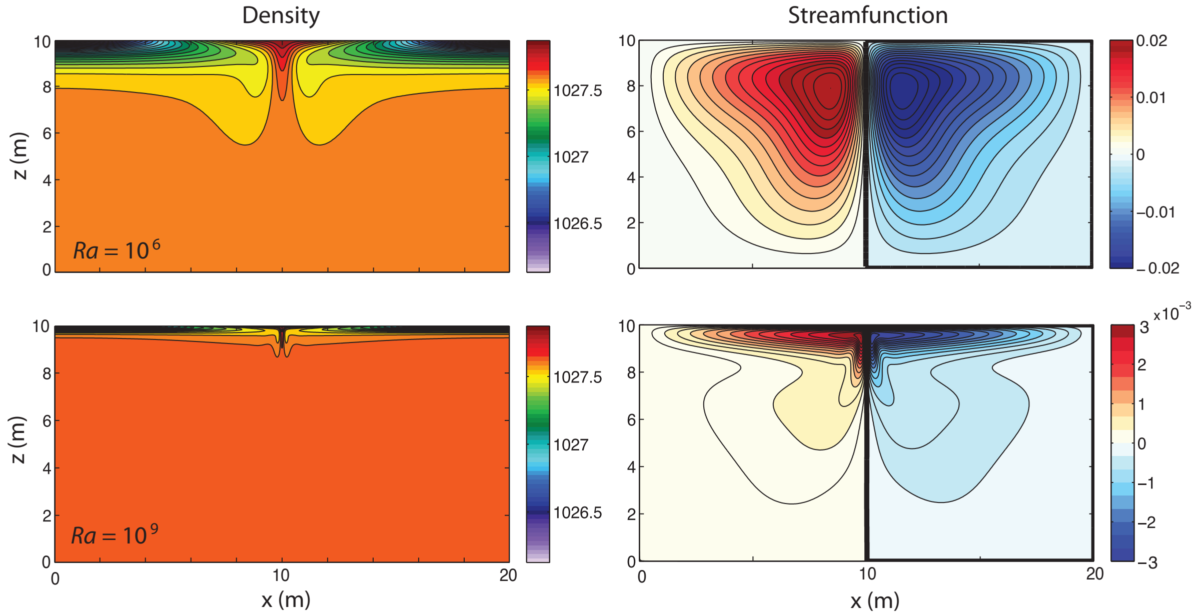

Fig. 21.4 The density and streamfunction in two numerical simulations of two-dimensional sideways convection, with Rayleigh numbers of 10^6 (top) and 10^9 (bottom).[8] The imposed temperature at the top linearly decreases from the center outwards, the side and bottom walls are insulating, and the Prandtl number is 10. The two density plots use the same colourmap, but the streamfunction plots do not. Note the weaker circulation and thinner thermocline at the higher Rayleigh number.

driven, and if it is larger there is nothing to balance it. Our demand can be satisfied only if the vertical scale of the motion adjusts appropriately and, for $\sigma = \mathcal{O}(1)$, this suggests the scalings:[7]

$$H = L\sigma^{-1/5} Ra^{-1/5} = \left(\frac{\kappa^2 L^2}{\Delta b}\right)^{1/5}, \qquad \Psi = Ra^{1/5}\sigma^{-4/5}\nu = (\kappa^3 L^3 \Delta b)^{1/5}. \tag{21.12a,b}$$

The vertical scale H arises as a consequence of the analysis, and the vertical size of the domain plays no direct role. [For $\sigma \gg 1$ we might expect the nonlinear terms to be small and if the buoyancy term balances the viscous term in (21.9) the right-hand sides of (21.12) are multiplied by $\sigma^{1/5}$ and $\sigma^{-1/5}$. For seawater, $\sigma \approx 7$ using the molecular values of κ and ν. If small scale turbulence exists, then the eddy viscosity will likely be similar to the eddy diffusivity and $\sigma \approx 1$.] Numerical experiments (an example is shownFig. 21.4) provide support for this scaling, and a few simple and robust points that have relevance to the real ocean emerge, as follows.

- ⋆ Most of the box fills up with the densest available fluid, with a boundary layer in temperature near the surface required in order to satisfy the top boundary condition. The boundary gets thinner with decreasing diffusivity, consistent with (21.12). This is a diffusive prototype of the oceanic thermocline.
- ⋆ The horizontal scale of the overturning circulation is large, being at or near the scale of the box.
- ⋆ The downwelling regions (the regions of convection) are of smaller horizontal scale than the upwelling regions, especially as the Rayleigh number increases.

Let us now try to explain some of the features in a simple and heuristic way, beginning with the scale of the motion.

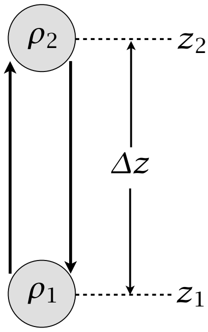

Fig. 21.5 Two fluid parcels, of density ρ_1 and ρ_2 and initially at positions z_1 and z_2 respectively, are interchanged. If $\rho_2 > \rho_1$ then the final potential energy is lower than the initial potential energy, with the difference being converted into kinetic energy.

21.1.2 The relative scale of convective plumes and diffusive upwelling

Why is the downwelling region narrower than the upwelling? The short answer is that high Rayleigh number convection is much more efficient than diffusive upwelling, so that the convective buoyancy flux can match the the diffusive flux only if the convective plumes cover a much smaller area than diffusion.[9] Suppose that the basin is initially filled with water of an intermediate temperature, and that surface boundary conditions of a temperature decreasing linearly from low latitudes to high latitudes are imposed. The deep water will be convectively unstable, and convection at high latitudes (where the surface is coldest) will occur, quickly filling the abyss with dense water. After this initial adjustment the deep, dense water at lower latitudes will slowly warmed by diffusion, but at the same time surface forcing will maintain a cold high latitude surface, thus leading to high latitude convection. A steady state or statistically steady state is eventually reached with the deep water having a slightly higher potential density than the surface water at the highest latitudes, and so with continual convection, but convection that takes place only at the highest latitudes.

To see this more quantitatively consider the respective efficiencies of the convective heat flux and the diffusive heat flux. Consider an idealized re-arrangement of two parcels, initially with the heavier one on top as illustrated in Fig. 21.5. The potential energy released by the re-arrangement, ΔP is given by

$$
\begin{aligned}
\Delta P &= P_{\text{final}} - P_{\text{initial}} && (21.13)\\
&= g\left[(\rho_1 z_2 + \rho_2 z_1) - (\rho_1 z_1 + \rho_2 z_2)\right] && (21.14)\\
&= g(z_2 - z_1)(\rho_1 - \rho_2) = \rho_0 \Delta b \Delta z && (21.15)
\end{aligned}
$$

where $\Delta z = z_2 - z_1$ and $\Delta b = g(\rho_1 - \rho_2)/\rho_0$.

The kinetic energy gained by this re-arrangement, ΔK is given by $\Delta K = \rho_0 w^2$ and equating this to (21.13) gives

$$
w^2 = -\Delta b \Delta z. \tag{21.16}
$$

If the heavier fluid is initially on top then $\rho_2 > \rho_1$ and, as defined, $\Delta b < 0$. The vertical convective buoyancy flux per unit area, B_c, is given by $B_c = w\Delta b$ and using (21.16) we find

$$
B_c = (-\Delta b)^{3/2}(\Delta z)^{1/2}. \tag{21.17}
$$

The upwards diffusive flux, B_d, per unit area is given by

$$
B_d = \kappa \frac{\Delta b}{H} \tag{21.18}
$$

where H the thickness of the layer over which the flux occurs. In a steady state the total diffusive flux must equal the convective flux so that, from (21.17) and (21.18),

$$
(-\Delta b)^{3/2}(\Delta z)^{1/2}\delta = \kappa \frac{\Delta b}{H}, \tag{21.19}
$$

where δ is the fractional area over which convection occurs. If we set $\Delta z = H$, and use (21.12a) we find

$$(-\Delta b)^{3/2}\left(\frac{\kappa^2 L^2}{\Delta b}\right)^{1/5}\delta = \kappa\frac{\Delta b}{(\kappa^2 L^2/\Delta b)^{1/5}}, \tag{21.20}$$

giving

$$\boxed{\delta = \left(\frac{\kappa^4}{\Delta b L^3}\right)^{1/5} = (Ra\,\sigma)^{-1/5}.} \tag{21.21}$$

For geophysically relevant situations this is a very small number, usually smaller than 10^{-5}. Although the details of the above calculation may be questioned (for example, the use of the same buoyancy difference and vertical scale in the convection and the diffusion), the physical basis for the result is clear: for realistic choices of the diffusivity the convection is much more efficient than the diffusion and so will occur over a much smaller area.

21.1.3 Phenomenology of the overturning circulation

No water can be denser (or, more accurately, have a greater potential density) than the densest water at the surface. If the surface water is densier that the water at depth then it will be convectively unstable and sink in a plume.[10] The plume slowly entrains the warmer water that surrounds it, and then spreads horizontally when it reaches the bottom or when its density becomes similar to that of its surroundings. The presence of water denser than its surroundings creates a horizontal pressure gradient, and the ensuing flow will displace any adjacent lighter fluid, and so the domain fills with the densest available fluid. This process is a continuous one: the plumes take cold water into the interior, where the water slowly warms by diffusion, and the source of cold water at the surface is continuously replenished. If diffusion is small, the end result is that the potential density of the fluid in the interior will be slightly less than that of the densest fluid formed at the surface. (Because diffusion can act only to reduce extrema, no fluid in the interior can be colder than the coldest fluid formed at the surface.)

However, the value at the surface is given by the boundary condition $b(x, y, z = 0) = f(x, y)$. Thus, the interior cannot fill all the way to the surface with this cold water and there must be a boundary layer connecting the cold, dense interior with the surface; its thickness δ is given by the height scale of (21.12a); that is $\delta \sim H = (\kappa^2 L^2/\Delta b)^{1/5}$. Such a strong boundary layer will not necessarily be manifest in the velocity field, however, because the no-normal flow boundary condition on the velocity field is satisfied by setting $\psi = 0$ as a boundary condition to the elliptic problem $\nabla^2\psi = \zeta$, where ζ is the prognostic variable in (21.5a), and this boundary condition has a global effect on the velocity field.

Why is the horizontal scale of the circulation large? The circulation transfers heat meridionally, and it is far more efficient to do this by a single overturning cell than by a multitude of small cells; hence, although we cannot entirely eliminate the possibility that some instability will produce such small scales of motion, it seems likely the horizontal scale of the mean circulation will be determined by the domain scale. (At low Rayleigh number we can in fact explicitly calculate an approximate analytic solution for the flow, demonstrating this.) For higher Rayleigh number perturbation approaches fail and we must resort to numerical solutions; these (e.g., Fig. 21.4), do show the circulation dominated by a single overturning circulation rather than many small convective cells.

Finally, it is important to realize that *even for large diffusion and viscosity there is no stationary solution:* as soon as we impose a temperature gradient at the top the fluid begins to circulate, a manifestation of the dictum that a baroclinic fluid is a moving fluid, encountered in section 4.2. Put simply, a temperature gradient leads a density gradient, which in turn leads to a pressure gradient. The pressure gradient leads to motion: viscosity cannot prevent that, for it can have an effect only if the velocity is non-zero.

21.2 THE MAINTENANCE OF SIDEWAYS CONVECTION

In most conventional convection problems the fluid is heated from below, becomes buoyant and rises, and is cooled at the top. In contrast, in sideways convection the heating and cooling occur at the same level, and the conditions under which a circulation can be maintained are by no means clear; our purpose here to make them clearer. The energetic derivations of this section are but an extension of section 2.4.3, but now with starring roles for friction, diffusion and the boundary conditions, and the reader may wish to review that section first. The derivations below are not difficult, but they lead to powerful and perhaps counter-intuitive results that provide important information about the overturning circulation of the real ocean. To begin, we rewrite the equations of motion, (21.1), in a slightly different way, namely

$$\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{f} + 2\boldsymbol{\omega}) \times \boldsymbol{v} = -\nabla B + b\mathbf{k} + \nu\nabla^2\boldsymbol{v}, \tag{21.22a}$$

$$\frac{\partial b}{\partial t} + \nabla \cdot (b\boldsymbol{v}) = \dot{Q} = J + \kappa\nabla^2 b, \tag{21.22b}$$

$$\nabla \cdot \boldsymbol{v} = 0, \tag{21.22c}$$

where $B = \boldsymbol{v}^2/2 + \phi$ is the Bernoulli function for Boussinesq flow and $\dot{Q}$ $(= \dot{b})$ is the total rate of heating (including diffusion, and absorbing constant factors such as heat capacity into its definition) with J its non-diffusive component. The fluid occupies a finite volume, and in a steady state $\langle \dot{Q} \rangle = 0$, where the angle brackets denote a volume and time integration.

21.2.1 The energy budget

To obtain an energy budget we follow the procedure of section 2.4.3. First take the dot product of (21.22a) with $\boldsymbol{v}$ to give

$$\frac{1}{2}\frac{\partial \boldsymbol{v}^2}{\partial t} = -\nabla \cdot (\boldsymbol{v}B) + wb + \nu\boldsymbol{v} \cdot \nabla^2\boldsymbol{v}. \tag{21.23}$$

Integrating over a domain bounded by stress-free rigid walls gives the kinetic energy equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\langle \frac{1}{2}\boldsymbol{v}^2 \right\rangle = \langle wb \rangle - \varepsilon, \tag{21.24}$$

where angle brackets denote (for the moment) just a volume average and ε is the average dissipation of kinetic energy ($\varepsilon = -\nu \langle \boldsymbol{v} \cdot \nabla^2\boldsymbol{v} \rangle = \nu \langle \boldsymbol{\omega}^2 \rangle$), a positive definite quantity. Thus, in a statistically steady state in which the left-hand side vanishes after time averaging, the dissipation of kinetic energy is maintained by the buoyancy flux; that is, by a release of potential energy with light fluid ascending and dense fluid descending.

We obtain a potential energy budget by using (21.22b) to write

$$\frac{\mathrm{D}bz}{\mathrm{D}t} = z\frac{\mathrm{D}b}{\mathrm{D}t} + b\frac{\mathrm{D}z}{\mathrm{D}t} = z\dot{Q} + bw, \tag{21.25}$$

and integrating this over the domain gives the potential energy equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle bz \rangle = \langle z\dot{Q} \rangle + \langle bw \rangle. \tag{21.26}$$

Subtracting (21.26) from (21.24) gives the energy equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\langle \frac{1}{2}\boldsymbol{v}^2 - bz \right\rangle = -\langle z\dot{Q} \rangle - \varepsilon. \tag{21.27}$$

21.2.2 Conditions for maintaining a thermally-driven circulation

In a statistically steady state the left-hand side of (21.27) vanishes and the kinetic energy dissipation is balanced by the buoyancy source terms; that is

$$\langle z\dot{Q}\rangle = -\varepsilon < 0. \tag{21.28}$$

The right-hand side (the term $-\varepsilon$) is negative definite, and to balance this the heating must be negatively correlated with height. (Recall that $\langle\dot{Q}\rangle = 0$, and the origin of the z-coordinate is then immaterial.) Thus, *in order to maintain a circulation in which kinetic energy is dissipated, the heating (including diffusive heating) must occur, on average, at lower levels than the cooling.* This result is related to, but not quite the same as, a postulate due to Sandström, discussed below.

In the ocean the non-diffusive heating occurs predominantly at the surface, except for the negligible effects of hydrothermal vents. Thus, $\langle Jz\rangle \approx 0$ and a kinetic-energy-dissipating circulation can *only* be maintained, in the absence of mechanical forcing, if the diffusion is non-zero — in that case heat may be diffused from the surface to depth so effectively providing a deep heat source. In the atmosphere, the heating is mostly at the surface and cooling is mostly in the mid-troposphere so that $\langle z\dot{Q}\rangle < 0$; thus, (21.28) is readily satisfied and the circulation is not restricted.

✦ *Maintaining a steady baroclinic circulation*

By a rather different method — and one closer to a suggestion of Sandström dating from 1908 — we can obtain a result that is different from but related to (21.28).[11] Using (4.37) and (21.22a), the circulation in a Boussinesq system obeys

$$C = \oint \boldsymbol{v}\cdot \mathrm{d}\boldsymbol{r}, \qquad \frac{\mathrm{D}C}{\mathrm{D}t} = \oint b\mathbf{k}\cdot \mathrm{d}\boldsymbol{r} + \oint \boldsymbol{F}\cdot \mathrm{d}\boldsymbol{r}, \tag{21.29a,b}$$

where C is the circulation, $\mathrm{d}\boldsymbol{r}$ is a path element, and $\boldsymbol{F}$ represents the frictional terms. (The Coriolis parameter plays no role in this argument because the Coriolis force does no work, and f may be set to zero without loss of generality.) Now we may write the rate of change of circulation in the form

$$\frac{\mathrm{D}C}{\mathrm{D}t} = \oint\left(\frac{\partial \boldsymbol{v}}{\partial t} + \boldsymbol{v}\cdot\nabla\boldsymbol{v}\right)\cdot \mathrm{d}\boldsymbol{r} = \oint\left(\frac{\partial \boldsymbol{v}}{\partial t} + \boldsymbol{\omega}\times\boldsymbol{v}\right)\cdot \mathrm{d}\boldsymbol{r}, \tag{21.30}$$

because the integral of the potential term that arises when going to the last expression vanishes. Let us assume the flow is steady, so that $\partial\boldsymbol{v}/\partial t$ vanishes. Let us further choose the path of integration to be a streamline, which since the flow is steady is also a parcel trajectory. The second term on the right-most expression of (21.30) then also vanishes and (21.29b) becomes

$$\oint b\,\mathrm{d}z = -\oint \boldsymbol{F}\cdot \mathrm{d}\boldsymbol{r} = -\oint \frac{\boldsymbol{F}}{|\boldsymbol{v}|}\cdot \boldsymbol{v}\,\mathrm{d}r, \tag{21.31}$$

where the last equality follows because the path is everywhere parallel to the velocity. Let us now assume that the friction retards the flow, and that $\oint \boldsymbol{F}\cdot\boldsymbol{v}/|\boldsymbol{v}|\,\mathrm{d}r < 0$. (One form of friction that has this property is linear drag, $\boldsymbol{F} = -C\boldsymbol{v}$ where C is a constant. The property is similar to, but not the same as, the property that the friction dissipates kinetic energy over the circuit.) Making this assumption, if we integrate the term on the left-hand side by parts we obtain

$$\oint z\,\mathrm{d}b < 0. \tag{21.32}$$

Now, because the integration circuit in (21.32) is a fluid trajectory, the change in buoyancy $\mathrm{d}b$ is proportional to the heating of a fluid element as it travels the circuit; in the notation of (21.22b),

$\mathrm{d}b = \dot{Q}\,\mathrm{d}t$, where the heating, $\dot{Q}$, includes diffusive effects. Thus, the inequality implies that the net heating must be negatively correlated with height: that is, *the heating must occur, on average, at a lower level than the cooling in order that a steady circulation may be maintained against the retarding effects of friction.*

A similar result can be obtained for a compressible fluid. From equations (4.43)–(4.45) on page 152 we write the baroclinic circulation theorem as

$$\frac{\mathrm{D}C}{\mathrm{D}t} = \oint p\,\mathrm{d}\alpha + \oint \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{r} = \oint T\,\mathrm{d}\eta + \oint \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{r}, \tag{21.33}$$

where η is the specific entropy. Then, by precisely the same arguments as led to (21.32), we are led to the requirements that

$$\oint T\,\mathrm{d}\eta > 0 \qquad \text{or equivalently} \qquad \oint p\,\mathrm{d}\alpha > 0. \tag{21.34a,b}$$

Equation (21.34a) means that parcels must gain entropy at high temperatures and lose entropy at low temperatures; similarly, from (21.32b), a parcel must expand ($\mathrm{d}\alpha > 0$) at high pressures and contract at low pressures.

For an ideal gas we can put these statements into a form analogous to (21.32) by noting that $\mathrm{d}\eta = c_p(\mathrm{d}\theta/\theta)$, where θ is potential temperature, and using the definition of potential temperature, (1.106). With these we have

$$\oint T\,\mathrm{d}\eta = \oint c_p \frac{T}{\theta}\,\mathrm{d}\theta = \oint c_p \left(\frac{p}{p_R}\right)^{\kappa} \mathrm{d}\theta, \tag{21.35}$$

and (21.34a) becomes

$$\boxed{\oint c_p \left(\frac{p}{p_R}\right)^{\kappa} \mathrm{d}\theta > 0.} \tag{21.36}$$

Because the path of integration is a fluid trajectory, $\mathrm{d}\theta$ is proportional to the heating of a fluid element. Thus, as with the Boussinesq result, (21.36) implies that *the heating (the potential temperature increase) must occur at a higher pressure than the cooling in order that a steady circulation may be maintained against the retarding effects of friction.*

These results may be understood by noting that the heating must occur at a higher pressure than the cooling in order that work may be done, the work being necessary to convert potential energy into kinetic energy to maintain a circulation against friction. Intuitively, if the heating is below the cooling, then the heated fluid will expand and become buoyant and rise, and a steady circulation between heat source and heat sink can readily be imagined. On the other hand, if the heating is above the cooling there is no obvious pathway between source and sink.

Intuitive as these results may be, the conditions required to prove (21.32) and (21.36) are much more restrictive than those needed to prove (21.28). To prove the former, we must *assume* that the flow is absolutely steady, *and* that streamlines form a closed path, *and* that the friction has retarding properties. The second of these conditions is not generally satisfied in three dimensions, even when the flow is steady. Furthermore, one cannot prove that Newtonian viscosity ($\nu\nabla^2\boldsymbol{v}$) will always act to retard the flow. On the other hand, (21.28) provides a condition for the maintenance of a statistically steady circulation, assuming only that the friction acts to dissipate kinetic energy. In any case, it is clear from all of the above results that the overturning circulation is greatly affected by the relative pressures at the locations of the heating and cooling, and this is called *Sandström's effect*. In all of these cases, the heating must be taken to include diffusive effects; if the molecular diffusivity is small and the heating is at the surface we can further constrain the flow, as we now see.

21.2.3 Surface fluxes and non-turbulent flow at small diffusivities

Suppose that the only heating to the fluid is via diffusion through the upper surface; that is $J = 0$ in (21.22b). Is a circulation possible? If the diffusivity is finite then heat can diffuse into the fluid, and thereby potentially provide an difference in altitude between the heating and the cooling. However, as $\kappa \to 0$ this mechanism ceases to operate and we therefore expect that the left-hand sides of (21.28) and (21.32) will go to zero, and the circulation will cease. In what follows we put this argument on a more rigorous footing: we will show that as $\kappa \to 0$ the kinetic energy dissipation also goes to zero, and therefore the flow is 'non-turbulent', the meaning of which will be made clearer below.

Assuming a statistically steady state, integrating (21.22b) horizontally gives

$$\frac{\partial \overline{bw}}{\partial z} = \kappa \frac{\partial^2 \overline{b}}{\partial z^2}, \tag{21.37}$$

where an overbar indicates a horizontal average. Integrating this equation up from the bottom (where there is no flux) to a level z gives

$$\overline{wb} - \kappa \overline{b}_z = 0 \tag{21.38}$$

at every level in the fluid. The two terms on the left-hand side together comprise the total buoyancy flux through the level z, and this vanishes because there is no buoyancy input except at the surface. If we integrate this vertically we have

$$\langle wb \rangle = H^{-1}\kappa \left[\overline{b}(0) - \overline{b}(-H) \right], \tag{21.39}$$

where the angle brackets denote an average over the entire volume. In the limit $\kappa \to 0$, the integrated advective buoyancy flux will vanish, because the term $\overline{b}(0) - \overline{b}(-H)$ remains finite. This follows because b is conserved on parcels, except for the effects of diffusion, which can only act to reduce the value of extrema in the fluid. Thus, $\overline{b}(0) - \overline{b}(-H)$ can only be as large as the temperature difference at the surface, which is set by the boundary conditions.

Now consider the kinetic energy budget. Using (21.24) and (21.39) we have in a statistically steady state

$$\boxed{\varepsilon = H^{-1}\kappa \left[\overline{b}(0) - \overline{b}(-H) \right].} \tag{21.40}$$

Because, as noted above, the buoyancy difference on the right-hand side is bounded, the kinetic energy dissipation must go to zero if the thermal diffusivity goes to zero; that is, $\varepsilon \to 0$ as $\kappa \to 0$ and in particular $\varepsilon < \kappa \Delta b / H$ where Δb is the maximum buoyancy difference at the surface. We may also consider the limit $(\kappa, \nu) \to 0$ with a fixed Prandtl number, $\sigma \equiv \nu/\kappa$, and in this limit the energy dissipation also vanishes with κ.

Finally, let us see how the surface buoyancy is related to the buoyancy flux, for any value of κ. Multiplying (21.22b) (with $J = 0$) by b and integrating over the domain gives the buoyancy variance equation

$$\frac{1}{2}\frac{\mathrm{d}\langle b^2 \rangle}{\mathrm{d}t} = \kappa \left[\overline{b \frac{\partial b}{\partial z}} \bigg|_{z=0} - \langle |\nabla b|^2 \rangle \right]. \tag{21.41}$$

We have assumed that the normal derivative of b vanishes on all surfaces except the top one ($z = 0$) and an overbar denotes a horizontal average. In a statistically steady state,

$$\overline{b \frac{\partial b}{\partial z}} \bigg|_{z=0} = \langle |\nabla b|^2 \rangle, \tag{21.42}$$

where the overbar and angle brackets now also imply a time average. The right-hand side is positive definite, and thus there must be a positive correlation between b and $\partial b/\partial z$, meaning that there is a heat flux into the fluid where it is hot, and a heat flux out of the fluid where it is cold. This result holds no matter whether the upper boundary condition is a condition on b or on $\partial b/\partial z$.

Interpretation

The result encapsulated by (21.40) means that, for a fluid forced only at the surface by buoyancy forcing, as the diffusivity goes to zero so does the energy dissipation. For a fluid of finite viscosity the vorticity in the fluid must then go to zero, because $\varepsilon = \nu \langle \boldsymbol{\omega}^2 \rangle$; this in turn means that the flow cannot be baroclinic, because baroclinicity generates vorticity, even in the presence of viscosity (section 4.2). An even more interesting result follows for a fluid with small viscosity. In turbulent flow, the energy dissipation at high Reynolds number is not a function of the viscosity; if the viscosity is reduced, the cascade of energy to smaller scales merely continues to still smaller scale, generating vorticity at these smaller scales, and the energy dissipation is unaltered, remaining finite even in the limit $\nu \to 0$. In contrast, for a fluid heated and cooled only at the upper surface, the energy dissipation *tends to zero* as $\kappa \to 0$, whether or not one is in the high-Reynolds number limit. This means that vorticity cannot be generated at the viscous scales by the action of a turbulent cascade, as that would lead to energy dissipation. Effectively, the result prohibits an ocean that is forced only at the surface by a buoyancy flux from having an 'eddy viscosity' that would enable the fluid to efficiently dissipate energy, and if there is no small scale motion producing an eddy viscosity there can be no eddy diffusivity either. Thus, such an ocean is *non-turbulent*. This is a rather different picture from that of the real ocean, where there is some dissipation of energy in the interior because of breaking gravity waves, and dissipation at the boundary in Ekman layers, and the eddy diffusivity is needed for there to be a non-negligible buoyancy-driven meridional overturning circulation.

Of course, thermal forcing in the ocean is in part an imposed flux, coming from radiation among other things, and this penetrates below the surface. However, this makes little physical difference to the argument, provided that this forcing remains confined to the upper ocean. If so, then for any level below this forcing we still have the result (21.38), and the final result (21.40) holds, assuming that the range of temperatures produced by the forcing is still finite.

21.2.4 The importance of mechanical forcing

The results of (21.28) and (21.40) do not, strictly speaking, prohibit there from being a thermal circulation, with fluid sinking at high latitudes and rising at low, even for zero diffusivity. However, in the absence of any mechanical forcing, this circulation must be laminar, even at high Rayleigh number, meaning that flow is not allowed to break in such a way that energy can be dissipated — a very severe constraint that most flows cannot satisfy. The scalings (21.12) further suggest that the magnitude of the circulation in fact scales (albeit nonlinearly) with the size of the molecular diffusivity, and if these scalings are correct the circulation will in fact diminish as $\kappa \to 0$. For small diffusivity, the solution most likely to be adopted by the fluid is for the flow to become confined to a very thin layer at the surface, with no abyssal motion at all, which is completely unrealistic vis-à-vis the observed ocean. Thus, the deep circulation of the ocean cannot be considered to be wholly forced by buoyancy gradients at the surface.

Suppose we add a mechanical forcing, $\boldsymbol{F}$, to the right-hand side of (21.22a); this might represent wind forcing at the surface, or tides. The kinetic energy budget becomes

$$\varepsilon = \langle wb \rangle + \langle \boldsymbol{F} \cdot \boldsymbol{v} \rangle = H^{-1}\kappa[\overline{b}(0) - \overline{b}(-H)] + \langle \boldsymbol{F} \cdot \boldsymbol{v} \rangle \,. \tag{21.43}$$

In this case, even for $\kappa = 0$, there is a source of energy and therefore turbulence (i.e., a dissipative circulation) can be maintained. The turbulent motion at small scales then provides a mechanism of mixing and so can effectively generate an 'eddy diffusivity' of buoyancy. *Given* such an eddy diffusivity, wind forcing is no longer necessary for there to be an overturning circulation. Therefore, it is useful to think of mechanical forcing as having two distinct effects.

(i) The wind provides a stress on the surface that may directly drive the large-scale circulation, including the overturning circulation. (An example of this is discussed in section 21.6.)

(ii) Both tides and the wind provide a mechanical source of energy to the system that allows the flow to become turbulent and so provides a source for an eddy diffusivity and eddy viscosity.

In either case, we may conclude that the presence of mechanical forcing is necessary for there to be an overturning circulation in the world's oceans of the kind observed. Let us first suppose that the most important effect of the wind is that it enables there to be an eddy diffusivity that is much larger than the molecular one; the eddy diffusivity enables large volumes of the ocean to become mixed, so allowing a buoyancy-driven overturning circulation (a 'thermohaline circulation') to exist.

21.2.5 The mixing-driven ocean

Before moving on to other matters, let's make a connection to the scaling for the overturning circulation discussed in section 20.4.1. In that section we considered the planetary-geostrophic equations,

$$\boldsymbol{v}\cdot\nabla b = \kappa\nabla^2 b, \qquad \nabla\cdot\boldsymbol{u} + \frac{\partial w}{\partial z} = 0, \qquad \boldsymbol{f}\times\boldsymbol{u} = -\nabla\phi, \qquad b = \frac{\partial\phi}{\partial z}. \tag{21.44a,b,c,d}$$

and obtained, after a little algebra, the scalings for vertical velocity and thermocline thickness,

$$W = \kappa^{2/3}\left(\frac{\beta\,\Delta b}{f^2 L}\right)^{1/3}, \qquad \delta = \kappa^{1/3}\left(\frac{f^2 L}{\beta\,\Delta b}\right)^{1/3}. \tag{21.45}$$

These are different in detail from the Rossby scalings, (21.12), but they also require a finite diffusivity to produce a circulation and a thermocline. The Sandström effect applies, as it has to, to oceanically relevant equations.

In order for the scales given in (21.45) to be representative of those observed in the real ocean, we must use an eddy diffusivity for κ. Using $f = 10^{-4}\,\mathrm{s^{-1}}$, $\beta = 10^{-11}\,\mathrm{m^{-1}\,s^{-1}}$, $L = 5\times10^6\,\mathrm{m}$, $g = 10\,\mathrm{m\,s^{-2}}$, $\kappa = 10^{-5}\,\mathrm{m^2\,s^{-1}}$, $\Delta b = -g\Delta\rho/\rho_0 = g\beta_T\Delta T$ and $\Delta T = 10\,\mathrm{K}$ we find the not unreasonable values of $\delta \approx 150\,\mathrm{m}$ and $W = 10^{-7}\,\mathrm{m\,s^{-1}}$, albeit δ is rather smaller than the thickness of the observed thermocline. However, if we take the molecular value of $\kappa \approx 10^{-7}\,\mathrm{m^2}$ the values of W and δ are unrealistically small (although still non-zero). Evidently, if the deep circulation of the ocean is buoyancy (or mixing) driven, it must take advantage of turbulence that enhances the small scale mixing and produces an eddy diffusivity.

21.3 ✦ SIMPLE BOX MODELS

This section is marked with a black diamond not because it is advanced; rather, it is a little peripheral to our main development. The purist may consider this section a diversion away from a consideration of the fluid dynamical properties of the ocean, and the content implied by the title of this book, but such box models have been quite fecund and an evident source of qualitative understanding, and thus find a place in our discussion, if not in our canon. Readers may skim this section without fear of disapprobation.

Even though they are far simpler than the real ocean, the fluid dynamical models of the previous sections are still quite daunting. The analysis that can be performed is either very specific and of little generality, for example the construction of solutions at low Rayleigh number, or it is a very general form such as scaling or energetic arguments. Models based on the fluid dynamical equations do not easily allow for the construction of explicit solutions in the parameter regime — high

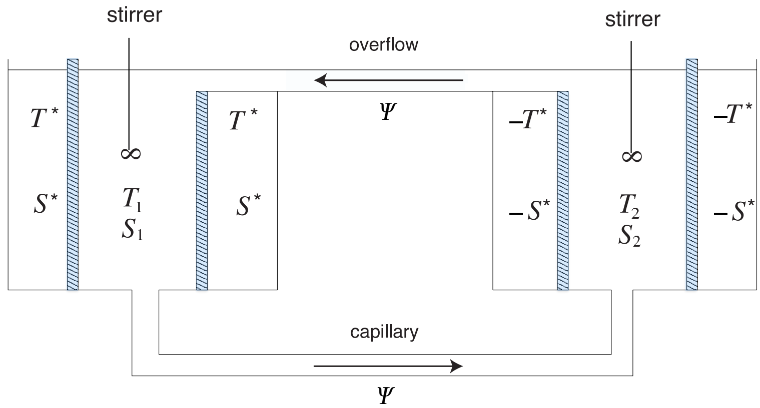

Fig. 21.6 A two-box model of the overturning circulation of the ocean. The shaded walls are porous, and each box is well mixed by its stirrer. Temperature and salinity evolve by way of fluid exchange between the boxes via the capillary tube and the overflow, and by way of relaxation with the two infinite reservoirs at $(+T^*, +S^*)$ and $(-T^*, -S^*)$.

Rayleigh and Reynolds numbers — of interest. It is therefore useful to consider an extreme simplification of the overturning circulation, namely *box models.* These are caricatures of the circulation, constructed by dividing the ocean into a small number of boxes with simple rules determining the transport of fluid properties between them.[12]

21.3.1 A two-box model

Consider two boxes as illustrated in Fig. 21.6. Each box is well-mixed and has a uniform temperature and salinity, T_1, T_2 and S_1, S_2. The boxes are connected with a capillary tube at the bottom along which the flow is viscous, obeying Stokes' Law. That is, the flow along the tube is proportional to the pressure gradient which, because the flow is hydrostatic, is proportional to the density difference between the two boxes. An overflow at the top keeps the upper surfaces of the two boxes at the same level. Thus, the circulation is given by

$$\Psi = A(\rho_1 - \rho_2), \tag{21.46}$$

where Ψ is proportional to the flow in the pipe, ρ_1 and ρ_2 are the densities of the fluids in the two boxes and A is a constant. The boxes are enclosed by porous walls beyond which are reservoirs of constant temperature and salinity, and we are at liberty to choose the origin of the temperature scale such that the two reservoirs are at $+T^*$ and $-T^*$, and similarly for salinity. Thus, heat and salt are transferred into and out of the boxes as represented by simple linear laws and we have

$$\begin{aligned}
\frac{\mathrm{d}T_1}{\mathrm{d}t} &= c(T^* - T_1) - |\Psi|(T_1 - T_2), & \frac{\mathrm{d}T_2}{\mathrm{d}t} &= c(-T^* - T_2) - |\Psi|(T_2 - T_1), \\
\frac{\mathrm{d}S_1}{\mathrm{d}t} &= d(S^* - S_1) - |\Psi|(S_1 - S_2), & \frac{\mathrm{d}S_2}{\mathrm{d}t} &= d(-S^* - S_2) - |\Psi|(S_2 - S_1).
\end{aligned} \tag{21.47}$$

The advective transfer is independent of the sign the circulation, because it occurs through both the capillary tube and the overflow. From these equations it is easy to show that the sum of the temperatures, $T_1 + T_2$ decays to zero and is uncoupled from the difference, and similarly for salinity. Defining $\widehat{T} = (T_1 - T_2)/(2T^*)$ and $\widehat{S} = (S_1 - S_2)/(2S^*)$, we obtain

$$\frac{\mathrm{d}\widehat{T}}{\mathrm{d}t} = c(1 - \widehat{T}) - 2|\Psi|\widehat{T}, \qquad \frac{\mathrm{d}\widehat{S}}{\mathrm{d}t} = d(1 - \widehat{S}) - 2|\Psi|\widehat{S}. \tag{21.48a,b}$$

Using a linear equation of state of the form $\rho = \rho_0(1 - \beta_T T + \beta_S S)$ (where the variables are dimensional) the circulation (21.46) becomes

$$\Psi = 2\rho_0 T^* \beta_T A \left(-\widehat{T} + \frac{\beta_S S^*}{\beta_T T^*} \widehat{S} \right). \tag{21.49}$$

Finally, non-dimensionalizing time using $\tau = ct$, the equations of motion become

$$\frac{\mathrm{d}\widehat{T}}{\mathrm{d}\tau} = (1 - \widehat{T}) - |\Phi|\widehat{T}, \qquad \frac{\mathrm{d}\widehat{S}}{\mathrm{d}\tau} = \delta(1 - \widehat{S}) - |\Phi|\widehat{S}, \qquad \Phi = -\gamma(\widehat{T} - \mu\widehat{S}), \tag{21.50a,b,c}$$

where $\Phi = 2\Psi/c$ and the three parameters that determine the behaviour of the system are

$$\gamma = \frac{4\rho_0 T^* \beta_T A}{c}, \qquad \delta = \frac{d}{c}, \qquad \mu = \frac{\beta_S S^*}{\beta_T T^*}. \tag{21.51}$$

The parameter γ measures the overall strength of the forcing in determining the strength of the circulation, and is the ratio of a relaxation time scale to an advective time scale. The parameter δ is the ratio of the reciprocal time constants of temperature and salinity relaxation, and μ is a measure of the ratio of the effect of the salinity and temperature forcings on the density. Salinity transfer will normally be much slower than heat transfer so that $\delta \ll 1$, whereas if salinity and temperature are both to play a role in the dynamics we need $\mu = \mathcal{O}(1)$. We might also expect both advection and relaxation to be important if $\gamma = \mathcal{O}(1)$, and this will depend on the properties of the capillary tube.

Interpretation

Although the above model describes a potentially real system, one that might be constructed in the laboratory, it is the analogy to aspects of the ocean circulation that interests us here. To make the analogy, we suppose that one box represents the entire high-latitude ocean and the other the entire low-latitude ocean, and the capillary tube and the overflow carry the overturning circulation between them. The reservoirs at $\pm T^*$ and $\pm S^*$ represent the atmosphere. Typically, we would choose the low latitudes to be both heated and salted (the latter because of the low rainfall and high evaporation in the subtropics) and the high latitudes to be cooled and freshened by rainfall. Thus, T^* and S^* have the same sign, and they force the circulation in opposite directions.

It is a common fluid-dynamical experience that the behaviour of a highly-truncated system has little resemblance to that of the corresponding continuous system, and so we expect the model to be only be a cartoon of the ocean circulation. For example, we have restricted the circulation to be of basin scale, and the parameterization of the intensity of the overturning circulation by (21.50c) must be regarded with caution, because it represents a frictionally controlled flow rather than a nearly inviscid geostrophic flow. Nevertheless, observations and numerical simulations do indicate that the overturning circulation does have a relatively simple vertical and horizontal structure: the circulation in the North Atlantic is similar to that of a single cell, for example, indicating that an appropriate low-order model may be useful.

One might also question the oceanic appropriateness of the linear relaxation terms. For temperature, the bulk aerodynamic formulae often used to parameterize air–sea fluxes do have a similar form, but the freshening of seawater by rainfall is more akin to an imposed negative flux of salinity, and evaporation is a function of temperature. An alternative might be to impose an effective salt flux so that

$$\frac{\mathrm{d}}{\mathrm{d}t}(S_1 - S_2) = 2E - 2|\Psi|(S_1 - S_2), \tag{21.52}$$

where E is an imposed, constant, effective rate of salt exchange with the atmosphere. After non-dimensionalization, using E/c to non-dimensionalize salt, (21.50b) is replaced by

$$\frac{\mathrm{d}S}{\mathrm{d}\tau} = 1 - |\Phi|S. \tag{21.53}$$

Another aspect of the model that is oceanographically questionable is that the model assumes that the water masses can be mixed below the surface. Thus, when water enters one box from the

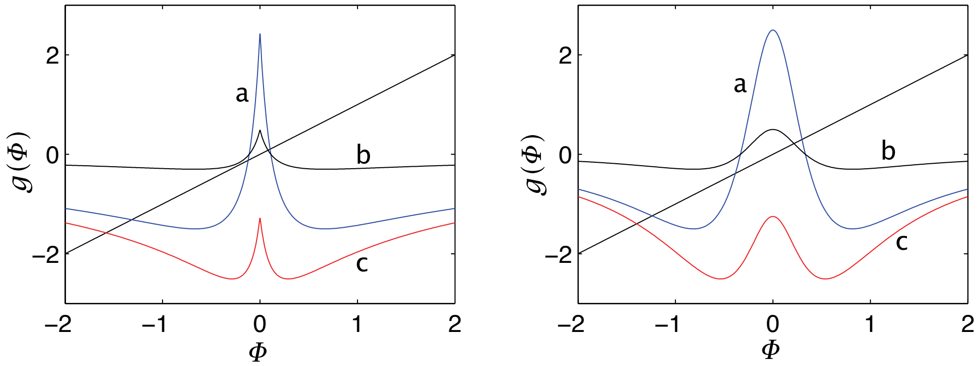

Fig. 21.7 Left panel: graphical solution of the two-box model. The straight line has unit slope and passes through the origin, and the curved lines plot the function $f(\Phi)$ as given by the right-hand side of (21.54). The intercepts of the two are solutions to the equation. The parameters for the three curves are: a, $\gamma = 5$, $\delta = 1/6$, $\mu = 1.5$; b, $\gamma = 1$, $\delta = 1/6$, $\mu = 1.5$; c, $\gamma = 5$, $\delta = 1/6$, $\mu = 0.75$. Right panel: the same except with Φ^2 in place of $|\Phi|$ on the rhs of (21.54).

other it immediately mixes with its surroundings. Without the stirrer this would not occur and the equations of the box model would not represent a real system. In the real ocean, most of the mixing of water masses happens near the surface (in the mixed layer) and near lateral boundaries or regions of steep topography. Elsewhere in the ocean, mixing is quite small, and probably far from sufficient to mix a large volume of water in the advective or relaxation times of the box model. Having noted all these objections, we will put them aside and continue with an analysis of the model.

Solutions

Equilibria occur when the time-derivatives vanish, and the circulation then satisfies

$$\Phi = g(\Phi) \equiv \gamma \left(\frac{-1}{1 + |\Phi|} + \frac{\mu}{1 + |\Phi|/\delta} \right). \tag{21.54}$$

A graphical solution of this is obtained as the intercept of the right-hand side with the left-hand side, the latter being a straight line through the origin at an angle of 45°, and this is plotted in Fig. 21.7. Perhaps the most interesting aspect of the solutions is that they exhibit *multiple equilibria;* that is, there are multiple steady solutions with the same parameters.

Evidently, for a range of parameters three solutions are possible, whereas for others only one solution exists. Although a fairly complete analysis of the nature of the steady solutions is possible, it is instructive to consider the special case with $\gamma \gg 1$ and $\delta \ll 1$ This corresponds to the situation in which the advective time scale is shorter than the diffusive one and temperature relaxation is much faster than salt relaxation. Using the graphical solution as a guide, two of the solutions are then close to the origin, with $\Phi \ll 1$, and satisfy

$$\Phi \approx \gamma \left(-1 + \frac{\mu\delta}{\delta + |\Phi|} \right), \tag{21.55}$$

giving, for small $|\Phi|$ and $\mu > 1$,

$$\Phi \approx \pm[\delta(\mu - 1)]. \tag{21.56}$$

The positive solution, with flow in the capillary tube from box 1 to box 2, is salinity driven — driven by the density gradient of the same sign as that caused by the salinity, with the density

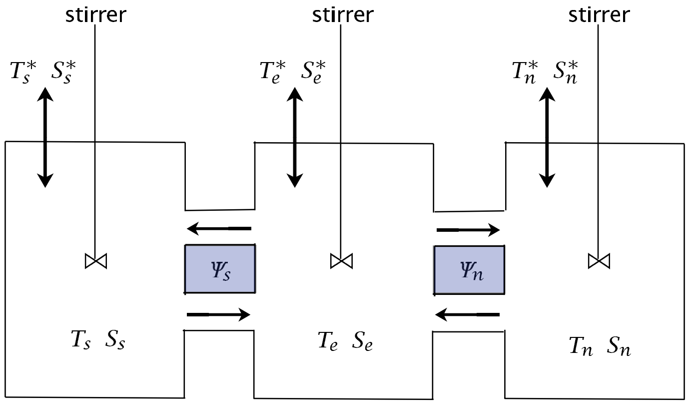

Fig. 21.8 A three-box model. Each box contains fluid with uniform values of temperature and salinity, each exchanges fluid with its neighbour, and in each the temperature and salinity are relaxed toward fixed values.

gradient due to temperature opposing the motion. That is, box 1 is denser than box 2 because it is more saline, even though it is also warmer. The negative solution is thermally driven, with the flow in the capillary tube going from the denser (cold and fresh) box 2 to the lighter (warm and salty) box 1. However, this solution is unstable, and any small perturbation will amplify and the system will move away from this solution. Solving for temperature and salinity we find that $T \approx 1$ (i.e., it is close to its relaxation value and hardly altered by advection), and $S \approx 1/\mu$.

The third solution has a circulation far from the origin, and the balance in (21.54) is between the left-hand side and the first term on the right. In the limiting case we find

$$\Phi \approx -\sqrt{\gamma}. \tag{21.57}$$

This solution has a density gradient dominated by the temperature effect: the temperature difference is $T \approx 1/\sqrt{\gamma}$ whereas the salinity difference is $S \approx \delta/\sqrt{\gamma}$, and thus its effect on density is much smaller.

21.3.2 ✦ More boxes

More boxes can be added in a variety of ways and, now forgoing an easy relevance to a laboratory apparatus, one such is illustrated in Fig. 21.8. The three boxes represent the mid- and high-latitude Northern Hemisphere, the mid- and high-latitude Southern Hemisphere, and the equatorial regions. Each of the three boxes can exchange fluid with its neighbour, and each is also in contact with a reservoir and subject to a relaxation to a fixed value of temperature and salinity, (T_s^*, S_s^*), (T_e^*, S_e^*) and (T_n^*, S_n^*). Then, with obvious notation, we infer the equations of motion for temperature

$$\frac{\mathrm{d}T_s}{\mathrm{d}t} = c(T_s^* - T_s) - |\Psi_s|(T_s - T_e), \qquad \frac{\mathrm{d}T_n}{\mathrm{d}t} = c(T_n^* - T_n) - |\Psi_n|(T_n - T_e),$$
$$\frac{\mathrm{d}T_e}{\mathrm{d}t} = c(T_e^* - T_e) - |\Psi_s|(T_e - T_s) - |\Psi_n|(T_e - T_n), \tag{21.58}$$

and similarly for salt, with flow rates given by the density differences

$$\Psi_s = A\rho_0[-\beta_T(T_s - T_e) + \beta_S(S_s - S_e)], \qquad \Psi_n = A\rho_0[-\beta_T(T_n - T_e) + \beta_S(S_n - S_e)]. \tag{21.59}$$

These equations may be non-dimensionalized and reduced to four prognostic equations for the quantities $T_e - T_n$, $T_e - T_s$, $S_e - S_n$, $S_e - S_s$. Not surprisingly, multiple equilibria can again be found. One interesting aspect is that stable asymmetric solutions arise with symmetric forcing ($T_s^* = T_n^*$, $S_s^* = S_n^*$). These effectively have a pole-to-pole circulation, illustrated in the upper row of Fig. 21.9. Such a circulation can be thought of as the superposition of a thermal circulation in

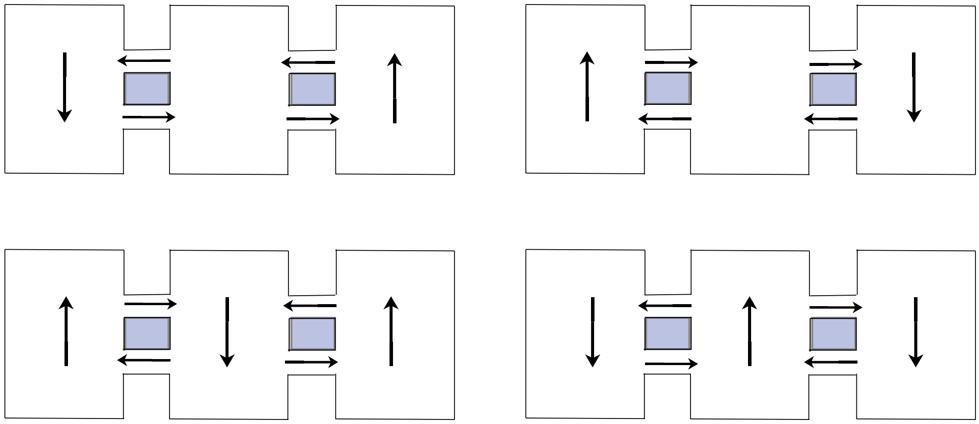

Fig. 21.9 Schematic of four solutions to the three-box model with the symmetric forcing $S_s^* = S_n^*$ and $T_n^* = T_s^*$. The two solutions on the top row have an asymmetric, 'pole-to-pole', circulation whereas the solutions on the bottom row are symmetric.[13]

one hemisphere and a salinity-driven circulation in the other.

The box models are useful because they are suggestive of behaviour that might occur in real fluid systems, and because they provide a means of interpreting behaviour that does occur in more complete numerical models and perhaps in the real world. However, they are by no means good approximations of the real equations of motion and without other supporting evidence the solutions found in box models should not be regarded as representing real solutions of the fluid equations for the world's oceans.[14] Indeed, the mechanism for the observed pole-to-pole circulation in the real ocean may be quite different from that of the box models — see the sections beginning with sec:windmoc.

21.4 A LABORATORY MODEL OF THE ABYSSAL CIRCULATION

We now return to a more fluid dynamical description of the deep ocean circulation, and consider two simple, closely related, models that are relevant to aspects of the deep circulation, still assuming it to be buoyancy- or mixing-driven. The first, which we consider in this section, is a laboratory model, originally envisioned as being a prototype for the deep circulation. The second model, considered in the following sections, is explicitly a model of the deep circulation. Both models are severe idealizations that describe only limited aspects of the circulation, but they are both very helpful tools that enable us to understand more complete models and, in part, the real circulation itself.

21.4.1 Set-up of the laboratory model

Let us consider flow in a rotating tank, as illustrated in Fig. 21.10. The fluid is confined by vertical walls to occupy a sector, and the entire tank rotates anticlockwise when viewed from above, like the Northern Hemisphere. When the fluid is stationary in the rotating frame, the fluid slopes up toward the outer edge of the tank and the balance of forces in the rotating frame is between a centrifugal force pointing outwards and the pressure gradient due to the sloping fluid pointing inwards. In the inertial frame of the laboratory itself, the pressure gradient pointing inwards provides a centripetal force that causes the fluid to accelerate toward the centre of the tank, resulting in a circular motion. (Recall that steady circular motion is always accompanied by an acceleration toward the centre of the circle.)

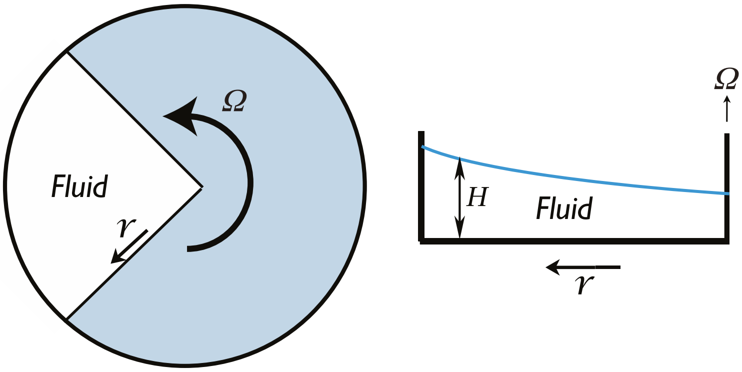

Fig. 21.10 Stommel–Arons–Faller rotating tank experiment. Left: A plan view, with the fluid in the sector at left. Right: Side view. The free surface of the fluid slopes up with increasing radius, giving a balance between the centrifugal force pointing outwards and the pressure force pointing inwards. Small pipes (not shown) provide mass sources and sinks.

This set-up, and the accompanying theory, has become known as the *Stommel–Arons–Faller* model.[15] The motivation of the construct is clear, in that the sector represents an ocean basin. However, rather than driving the fluid with wind or by differential heating, we drive it with localized mass sources and sinks, for example from small pipes inserted into the tank. If an oceanic analogy is desired, the mass source might be thought of as representing a sources of deep abyssal water due to deep convection. The oceanic analogy is not perfect but it helps build intuition about the real ocean.

21.4.2 Dynamics of flow in the tank

Let us assume that the motion of the fluid in the tank is sufficiently weak that its Rossby number is small, and that it obeys the shallow water planetary-geostrophic equations, namely

$$\boldsymbol{f}_0 \times \boldsymbol{u} = -g\nabla h + \Omega^2 r\,\hat{\boldsymbol{r}} + \boldsymbol{F}, \tag{21.60a}$$

$$\frac{\partial h}{\partial t} + \nabla\cdot(\boldsymbol{u}h) = S, \tag{21.60b}$$

where $\boldsymbol{u} = (v^r, v^\theta)$ is the horizontal velocity in cylindrical (r,θ) coordinates, $\hat{\boldsymbol{r}}$ is a unit vector in the direction of increasing r, $\boldsymbol{F}$ represents frictional terms (which we will suppose are small except in boundary layers) and S represents mass sources. These two equations yield the potential vorticity equation,

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{f_0}{h}\right) = \frac{\mathrm{curl}_z\boldsymbol{F}}{h} - \frac{f_0 S}{h^2}. \tag{21.61}$$

Let us write the height field as

$$h = H(r,t) + \eta(r,\theta,t) \tag{21.62}$$

where $H(r,t)$ is the height field corresponding to the rest state of the fluid (in the rotating frame) and η the perturbation. Thus, from (21.60a)

$$0 = -g\nabla H + \Omega^2 r\,\hat{\boldsymbol{r}}, \tag{21.63}$$

which gives

$$H = \frac{\Omega^2 r^2}{2g} + \widehat{H}(t), \tag{21.64}$$

where $\widehat{H}$ is a measure of the overall mass of the fluid. Its rate of change is determined by the mass source

$$\frac{\mathrm{d}\widehat{H}}{\mathrm{d}t} = \langle S\rangle\,, \tag{21.65}$$

the angle brackets indicating a domain average. The equations of motion (21.60) become

$$\boldsymbol{f}_0 \times \boldsymbol{u} = -g\nabla\eta + \boldsymbol{F}, \tag{21.66a}$$

$$\frac{\partial}{\partial t}(\eta + H) + \nabla \cdot [\boldsymbol{u}(\eta + H)] = 0. \tag{21.66b}$$

Equation (21.66a) tells us that, away from frictional regions, the velocity is in geostrophic balance with the pressure field due to the perturbation height η.

Let us now suppose $|\eta| \ll H$, which holds if the mass source is small and gentle enough. Then (21.66b) may be written

$$\frac{\partial H}{\partial t} + \nabla \cdot (\boldsymbol{u}H) = 0. \tag{21.67}$$

In this approximation, the potential vorticity equation (21.61) becomes, away from friction and mass sources,

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{f_0}{H}\right) = 0 \qquad \text{or} \qquad \frac{\mathrm{D}H}{\mathrm{D}t} = 0, \tag{21.68a,b}$$

where the second equation follows because f_0 is a constant. [This equation also follows directly from (21.67), because the velocity is geostrophic and divergence-free where friction is absent; however, it is better thought of as a potential vorticity equation, not a mass conservation equation.] Equation (21.68b) means that fluid columns change position in order to keep the same value of H. Further, because H only varies with r, (21.68b) becomes

$$\frac{\partial H}{\partial t} + v^r \frac{\partial H}{\partial r} = 0, \tag{21.69}$$

which, using (21.64) and (21.65), gives

$$\boxed{v^r = -\frac{g}{\Omega^2 r}\langle S \rangle .} \tag{21.70}$$

This is a remarkable result, for it implies that, if $\langle S \rangle$ is positive, the flow is *toward* the apex of the dish, except at the location of the mass sources and in frictional boundary layers, *no matter where the mass source is actually located*. The explanation of this counter-intuitive result is simple enough. If $\langle S \rangle > 0$ the overall height of the fluid is increases with time; thus, in order that a given material column of fluid keep its height fixed, it must move toward the apex of the dish. The full velocity may be obtained, away from the frictional regions, using the divergence-free nature of the velocity:

$$\nabla \cdot \boldsymbol{u} = \frac{1}{r}\left[\frac{\partial (r v^r)}{\partial r} + \frac{\partial v^\theta}{\partial \theta}\right] = 0. \tag{21.71}$$

Then, using (21.70), $\partial v^\theta / \partial \theta = 0$ except at a source or sink, or in a frictional boundary layer. Assuming there is only one frictional boundary layer, $v^\theta = 0$ except at those latitudes (i.e., values of r) that contain a mass source or sink.

To balance the flow toward the apex there must, then, be a *boundary layer* in which the flow has the opposite sense, and therefore in which frictional effects are important. To determine where the boundary layer is — on the east or west side of the domain — we need some vorticity dynamics. Away from the mass source, but including friction, the potential vorticity equation is

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{f_0}{H}\right) = \frac{\mathrm{curl}_z \boldsymbol{F}}{H} \qquad \text{or} \qquad -\frac{f_0}{H^2}\frac{\mathrm{D}H}{\mathrm{D}t} = \mathrm{curl}_z \boldsymbol{F}, \tag{21.72a,b}$$

and the free surface of the water slopes downwards toward the apex, as illustrated in Fig. 21.10. Now, suppose that there are mass source and a sink of equal magnitudes, with the source further

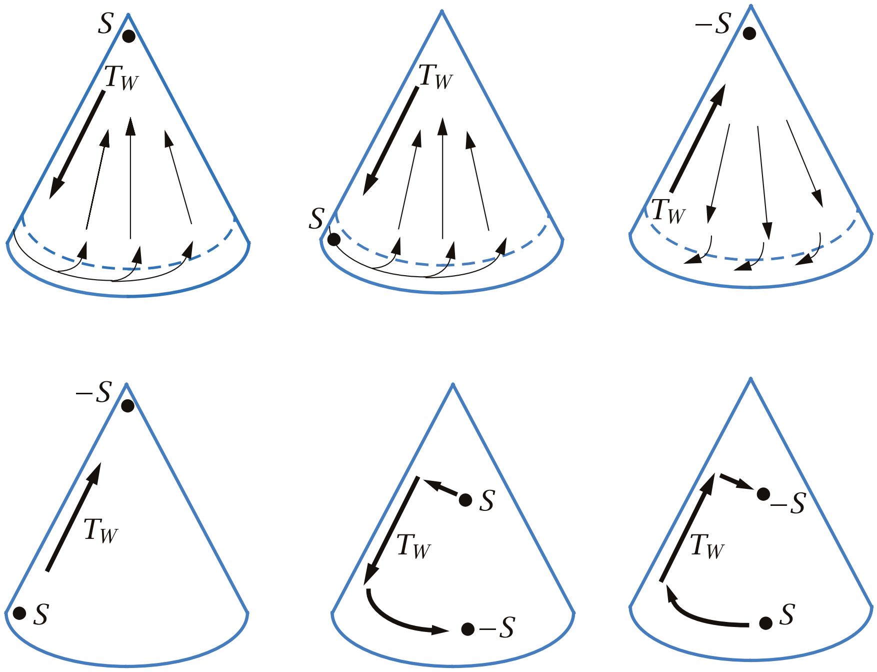

Fig. 21.11 Idealized examples of the flow in the rotating sector experiments, with various locations of a source (S) or sink ($-S$) of mass.

from the apex than the sink, as in the panel at the bottom right of Fig. 21.11. The flow from source to sink must be along either the left or right boundary of the container. To see which, note that the flow is toward smaller values of H, and therefore the left-hand side of (21.72a) is positive. To balance this, the friction in the boundary current must import a positive vorticity to the flow (i.e., $\mathrm{curl}_z \boldsymbol{F} > 0$), which means in general that the flow itself must have negative vorticity, and the flow is clockwise. (For example, if $\boldsymbol{F} = -\lambda \boldsymbol{u}$ the right hand side of (21.72) is $-(\lambda/H)\mathrm{curl}_z \boldsymbol{u}$ and this is positive if the flow is clockwise.) Clockwise flow implies a *western* boundary layer, on the left of the container. A western boundary layer is general feature, not dependent on the placement of mass sources or sinks. For suppose there is a single source of mass, as for example in the upper left example of Fig. 21.11; the interior mass flow will then be toward the apex and the flow in the boundary layer away from the apex. The left-hand side of (21.72a) is then negative, and so $\mathrm{curl}_z \boldsymbol{F}$ must be negative. The flow must then have an anticlockwise sense, again requiring a western boundary layer to achieve a balance in the potential vorticity equation.

The flow is in some ways analogous to flow on the β-plane, and in particular:

(i) the r-dependence of the height field provides a background potential vorticity gradient, analogous to the β-effect;

(ii) the time-dependence of H is analogous to a wind curl, since it is this that ultimately drives the fluid motion.

The analogies are drawn out explicitly in the shaded box on page 823; the box also includes a column for abyssal flow in the ocean, discussed in the next two sections.

21.5 A MODEL FOR OCEANIC ABYSSAL FLOW

We will now extend the reasoning applied to the rotating tank to the rotating sphere, and so construct a model — the *Stommel–Arons model* — of the abyssal flow in the ocean.[16] The basic idea

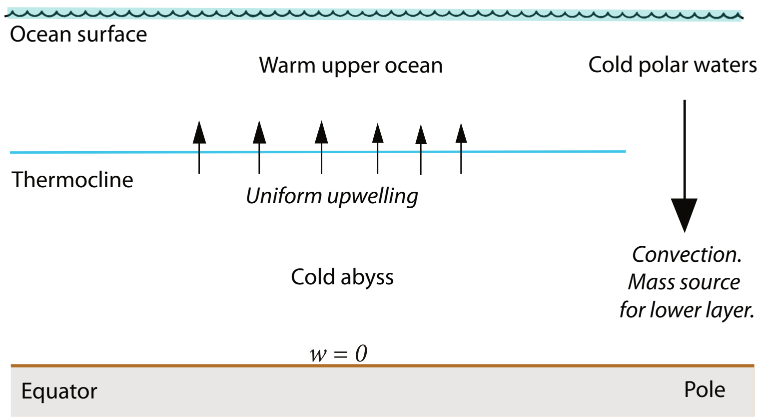

Fig. 21.12 The structure of a Stommel–Arons ocean model of the abyssal circulation. Convection at high latitudes provides a localized mass-source to the lower layer, and upwelling through the thermocline provides a more uniform mass sink.

is simple: we will model the deep ocean as a single layer of homogeneous fluid in which there is a localized injection of mass at high latitudes, representing convection (Fig. 21.12). However, unlike the rotating dish, mass is extracted from this layer by upwelling into the warmer waters above it, keeping the average thickness of the abyssal layer constant. We assume that this upwelling is nearly uniform, that the ocean is flat-bottomed, and that a passive western boundary current may be invoked to satisfy mass conservation, and which does not affect the interior flow. Obviously, these assumptions are very severe and the model can at best be a conceptual model of the real ocean. Given that, we will work in Cartesian coordinates on the β-plane, and use the planetary-geostrophic approximation.

The momentum and mass continuity equations are

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla_z \phi \quad \text{and} \quad \nabla_z \cdot \boldsymbol{u} = -\frac{\partial w}{\partial z}, \tag{21.73a,b}$$

where $\boldsymbol{f} = (f_0 + \beta y)\mathbf{k}$. On elimination of ϕ, (21.73) yields the now-familiar balance,

$$\beta v = f \frac{\partial w}{\partial z}. \tag{21.74}$$

Except in the localized regions of convection, the vertical velocity is, by assumption, positive and uniform at the top of the lower layer, and zero at the bottom. Thus (21.74) becomes

$$v = \frac{f}{\beta}\frac{w_0}{H}, \tag{21.75}$$

where w_0 is the uniform upwelling velocity and H the layer thickness. Thus, the flow is *polewards* everywhere (including the Southern Hemisphere), vanishing at the equator.

21.5.1 Completing the solution

Since $v = f^{-1}(\partial \phi / \partial x)$, the pressure is given by

$$\phi = \int_{x_0}^{x} \left(\frac{f^2 w_0}{\beta H} \right) \mathrm{d}x', \tag{21.76}$$

where x_0 is a constant of integration, to be determined by the boundary conditions. Because there no flow into the Eastern boundary, x_E, we set ϕ = constant at $x = x_E$, and because this is a one-layer

Analogies Between a Rotating Dish, Wind-Driven and Abyssal flows

Consider homogeneous models of (i) rotating dish, (ii) wind-driven flow on the β-plane, and (iii) abyssal flow on the β-plane. We model them all with a single layer of homogeneous fluid satisfying the planetary geostrophic equations. In (i) the mass source, $\langle S \rangle$, is localized and the total depth of the fluid layer changes with time; fluid columns move to keep their depth constant. In (ii) there is no mass source and the depth of the fluid layer is constant; the fluid motion is determined by the wind-stress curl, $\mathrm{curl}_z \boldsymbol{\tau}$, and by β. In (iii) the fluid source (convection) is localized at high latitudes and exactly balanced by a mass loss, S_u, due to upwelling everywhere else, so that layer depth is constant and S_u is uniform and negative nearly everywhere. The equations below then apply away from frictional boundary layers and localized mass sources.

	(i) Rotating dish	*(ii) Wind-driven flow*	*(iii) Abyssal flow*
PV Conservation	$\dfrac{\mathrm{D}}{\mathrm{D}t}\left(\dfrac{f_0}{H}\right) = 0$	$\dfrac{\mathrm{D}}{\mathrm{D}t}\left(\dfrac{f}{H_0}\right) = \dfrac{1}{H_0}\mathrm{curl}_z \boldsymbol{\tau}$	$\dfrac{\mathrm{D}}{\mathrm{D}t}\left(\dfrac{f}{h}\right) = -\dfrac{f S_u}{h^2}$
This leads to	$v^r \dfrac{\partial H}{\partial r} = -\dfrac{\partial H}{\partial t}$	$\dfrac{v}{H_0}\dfrac{\partial f}{\partial y} = \dfrac{1}{H_0}\mathrm{curl}_z \boldsymbol{\tau}$	$\dfrac{v}{h}\dfrac{\partial f}{\partial y} = -\dfrac{f S_u}{h^2}$
and	$v^r = -\dfrac{g}{\Omega^2 r}\langle S \rangle$	$v = \dfrac{1}{\beta}\mathrm{curl}_z \boldsymbol{\tau}$	$v = -\dfrac{f S_u h}{\beta}$
	$\langle S \rangle$ is localized mass source	$\mathrm{curl}_z \boldsymbol{\tau}$ is wind stress curl	S_u is upwelling mass loss

Meridional mass flow away from boundaries is thus determined by:

sign (and not location) of localized mass source, $\langle S \rangle$,	sign of wind-stress curl, $\mathrm{curl}_z \boldsymbol{\tau}$,	upwelling and sign of f, so polewards if $S_u < 0$ (upwelling).

model we are at liberty to set that constant equal to zero. Thus,

$$\phi(x) = -\int_x^{x_E} \left(\frac{f^2 w_0}{\beta H}\right) \mathrm{d}x' = -\frac{f^2}{\beta H} w_0 (x_E - x). \tag{21.77}$$

The zonal velocity follows using geostrophic balance,

$$u = \frac{1}{f}\frac{\partial \phi}{\partial y} = \frac{2}{H} w_0 (x_E - x), \tag{21.78}$$

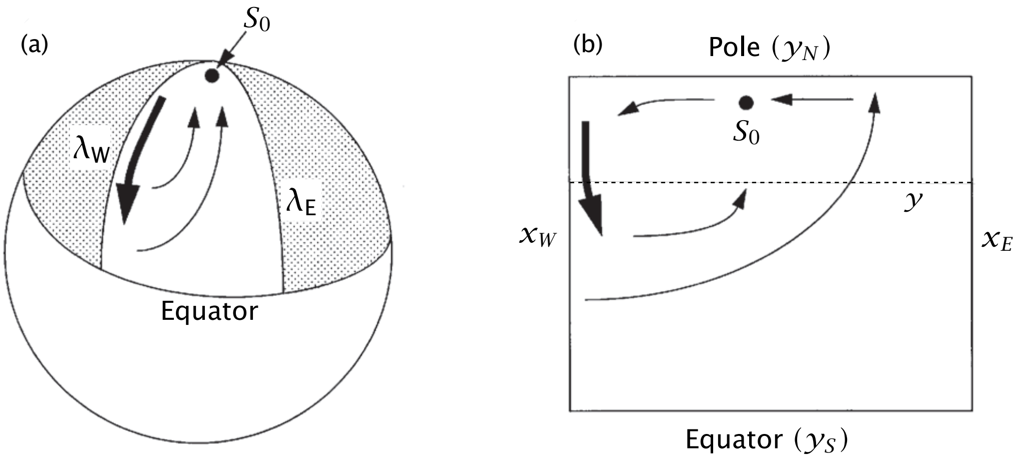

Fig. 21.13 Abyssal circulation in a spherical sector (left) and in a corresponding Cartesian rectangle (right).

where we have also used $\partial f/\partial y = \beta$ and $\partial\beta/\partial y = 0$. Thus the velocity is eastwards in the interior, and independent of f and latitude, provided x_E is not a function of y.

Using (21.75) and (21.78) we can confirm mass conservation is indeed satisfied:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = -\frac{2w_0}{H} + \frac{w_0}{H} + \frac{w_0}{H} = 0. \tag{21.79}$$

21.5.2 Application to the ocean

Let us consider a rectangular ocean with a mass source at the northern boundary, balanced by uniform upwelling (see Figs. 21.13 and 21.14). Since the interior flow will be northwards, we anticipate a southwards flowing western boundary current to balance mass. Conservation of mass in the area polewards of the latitude y demands that

$$S_0 + T_I(y) = T_W(y) + U(y), \tag{21.80}$$

where S_0 is the strength of the source, T_W the equatorwards transport in the western boundary current, T_I the poleward transport in the interior, and U is the integrated loss due to upwelling polewards of y. Then, using (21.75),

$$T_I = \int_{x_W}^{x_E} vH\,\mathrm{d}x = \int_{x_W}^{x_E} \frac{fw_0}{\beta}\,\mathrm{d}x = \frac{f}{\beta}w(x_E - x_W). \tag{21.81}$$

The upwelling loss is given byy

$$U = \int_{x_W}^{x_E}\int_{y}^{y_N} w\,\mathrm{d}x = w_0(x_E - x_W)(y_N - y), \tag{21.82}$$

where y_N denotes the northern (polar) boundary. Assuming the source term is known, then using (21.80) we obtain the strength of the western boundary current,

$$T_W(y) = S_0 + T_I - U = S_0 + \frac{f}{\beta}w(x_E - x_W) - w_0(x_E - x_W)(y_N - y). \tag{21.83}$$

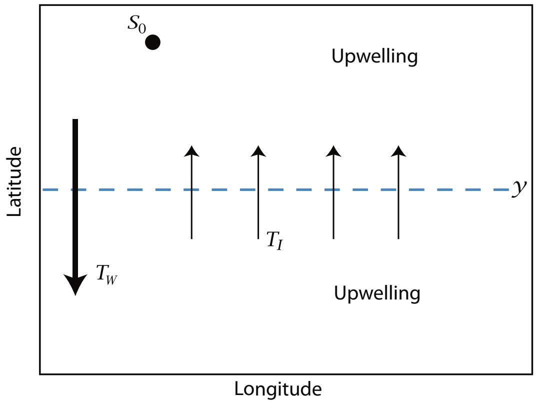

Fig. 21.14 Mass budget in an idealized abyssal ocean. Polewards of some latitude y, the mass source (S_0) plus the poleward mass flux across y (T_I) are equal to the sum of the equatorward mass flux in the western boundary current (T_W) and the integrated loss due to upwelling (U) polewards of y. See (21.80).

To close the problem we use the fact that over the entire basin mass must be balanced, which gives a relationship between w_0 and S_0,

$$S_0 = w_0 \Delta x \Delta y, \tag{21.84}$$

where $\Delta x = x_E - x_W$ and $\Delta y = y_N - y_S$, with y_S being the southern boundary of the domain. Using (21.84), (21.83) becomes

$$T_W(y) = -w_0 \left(\Delta x (y_N - y) - \frac{f}{\beta} \Delta x - \Delta x \Delta y \right) = w_0 \Delta x \left(y - y_S + \frac{f}{\beta} \right). \tag{21.85}$$

With no loss of generality we will take $y_S = 0$ and $f = f_0 + \beta y$. Then (21.85) becomes

$$T_W(y) = w_0 \Delta x \left(2y + f_0/\beta \right) \tag{21.86}$$

or, using $S_0 = w_0 \Delta x y_N$,

$$T_W(y) = \frac{S_0}{y_N} \left(2y - \frac{f_0}{\beta} \right). \tag{21.87}$$

With a slight loss of generality (but consistent with the spirit of the planetary-geostrophic approximation) we take $f_0 = 0$, which is equivalent to supposing that the equatorial boundary of the domain is at the equator, and finally obtain

$$\boxed{T_W(y) = 2S_0 \frac{y}{y_N}.} \tag{21.88}$$

At the northern boundary this becomes

$$T_W(y) = 2S_0, \tag{21.89}$$

which means that the flow southwards from the source is *twice* the strength of the source itself! We also see that:

(i) the western boundary current is equatorwards everywhere;
(ii) at the northern boundary the equatorward transport in the western boundary current is equal to *twice* the strength of the source;
(iii) the northward mass flux at the northern boundary is equal to the strength of the source itself.

We may check this last point directly: from (21.81)

$$T_I(y_N) = \frac{\beta y_N}{\beta} w_0 \Delta x = S. \tag{21.90}$$

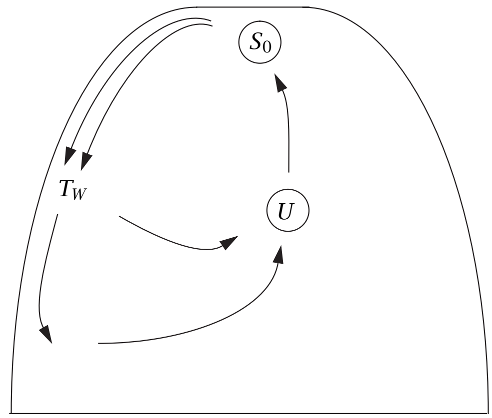

Fig. 21.15 Schematic of a Stommel–Arons circulation in a single sector. The transport of the western boundary current is greater than that provided by the source at the apex, illustrating the property of *recirculation*. The transport in the western boundary current T_W decreases in intensity equatorwards, as it loses mass to the polewards interior flow, and thence to upwelling. The integrated sink, due to upwelling, U, exactly matches the strength of the source, S.

The fact that convergence at the pole balances T_W and S_0 does not of course depend on the particular choice we made for f and y_S.

The flow pattern evidently has the property of *recirculation* (see Fig. 21.15): this is one of the most important properties of the solution, and one that is likely to transcend all the limitations inherent in the model. This single-hemisphere model may be thought of as a crude model for aspects of the abyssal circulation in the North Atlantic, in which convection at high latitudes near Greenland is at least partially associated with the abyssal circulation. In the North Pacific there is, in contrast, little if any deep convection to act as a mass source. Rather, the deep circulation is driven by mass sources in the opposite hemisphere, and we now consider a simple model of this.

21.5.3 A two-hemisphere model

Our treatment now is even more obviously heuristic, since our domain crosses the equator yet we continue to use the planetary-geostrophic equations, invalid at the equator. We also persist with Cartesian geometry, even for these global-scale flows. In our defence, we remark that the value of the solutions lies in their qualitative structure, not in their quantitative predictions. Let us consider a situation with a source in the Southern Hemisphere but none in the Northern Hemisphere. For later convenience we take the Southern Hemisphere source to be of strength $2S_0$, and we suppose the two hemispheres have equal area. As before, the upwelling is uniform, so that to satisfy global mass balance

$$S_0 = w_0 \,\Delta x\, \Delta y, \tag{21.91}$$

where $\Delta x \Delta y$ is the area of each hemisphere. Then, given w_0, the zonally integrated poleward interior flow in each hemisphere, away from the equator, follows from Sverdrup balance,

$$T_I(y) = \frac{f}{\beta} w_0 (x_E - x_W) = S_0 \frac{y}{y_p}, \tag{21.92}$$

where y_p is either y_N (the northern boundary) or y_S. The western boundary current is assumed to 'take up the slack', that is to be able to adjust its strength to satisfy mass conservation. Thus, since $T_I(y_N) = S_0$, where S_0 is half the strength of the source in the *Southern* Hemisphere, it is plain that there must be a southwards flowing western boundary current near the northern end of the Northern Hemisphere, even in the absence of any deep water formation there!

In the northern hemisphere, the total loss due to upwelling polewards of a latitude y is given by

$$U(y) = w_0 \Delta x |y_N - y|. \tag{21.93}$$

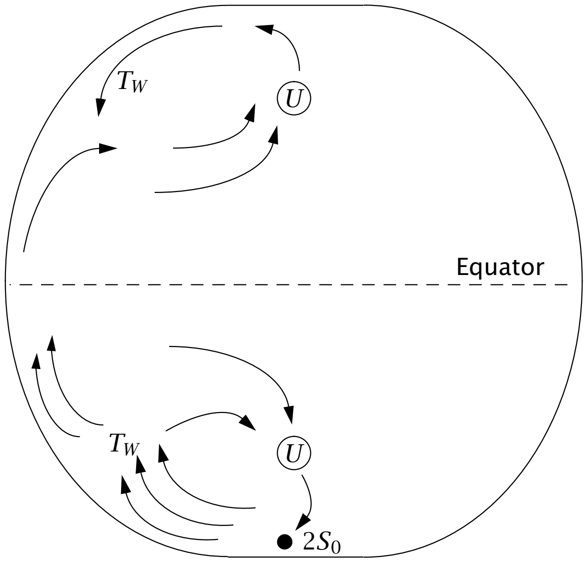

Fig. 21.16 Schematic of a Stommel–Arons circulation in a two-hemisphere basin. There is only one mass source, and this is in the Southern Hemisphere and for convenience it has a strength of 2. Although there is no source in the Northern Hemisphere, there is still a western boundary current and a recirculation. The integrated sinks due to upwelling exactly match the strength of the source.

The strength of the western boundary current is then given by, with southward flow positive,

$$T_W(y) = T_I - U = \frac{f}{\beta} w_0 \Delta x - w_0 \Delta x(y_N - y) = -w_0 \Delta x(y_N - 2y), \tag{21.94}$$

using $f = \beta y$. The boundary current thus changes sign halfway between equator and North Pole, at $y = y_N/2$. [In spherical coordinates, the analogous latitude turns out to be at $\theta = \sin^{-1}(1/2)$.] At the North Pole $y = y_N$ and we have

$$T_W(y_N) = w_0 \Delta x y = S. \tag{21.95}$$

The solution is illustrated schematically in Fig. 21.16. We can (rather fancifully) imagine this to represent the abyssal circulation in the Pacific Ocean, with no source of deep water at high northern latitudes.

21.5.4 Summary remarks on the Stommel–Arons model

If we were given the location and strength of the sources of deep water in the real ocean, the Stommel–Arons model could give us a global solution for the abyssal circulation. The solution for the Atlantic, for example, resembles a superposition of Fig. 21.15 and Fig. 21.16 (with deep water sources in the Weddell Sea and near Greenland), and that for the Pacific resembles Fig. 21.16 (with a deep water source emanating from the Antarctic Circumpolar Current). Perhaps the greatest success of the model is that it introduces the notions of deep western boundary currents and recirculation — enduring concepts of the deep circulation that remain with us today. For example, the North Atlantic ocean does have a well-defined deep western boundary current running south along the eastern seaboard of Canada and the United States, as seen in Fig. 21.17.[17] However, in other important aspects the model is found to be in error, in particular it is found that there is little upwelling through the main thermocline — much of the water formed by deep convection in the North Atlantic in fact upwells in the Southern Hemisphere.[18] Are there fundamental problems with the model, or just discrepancies in details that might be corrected with a slight reformulation? To help answer that we summarize the assumptions and corresponding predictions of the model, and distinguish the essential aspects from what is merely convenient.

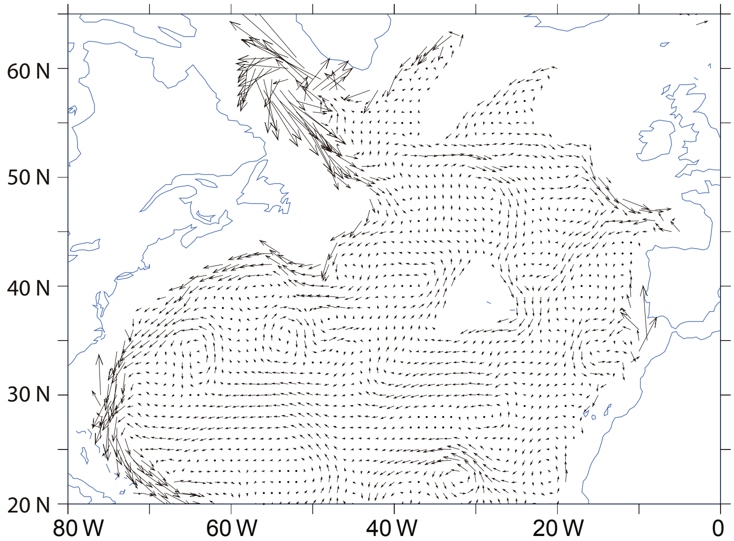

Fig. 21.17 The ocean currents at a depth of 2500 m in the North Atlantic, obtained using a combination of observations and model (as in Fig. 19.3). Note the southwards flowing *deep western boundary current.*

(i) A foundational assumption is that of linear geostrophic vorticity balance in the ocean abyss, represented by $\beta v \approx f \partial w / \partial z$, or its shallow water analogue.

— The effects of mesoscale eddies are thereby neglected. As discussed in chapter 12, in their mature phase mesoscale eddies seek to barotropize the flow, and so create deep eddying motion that might dominate the deep flow.

(ii) A second important assumption is that of uniform upwelling, across isopycnals, into the upper ocean, and that $w = 0$ at the ocean bottom. When combined with *(i)* this gives rise to a poleward interior flow, and by mass conservation a deep western boundary current. The upwelling is a consequence of a finite diffusion, which in turn leads to deep convection as in the model of sideways convection of section 21.1.

— The uniform-upwelling assumption might be partially relaxed, while remaining in the Stommel–Arons framework, by supposing (for example) that the upwelling occurs near boundaries, or intermittently, with corresponding detailed changes to the interior flow.

— If bottom topography is important, then $w \neq 0$ at the ocean bottom. This effect may be most important if mesoscale eddies are present, for then in an attempt to maintain its value of potential vorticity the abyssal flow will have a tendency to meander nearly inviscidly along contours of constant topography. In the presence of a mid-ocean ridge, some of the deep western boundary current might travel meridionally along the eastern edge of the ridge instead of along the coast.

— In reality, the deep water might not upwell across isopycnals at all, but might move along isopycnals that intersect the surface (or are connected to the surface by convection). If so, then in the presence of mechanical forcing a deep circulation could be maintained even in the absence of a diapycnal diffusivity. The circulation might then be qualitatively different from the Stommel–Arons model, although a linear vorticity balance might still hold, with deep western boundary currents. This is discussed in section 21.6.

Even if the Stommel–Arons picture were to be essentially correct, we should not consider the deep flow as being driven by deep convection at the source regions. It is a *convenience* to specify the strength of the source term in these regions for the calculations but, just as in the models of sideways convection considered in section 21.1, the overall strength of the circulation (insofar as it is

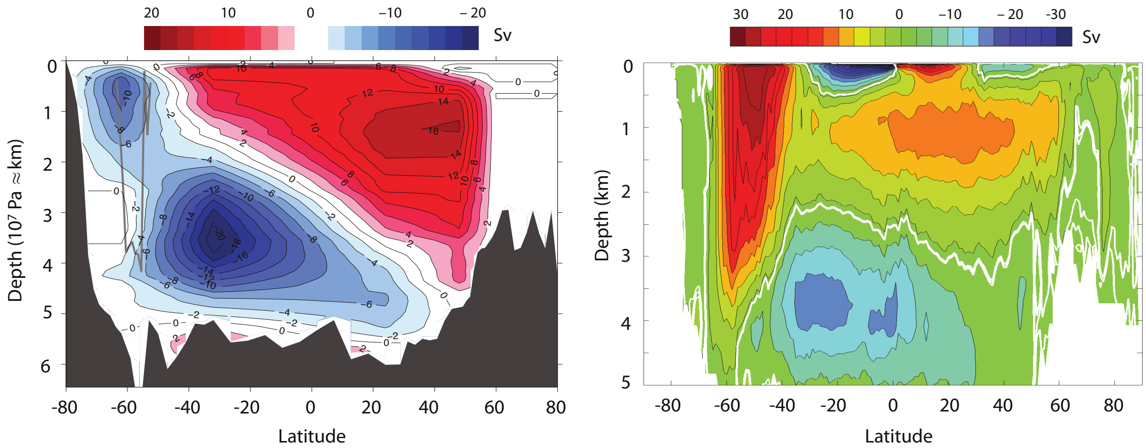

Fig. 21.18 Two independent estimates of the overturning circulation of the world's ocean. The left panel is from an inverse model that mainly uses hydrographic observations, and shows the residual circulation. The right panel is a state estimation that makes explicit use of a numerical model, and shows the Eulerian circulation.[20]

buoyancy driven) is a function of the size of the diffusivity and the meridional buoyancy gradient at the surface.

21.6 † A MODEL OF DEEP WIND-DRIVEN OVERTURNING

We have previously noted that, with values of the diapycnal diffusivity that are measured in the main thermocline, the theoretical predictions of the MOC are rather weaker than observations suggest. There are two possible resolutions to this problem. One is that the measured diapycnal diffusivity is in fact large in some parts of the ocean (e.g., in the abyss over steep topography), and if this were sufficient to produce the measured overturning and stratification the issue would be resolved. However, such a calculation would likely be fraught with uncertainty. A second and more straightforward resolution would arise if a deep circulation, and deep stratification, could be maintained by a mechanism that was *independent* of the diapycnal diffusivity. This second approach is the one we shall take in for the remainder of this chapter. Specifically, our goal is to construct and explore models of the overturning circulation of the ocean, and the concomitant deep stratification, that have a 'wind-driven' component that persists even as the diffusivity goes to zero.[19]

21.6.1 Observations and physical principles

We are motivated by the observation that the MOC is in large part *interhemispheric,* with water sinking at high northern latitudes and upwelling in the Antarctic Circumpolar Current (ACC), as seen for for the global circulation in Fig. 21.18 (where the left panel better shows the trajectory of water parcels) and in the Atlantic in Fig. 21.29. The Atlantic MOC (which is the dominant contributor to the global MOC) is dominated by two cells, an upper cell of North Atlantic Deep Water (NADW) with water sinking at about 60° N, moving southwards largely along isopycnals and upwelling in the south. Beneath this cell lies Antarctic Bottom Water (AABW), with sinking at high southern latitudes followed by a deep cross hemispheric circulation and upwelling again in the ACC. Our goal is this section is to construct a purely wind-driven model that shows these features as simply as possible.

In the absence of a diapycnal diffusivity no upwelling can occur *through* the stratification, because that is a diabatic process. Rather, if there is deep stratification, the deep water must be directly

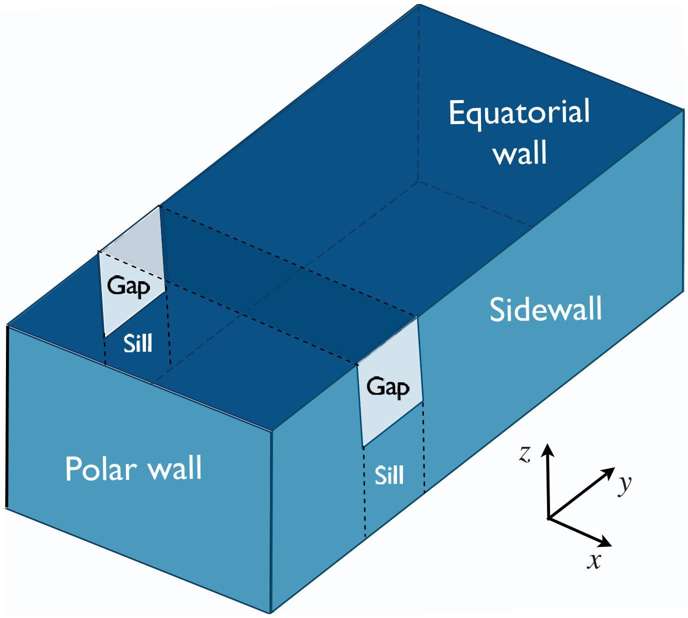

Fig. 21.19 Idealized geometry of the Southern Ocean: a re-entrant channel, partially blocked by a sill, is embedded within a closed rectangular basin; thus, the channel has periodic boundary conditions. The channel is a crude model of the Antarctic Circumpolar Current, with the area over the sill analogous to the Drake Passage.

connected to the surface along isopycnals or via a convective pathway, for convection, although diabatic, does not rely on a finite eddy diapycnal diffusivity. Let us recall two *de facto* principles about deep circulation.

(i) A basin will, in the absence of mechanical forcing, tend to fill with the densest available fluid.

(ii) Light fluid forced down by wind may displace the cold fluid, so producing stratification.

A completely closed ocean thus fills completely with dense polar water, except in the upper several hundred metres where the main thermocline forms, and whose thickness is the sum of wind-driven and diffusive components. However, suppose that the polewards part of the basin is not fully enclosed but is periodic, as illustrated in Fig. 21.19, with a sill across it at mid-depth, and suppose too that the surface boundary conditions are such that the surface temperature decreases monotonically polewards. A fully enclosed basin exists only beneath the level of the sill, and we may expect the densest water in the basin, formed at the polewards edge of the domain, to fill the basin only below the level of the sill, and that above this may lie warmer water with origins at lower latitudes. Furthermore, suppose that an eastward wind blows over the channel that produces an equatorial flow in the Ekman layer. Then mass conservation demands that there must exist a *subsurface* return flow, and thus a meridional overturning circulation is set up. Note the essential role of the channel in this: if the gap were closed, then the return flow could take place at the surface via a western boundary current, as in a conventional subpolar gyre, and no overturning circulation need be set up. But in a zonally-periodic channel, an eastward wind produces a northward Ekman flow that can only be balanced by a return flow at depth — that is, a meridional overturning circulation.

21.6.2 A single-hemisphere model

We consider first a single-hemisphere basin with a periodic channel near its poleward edge. We suppose it to be in the southern hemisphere, so the channel represents the Antarctic Circumpolar Current (ACC), and that the dynamics are Boussinesq and planetary-geostrophic. We will choose extremely simple forms of wind and buoyancy forcing to allow us to obtain an analytic solution, and then later discuss how the qualitative forms of these solutions might more apply generally.

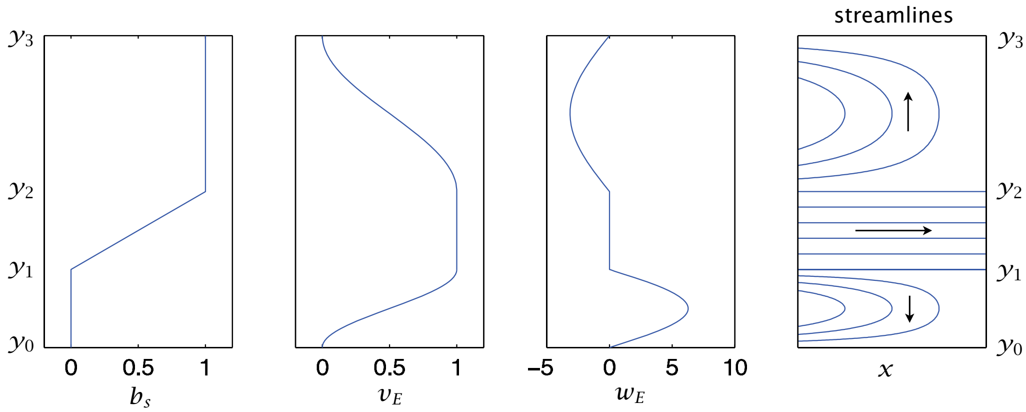

Fig. 21.20 The surface buoyancy b_s, meridional Ekman velocity v_E, vertical Ekman velocity w_E and the solution streamlines for the geostrophic horizontal flow, omitting the western boundary currents. The ordinate in all plots is latitude, with the pole at the bottom, and the four fields are given by, respectively, (21.96), (21.97a), (21.97b) and (21.98), with purely zonal flow given by (21.101) in the channel.

Wind and buoyancy forcing

Thermodynamic forcing is imposed by fixing the surface buoyancy, b_s. (In the discussion following salinity is absent, and buoyancy is virtually equivalent to temperature.) South of the gap we suppose the buoyancy to be constant, then that it increases linearly across the gap, and is constant again polewards of the gap. Thus, there is no temperature gradient across the subtropical gyre, focusing attention on the influence of the channel. Thus, referring to Fig. 21.20 or Fig. 21.21 for the definitions of the geometric factors,

$$b_s = \begin{cases} b_1, & y_0 \le y \le y_1 \\ b_1 + \dfrac{(b_2 - b_1)(y - y_1)}{y_2 - y_1}, & y_1 \le y \le y_2 \\ b_2, & y \ge y_2, \end{cases} \tag{21.96}$$

where $b_2 > b_1$, and both are constants, and we may take $b_1 = 0$ and $y_0 = 0$.

The wind forcing is purely zonal, and it is convenient to express this in terms of the Ekman transport and associated pumping (refer to section 5.7). In the channel the Ekman transport is chosen to be (realistically) equatorwards and (less realistically) constant, a simplification that avoids complications of wind-driven upwelling in the channel. South (polewards) of the channel there is a conventional subpolar gyre, with an Ekman upwelling and an equatorward Ekman transport that joins smoothly to that of the channel. Equatorwards of the channel there is conventional subtropical gyre, with Ekman downwelling. All this may be achieved by specifying:

$$v_E = \begin{cases} \dfrac{V}{2}\left[1 - \cos\left(\dfrac{\pi y}{\Delta y_1}\right)\right] \\ V \\ \dfrac{V}{2}\left[1 + \cos\left(\dfrac{\pi(y - y_2)}{\Delta y_2}\right)\right] \end{cases} \qquad w_E = \begin{cases} W_1 \sin\left(\dfrac{\pi y}{\Delta y_1}\right) & 0 \le y < y_1 \\ 0 & y_1 \le y < y_2 \\ -W_2 \sin\left(\dfrac{\pi(y - y_2)}{\Delta y_2}\right) & y_2 \le y < y_3, \end{cases} \tag{21.97a,b}$$

where $\Delta y_1 = y_1$, $\Delta y_2 = y_3 - y_2$, and V is a constant that determines the magnitude of the meridional Ekman flow. The meridional Ekman transport, v_E, is related to the Ekman pumping by $w_E/\delta_E = \partial v_E/\partial y$, so that $W_i = \delta_E \pi V/(2\Delta y_i)$, where δ_E is the Ekman layer thickness. If f were constant,

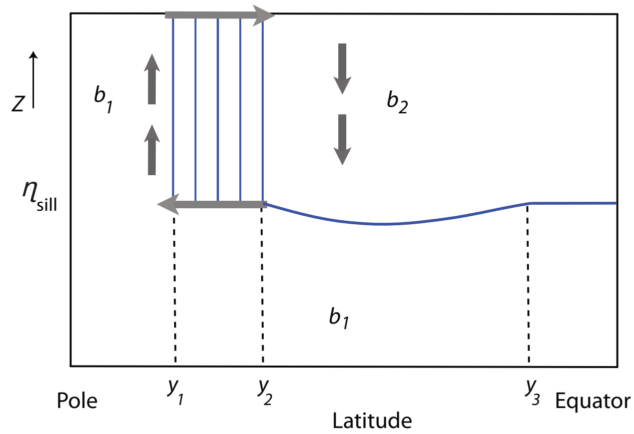

Fig. 21.21 Cross-section of the structure of the single-hemisphere ocean model described in section 21.6.2. The domain is zonally closed equatorwards of y_2 and polewards of y_1, with a zonally periodic channel between latitudes y_1 and y_2 and above the sill, which has height η_{sill}. The arrows indicate the fluid flow driven by the equatorward Ekman transport in the channel, and the solid lines are isopycnals.

the wind-stress curl would be proportional to the w_E field above. The precise details of the forcing do not affect the qualitative form of the solution — they merely allow an analytic solution to be obtained — but there are two essential aspects to it.

(i) The surface is cold south of the channel, warm north of the channel, and there is a temperature gradient across the channel.

(ii) The Ekman flow is equatorwards within the channel, with conventional gyres to either side.

The meridional extent of the region south of the channel and the wind forcing within it are relatively unimportant, and the region could be shrunk to nearly zero.

Solution in the gyres

Below the depth of the sill the basin is fully enclosed, and therefore up to that level the basin will fill with the densest available water (much as described in section 21.1), except where it may be displaced by warmer fluid equatorward of the gap that is pumped down below the level of the sill by the wind (Fig. 21.21). Thus, all of the domain south of the channel, and nearly everywhere below the sill, the water has buoyancy b_1. Polewards of the channel, then, the fluid is barotropic and its vertically integrated horizontal circulation is given by Sverdrup balance, $\beta V = f w_E/H$, where V is the vertically integrated flow. With the wind stress of (21.97) we get a conventional barotropic subpolar gyre (and associated western boundary current) by the same methods that we employed in chapter 19.

Above the sill, net meridional geostrophic transfer is forbidden in the channel region, because by geostrophic balance $f\overline{v}_g = \partial\overline{\phi}/\partial x = 0$, where ϕ is the pressure and the overbar denotes a zonal average. Equatorward of the channel the region above the sill will therefore tend to fill with the densest water available to it, and this is water with buoyancy equal to b_2 (which is the buoyancy of the water as it emerges from the channel). However because of the presence of wind forcing, the base of this layer is not flat; rather, this fluid obeys the dynamics of the reduced-gravity single-layer ventilated thermocline model discussed in section 20.6.1. In such a model the depth of the fluid on the eastern boundary is constant, and this must be specified. Here, this is given by the height of the sill, and therefore $h(x = x_e, y) = h_e = H - \eta_{sill}$, where H is the total depth of the basin and η_{sill} is the sill height. Then, using (20.93) and (21.97), the thickness of the moving layer equatorward of the sill is given by, for $y_2 < y < y_3$,

$$h^2 = D^2(x, y) + h_e^2 \tag{21.98}$$

where

$$D^2 = -\frac{2f^2}{g'\beta}\int_x^{x_e} w_E \,\mathrm{d}x' = \frac{2f^2}{g'\beta}W_2(x_e - x)\sin\left(\frac{\pi(y-y_2)}{\Delta y_2}\right) \tag{21.99}$$

with $g' = b_2 - b_1$. The solution is closed by the addition of a western boundary current. Note that because $h > h_e$, the light fluid is pushed below the level of the sill in the subtropical gyre.

Solution in the channel

In the channel, the fluid in the Ekman layer flows equatorward, and therefore there must be a compensating poleward flow at depth. This will occur at and just below the level of the sill: it cannot be deeper, because here the basin is full of denser, b_1 fluid, and in the absence of eddying or ageostrophic flow it cannot be shallower because of the geostrophic constraint. Now, because of the temperature gradient across the channel the polewards flowing fluid is warmer than the fluid at the surface, and therefore convectively unstable. Convection ensues, the result of which is the entire column of fluid between the top of the sill and the surface mixes and takes on the temperature of the surface. Thermal wind demands that there be a zonal flow associated with this meridional temperature gradient, so this temperature distribution is advected eastwards into the interior of the channel. Because the interior is presumed to be adiabatic, this temperature fields extends zonally throughout the channel. Thus, in steady state, the temperature *everywhere* in the channel above the level of the sill is given by

$$b(x,y,z) = b_s(y) = b_1 + \frac{(b_2-b_1)(y-y_1)}{y_2-y_1}, \qquad y_1 \le y \le y_2, \;\; z > \eta_{sill.} \tag{21.100}$$

Convective mixing does not rely on a diapycnal diffusivity other than a molecular one: convective plumes are generally turbulent, generating small scales in the fluid interior where mixing and entrainment may occur; failing that, the lighter fluid is displaced to the surface where it cools by way of interaction with the atmosphere. The zonal velocity within the channel is then given by thermal wind balance, so that

$$u(x,y,z) = -\frac{1}{f}\left(\frac{b_2-b_1}{y_2-y_1}\right)(z-\eta_{sill}), \tag{21.101}$$

and since $f < 0$ so that the shear is positive.

Regarding the depth-integrated zonal momentum budget, the wind stress at the surface is balanced by a pressure force against the sill walls. This pressure gradient arises through the meridional circulation, as the southward return flow just below the level of the sill is associated with a zonal pressure gradient that is exactly equal, but opposite, to the stress exerted by the wind. That is to say, in the Ekman layer the wind stress is balanced by the Coriolis force on the equatorward flow in the Ekman layer, which by mass conservation is equal and opposite to the Coriolis force on the deep poleward flow, which by geostrophy is equal to the net pressure force on the sill walls. The wind stress plays no role in determining the zonal transport of the channel: if the wind increases, the meridional overturning and the pressure force increase but with no change to the transport. This is a somewhat unrealistic feature of the model, for in reality the form stress induced by the flow over bottom topography (and that balances the wind stress) is likely to be a function of the zonal transport as well as the meridional transport.

A qualitative summary

The circulation of the model may be described as follows. The entire basin polewards of the channel fills with dense, b_1, water. Below the sill this fluid extends equatorward, filling the lower part of the channel and subtropical basin, up to the level of the sill. Now, Ekman pumping in the channel forces near-surface fluid equatorward, which warms as it goes, entering the subtropical basin with buoyancy b_2. This fluid fills the basin down to the level of the sill, where it encounters the dense, b_1, fluid. The subtropical basin is wind-driven, and it forms a subtropical gyre with a single moving layer. Its dynamics are completely determined by specifying the wind, the reduced gravity ($g' = b_2 - b_1$), and the depth of the fluid at the eastern boundary (the sill depth). Because of the requirements of mass conservation, there must be a poleward return flow at depth, and so at the level of the sill warm water flows polewards. This flow is convectively unstable (because the water is lighter than that at the surface), and so the entire column of fluid mixes and its density takes on the value at the surface. The meridional temperature gradient gives rise to an eastward flow, and this temperature field is advected zonally, and in steady state the temperature distribution is zonally symmetric and given by (21.100). The overturning circulation within the ACC is known as the Deacon Cell, and this is a crude model of it. It is considered further in section 21.7.

If the diapycnal diffusivity were non-zero, the sharp boundary between the two fluid masses at the sill height would be diffused to a front of finite thickness, with some upwelling and water mass transformation occurring across the front. This diffusive loss of dense fluid would be compensated by water-mass formation at the surface, polewards of the channel, leading to a deep, diffusively-driven circulation. That is, the deep water mass of b_1 fluid would circulate: this is a crude model of the 'Antarctic Bottom Water' cell.

Suppose now that the wind were everywhere zero, and the diffusivity small but non-zero. The cold, b_1 fluid would quickly completely fill the basin polewards of the channel, and would also fill the basin equatorward of the channel up to the level of the sill. However, with no wind to drive an overturning circulation dense b_1 water would slowly drift ageostrophically across the channel, displacing any warmer water until the *entire basin* were filled with the dense, b_1, fluid, except for a thin boundary layer at the top needed to satisfy the upper boundary condition. The final state would be one of no motion, and no stratification, below this boundary layer.

The important overall conclusion to be drawn is the following: *a deep meridional circulation and a deep stratification can be maintained, even as the diapycnal diffusivity goes to zero, in the presence of a wind forcing and a circumpolar channel.* Of course there are a number of idealized or unrealistic aspects to this model, perhaps the most egregious being the ones following.

- The vertical isopycnals in the channel will be highly baroclinically unstable. This will cause the isopycnals to slump and will potentially set up an eddy-induced circulation. We consider this in section 21.7.
- This model has no surface temperature gradient across the subtropical gyre. If one were present, it would lead to the formation of a 'main' subtropical thermocline, a full treatment of which would require determining its eastern boundary conditions. This would not qualitatively affect the presence of a deep, wind-driven overturning circulation.
- The wind stress in the model channel is chosen so that the meridional Ekman transport is constant. (This means the wind stress is chosen to vary in the same fashion as the Coriolis parameter, and if f were constant, the wind-stress curl would vanish.) Thus, there is no wind-driven downwelling or upwelling in the channel, and this simplifies the solution. Numerical simulations suggest that this choice does not affect the qualitative nature of the overturning circulation or temperature distribution.

21.6.3 A cross-equatorial wind-driven deep circulation

We qualitatively and heuristically extend the above model to consider flow across the equator. Thus, we suppose that the ocean basin extends to high northern latitudes, where there is, potentially, another source of cold deep water. To keep the model simple and tractable we will assume a very

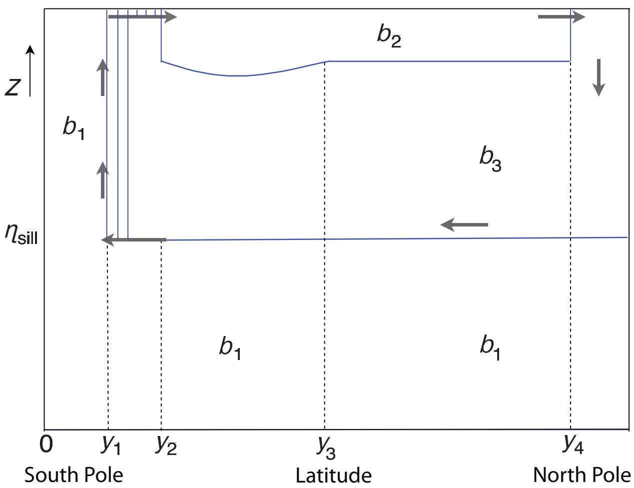

Fig. 21.22 As for Fig. 21.21, but now for a two-hemisphere ocean with a source of dense water, b_3, at high northern latitudes. The solid lines are isopycnals, and here the wind is zero in the Northern Hemisphere.

simple buoyancy structure:

$$b_s = \begin{cases} b_1, & 0 \le y \le y_1 \\ b_1 + \dfrac{(b_2 - b_1)(y - y_1)}{y_2 - y_1}, & y_1 \le y \le y_2 \\ b_2, & y_2 \le y \le y_4 \\ b_3 & y > y_4, \end{cases} \tag{21.102}$$

where the geometry is illustrated in Fig. 21.22. Given that $b_2 > b_1$, there are three cases to consider.

(i) $b_3 > b_2$. This is not oceanographically relevant to today's climate, nor does it provide another potential deep water source.

(ii) $b_3 < b_1$. The northern water is now the densest in the ocean, and would fill up the entire basin north of the channel (except near the surface in regions where some b_2 water is pushed down by the wind), and so provide no mid-depth stratification.

(iii) $b_1 < b_3 < b_2$. This is the most interesting and relevant case, and the only one we explore further.

As regards the wind, we will assume that south of the equator this is given by (21.97). North of the equator the wind forcing does not affect the qualitative nature of the overturning circulation, and may be taken to be zero.

Descriptive solution

In case *(iii)*, the entire basin below the sill fills with b_1 water, except where wind forcing forces warmer fluid below the sill level, as before. However, unlike the earlier case, the fluid above the sill is predominantly b_3 water from high northern latitudes. This forms in high polar latitudes and fills most of the basin above the sill, from the basin boundary in the north to the channel in the south (as discussed more below). However, except at latitudes where the b_3 is formed, it does not reach the surface because of the presence of b_2 water. That water is pushed down by the wind in the southern hemisphere to some as yet undetermined depth (discussed below), the boundary between b_2 and b_3 water then forming the upper ocean thermocline.

These water masses circulate because of the wind forcing in the channel. As in the single-hemisphere case, northwards flowing water emerges from the channel with buoyancy b_2. This emerges into a region of Ekman downwelling, with a northward transport carried by a western boundary current. This transport crosses the equator finally reaching the latitudes where b_3 water is formed where it sinks and returns equatorwards, again in a western boundary current. (Away from the western boundary layer there is no meridional flow in the absence of diffusion, because the flow satisfies $\beta v = f\partial w/\partial z$ and there is no upwelling.) This water then crosses the sill. However, unlike the single-hemisphere case, in the northern part of the sill this water is denser than the surface water; no convection occurs and so the b_3 water extends upwards to the surface, where it warms by contact with the atmosphere and advected equatorwards to become b_2 water. Further south the surface buoyancy in the channel is less than b_3, and the column now mixes convectively, much as in the single-hemisphere case. The solution is completed by specifying the thickness of the layer of b_2 water at the surface. Now, if the circulation is in steady state, the meridional transport between the gyres must equal that of the northward Ekman flow at the northern edge of the circumpolar channel, and given the wind forcing, this is determined by the depth of the layer at the eastern boundary, a constant. Thus, in this model, global constraints determine the depth of the eastern boundary of the thermocline.

Suppose that the wind were everywhere zero. Then, as in the single-hemisphere case, the circulation would eventually die. Again, though, slow ageostrophic motion across the channel would first allow the entire basin, within and on both sides of the channel, to fill with the densest available water, and in the final steady state there would be no stratification (and no motion) below a thin surface layer.

Suppose, on the other hand, that a small amount of diffusion were added to the wind-forced model above. Then there would be mass exchange between the layers and, in particular, the deep cell of b_1 water would begin to circulate diffusively. In addition, the mid-depth cell would begin to upwell through the b_2–b_3 interface, and develop a diffusively driven circulation, in much the same way as is illustrated in Fig. 20.15.

Summary remarks

The key result of this model is that, even as the diffusivity falls and the interior of the ocean becomes more and more adiabatic, a meridional cross-hemispheric circulation can be maintained, provided that the wind across the circumpolar channel remains finite. The diabatic water mass transformations all occur at the surface or in convection: these processes require a non-zero diffusivity, but this can be the molecular diffusivity because the associated mixing involves turbulence, which can generate arbitrarily small scales. (Note also that the convection that occurs in the circumpolar channel reduces the potential energy of the column, and requires no mechanical input of energy.) Aside from the region of the ACC, the meridional transport will occur (in this model) in western boundary layers. Indeed, we may still expect to see a southwards flowing deep western boundary current south of y_4 and below the b_2 water in Fig. 21.22, just as in the implicitly diffusive Stommel–Arons model. In the ACC itself, the meridional transport occurs in a subsurface current, nestled against the sill. Although the overturning circulation in this model is 'wind-driven', the possibility that it may be cross-equatorial depends upon the thermodynamic forcing; in particular, if there is no source of dense water in the northern hemisphere, then the basin above the sill simply fills with b_2 water, as in the model of section 21.6.2, and there need be little or no interhemispheric flow. We emphasize, too, that our model of interhemispheric flow is quite heuristic: we have essentially *posited* that b_2 water may continuously flow across the equator, without examining the equatorial dynamics in any detail.

The ACC plays a key role in the above description,ut we have grossly oversimplified it. In particular, the nearly vertical isopycnals of the model will be highly baroclinically unstable, and this provides a connvenient segue into our next topic.

21.7 THE ANTARCTIC CIRCUMPOLAR CURRENT

We now take a closer look at the Antarctic Circumpolar Current (ACC) itself, with a focus on its own internal dynamics; we come back to the connection with the rest of the world's oceans in

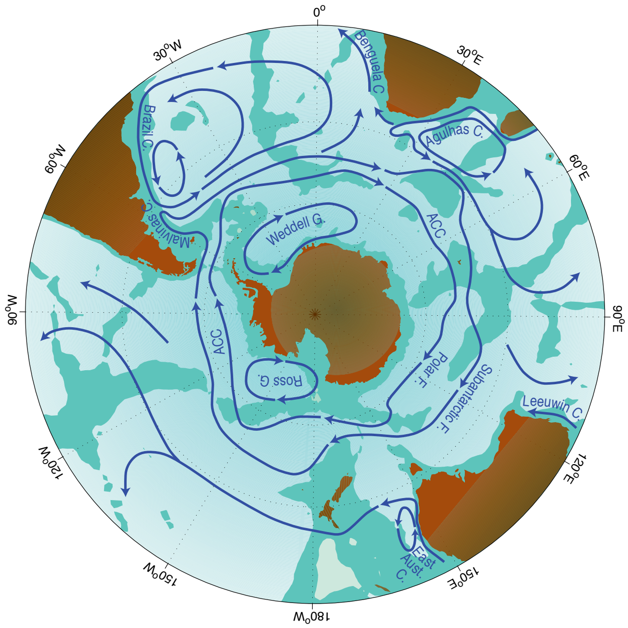

Fig. 21.23 Sketch of the major currents in the Southern Ocean. Shown are the South Atlantic subtropic gyre, and the two main cores of the ACC, associated with the Polar front and the sub-Antarctic front.[21]

section 21.8. The ACC system, sketched in Fig. 21.23, differs from other oceanic regimes primarily in that the flow is, like that of the atmosphere, predominantly zonal and re-entrant. The two obvious influences on the circulation are the strong, eastward winds (the 'roaring forties' and the 'furious fifties') and the buoyancy forcing associated with the meridional gradient of atmospheric temperature and radiative effects that cause ocean cooling at high latitudes and warming at low ones. Providing a detailed description of the resulting flow is properly the province of numerical models, and here our goals are much more modest, namely to describe and understand some of the fundamental dynamical mechanisms that determine the structure and transport of the system, with a view to then connecting the ACC to the rest of the world's ocean.[22]

21.7.1 Steady and eddying flow

Consider again the simplified geometry of the Southern Ocean as sketched in Fig. 21.19. The ocean floor is flat, except for a ridge (or 'sill') at the same longitude as the gyre walls; this is a crude representation of the topography across the Drake Passage, that part of the ACC between the tip of South America and the Antarctic Peninsula. In the planetary-geostrophic approximation, the steady response is that of nearly vertical isopycnals in the area above the sill, as illustrated in Fig. 21.21. Below the sill a meridional flow can be supported and the isotherms spread polewards, as illustrated in the left panel of numerical solutions using the primitive equations (Fig. 21.24).[23]

The stratification of the non-eddying simulation is similar to that predicted by the idealized model illustrated in Fig. 21.21. However, the steep isotherms within the channel contain a huge amount of available potential energy (APE), and the flow is highly baroclinically unstable. If baroclinic eddies are allowed to form, the solution is dramatically different: the isotherms slump, releasing that APE and generating mesoscale eddies that exercise control over much of the circulation. An important conclusion is that *baroclinic eddies are of leading-order importance in the dynamics of the ACC.* A dynamical description of the ACC without eddies would be *qualitatively* in error, in much the same way as would a similar description of the mid-latitude troposphere (i.e., the Ferrel Cell). These eddies transfer both heat and momentum, and much of the rest of our description will focus on their effects.

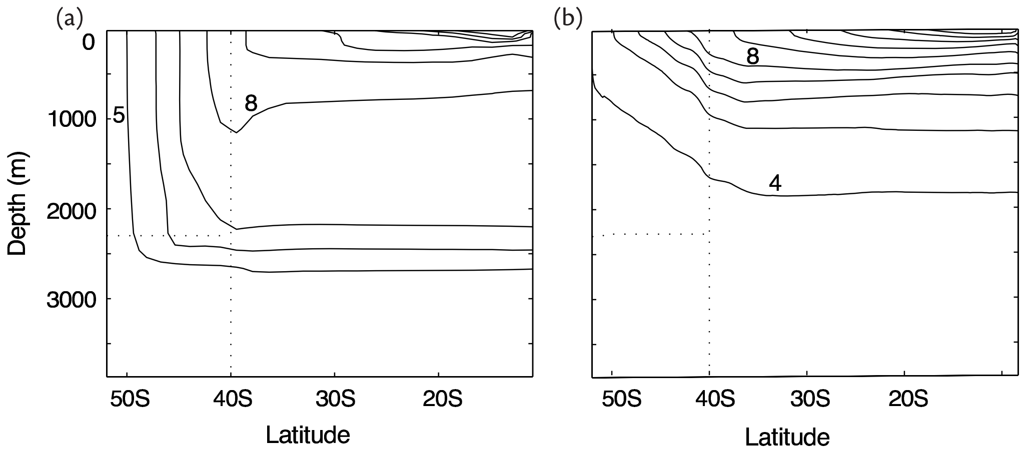

Fig. 21.24 The zonally averaged temperature field in numerical solutions of the primitive equations in a domain similar to that of Fig. 21.19 (except that here the channel and sill are nestled against the poleward boundary). Panel (a) shows the steady solution of a diffusive model with no baroclinic eddies, and (b) shows the time averaged solution in a higher-resolution model that allows baroclinic eddies to develop. Two contour values in each panel are labelled. The dotted lines show the channel boundaries and the sill.[24]

21.7.2 Vertically integrated momentum balance

The momentum supplied by the strong eastward winds must somehow be removed. Presuming that lateral transfers of momentum are small the momentum must be removed by fluid contact with the solid Earth at the bottom of the channel. Thus, let us first consider the vertically integrated momentum balance in a channel, without regard to how the momentum might be vertically transferred. We begin with the frictional–geostrophic balance, namely

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi + \frac{\partial \widetilde{\boldsymbol{\tau}}}{\partial z}, \tag{21.103}$$

where $\widetilde{\boldsymbol{\tau}}$ is the kinematic stress, $\boldsymbol{\tau}/\rho_0$, and $\phi = p/\rho_0$. Integrating over the depth of the ocean and using Leibniz's rule [cf. (19.95), that is $(\nabla \int_{\eta_B}^{0} \phi \, \mathrm{d}z = \int_{\eta_B}^{0} \nabla\phi \, \mathrm{d}z - \phi_B \nabla \eta_B)$] gives

$$\boldsymbol{f} \times \widehat{\boldsymbol{u}} = -\nabla\widehat{\phi} - \phi_b \nabla \eta_b + \widetilde{\boldsymbol{\tau}}_w - \widetilde{\boldsymbol{\tau}}_f, \tag{21.104}$$

where $\widetilde{\boldsymbol{\tau}}_w$ is the stress at the surface (due mainly to the wind) and $\widetilde{\boldsymbol{\tau}}_f$ is the frictional stress at the bottom, a hat denotes a vertical integral and ϕ_b is the pressure at $z = \eta_b$, where η_b is the z-coordinate of the bottom topography. The x-component of (21.104) is

$$-f\widehat{v} = -\frac{\partial \widehat{\phi}}{\partial x} - \phi_b \frac{\partial \eta_b}{\partial x} + \widetilde{\tau}_w^x - \widetilde{\tau}_f^x, \tag{21.105}$$

and on integrating around a line of latitude the term on the left-hand side vanishes by mass conservation and we are left with

$$-\overline{\phi_b \frac{\partial \eta_b}{\partial x}} + \overline{\widetilde{\tau}_w^x} - \overline{\widetilde{\tau}_f^x} = 0, \tag{21.106}$$

where overbars denote zonal averages. The first term is the bottom, or topographic, form drag, encountered in sections 3.6 and 19.6, and observations and numerical simulations indicate that it is this, rather than the frictional term $\widetilde{\tau}_f^x$, that predominantly balances the wind stress.[25] We address the question of *why* this should be so in section 21.7.5.

The vorticity balance is similarly dominated by a balance between bottom form-stress curl and wind-stress curl. Taking the curl of (21.104), noting that $\nabla \cdot \widehat{\boldsymbol{u}} = 0$, gives

$$\beta \widehat{v} = -\mathbf{k} \cdot \nabla \phi_b \times \nabla \eta_b + \mathrm{curl}_z \widetilde{\boldsymbol{\tau}}_w - \mathrm{curl}_z \widetilde{\boldsymbol{\tau}}_f. \tag{21.107}$$

Now, on integrating over an area bounded by two latitude circles and applying Stokes' theorem the β-term vanishes by mass conservation and we regain (21.106). This means that *Sverdrup balance, in the usual sense of* $\beta v \approx \mathrm{curl}_z \boldsymbol{\tau}_w$, *cannot hold in the zonal average:* the left-hand side vanishes but the right-hand side does not. The same could be said for the zonal integral of (21.107) across a gyre, but the two cases do differ: In a gyre Sverdrup balance can (in principle) hold over most of the interior, with mass balance being satisfied by the presence of an intense western boundary current. In contrast, in a channel where the dynamics are zonally homogeneous then v must be, on average, zero at *all* longitudes and form drag and/or frictional terms must balance the wind-stress curl in a given water column. Sverdrup balance is thus a less useful foundation for channel dynamics — at least zonally homogeneous ones — than it is for gyres. Of course, the real ACC is *not* zonally homogeneous, and may contain regions of poleward Sverdrup flow balanced by equatorward flow in boundary currents along the eastern edges of sills and continents, and the extent to which Sverdrup flow is a leading-order descriptor of its dynamics is a matter of geography (and debate!). See also section 21.7.6.

Even though topographic drag may be dominant in removing momentum, non-conservative frictional terms cannot be neglected, for two reasons. First, they are the means whereby kinetic energy is dissipated. Second, if there is a contour of constant orographic height encircling the domain (i.e., encircling Antarctica) then the form drag will vanish when integrated along it. However, the same integral of the wind stress will not vanish, and therefore must be balanced by something else. To see this explicitly, write the vertically integrated vorticity equation, (21.107), in the form

$$\beta \widehat{v} + J(\phi_b, \eta_b) = \mathrm{curl}_z \widetilde{\boldsymbol{\tau}}_w - \mathrm{curl}_z \widetilde{\boldsymbol{\tau}}_f. \tag{21.108}$$

If we integrate over an area bounded by a contour of constant orographic height (i.e., constant η_b) then both terms on the left-hand side vanish, and the wind stress along that line must be balanced by friction. In the real ocean there may be no such contour that is confined to the ACC— rather, any such contour would meander through the rest of the ocean; indeed, no such confined contour exists in the idealized geometry of Fig. 21.19.

21.7.3 Form drag and baroclinic eddies

How does the momentum put in at the surface by the wind stress make its way to the bottom of the ocean where it may be removed by form drag? We saw in section 21.6.2 that one mechanism is by way of a mean meridional overturning circulation, with an upper branch in the Ferrel Cell and a lower branch at the level of the sill, with no meridional flow between. However, the presence of baroclinic eddies allows an eddy form drag can pass momentum vertically within the fluid. Let's see how that works.

We model the channel as a finite number of fluid layers, each of constant density and lying one on top of the other — a 'stacked shallow water' model, and one equivalent to a model expressed in isopycnal coordinates. The wind provides a stress on the upper layer, which sets it into motion, and this in turn, via the mechanism of form drag, provides a stress to the layer below, and so on until the bottom is reached. The lowest layer then equilibrates via form drag with the bottom topography or via Ekman friction, and the general mechanism is illustrated in Fig. 21.25.

Recalling the results of section 3.6, the zonal form drag at a layer interface is given by

$$\tau_i = -\overline{\eta_i \frac{\partial p_i}{\partial x}} = -\rho_0 f \overline{\eta_i v_i}, \tag{21.109}$$

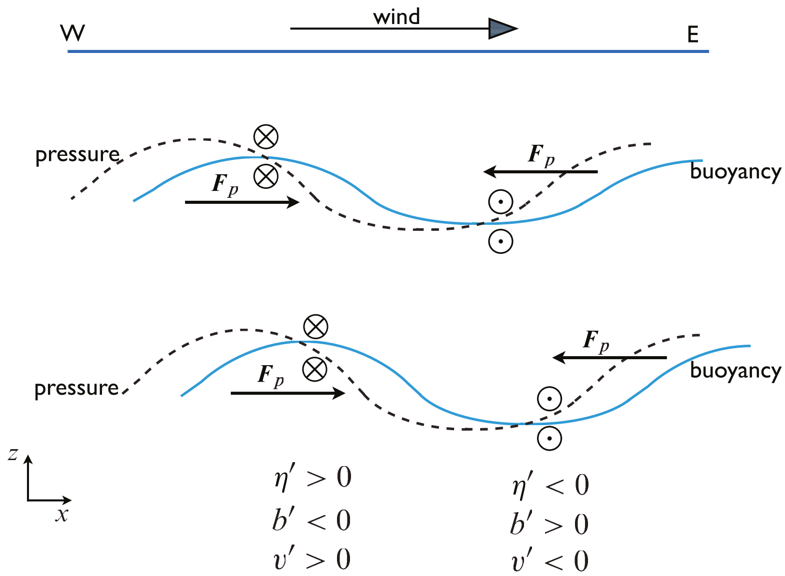

Fig. 21.25 Eddy fluxes and form drag in a Southern Hemisphere channel, viewed from the south. In this example, cold (less buoyant) water flows equatorwards and warm water polewards, so that $\overline{v'b'} < 0$. The pressure field associated with this flow (dashed lines) provides a form drag on the successive layers, $\boldsymbol{F}_p$, shown. At the ocean bottom the westward form drag on the fluid arising through its interaction with the orography of the sea-floor is equal and opposite to that of the eastward wind stress at the top. The mass fluxes in each layer are given by $\overline{v'h'} \approx -\partial_z(\overline{v'b'}/N^2)$. If the magnitude of buoyancy displacement increases with depth then $\overline{v'h'} < 0$.

where p_i is the pressure and η_i is the displacement at the ith interface (i.e., between the ith and $(i+1)$th layer as in Fig. 21.26), and the overbar denotes a zonal average. If we define the averaged meridional transport in each layer by

$$V_i = \int_{\eta_i}^{\eta_{i-1}} \rho_0 v \, \mathrm{d}z, \tag{21.110}$$

then, neglecting the meridional momentum flux divergence (for reasons given in the next subsection), the time and zonally averaged zonal momentum balance for each layer of fluid are:

$$-f\overline{V}_1 = \tau_w - \tau_1 = \overline{\eta_1 \frac{\partial p_1}{\partial x}} + \tau_w, \tag{21.111a}$$

$$-f\overline{V}_i = \tau_{i-1} - \tau_i = -\overline{\eta_{i-1} \frac{\partial p_{i-1}}{\partial x}} + \overline{\eta_i \frac{\partial p_i}{\partial x}}, \tag{21.111b}$$

$$-f\overline{V}_N = \tau_{N-1} - \tau_N = -\overline{\eta_{N-1} \frac{\partial p_{N-1}}{\partial x}} + \overline{\eta_b \frac{\partial p_b}{\partial x}} - \tau_f, \tag{21.111c}$$

where the subscripts 1, i and N refer to the top layer, an interior layer, and the bottom layer, respectively. Also, η_b is the height of bottom topography and τ_w is the zonal stress imparted by the wind which, we assume, is confined to the uppermost layer. The term τ_f represents drag at the bottom due to Ekman friction, but we have neglected any other viscous terms or friction between the layers.

The vertically integrated meridional mass transport must vanish, and thus summing over all the layers (21.111) becomes

$$0 = \tau_w - \tau_f - \tau_N, \tag{21.112}$$

or, noting that $\tau_N = -\overline{\eta_b \partial p_b / \partial x}$,

$$\tau_w = \tau_f - \overline{\eta_b \frac{\partial p_b}{\partial x}}. \tag{21.113}$$

Thus, the stress imparted by the wind (τ_w) may be communicated vertically through the fluid by form drag, and ultimately balanced by the sum of the bottom form stress (τ_N) and the bottom friction (τ_f).

Momentum dynamics in height coordinates

We now look at these same dynamics in height coordinates, using the quasi-geostrophic TEM formalism, and it may be helpful to review section 10.3 before proceeding. As in (10.61), we write the zonally averaged momentum equation in the form

$$-f_0 \overline{v}^* = \nabla_m \cdot \boldsymbol{\mathcal{F}} + \frac{\partial \widetilde{\tau}}{\partial z} \tag{21.114}$$

where $\overline{v}^* = \overline{v} - \partial_z(\overline{v'b'}/\overline{b}_z)$ is the residual meridional velocity, $\widetilde{\tau}$ is the zonal component of the kinematic stress (wind-induced and frictional, and typically important only in an Ekman layer at the surface and in a frictional layer at the bottom) and $\boldsymbol{\mathcal{F}}$ is the Eliassen–Palm flux, which satisfies

$$\nabla_m \cdot \boldsymbol{\mathcal{F}} = -\frac{\partial}{\partial y}\overline{u'v'} + \frac{\partial}{\partial z}\left(\frac{f_0}{N^2}\overline{v'b'}\right) = \overline{v'q'}. \tag{21.115}$$

Now, if the horizontal velocity and buoyancy perturbations are related by $v' \sim b'/N$ (meaning available potential energy and kinetic energy are roughly similar, see also section 12.4), then the two terms comprising the potential vorticity flux scale as

$$\frac{\partial}{\partial y}\overline{u'v'} \sim \frac{v'^2}{L_e}, \qquad \frac{\partial}{\partial z}\left(f_0 \frac{\overline{v'b'}}{\overline{b}_z}\right) \sim \frac{v'^2}{L_d}, \tag{21.116}$$

where L_e is the scale of the eddies and L_d is the deformation radius. If the former is much larger than the latter, as we might expect in a field of developed geostrophic turbulence (and as is observed in the ACC), then the potential vorticity flux is dominated by the buoyancy flux [so also justifying the neglect of the lateral momentum fluxes in (21.111)] and (21.114) becomes

$$-f_0 \overline{v}^* \approx \frac{\partial \widetilde{\tau}}{\partial z} + \frac{\partial}{\partial z}\left(f_0 \frac{\overline{v'b'}}{\overline{b}_z}\right). \tag{21.117}$$

In the ocean interior the frictional terms, $\partial \widetilde{\tau}/\partial z$, are small, and (21.117) represents a balance between the Coriolis force on the residual flow and the form stress associated with the vertical component of the EP flux (an association further explained in section 10.4.3).

If we integrate (21.117) equation over the depth of the channel the term on the left-hand side vanishes and we have

$$\widetilde{\tau}_w = \widetilde{\tau}_f - \left[f_0 \frac{\overline{v'b'}}{\overline{b}_z}\right]_{-H}^{0}, \tag{21.118}$$

where $\widetilde{\tau}_w$ is the wind stress and $\widetilde{\tau}_f$ is the frictional stress at the bottom (both divided by ρ_0). Equation (21.118) expresses essentially the same momentum balance as (21.113). Thus, the EP flux expresses the passage of momentum vertically through the water column, the momentum being removed at the bottom through frictional stresses and/or form drag with the orography.

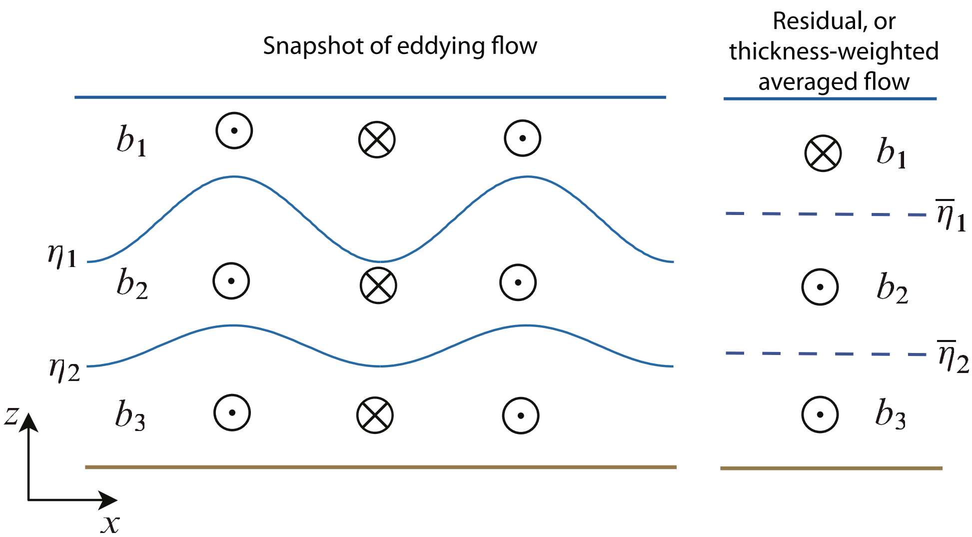

Fig. 21.26 An example of the meridional flow in an eddying channel. The eddying flow may be organized such that, even though at any given level the Eulerian meridional flow may be small, there is a net flow in a given isopycnal layer. The residual ($\overline{v}^*$) and Eulerian ($\overline{v}$) flows are related by $\overline{v}^* = \overline{v} + \overline{v'h'}/\overline{h}$; thus, the thickness-weighted average of the eddying flow on the left gives rise to the residual flow on the right, where $\overline{\eta}_i$ denotes the mean elevation of the isopycnal interface η_i.

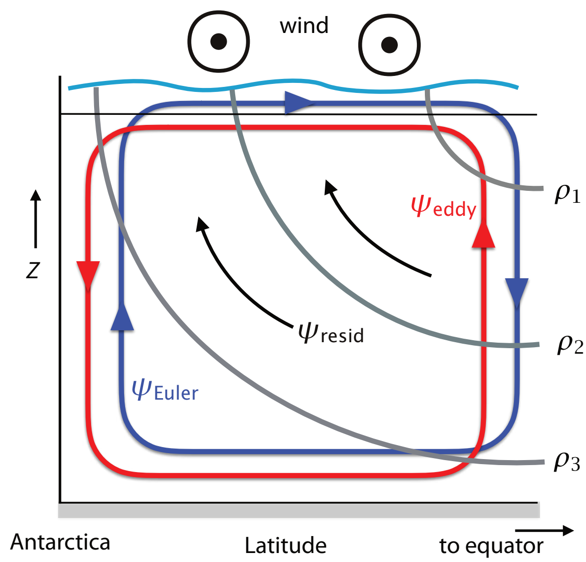

Fig. 21.27 Idealization of the Eulerian, eddy-induced (or bolus), and resisdual streamfunctions in a circumpolar channel. The clockwise Eulerian circulation is forced by the eastward winds, the bolus circulation opposes it, and the net, or residual, circulation is nearly along isopycnals.[26]

Mass fluxes and thermodynamics

Associated with the form drag is a meridional mass flux in each layer, which in the layered model appears as V_i (a thickness flux) in each layer. The satisfaction of the momentum balance at a particular latitude goes hand-in-hand with the satisfaction of the mass balance. Above any topography the Eulerian mean momentum equation is, with quasi-geostrophic scaling and neglecting eddy momentum fluxes,

$$f_0\overline{v} = \frac{\partial\widetilde{\tau}}{\partial z} \tag{21.119}$$

where $\overline{v}$ is the zonally averaged meridional velocity and $\widetilde{\tau}$ is the zonal component of the kinematic stress. The zonally averaged meridional flow is thus purely ageostrophic and since the stress, $\widetilde{\tau}$, is fairly constant in the interior, the mean meridional flow non-zero only near the surface (i.e., equatorward Ekman flow) and near the ocean bottom, where the flow can be supported by friction and/or form drag. Even in an eddying flow, the Eulerian circulation is primarily confined to the upper Ekman layer and a frictional or topographically interrupted layer at the bottom, as sketched in Fig. 21.27. This is a perfectly acceptable description of the flow, and is not an artifact in any way. However, and analogously to the atmospheric Ferrel Cell (sections 14.7 and 15.2.2), if the flow is unsteady this circulation does not necessarily represent the flow of water parcels, nor does it imply that water parcels cross isopycnals, as might be suggested by the dark blue circulation (ψ_{Euler}) in Fig. 21.27. The flow of parcels is better represented by the *residual,* or *thickness-weighted,* flow, and as sketched in Fig. 21.26 and Fig. 21.27 there can be a net meridional residual flow in a given layer (i.e., of a given water mass type) *even when the net meridional Eulerian flow at the level of mean height of the layer is zero.*

The vertically integrated residual mass flux must vanish, and even though one component of this — the equatorward Ekman flow — is determined mechanically, the overall sense of the residual circulation is not determined by the momentum balance alone: thermodynamic effects play a role. The zonally averaged thermodynamic equation may be written in TEM form as

$$\frac{\partial\overline{b}}{\partial t} + J(\psi^*, \overline{b}) = Q[b] \tag{21.120}$$

where $J(\psi^*, \overline{b}) = (\partial_y\psi^*)(\partial_z b) - (\partial_z\psi^*)(\partial_y b) = \overline{v}^*\partial_y b + \overline{w}^*\partial_z b$, ψ^* is the streamfunction of the residual flow and $Q[b]$ represents heating and cooling, which occur mainly at the surface. In the ocean interior and in a statistically steady state we therefore have

$$J(\psi^*, \overline{b}) = 0, \tag{21.121}$$

the general solution of which is $\psi^* = G(\overline{b})$, where G is an arbitrary function. That is, the interior residual flow is along isopycnals (Fig. 21.27. At the surface, however, the flow is generally not adiabatic, because of heat exchange with the atmosphere, and so the residual flow can be across isopycnals. The sense of the subsurface circulation determines how the form drag varies with depth; if the residual flow were zero, for example, then, either from (21.111) or from (21.117), we see that the form drag must be constant with depth.

21.7.4 † An idealized adiabatic model

We finally consider a simple but rather illuminating model of the ACC.[27] The simplifying assumption we make is that the flow is adiabatic everywhere; it then follows that the net overturning, as given by the residual circulation, is zero. We can see this by first noting that in a statistically steady state the flow satisfies (21.121), implying that the residual flow is along isopycnals. However, if there is a meridional buoyancy gradient at the surface (where isopycnals outcrop) there can be no surface residual flow (because this would be cross-isopycnal); it then follows that there can be no net flow along isopycnals in the interior, because if these outcrop there would be a net fluid source, and hence diapycnal flow, at the surface. This idealized limit has thus led to the 'vanishing of the Deacon Cell'. In reality the flow is not completely adiabatic and the residual flow will not vanish, but it is likely to be weaker than either the Eulerian or the eddy-induced flow (as sketched in Fig. 21.27).

The zonal momentum equation in this limit follows from (21.117), which with $\overline{v}^* = 0$ gives

$$\frac{\partial \widetilde{\tau}}{\partial z} \approx -\frac{\partial}{\partial z}\left(f_0 \frac{\overline{v'b'}}{\overline{b}_z} \right). \tag{21.122}$$

The equivalent balance for the Eulerian flow is, using the definition of $\overline{v}^*$,

$$-f_0\overline{v} = \frac{\partial \widetilde{\tau}}{\partial z} \qquad \rightarrow \qquad f_0\overline{v} = f_0 \frac{\partial}{\partial z}\left(\frac{\overline{v'b'}}{\overline{b}_z} \right). \tag{21.123a,b}$$

These equations represent *dynamical* balances; they do not follow from the momentum equation without making addition assumptions, in this case that $\overline{v}^* = 0$. In the residual equation, (21.122), the wind stress is balanced by the divergence of the Eliassen–Palm flux, which is dominated by the contribution from the buoyancy flux, and in the ocean interior where the stress is small the meridional buoyancy flux will be constant with height. Equation (21.123b) represents a balance between the Coriolis force on the equatorward flow and form drag. If the frictional stress is small in the interior then the form drag does not vary in the vertical and the right-hand side of (21.123) is small. The zonally averaged meridional flow in the interior is then also small, and the equatorial flow in the top Ekman layer is balanced by a return flow at the bottom of the ocean involving topographic form stress or a bottom Ekman layer.

Integrating (21.122) from the surface (where $\widetilde{\tau} = \widetilde{\tau}_w$) to a stress-free level in the interior (where $\widetilde{\tau} = 0$) gives

$$\widetilde{\tau}_w = f_0 \frac{\overline{v'b'}}{\overline{b}_z}, \tag{21.124}$$

if the buoyancy flux at the surface is small. If we are now willing to parameterize the eddy fluxes in terms of the mean flow, then we can predict the stratification. Thus, let $\overline{v'b'} = -\kappa\, \partial\overline{b}/\partial y$, where κ is an eddy diffusivity, and noting that $s = -\overline{b}_y/\overline{b}_z$ is the slope of the isopycnals, we find

$$\widetilde{\tau}_w = \kappa f_0 s = \kappa \frac{f_0^2}{\overline{b}_z}\frac{\partial \overline{u}}{\partial z}, \tag{21.125}$$

where the second equality uses thermal wind balance. Thus, given κ, we can predict the isopycnal slope $[s = \widetilde{\tau}_w/(\kappa f_0)]$ and, potentially, the total baroclinic transport of the ACC as a function of the wind stress. The sense of the residual circulation can be inferred if the diabatic fluxes at the surface are known, but at the same time these fluxes depend in a complicated way on both the lateral eddy fluxes and the general circulation itself. We come back to this in sections 21.8 and 21.9.

21.7.5 Form stress and Ekman stress at the ocean bottom

Earlier, we noted that the stress at the ocean bottom is observed to be dominated by form stress, rather than Ekman friction, in the ACC. A simple scaling argument helps understand why this should be. The form stress scales like

$$\tau_{form} \sim \eta_b \frac{\partial p_b}{\partial x} \sim \eta_b \rho_0 U f, \tag{21.126}$$

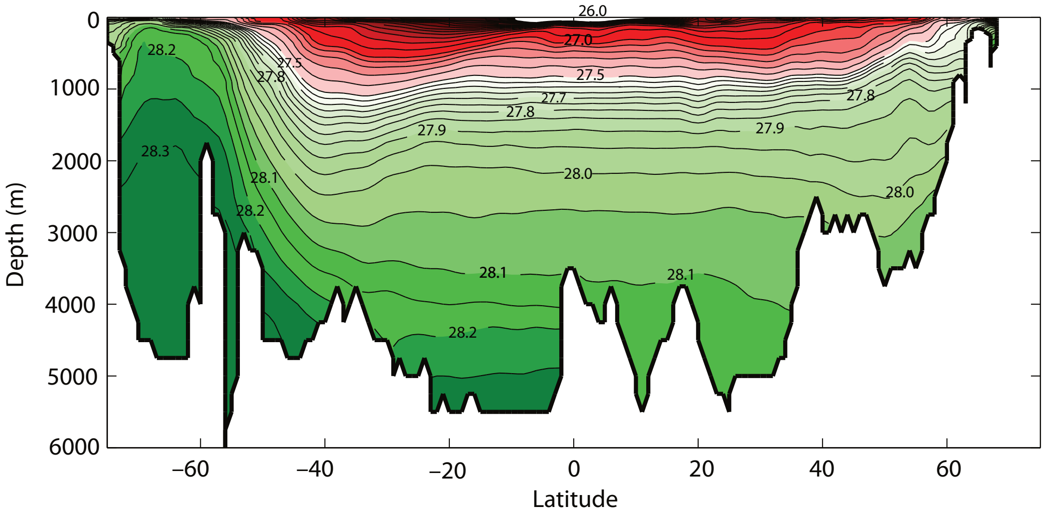

Fig. 21.28 Neutral density in the Atlantic at 25° W from WOCE. A weakly stratified water mass at middepth, roughly between the 28 and 28.1 isopycnals, is associated with the inflow of NADW in the Atlantic Ocean. The contour intervals are 0.1 and 0.05 kg m^{-3} for isopycnals greater and lower than 27.5 kg m^{-3}, coloured green and red, respectively.

where we have used geostrophic balance and U is a scaling for the horizontal velocity. The frictional stress due to an Ekman layer (section 5.7) scales like

$$\tau_{Ekman} \sim \rho_0 A \frac{\partial u}{\partial z} \sim \frac{\rho_0 A U}{d} \sim \rho_0 d U f \tag{21.127}$$

where A is the eddy kinematic viscosity and $d = \sqrt{A/f}$ is the Ekman layer thickness. The ratio of these two stresses thus scales as

$$\frac{\tau_{form}}{\tau_{Ekman}} \sim \frac{\eta_b}{d}. \tag{21.128}$$

We therefore expect the form stress to dominate the Ekman stress if the variations in topography are greater than the Ekman layer thickness, and if the flow goes *over* the topography rather than around it. In the ACC the topography is hundreds or even thousands of metres high whereas the bottom Ekman layer may be of order tens of metres, and furthermore the predominantly eastward flow must (unlike the situation in gyre circulations) go *over* the topography. Thus, form stress dominates the frictional, Ekman layer, stress at the bottom of the ACC.

21.7.6 Differences between gyres and channels

In the dynamics of the ACC, the wind stress itself seems to play an important role, whereas in our discussion of gyres in chapter 19 the wind stress *curl* was dominant. What is the root of this difference?[28] Suppose that we change the wind stress, but not its curl, in a closed basin. The vertically integrated gyral flow, as given for example by the Stommel solution (19.39) or its two-gyre counterpart, does not change at all. However, the vertical structure of this flow will in general change; for example, if the wind is made uniformly more eastward, there will be a corresponding increase in the equatorward flux in the Ekman layer that must return polewards at depth (assuming that the western boundary current balances only the Sverdrup flow). At the same time, the added force from the wind must be balanced by an increased pressure difference between the western and eastern boundaries. This may be achieved if the the sea-surface tilts upwards to the east, so

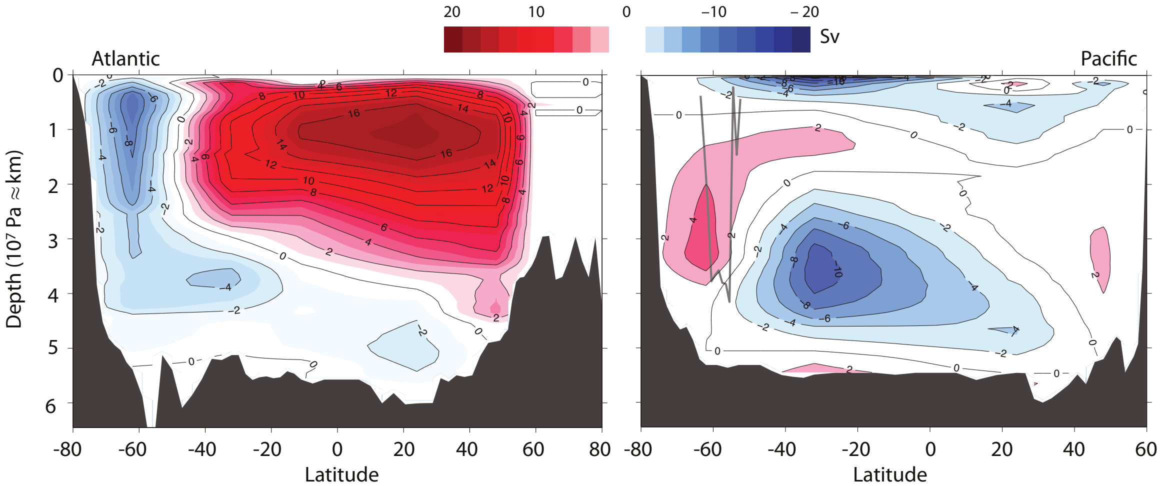

Fig. 21.29 Overturning circulations in the Atlantic and Pacific Oceans as determined by an inverse calculation. South of about 35° the circulation is not a true streamfunction, because of the open boundaries, and this may lead to errors, especially in the Pacific. (See Fig. 21.1 for another Atlantic estimate.)

producing a net (vertically integrated) poleward geostrophic flow. The subsurface isopycnal slopes may then adjust in order to reduce this flow to near zero in the abyss. The added force provided by the basin walls on the fluid in the basin is a kind of form drag (rather like the force provided by the sill in section 21.6), and integrated around the basin this force must be equal and opposite to the force supplied by the wind. In contrast, in a channel adding a constant wind produces a direct change in its zonal transport. This is because the wind stress is balanced by form drag and bottom friction, and both of these depend on the zonal flow at the channel bottom.

21.8 † A DYNAMIC MODEL OF THE MERIDIONAL OVERTURNING CIRCULATION

In the last section it became clear that the ACC is a region of strongly eddying activity, and one effect of these eddies is to reduce the slope of the isopycnals, so reducing the available potential energy of the flow. Thus, the sketches of Fig. 21.21 and Fig. 21.22 do not properly represent the state of the channel region: not only do the isopycnals slope, but the southward flowing water parcels can enter the channel region above any sill. That is, the zonally averaged *residual* meridional flow can be non zero, and the deep stratification can be nonzero, even without topography. In this section we seek to combine our view of the ACC, as described in section 21.7, with the view of the partially wind-driven overturning circulation described in section 21.6.

Our model is partially motivated by the plot of the stratification shown in Fig. 21.28 and the overturning circulations of Fig. 21.29, as abstracted in Fig. 21.30. Although nominally a sketch of the global MOC, of the individual basins it most resembles the circulation of the Atlantic where there are two main circulating masses of water, North Atlantic Deep Water (NADW) and Antarctic Bottom Water (AABW), as seen in Fig. 21.29. The NADW outcrops in high northern latitudes and high southern latitudes, and AABW just at high Southern latitudes. The Pacific overturning circulation (Fig. 21.29) is rather different, for here there is really no mid-depth cell corresponding to NADW; there is essentially *only* a bottom cell of Antarctic bottom water spreading northward. Finally, we note that isopycnals are flat over most of the ocean, but have a fairly uniform slope in the Southern Ocean. (Figure 21.28 shows the Atlantic; the situation is similar for the Pacific.) We will now construct a dynamical model that attempts to describe these features.[29]

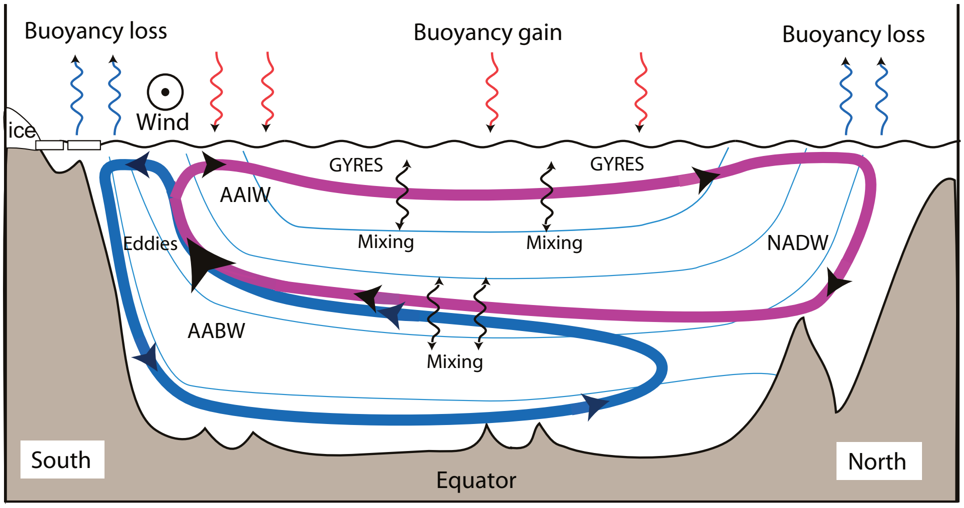

Fig. 21.30 A sketch of overturning circulation of the ocean and the main processes that produce it — winds, mixing, baroclinic eddies and surface buoyancy fluxes. The sketch is most representative of the Atlantic, but this dominates the global average. An observational view is given in Fig. 21.18.

Let us first imagine there is a wall at the equator, and make a model of the circulation in the Southern Hemisphere, that is, essentially of AABW. There's an obvious connection to the Indian Ocean and, if a little less obvious at the moment, to the Pacific.

21.8.1 Model Phenomenology

We divide the basin into two regions, a *Southern Channel* and a *basin*, as in Fig. 21.31 (see also the shaded box on page 849). In the channel the isopycnals slope, and we anticipate a balance between wind effects and baroclinic instability: in the absence of eddies the isopycnals are vertical, as in Fig. 21.21, and baroclinic activity causes the isopycnals to slump. In the basin region we invoke an ansatz that the isopycnals are flat — the model applies below the thermocline where wind effects cause stratification. Wind over the channel induces a northwards Ekman flux, and the return flow occurs at the bottom of the channel, as in the thick arrows in Fig. 21.31, because the in the interior the flow is nearly geostrophic and the zonally-averaged geostrophic meridional flow is zero. However, it is the residual circulation that carries water properties and that will connect to the basin flow, and thermodynamic considerations suggest that the flow will circulate along the dashed lines in Fig. 21.31. In the basin there will be an advective-diffusive balance in the vertical, and the flow will be non-zero only if the diffusivity is non- zero. This flow should connect smoothly to the more adiabatic flow in the channel. Let us see how the equations allow this to be accomplished, and if we can obtain estimates for the strength and structure of the flow.

21.8.2 Equations of motion

We will use zonally-averaged equations of motion and write them in residual, or TEM, form because the treatment of mesoscale eddies is more convenient and the equations direclty predict the velocities that advect the tracers. Thus, following the methodology of section 10.3, we define a *residual flow* such that

$$\overline{v}^* = \overline{v} - \frac{\partial}{\partial z}\left(\frac{1}{N^2}\overline{v'b'}\right), \qquad \overline{w}^* = \overline{w} + \frac{\partial}{\partial y}\left(\frac{1}{N^2}\overline{v'b'}\right), \tag{21.129}$$

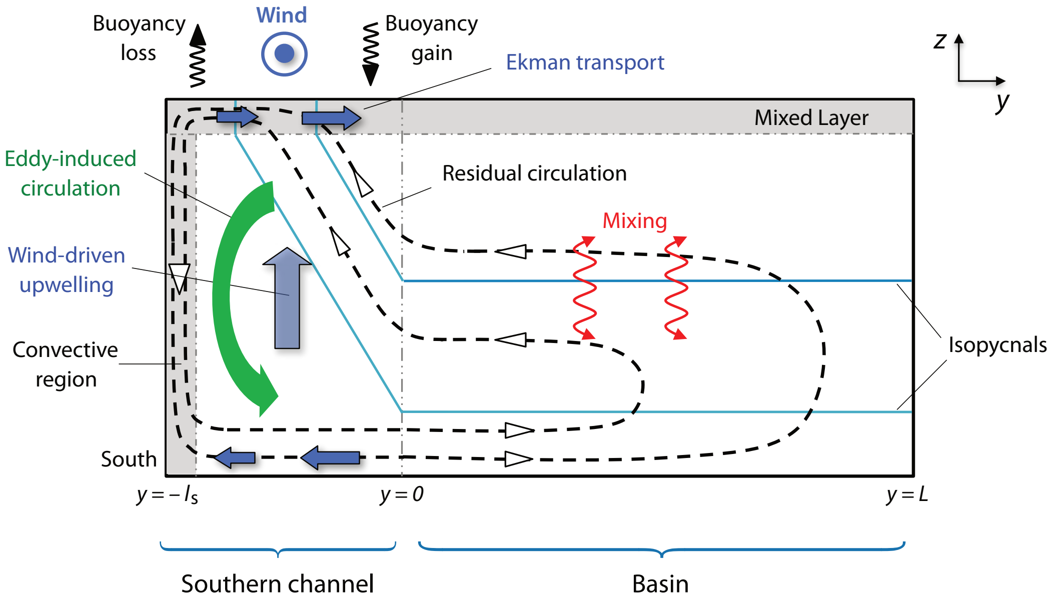

Fig. 21.31 A model of the single hemisphere meridional overturning circulation, crudely representing AABW in the Pacific or Indian Oceans. Thin solid lines are the isopycnals, the dashed black line is a residual overturning streamfunction. The thick dark blue arrows are the Eulerian circulation, namely the Ekman transport and the wind-driven upwelling.

where $N^2 = \partial\overline{b}/\partial z$, which is assumed to vary only very slowly. The residual velocities $\overline{v}^*$ and $\overline{w}^*$ more nearly represent the trajectories of fluid parcels than the Eulerian velocities, $\overline{v}$ and $\overline{w}$. There are no fluxes in the buoyancy equation and that only the potential vorticity flux, $\overline{v'q'}$, need be parameterized.

We will further suppose that the large scale flow satisfies planetary-geostrophic scaling, and so we drop the time derivative in the momentum equation and assume the zonal flow is in geostrophic wind balance. Including forcing and dissipation terms, our equations of motion become

$$-f\overline{v}^* = \overline{v'q'} + \frac{\partial \tau}{\partial z}, \qquad \frac{\partial \overline{b}}{\partial t} + \overline{v}^*\frac{\partial \overline{b}}{\partial y} + \overline{w}^*\frac{\partial \overline{b}}{\partial z} = \kappa_v \frac{\partial^2 \overline{b}}{\partial z^2}. \tag{21.130a,b}$$

The velocities are non-divergent and may be represented by a streamfunction so that $(\overline{v}^*, \overline{w}^*) = (-\partial\psi/\partial z, \partial\psi/\partial y)$, and we will assume that the residual veoclities themselves satisfy the boundary conditions of no normal flow. The zonal wind, $\overline{u}$, may be obtained thermal wind balance, $f\partial\overline{u}/\partial z = -\partial\overline{b}/\partial y$, and the stress, τ, is only non-zero near the top (wind-stress) and bottom (Ekman drag). We will henceforth drop the * notation, and all variables are understood to be residuals and zonal averages. These equations apply in both the channel and basin regions, but with different dominant balances.

Equations in the channel

The right-hand side of (21.130a) contains the eddy flux of potential vorticity which we parameterize using an eddy diffusivity,

$$\overline{v'q'} = -K_e\frac{\partial \overline{q}}{\partial y}. \tag{21.131}$$

A Model of the Meridional Overturning Circulation

The essential features of the model of the MOC of sections 21.8 and 21.9 are as follows.

Formulation

- The model is zonally averaged, in a single basin, with simple geometry: a zonally re-entrant channel at high latitudes, with an enclosed basin between it and the northern boundary. The effects of wind-driven gyres are neglected; the dynamics in the enclosed basin can be regarded as being below the main thermocline.
- The equations solved are the planetary geostrophic equations, in a transformed Eulerian mean form, with the effects of mesoscale eddies being parameterized with a simple down-gradient buoyancy flux scheme in the momentum equation.
- The ocean is divided into three regions: a northern convective region, a cross-equatorial basin region, and a southern channel.
- The dynamics are treated separately in these three regions, and the solutions matched at the boundaries.
  - *(i)* In the southern channel there is a balance between wind stress (causing isopycnals to steepen) and the mesoscale eddies, which have a flattening effect. Buoyancy satisfies the full nonlinear advective-diffusive equation.
  - *(ii)* In the basin region the isopycnals are assumed flat, and by integrating meridionally over the basin this region essentially becomes a boundary condition for the southern channel. The buoyancy equation reduces to a vertical advective-diffusive balance ($w\,\partial b/\partial z = \kappa\,\partial^2 b/\partial z^2$).
  - *(iii)* In the northern convective region the isopycnals are vertical, descending sufficiently far down to connect to the corresponding horizontal isopycnals of the basin. If the northern region is too warm to allow this, the basin isopyncals will extend all the way to the northern boundary.

Properties and Predictions

- If the surface boundary conditions on buoyancy permit, there is an isopycnal pathway from the northernconvective region to the southern channel. For small values of diffusivity flow can then circulate, largely adiabatically, from high northern latitudes to high southern latitudes, mechanically pumped by the wind over the southern channel. This may roughly correspond to flow in the Atlantic.
- If the boundary conditions are such that the surface of the northern region is too buoyant, then there is no interhemispheric wind-driven mid-depth circulation and no northern convection, roughly correspond to flow in the Pacific and Indian Ocean.
- For larger values values of diffusivity, the flow sinks at high latitudes and upwells in low latitudes, as in a conventional bouyancy/mixing-driven circulation.
- Beneath the wind-driven mid-depth cell, a diffusive cell corresponding to Antarctic Bottom Water forms. Its strength is determined by the diapycnal diffusivity and surface meridional buoyancy gradients.
- The southern channel there is convection at the southern end and elsewhere the isopycnal slope is determined by a balance between wind forcing and eddy effects.
- In the model as presented here, the effects of mesoscale eddies are parameterized by an eddy diffusivity. The method is far from perfect, but the overall model framework is not fundamentally dependent on that.

where K_e is the eddy diffusivty. (It is more-or-less a 'Gent–McWilliams' coefficient, as in section 13.6.) The Coriolis parameter is almost constant in the channel, and we denote it f_S. For the large-scale ocean the potential vorticity is given by

$$\overline{q} \approx f_S \frac{\partial}{\partial z}\left(\frac{\overline{b}}{\overline{b}_z}\right), \quad \text{so that} \quad \frac{\partial \overline{q}}{\partial y} \approx f_S \frac{\partial}{\partial z}\left(\frac{\overline{b}_y}{\overline{b}_z}\right) = -f_S \frac{\partial S}{\partial z} \tag{21.132}$$

where $S = -\overline{b}_y/\overline{b}_z$ is the slope of the isopycnals (and the similarity with the Gent–McWilliams scheme is now clear). The potential vorticity flux is then given by

$$\overline{v'q'} \approx f_S K_e \frac{\partial S}{\partial z}. \tag{21.133}$$

and the momentum equation becomes

$$-f_S \overline{v} = f_S K_e \frac{\partial S_b}{\partial z} + \frac{\partial \tau}{\partial z}. \tag{21.134}$$

Since $\overline{v}^* = -\partial\psi/\partial z$ we integrate this from the top to a level z and obtain

$$\boxed{\psi = -\frac{\tau_w}{f_S} + K_e S,} \tag{21.135}$$

where both f_S and S are negative and τ_S is the surface kinematic stress in the channel. We have assumed $\psi = 0$, $S = 0$ at the surface (the base of mixed layer) and $\tau = 0$ in the interior.

The buoyancy equation in terms of streamfunction is

$$\overline{\boldsymbol{v}} \cdot \nabla \overline{b} = \kappa_v \frac{\partial^2 \overline{b}}{\partial z^2} \quad \text{or} \quad \frac{\partial \psi}{\partial y}\frac{\partial \overline{b}}{\partial z} - \frac{\partial \psi}{\partial z}\frac{\partial \overline{b}}{\partial y} = \kappa_v \frac{\partial^2 \overline{b}}{\partial z^2}, \tag{21.136}$$

which can be written as

$$\boxed{\frac{\partial \psi}{\partial y} + S\frac{\partial \psi}{\partial z} = \kappa_v \frac{\partial_z^2 \overline{b}}{\partial_z \overline{b}}.} \tag{21.137}$$

The boundary condition on ψ for this will be supplied by the basin! The other boundary condition we will need is the buoyancy distribution at the top, and so we specify

$$\overline{b}(y, z = 0) = b_0(y). \tag{21.138}$$

Equations in the basin

In the basin the slope of the isopycnals is assumed zero and (21.137) becomes the conventional upwelling diffusive balance,

$$w\frac{\partial \overline{b}}{\partial z} = \kappa \frac{\partial^2 \overline{b}}{\partial z^2} \quad \text{or} \quad \frac{\partial \psi}{\partial y}\frac{\partial \overline{b}}{\partial z} = \kappa_v \frac{\partial^2 \overline{b}}{\partial z^2}. \tag{21.139}$$

If we integrate this from the edge of the channel, $y = 0$, to the northern edge, $y = L$, we obtain

$$\boxed{\psi|_{y=0} = -\kappa_v L \frac{\overline{b}_{zz}}{\overline{b}_z}.} \tag{21.140}$$

This equation then becomes the needed boundary condition for the equations in the channel.

21.8.3 Scaling

The above equations do not give up analytic solutions, but we can use them to obtain estimates of the flow strength and structure. Let us scale the equations by letting

$$z = h\hat{z}, \qquad y = l\hat{y}, \qquad \tau_S = \tau_0\hat{\tau}_S, \qquad f_S = \overline{f}_S\hat{f}, \qquad \psi = \frac{\tau_0}{\overline{f}_S}\hat{\psi}, \qquad S = \frac{h}{l}\hat{S}, \tag{21.141}$$

where $\overline{f}_S = |f_S|$, a hat denotes a nondimensional value and h is a characteristic vertical scale such that $S \sim h/l$, and *this will emerge as part of the solution*.

If we have scaled properly then variables with hats on are of order one. The nondimensional equations of motion are then

$$\text{Buoyancy evolution:} \qquad \partial_{\hat{y}}\hat{\psi} + \hat{S}\,\partial_{\hat{z}}\hat{\psi} = \epsilon\left(\frac{l}{L}\right)\frac{\partial_{\hat{z}\hat{z}}\hat{b}}{\partial_{\hat{z}}\hat{b}}, \tag{21.142a}$$

$$\text{Momentum balance:} \qquad \hat{\psi} = -\frac{\hat{\tau}_S}{\hat{f}} + \Lambda\hat{S}, \tag{21.142b}$$

$$\text{Boundary condition:} \qquad \hat{\psi}|_{\hat{y}=0} = -\epsilon\frac{\partial_{\hat{z}\hat{z}}\hat{b}}{\partial_{\hat{z}}\hat{b}}, \tag{21.142c}$$

where

$$\Lambda = \frac{\text{Eddies}}{\text{Wind}} = \frac{K_e}{\tau_0/\overline{f}_S}\frac{h}{l}, \qquad \epsilon = \frac{\text{Mixing}}{\text{Wind}} = \frac{\kappa_v}{\tau_0/\overline{f}_S}\frac{L}{h}. \tag{21.143a,b}$$

These are two important nondimensional numbers, and we can obtain estimates of their values by using some observed values for the other parameters. Let us take

$$\begin{gathered} h = 1\,\text{km}, \ \kappa_v = 10^{-5}\,\text{m}^2\,\text{s}^{-1}, \ K_e = 10^3\,\text{m}^2\,\text{s}^{-1}, \ \rho_0 = 10^3\,\text{kg}\,\text{m}^{-3}, \\ \tau_0 = 0.1\,\text{N}\,\text{m}^{-2}/\rho_0 = 10^{-4}\,\text{N}\,\text{m}\,\text{kg}^{-1}, \ \overline{f}_S = 10^{-4}\,\text{s}^{-1}, \ L = 10\,000\,\text{km}, \ l_s = 1000\,\text{km}, \end{gathered} \tag{21.144}$$

and we find

$$\Lambda \approx 1, \qquad \epsilon \approx 0.1. \tag{21.145}$$

These values come with large error bars: the diffusivity, κ_v, may be much larger in the abyss, and the eddy coefficient K_e is very poorly constrained (indeed, it is a property of the flow itself, not the fluid). Finally, note that Λ and ϵ are not independent of each other for they both depend on the vertical scale of stratification h which is a part of the solution. To obtain some theoretical estimates of h we look at some limiting cases.

The small diffusion limit

Suppose that mixing is small and that $\epsilon \ll 1$. We can then *require* that $\Lambda = 1$ in order that the eddy-induced circulation nearly balance the wind-driven circulation (because the diffusive term is small), whence the vertical scale h is given by

$$\boxed{\frac{h}{l} = \frac{\tau_0/f_S}{K_e}.} \tag{21.146}$$

As K_e diminishes h becomes larger, meaning that the isopycnals are near vertical. Using (21.146) in (21.143b) gives

$$\epsilon = \frac{\kappa_v K_e}{(\tau_0/f_S)^2}\frac{L}{l}. \tag{21.147}$$

This is an appropriate nondimensional measure of the strength of the diapycnal diffusion in the ocean. Using (21.142c) we see that $\widehat{\psi} \sim \epsilon$ so that the dimensional strength of the circulation goes as

$$\Psi = \epsilon \frac{\tau_0}{f_0} = \kappa_v \frac{K_e}{\tau_0/f_S} \frac{L}{l}. \tag{21.148}$$

Another way to obtain this is to use the fact that for weak diffusion the balance in the dimensional momentum equation is between wind forcing and eddy effects (because they must nearly cancel) so that

$$\frac{\tau_w}{f} \sim K_e S, \qquad \text{or equivalently} \qquad \frac{h}{l} \sim \frac{\tau_w}{K_e f_S}. \tag{21.149a,b}$$

Advective-diffusive balance in the basin gives

$$\frac{\partial \psi}{\partial y}\frac{\partial \overline{b}}{\partial z} = \kappa_v \frac{\partial^2 \overline{b}}{\partial z^2} \qquad \text{whence} \qquad \Psi = \frac{\kappa_v L}{h} \tag{21.150a,b}$$

and (21.149b) and (21.150b) together give (21.148).

The high diffusion limit

To explore the high diffusion limit we take $\epsilon \gg 1$. The nondimensional strength of the circulation is given by

$$\widehat{\psi} = \mathcal{O}(\epsilon) \gg 1. \tag{21.151}$$

The circulation is now 'strong', since $\widehat{\psi} \neq \mathcal{O}(1)$. Dimensionally we still have that

$$\Psi = \epsilon \frac{\tau_0}{f_S} \qquad \text{or} \qquad \Psi = \frac{\kappa_v L}{h} \tag{21.152a,b}$$

but h and ϵ will be different than in the low diffusion limit. Now, if $\widehat{\psi} \sim \epsilon \gg 1$ the diffusion driven circulation in the basin cannot be matched by a purely wind-driven circulation in the channel, since the latter is $\mathcal{O}(1)$. Put more physically, as we increase diffusivity the circulation increases in strength, but this cannot connect smoothly to the flow in the channel unless the eddy-driven circulation changes because since the wind-driven circulation is externally fixed. We thus match the basin circulation to an eddy-driven channel circulation and require

$$\Lambda = \mathcal{O}(\epsilon). \tag{21.153}$$

In particular, if we set $\Lambda = \epsilon$ then

$$\epsilon = \Lambda = \sqrt{\frac{K_e \kappa_v L}{(\tau_0/f_S)^2 l}}. \tag{21.154}$$

This is the square root of the expression for ϵ in the weak diffusion limit. Using (21.154) and (21.152a) we find

$$\frac{h}{l} = \sqrt{\frac{\kappa_v L}{K_e l}}, \qquad \Psi = \sqrt{\frac{K_e \kappa_v L}{l}}. \tag{21.155a,b}$$

These are expressions for the characteristic depth and strength of the circulation in a strong mixing regime.

Meaning of the limits

If diffusion is weak the stratification is set by a trade-off between the eddies and wind and this determines h, as in (21.146), and this does not involve diapycnal diffusion at all. However, the strength of the circulation is determined by an upwelling-diffusion balance which gives the estimate $\psi \sim \kappa_v L/h$. Since h is independent of diffusivity we obtain a circulation strength that is linearly proportional to diffusivity, as in (21.148). Since diffusion is, in this limit, small then the circulation is weak, even in the southern channel. The weakness arises because there is a cancellation between the wind stress and eddy terms in (21.135), and a total cancellation would lead to the so-called 'vanishing of the Deacon Cell' — the Deacon Cell being the residual overturning in the southern channel. It is interesting that the circulation gets weaker as the wind gets stronger; this counter-intuitive effect arises because the wind steepens the isopycnals and deepens the stratification, so that the diffusive term $(\kappa_v \overline{b}_{zz})$ in the basin gets weaker. From an asymptotic perspective, in the small ϵ limit the residual circulation is zero to lowest order, and at the next order the flow is parallel to the isopycnals in the channel (except in the mixed layer). We will see in the next section that the Deacon cell need not vanish, even in the limit of weak diffusion, if there is a northern source of water.

In the strong diffusion case the diffusivity itself directly affects the stratification, and consequently we get a weaker dependence of the circulation strength on κ_v. In this limit diapycnal mixing deepens the isopycnals in the basin away from the channel, and this deepening in turn means that the diffusion has a weaker effect. Thus, although the circulation is stronger than in the weak diffusion case it has a weaker dependence on diffusivity, to the one half power in fact (21.155b). The second consequence of the deepening is that the isopycnals are steeper in the channel, with the steepening being balanced by the the enhanced slumping effects of baroclinic instability, with the wind then only having a secondary effect.

Finally, isnstead of varying diffusivity we can think of the wind changing. In the weak wind limit the circulation is diffusively driven and independent of the wind strength, as in (21.155b). In the strong wind limit the circulation, as noted above, actually decreases as the wind increases, but still remains proportional to the diapycnal diffusivity κ_v.

21.9 † AN INTERHEMISPHERIC CIRCULATION

We now introduce another 'water mass' into the mix — *North Atlantic Deep Water*, or NADW. We thus divide the ocean into three regions as sketched in Fig. 21.32 and as follows.

1. A southern channel (south of about 50° S) where, as before, we expect a balance between eddy effects and wind effects.
2. A basin region (from about 50° S to, say, 60° N), where the isopycnals are fairly flat.
3. A northern convective region (north of 60° N) in which convection produces vertical isopycnals that connect with those in the basin.

Although the dynamics of all three regions are locally different, they must act in concert to produce a dynamically consistent circulation. The main difference, and it is an important one, between this model and the previous one is the presence of an interhemispheric cell that, we will find, is primarily wind driven, and that (for realistic parameter values) sits on top of the lower cell. In the presentation that follows we focus on the Northern convective region and the upper cell, for the dynamics of the lower cell are very similar to those of the previous section. Further, we seek only scaling relations rather than full analytic or semi-analytic solutions.[30] We will use lower case letters (e.g., h, ψ) to denote variables and upper case symbols (e.g., H, Ψ) for representative values.

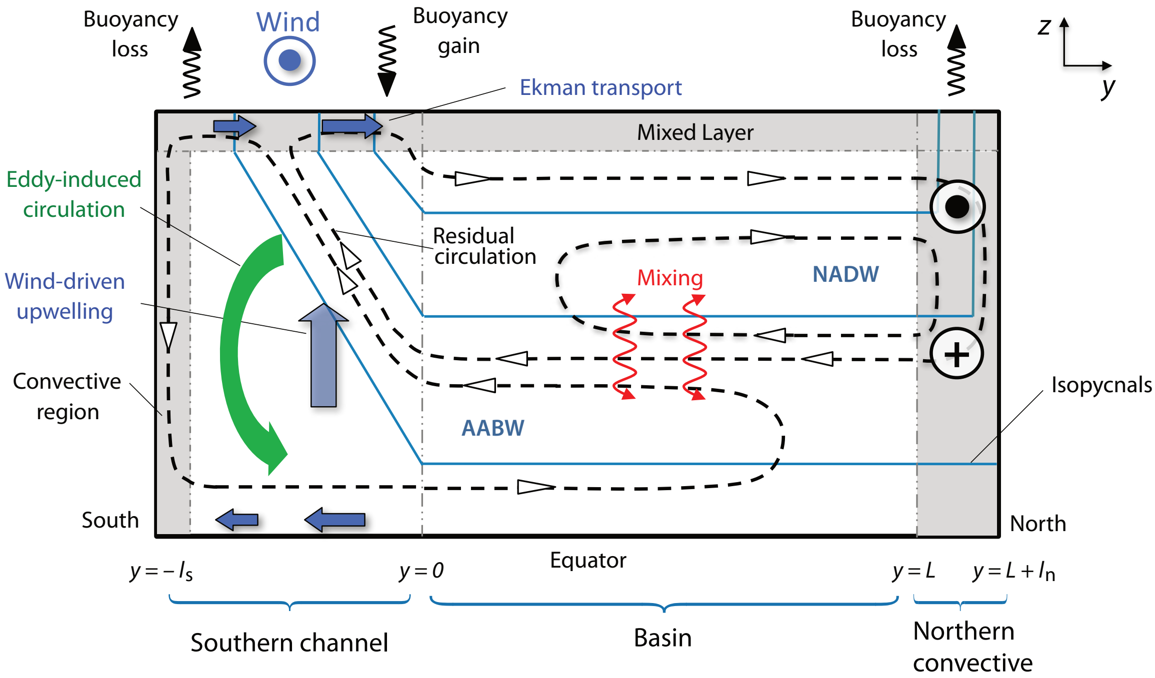

Fig. 21.32 Schematic of an interhemispheric MOC, representing an Atlantic or possibly a global circulation. The solid blue lines are the isopycnals, the dashed lines with arrows are the streamlines, the dashed vertical lines are the boundaries between adjacent regions, shaded grey areas are the convective regions at high latitudes and the surface mixed layer,and the red arrow represents downward diffusive heat flux. Labels 1, 2, and 3 (in circles) denote the circumpolar channel, ocean basin, and isopycnal outcrop regions.

21.9.1 Model phenomenology

The upper cell has similar characteristics to the wind-driven cell sketched in Fig. 21.22, but we now require the flow in the basin to connect smoothly to an eddy-rich southern channel region, in a similar manner to that described in the previous section. We also suppose that in the northern region the interior flow connects to the surface by way of convection. To see how this occurs, consider a given isopycnal, b_0 say, that outcrops at the surface in the southern channel, slopes down in the channel and becomes horizontal in the basin. If the surface values of b in northern convecitve region are all larger (warmer) than b_0 then the isopycnal never outcrops in the north; rather, it continues northward until it intersects the northern wall. If, on the other hand, at some latitude there is a latitude, y_n say, at which the surface values become lower than b_0 convection will occur and the b_0 isopycnal becomes vertical. There is then an isopycnal pathway from the surface at y_n through the interior to the southern channel. A parcel of water may move along this pathway even in the absence of diffusion; that is, there can be an interhemispheric mid-depth adiabatic circulation.[31] Below this cell (which we associate with NADW) there can be a bottom call of AABW that is diffusively driven, as in the previous section.

21.9.2 Dynamics in the northern convective region

In the northern region the values of buoyancy at the surface (i.e., $b_3(y, z = 0)$) are mapped on to the flat isopycnals in the interior (i.e, $b_2(z)$), and the simplest assumption to make is that the matching occurs by convection. That is, the surface waters convect downward to the level of neutral buoyancy, producing vertical isopyncnals, and then move meridionally. By thermal wind the

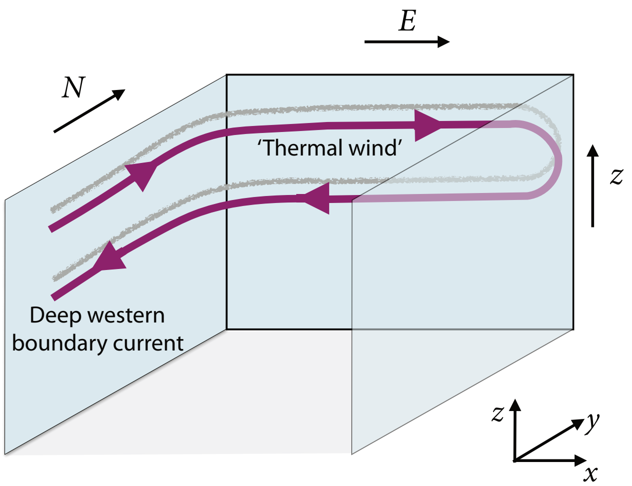

Fig. 21.33 Sketch of the mean flow in the northern region. The north-south temperature gradient induces a zonal 'thermal wind', which is supplied by and feeds the deep western boundary current, as shown.

vertical isopycnals give rise to a zonal flow, with the total zonal transport being determined by the meridional temperature gradient and the depth, h, to which flow convects. The zonal flow is thus

$$u_N(y,z) = -\frac{1}{f}\int_{-h}^{z} \frac{\partial b_N}{\partial y}\,\mathrm{d}z' + \text{constant}, \tag{21.156}$$

where the constant is determined by the requirement that $\int_{-h}^{0} u_3\,\mathrm{d}z = 0$, and there are boundary layers in both east and west to bring the flow to zero. When the relatively shallow eastward moving zonal flow collides with the eastern wall it subducts and returns, as sketched in Fig. 21.33, and when the flow reaches the western wall it then moves equatorward in the deep western boundary current. Similarly, it is the upper, northward moving branch of the western boundary current that feeds the eastward moving flow. The total volume transport ($\mathrm{m^3\,s^{-1}}$) in these thermally-induced zonal flows thus translates to a meridional streamfunction given by

$$\int_{x_W}^{x_E} \psi_N\,\mathrm{d}x = \int_{-h}^{z} \mathrm{d}z' \int_{L_y}^{L_n} u_N\,\mathrm{d}y, \tag{21.157}$$

where L_x is the zonal extent of the region, L_y is the latitude of the southern edge of the convecting region and $L_n = L_y + l_n$ is the northern edge of the domain (Fig. 21.32). Using (21.156) gives an estimate for the value of this streamfunction as

$$\Psi_N = \frac{\Delta b\, H^2}{L_x f_N}, \tag{21.158}$$

where Δb is the surface buoyancy difference across the northern convective region, which in the theory we are describing is an external parameter, and f_N is the Coriolis parameter in the northern region. The streamfunction ψ_N is function of space and Ψ_N is a representative value of it, and H is a representative value of h, the depth to which the convection reaches and so of the stratification.

21.9.3 Connection to other regions

In the basin region we posit flat isopycnals and an upwelling diffusive balance, whence

$$w_B\frac{\partial b}{\partial z} = \kappa_v\frac{\partial^2 b}{\partial z^2} \qquad \text{giving} \qquad \frac{\Psi_B}{L_y} = \frac{\Psi_N - \Psi_S}{L_y} = \frac{\kappa_v}{H} \tag{21.159}$$

In the channel region the residual circulation arise from a balance between the wind and eddy effects and is given by (21.135), which we can write as

$$\Psi_S = \left(\frac{\tau_0}{\overline{f}_S} - K_e \frac{H}{l_s}\right), \tag{21.160}$$

with $\overline{f}_S = |f_S|$ as before. Collecting the various expressions above for streamfunction we have

$$\Psi_S = \left(\frac{\tau_0}{\overline{f}_S} - K_e \frac{H}{l_s}\right) \qquad \Psi_N - \Psi_S = \frac{\kappa_v}{H} L_y \qquad \Psi_N = \frac{\Delta b\, H^2}{f_N L_x}, \tag{21.161a,b,c}$$

with unknowns Ψ_S, Ψ_N and H. The various parameters have approximate values as given in (21.144) as well as

$$L_x = 5000\,\mathrm{km}, \quad L_y = 10\,000\,\mathrm{km}, \quad \Delta b = 10^{-2}\,\mathrm{m\,s^{-2}}, \quad f_N = 10^{-4}\,\mathrm{s^{-1}}. \tag{21.162}$$

The streamfunctions, ψ, are properly functions of space and should be matched at the region boundaries. However, for scaling purposes we regard them as numbers (and refer to them as Ψ_N, Ψ_S, Ψ_B), in which case (21.161) reduces to

$$\frac{\Delta b H^2}{f_N} - \left(\frac{\tau_0}{\overline{f}_S} - K_e \frac{H}{l_s}\right) L_x = \frac{\kappa_v}{H} L_x L_y, \tag{21.163}$$

which is a cubic equation for the the characteristic depth, H, of the upper, NADW, cell in Fig. 21.32. Although there are analytic solutions to cubic equations it is more instructive to consider limiting cases, first with either the northern region or the southern channel absent and then with low or high diffusivity.

No northern source.

Suppose that $\Delta b = 0$ and that there is no deep water formation in the North. If κ_v is small then we obtain $H/l_s = (\tau_0/\overline{f}_S)/K_e$ a, equivalent to (21.146), and $\Psi_S = \Psi_B = 0$. If κ_v is large then we find

$$H^2 = \frac{\kappa_v L l_s}{K_e}, \tag{21.164}$$

so recovering (21.155a). Thus, the dynamics are essentially those section 21.8, and there is a single deep, and rather weak, diffusively-driven, cell. The same situation arises if the northern region is too buoyant (e.g., too warm) for then there is no isopycnal pathway between high northern hemispheres and the southern channel, and deep convection does not occur. This case may have relevance to the Pacific Ocean, where the surface at high northern latitudes is insufficiently dense and there is no Pacific equivalent of NADW.

No southern channel.

If there is no southern channel then $\Psi_S = 0$ and we have

$$\frac{\Delta b H^2}{f_N} = \frac{\kappa_v}{H} L_x L_y \tag{21.165}$$

giving

$$H^3 = \kappa_v \left(\frac{f_N L_x L_y}{\Delta b}\right) \quad \text{and} \quad \Psi_N = \Psi_B = (\kappa_v L_y)^{2/3} \left(\frac{\Delta b}{L_x f_N}\right)^{1/3}. \tag{21.166}$$

These are the classical expressions for the thickness of a diffusive thermocline and the strength of a diffusively-driven overturning circulation, essentially as obtained in sections 21.2.5 and 20.4.1.

Let us now look at the case with all three regions, in the limits of weak and strong diffusivity.

Low diffusivity limit

In this case the upwelling is weak and $|\Psi_N| \approx |\Psi_S|$ and

$$\frac{\Delta b H^2}{f_N} - \left(\frac{\tau_0}{\overline{f}_S} - K_e \frac{H}{l_s}\right) L_x = 0. \tag{21.167}$$

In this case the basin is just a 'pass-through' region: water formed in the North Atlantic just passes through the basin without change, and upwells in the Southern Ocean. For the above expression to be physical, τ_0 must be non-zero; this requirement is a manifestation of Sandström's effect, that in the absence of diffusivity a mechanical forcing is needed, and if $\tau_0 = 0$ in the above then the only physical solution is $h = 0$.

If we assume that K_e is small then we obtain

$$H = \left(\frac{\tau_0 f_N L_x}{\overline{f}_S \Delta b}\right)^{1/2}, \qquad \Psi_S = \Psi_N = \frac{\tau_0 L_x}{\overline{f}_S}, \tag{21.168}$$

which with the parameters chosen earlier gives $H \sim 320$ m and $\Psi \sim 10$ Sv. The residual circulation *does not vanish in the limit of small diffusivity;* rather, it is wind driven, adiabatic, interhemispheric and independent of diffusivity. This is to be contrasted with the case in which deep waters are not produced in the north, as observed in the Pacific Ocean, where in low diffusivity limit the eddy-induced circulation nearly cancels the wind-driven circulation resulting in small residual circulation, dependent on that diffusivity.

In the more general case we solve (21.167) to give

$$H = \left(\frac{\tau_0 f_N L_x}{\overline{f}_S \Delta b}\right)^{1/2} \left(-\alpha + \sqrt{1+\alpha^2}\right) \tag{21.169}$$

where α is a nondimensional number giving the ratio of eddy to wind effects,

$$\alpha = \frac{1}{2}\frac{K_e}{l_s}\left(\frac{L_x \overline{f}_S f_N}{\tau_0 \Delta b}\right)^{1/2} = \frac{1}{2}\frac{\Psi_{eddy}}{\Psi_{wind}}. \tag{21.170}$$

where

$$\Psi_{wind} = \frac{\tau_0}{\overline{f}_S} \quad \text{and} \quad \Psi_{eddy} = \frac{K_e}{l_s}\left(\frac{\tau_0 f_N L_x}{f_S \Delta b}\right)^{1/2}. \tag{21.171}$$

Putting in values from (21.144) and (21.162) gives $\alpha \sim 0.1$, $\Psi_{eddy} \sim 1.6$ Sv and $\Psi_{wind} \sim 10$ Sv suggesting that wind effects are dominant, but there is considerable uncertainty because the value of K_e is poorly known.

High diffusivity limit

In the high diffusivity limit the wind-driven upwelling in the Southern Hemisphere is small compared to the mixing-driven upwelling in the ocean basin, and although it seems to be not relevant for today's circulation it may have been important in glacial climates. Equation (21.163) simply becomes (21.165), which as already noted gives us the classical scaling for a diffusively-driven circulation. The upper cell thus fades out before reaching the southern channel but there remains a lower, AABW, cell that connects to the flow in the southern channel as described in section 21.8.

21.9.4 Final remarks and relevance to the ocean

Over the last several pages we've described a conceptual, but quantitative, theoretical model of the overturning circulation in the ocean. Aside from its idealizations (e.g.., simplified geometry) the model has two main shortcomings: it uses an eddy-diffusivity parameterization for the effects of mesoscale eddies, and it treats the ocean one basin at a time. Putting these aside, what does the model tell us?

In the limit of weak diapycnal mixing, which seems relevant to the present mid-depth ocean, and with a northern source of deep water, then the model produces a circulation relevant to the Atlantic — Fig. 21.32 is an idealization of the Atlantic panel in Fig. 21.29. The strength of the mid-depth overturning circulation is then largely determined by the Ekman transport in the Southern Ocean and, secondarily, eddy effects. The rest of the ocean is essentially forced to adjust and produce the amount of deep water demanded by the Ekman transport and the associated wind-driven upwelling in the Southern Ocean. Beneath this mechanically-forced mid-depth cell lies a diffusively-driven deep cell, and this is the model representation of AABW.

The Pacific Ocean is insufficiently dense at high northern latitudes to produce deep water — compared to the Atlantic it is relatively fresh. That is, there is no isopycnal pathway from the surface waters at high latitudes to the Southern channel; and consequently there is no wind-driven mid-depth cell comparable to that of the Atlantic. The model then produces a circulation similar to that of Fig. 21.31, where the northern wall is at a high latitudes in the Northern Hemisphere, and this is an idealization of the Pacific panel in Fig. 21.29. Since the diffusivity is weak the circulation is weak, particularly in the Northern Hemisphere. The flow in the world's ocean is much more interconnected than these simple ideas suggest, and in reality the flow travels from basin-to-basin on what has been metaphorically called a conveyor belt.[32] But our own story ends here, for now. How the models above might be extended to produce a global flow is a chapter for another day.

Notes

1 Adapted from Wunsch (2002). The figure shows a 'state estimate' — a combination of models and observation, similar to an atmospheric reanalysis.

2 Figure kindly prepared by Neven Stjepan Fučkar, using the climatology of Conkright *et al.* (2001).

3 Warren (1981) provides a review and historical background and Schmitz (1995) surveys the observations and provides an interpretation of the deep global circulation.

4 The word 'driven' is fraught with ambiguity, even when the subject matter is well understood. Does it refer to the proximate mechanical forces producing the motion, or to the controlling device? The former (which is quite common in physical science) suggests that an engine drives a car, for that is what makes the wheels go round, whereas the latter suggests that, in fact, the driver drives the car. For the less well-understood ocean there is scope for still more confusion, and the context in which the word is used becomes important. What we in this chapter sometimes call buoyancy-driven might be better called mixing-driven, since it is the mixing of fluid parcels that makes potential energy available for the circulation. *Caveat lector.*

5 As in Haney (1971). The value of C, which is not necessarily related to that of κ, is often taken to be such that the heat flux is of order $30\,\mathrm{W\,m^{-2}\,K^{-1}}$, but it is certainly not a universal constant.

6 Adapted from Paparella & Young (2002).

7 Rossby (1965).

8 Adapted from Ilicak & Vallis (2012).

9 I am grateful to Tom Haine for pointing out this argument. See also Haine & Marshall (1998) and Hughes & Griffiths (2008).

10 Ocean convection is also reviewed by Marshall & Schott (1999).

11 Sandström (1908, 1916). Sandström's discussion was rather qualitative and generally thermodynamic in nature, with friction playing only an implicit role. Since then a number of related statements with varying degrees of generality and preciseness have been given (e.g., Dutton 1986, Huang 1999, Paparella & Young 2002). Section 21.2.3 follows Paparella and Young.

12 The original box model is due to Stommel (1961), and many studies with variations around this have followed. Rooth (1982) developed the idea of a buoyancy-driven pole-to-pole overturning circulation, and Welander (1986) discussed, among other things, the role of boundary conditions on temperature and salinity at the ocean surface. Thual & McWilliams (1992) systematically explored how box models compare with two-dimensional fluid models of sideways convection, Quon & Ghil (1992) explored how multiple equilibria arise in related fluid models, and Dewar & Huang (1995) discussed the problem of flow in loops. Cessi & Young (1992) tried to derive simple models systematically from the equations of motion, obtaining various nonlinear amplitude equations. Our discussion is just a fraction of all this — see also Whitehead (1995) and Cessi (2001) for reviews.

13 Adapted from Welander (1986).

14 Having said this, Bryan (1986), Manabe & Stouffer (1988) and Marotzke (1989) did find evidence of multiple equilibria in various three-dimensional numerical models, motivated in part by the solutions of box models.

15 After Stommel *et al.* (1958).

16 Following Stommel & Arons (1960).

17 A global Stommel–Arons-like solution was presented by Stommel (1958). The discovery of deep western boundary currents by Swallow & Worthington (1961) was motivated by the theoretical model. Using neutrally-buoyant floats underneath the Gulf Stream they found an equatorward-flowing undercurrent with typical speeds of 10–20 cm s^{-1}. Some relevant observations of the deep circulation are summarized by Hogg (2001).

18 For example, Toggweiler & Samuels (1995).

19 Drawing from the various numerical, conceptual and analytic models of Toggweiler & Samuels (1995, 1998) Döös & Coward (1997), Gnanadesikan (1999), Vallis (2000), Webb & Suginohara (2001), Nof (2003), Samelson (1999a, 2004), Wolfe & Cessi (2011), and Nikurashin & Vallis (2011, 2012). The notion of a deep interhemispheric circulation driven by winds in the ACC was earlier proposed by Eady (1957), albeit informally.

20 Loic Jullion graciously provided the inverse calculations, which are similar to those described in Lumpkin & Speer (2007). Patrick Heimback kindly provided the state estimates, which are from the ECCO suite of calculations.

21 From Rintoul *et al.* (2001).

22 See Rintoul *et al.* (2001) and Olbers *et al.* (2004) for ACC reviews.

23 These simulations, described in Henning & Vallis (2005), solve the primitive equations in a domain similar to Fig. 21.19. The wind forcing produces a poleward Ekman drift across the channel, as well as a subtropical gyre, and there is a meridional temperature gradient across the whole domain, so giving rise to a subtropical thermocline.

24 Adapted from Henning & Vallis (2005).

25 Munk & Palmén (1951), Gille (1997) and Stevens & Ivchenko (1997).

26 Adapted from a figure in Burke *et al.* (2015).

27 Models of this ilk stem from Johnson & Bryden (1989). Straub (1993), Hallberg & Gnanadesikan (2001) Karsten *et al.* (2002), Marshall & Radko (2003), Henning & Vallis (2005), consider related issues and extensions.

28 See also Munk & Palmén (1951), Warren *et al.* (1996), Olbers (1998) and Hughes (2002).

29 Largely following Nikurashin & Vallis (2011, 2012). For related numerical simulations see Vallis (2000) and Wolfe & Cessi (2010, 2011).

30 A full description may be found in Nikurashin & Vallis (2012). The scalings for the upper cell are similar to those of Gnanadesikan (1999).

31 A continuous, unique, pole-to-pole isopycnal pathway is a chimera, because of the nonlinearity and saline dependence of the equation of state. But at the level of our theory there is an approximate one.

32 For variations involving multiple basins that are relevant to the work described here see Ferrari *et al.* (2014) and/or Thompson *et al.* (2016).

In the afternoon they came unto a land
In which it seemed always afternoon.
All round the coast the languid air did swoon,
Breathing like one that hath a weary dream.
Alfred Tennyson, *The Lotus Eaters*, 1832.

CHAPTER 22

Equatorial Circulation and El Niño

EQUATORIAL OCEANOGRAPHY DECEIVES US, hiding fascinating, non-intuitive dynamics beneath the languorous tropical air. The midatitudes give us the great gyres with their intense western boundary currents and mesoscale eddies and by comparison the equatorial currents may seem, on the surface, featureless and vapid. Yet the equatorial regions are home to the willful dynamics of the equatorial undercurrents that rush across the basins, opposite in bearing to the winds that drive them. And the equatorial ocean and atmosphere — in a collaboration that is more tango than waltz — give rise to the marvelous phenomenon that is El Niño, the most dramatic example of climate variability on human timescales that this planet has to offer. Such phenomena are the subjects of this chapter.

The defining feature of equatorial dynamics is that the Coriolis parameter becomes small, at least by comparison with the midlatitudes, and balanced and unbalanced dynamics become intertwined, as we encountered in chapter 8. Yet if we move more than a few degrees away from the equator the Rossby number again becomes quite small, suggesting that familiar ways of investigating the dynamics — Sverdrup balance for example — might yet play a role. Let's first see what we are trying to understand and if the observations can give us some intuition.

22.1 OBSERVATIONAL PRELIMINARIES

In midlatitudes the gyres are very robust features, existing in all the basins, and may be understood as the direct response to the curl of the wind stress. In the equatorial regions the currents also display some robust and distinctive features, illustrated in Fig. 22.1 and the top panel of Fig. 22.2, but their relation to the winds is less obvious. The main features are as follows.

1. A shallow westward[1] flowing surface current, typically confined to the upper 50 m or less, strongest within a few degrees of the equator, although not always symmetric about the equator. Its speed is typically a few tens of centimetres per second.

2. A strong coherent eastward undercurrent extending to about 200 m depth, confined to within a few degrees of the equator. Its speed is up to a metre per second or a little more, and it is this current that dominates the vertically integrated transport at the equator. Beneath the undercurrent the flow is relatively weak.

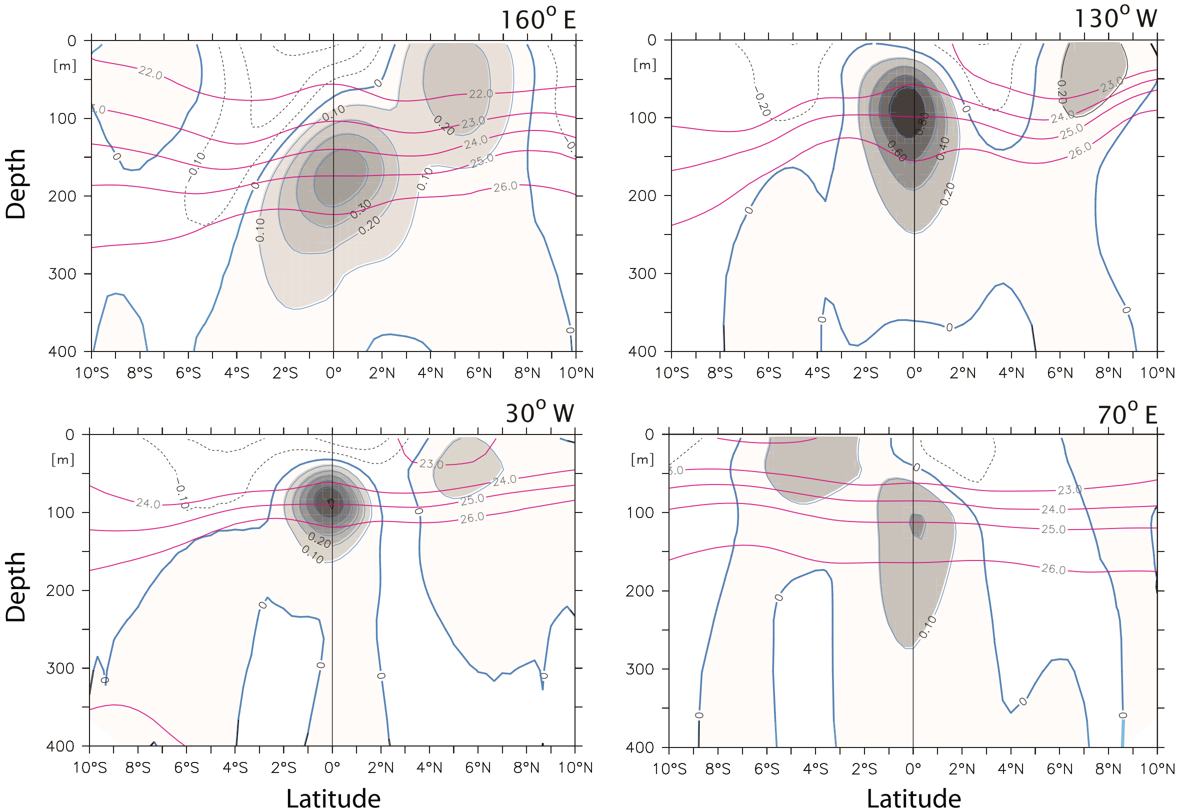

Fig. 22.1 Sections of the observed mean zonal current (shading and associated contours) at two longitudes in the Pacific (upper panels), in the Atlantic (lower left) and in the Indian Ocean (lower right). The contours are every 20 cm s^{-1} in the upper two panels and every 10 cm s^{-1} in the lower panels. Note the well-defined eastward undercurrent at the equator in all panels, and a weaker eastward countercurrent at about 6° N and/or 6° S. The red, more horizontal, lines are isolines of potential density.[2]

3. Westward flow on either side of the undercurrent, with eastward countercurrents poleward of this. In the Pacific the countercurrent is strongest in the Northern Hemisphere where it reaches the surface.

22.2 DYNAMICAL PRELIMINARIES

In mid-latitudes the large scale currents system may be understood using the planetary geostrophic equations of motion. Applying these allows us to understand formation of the great wind-driven gyres, with Sverdrup balance providing a solid foundation on which to build. As we approach lower latitudes the Coriolis parameter, f decreases and the Rossby number increases and one might expect that dynamics based on geostrophic balance will ultimately fail. A little surprisingly, it is only very close to the equator that the Rossby number exceeds unity: if we take a velocity of 0.5 m s^{-1}and a length scale of 500 km then the Rossby number at 5° latitude is 0.08, at 2°, 0.2 and at 1°, 0.4. These numbers suggest that until we are virtually at the equator (where the Rossby number is infinite) we can use some of the familiar tools from the midlatitude dynamics. At the equator the Coriolis parameter switches sign and this leads to some interesting features. The vertical structure is also a little complex so let us first see the extent to which the familiar Sverdrup balance can

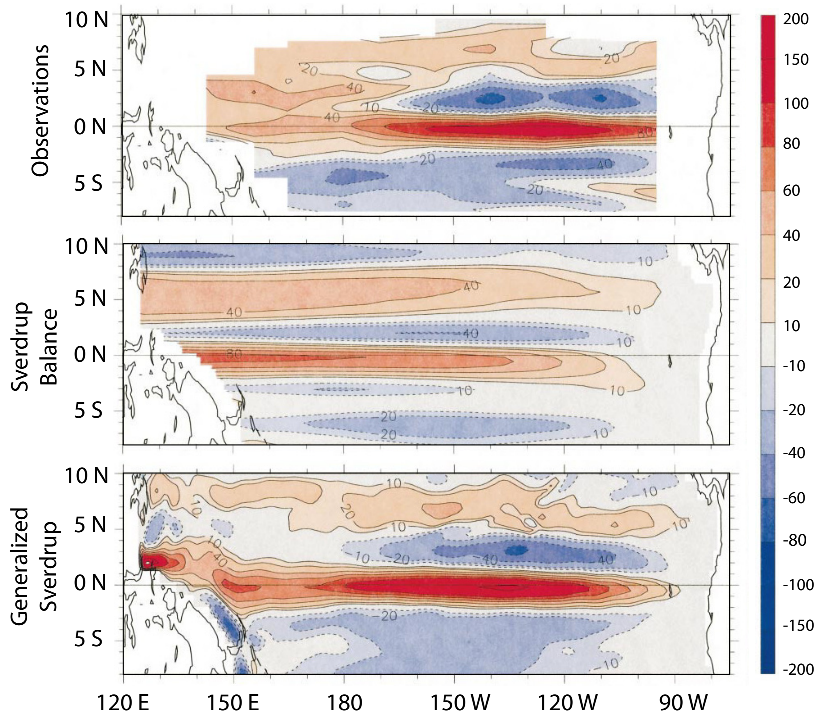

Fig. 22.2 Vertically integrated zonal transport in the Pacific. Red colours indicate eastward flow, blue colours westward. The top panel shows the observed flow, the middle panel shows the flow calculated using Sverdrup balance with the observed wind, and the bottom panel shows the flow calculated with a 'generalized' Sverdrup balance that includes the nonlinear terms in a diagnostic way.[3]

explain the vertically integrated flow.

22.2.1 The vertically integrated flow and Sverdrup balance

The horizontal momentum may be written

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} + \boldsymbol{f} \times \boldsymbol{u} = -\nabla \phi + \frac{1}{\rho_0} \frac{\partial \boldsymbol{\tau}}{\partial z} \tag{22.1}$$

where $\boldsymbol{\tau}$ is the stress on the fluid. The mass conservation equation is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \tag{22.2}$$

which on vertical integration over the depth of the ocean

$$\frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} = 0, \tag{22.3}$$

where U and V are the vertically integrated zonal and meridional velocities (e.g., $U = \int u \, \mathrm{d}z$) and we assume the ocean has a flat bottom and a rigid lid at the top. If we assume the flow is steady and integrate (22.1) vertically, then take the curl and use (22.3) we obtain

$$\beta V = \mathrm{curl}_z(\widetilde{\boldsymbol{\tau}}_T - \widetilde{\boldsymbol{\tau}}_B) + \mathrm{curl}_z \boldsymbol{N}, \tag{22.4}$$

where $\widetilde{\boldsymbol{\tau}}$ is the kinematic stress ($\widetilde{\boldsymbol{\tau}} = \boldsymbol{\tau}/\rho_0$ where ρ_0 is the reference density of seawater) with the subscripts T and B denoting top and bottom, $\boldsymbol{N}$ represents all the nonlinear terms and curl_z is defined by $\mathrm{curl}_z \boldsymbol{A} \equiv \partial A^y/\partial x - \partial A^x/\partial y = \mathbf{k} \cdot \nabla_3 \times \boldsymbol{A}$. Equations (22.4) and (22.3) are closed equations for the vertically averaged flow. In oceanography we nearly always deal with the kinematic stress

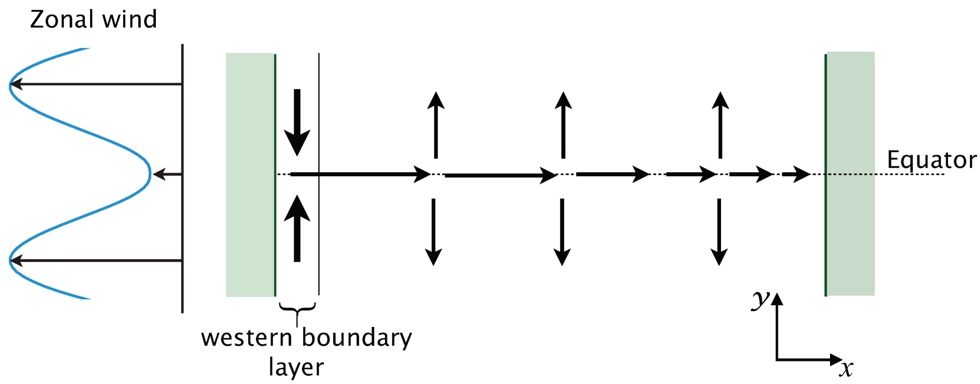

Fig. 22.3 Schema of Sverdrup flow at the equator between two meridional boundaries. The mean winds are all westward, but with a minimum in magnitude at the equator. By Sverdrup balance, (22.5), the wind stress produces the divergent meridional flow shown, which in turn induces an eastward equatorial zonal flow, strongest in the western part of the basin.

rather than the stress itself, so henceforth we will drop the tilde over the τ symbol as well as the adjective 'kinematic'. In the cases that we need to refer to the actual stress we will denote this by τ^*; thus, $\tau = \tau^*/\rho_0$.

If we neglect the nonlinear terms and the stress at the bottom (we'll come back to these terms later) then (22.4) becomes

$$\beta V = \mathrm{curl}_z \boldsymbol{\tau}_T. \tag{22.5}$$

This is just Sverdrup balance, which first appeared in chapter 14, eq. (19.20). The zonal transport is obtained by differentiating (22.5) with respect to y, using (22.3) to replace $\partial_y V$ with $\partial_x U$, and then integrating from the eastern boundary (x_E). This procedure gives

$$U = -\frac{1}{\beta}\int_{x_E}^{x} \frac{\partial}{\partial y}\mathrm{curl}_z \boldsymbol{\tau}_T \,\mathrm{d}x' + U(x_E, y). \tag{22.6}$$

We don't integrate from the western boundary because a boundary layer can be expected there, whereas the value of U at the eastern boundary will be small.

If $U(x_E, y) = 0$ and the wind is zonally uniform then (22.6) becomes

$$U(x, y) = \frac{1}{\beta}(x - x_E)\frac{\partial^2 \boldsymbol{\tau}_T}{\partial y^2}. \tag{22.7}$$

That is, the depth integrated flow is proportional to the second derivative of the zonal wind stress, and because $x < x_E$ we have $U \propto -\partial^2 \boldsymbol{\tau}_T/\partial y^2$. Evidently, the result will depend rather sensitively on the wind pattern. Although the zonal wind is generally westward in the tropics there is a minimum in the magnitude of that wind near the equator (that is, a local maximum as schematized in Fig. 22.3) so that $\partial^2 \boldsymbol{\tau}/\partial y^2$ is negative. Thus, using (22.7), U will generally be positive at the equator. Using the observed wind field the Sverdrup flow — that is, the solution of (22.6) with $U(x_E, y) = 0$ — can be calculated and this is plotted in the middle panel of Fig. 22.2. There is a good but not perfect agreement with the observations: the observed flow has its maximum further east. Further, in the western equatorial Pacific the observed eastward flow is quite broad whereas the eastward Sverdrup flow is narrow, flanked on either side by westward flow. Some of the discrepancy can be attributed to the role of the nonlinear and frictional terms, as illustrated in the bottom panel of Fig. 22.2. To obtain the flow illustrated, the calculation proceeds from (22.4) in the same way

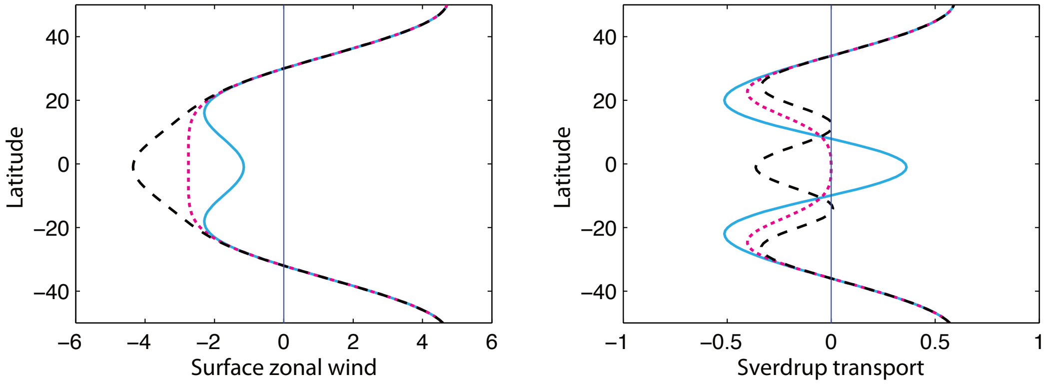

Fig. 22.4 The left panel shows three putative surface zonal (atmospheric) winds, u, all with westward winds in the tropics and with the solid line being the most realistic. The right panel shows the corresponding negative of the second derivative, $-\partial^2 u/\partial y^2$, proportional to the (oceanic) Sverdrup transport, in arbitrary units. The wind represented by solid (blue) line gives an eastward transport at the equator, as is observed, with the others differing markedly.

as before, but now includes the nonlinear terms and a representation of frictional effects in a diagnostic fashion. Thus, for example, the nonlinear terms of the form $\mathrm{curl}_z(\int \boldsymbol{u}\cdot\nabla\boldsymbol{u}\,\mathrm{d}z)$ are evaluated and used to calculate a generalized Sverdrup flow, where the velocities are taken from a nonlinear model forced by the observed winds. Of the nonlinear terms, the largest ones involve the meridional derivatives of the zonal flow, for example $\partial_y(uu_x)$. The effect of the nonlinear terms is to decelerate the eastward flow in the eastern Pacific, with friction tending to damp the flow especially in the central Pacific, and the resulting flow is evidently closer to the observations than is linear Sverdrup balance. Of course the full solution (22.4) must give a vertically integrated flow that closely resembles the observations, because there are only very weak approximations made in deriving it. The success of the Sverdrup theory lies in the extent to which the vertically integrated flow can be satisfied by the simple linear balance (22.5), and then improved by adding nonlinear and dissipative terms in a diagnostic fashion.

22.2.2 Delicacy of the Sverdrup flow

Although the calculations of Sverdrup flow do show good agreement with observations, the calculation — and, most likely, the observed flow — is rather sensitive to the precise form of the winds, as in Fig. 22.4. The figure shows three surface zonal wind distributions, with the 'w' shaped solid line having a minimum in the westward flow (i.e., a minimum in the trade winds) at the equator and so being the most realistic. The right-hand panel shows the negative of the second derivative of the winds which is proportional to the zonal Sverdrup flow. Only in the one case does the wind produce an eastward Sverdrup flow. In fact, in the case illustrated with the dashed lines, the small changes in the meridional gradient of the wind between 15° and 20° produce large variations in the Sverdrup transport. Given this delicacy, the small difference in the latitudinal variation of the Sverdrup flow and the observed flow, illustrated in the top and middle panels of Fig. 22.2, is not surprising and cannot be considered a major failure of the theory. However, the difference in the longitudinal structure of the two fields is indicative of the importance of other terms in the vorticity balance.

22.3 A LOCAL MODEL OF THE EQUATORIAL UNDERCURRENT

The most conspicuous feature of the ocean current system at low latitudes is the equatorial undercurrent, and we now consider its dynamics.[4] The physical picture that we first discuss is a 'local' one, and is essentially the following.[5] The mean winds are westward and provide a stress on the upper ocean, pushing the near-surface waters westward. Given that there is a boundary in the west, the water piles up there so creating a pressure-gradient force that pushes fluid eastward. To some degree the pressure gradient and the wind stress compensate each other leading to a state of no motion. However, the compensation is not perfect. Close to the surface the stress is dominant and a westwards surface current results. Below the surface the pressure gradient dominates, resulting in an eastward flowing undercurrent, as in the observations in Fig. 22.1.

The above description makes no mention of the Coriolis parameter or Sverdrup balance or the wind-stress curl. On the one hand that suggests that the dynamics are likely to be robust and will not depend in a delicate way on the wind pattern (a the Sverdrup flow does). On the other hand, given the usefulness of the Sverdrupian concept, such a description is also likely to be incomplete. To proceed further we'll construct a small hierarchy of mathematical models of the equatorial current system, beginning with the very simplest model of a homogeneous fluid subject to a uniform westward stress at the surface. Following that we will discuss a more inertial and non-local physical picture, in which the undercurrent may be thought of as being pushed by a pressure head that begins in extra-equatorial regions. In the extreme limiting case of this picture, the winds at the equator have no effect on the undercurrent. The real equatorial undercurrent likely involves a combination of local and inertial dynamics, and remains a topic of research.

22.3.1 Response of a homogeneous layer to a uniform zonal wind

Let us first consider the simple case of the response of a layer of homogeneous fluid to a steady zonal wind that is uniform in the y-direction. With our usual notation the equations of motion in the presence of momentum and mass forcing are

$$\frac{\mathrm{D}u}{\mathrm{D}t} - fv = -g'\frac{\partial \eta}{\partial x} + \frac{\tau^x}{H} \tag{22.8a}$$

$$\frac{\mathrm{D}v}{\mathrm{D}t} + fu = -g'\frac{\partial \eta}{\partial y} + \frac{\tau^y}{H} \tag{22.8b}$$

$$\frac{\mathrm{D}h}{\mathrm{D}t} + h\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = M. \tag{22.8c}$$

where (τ^x, τ^y) are the zonal and meridional kinematic stresses on the fluid, H is the depth of the fluid and M is a mass source, which for now we take to be zero. For steady flow and neglecting the nonlinear terms the equations become

$$-fv = -g'\frac{\partial \eta}{\partial x} + \frac{\tau^x}{H}, \qquad +fu = -g'\frac{\partial \eta}{\partial y}, \qquad H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0. \tag{22.9a,b,c}$$

If we take the y-derivative of (22.9a) and subtract it from the x-derivative of (22.9b), and noting that $\partial\tau^x/\partial y = 0$, we obtain

$$\beta v = 0. \tag{22.10}$$

Thus, using the continuity equation (22.9c), we have $\partial u/\partial x = 0$. That is, the zonal velocity is uniform. If there is a zonal boundary at which $u = 0$ then the zonal flow is zero everywhere and the complete solution is

$$u = 0, \qquad v = 0, \qquad g\frac{\partial \eta}{\partial x} = \frac{\tau^x}{H}, \qquad \frac{\partial \eta}{\partial y} = 0. \tag{22.11}$$

That is to say, the ocean is motionless and the wind stress is balanced by a pressure gradient. If the wind is westward, as it is on the equator, then $\partial\eta/\partial x < 0$ and the thermocline slopes down and deepens toward the west. The fact that there is no flow might have been anticipated from Sverdrup balance in the absence of a wind-stress curl. Although the real ocean is not as simple as our model of it, the analysis exposes a truth with some generality: *the wind stress is largely opposed by a pressure gradient* rather than inducing a large westward acceleration that is halted by friction.

22.3.2 An unstratified local model of the equatorial undercurrent

Let us now consider a model with some vertical structure, thereby allowing the wind stress to be taken up in the upper ocean. The wind will still push near-surface water westwards and create a zonal pressure gradient. The deeper water will feel the pressure-gradient force — because the pressure is hydrostatic — but not the wind stress, and so flows eastwards. A simple model that can capture these effects begins with the three-dimensional momentum equations, namely

$$-fv = -\frac{\partial\phi}{\partial x} + \nu_z\frac{\partial^2 u}{\partial z^2} + \nu_h\nabla^2 u, \tag{22.12a}$$

$$fu = -\frac{\partial\phi}{\partial y} + \nu_z\frac{\partial^2 v}{\partial z^2} + \nu_h\nabla^2 v, \tag{22.12b}$$

The parameters ν_z and ν_h are eddy viscosities acting on vertical and horizontal shear, respectively, and the ∇ operator is purely horizontal, so that $\nabla^2 u = \partial^2 u/\partial x^2 + \partial^2 u/\partial y^2$. Dealing with a horizontal viscosity requires a more mathematically cumbersome treatment that we defer to section 22.3.3; instead, we will first invoke a linear drag whence the momentum equations, along with the mass continuity equation, become

$$-fv = -\frac{\partial\phi}{\partial x} + \nu_z\frac{\partial^2 u}{\partial z^2} - ru, \qquad fu = -\frac{\partial\phi}{\partial y} + \nu_z\frac{\partial^2 v}{\partial z^2} - rv, \tag{22.13a,b}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0. \tag{22.13c}$$

The drag terms are presumed to act throughout the depth of the fluid and so are a little arbitrary, but their presence enables us to construct a simple and very illuminating model. We should also remember that almost *any* frictional terms in a model of the large-scale circulation are to some degree ad hoc: the viscosities (ν_z, ν_z) are certainly not molecular viscosities and there is no proper justification for the use of eddy viscosities on momentum.

The vertical friction terms $(\partial^2 u/\partial z^2, \partial^2 v/\partial z^2)$ enable the wind's influence to be felt in the upper ocean via the boundary conditions, namely

$$\nu_z\frac{\partial u}{\partial z} = \tau^x, \qquad \nu_z\frac{\partial v}{\partial z} = \tau^y, \qquad \text{at } z = 0, \tag{22.14a}$$

$$\nu_z\frac{\partial u}{\partial z} = 0, \qquad \nu_z\frac{\partial v}{\partial z} = 0, \qquad \text{at } z = -H, \tag{22.14b}$$

where (τ^x, τ^y) is the kinematic wind stress. With boundary conditions of $w = 0$ at top and bottom the vertical integral of (22.13c) is

$$\frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} = 0, \tag{22.14c}$$

where $(U, V) = \int (u, v)\,\mathrm{d}z$ is the vertically integrated flow. Equation (22.14c) allows for the introduction of a streamfunction ψ such that $U = -\partial\psi/\partial y$ and $V = \partial\psi/\partial x$. Cross-differentiating (22.13a,b) and vertically integrating then gives

$$r\nabla^2\psi + \beta\frac{\partial\psi}{\partial x} = \mathrm{curl}_z\tau. \tag{22.15}$$

This is the equation of Stommel's model, as in (19.6), and in the absence of the frictional term the vertically integrated flow is given by Sverdrup balance. If the wind has no curl the vertically integrated flow is zero, as before. However, the flow is not zero at each vertical level as we now see.

Let us now assume the flow is unstratified and that the buoyancy b is a constant, which we take to be zero. The hydrostatic relation is $\partial\phi/\partial z = b = 0$ so that ϕ is uniform with height. From (22.13a,b) the vertically integrated momentum equations are then

$$H\frac{\partial\phi}{\partial x} = \tau^x - rU + fV, \qquad H\frac{\partial\phi}{\partial y} = \tau^y - rV - fU. \tag{22.16a,b}$$

Let us further suppose that the stress (i.e., $\boldsymbol{\tau} = \nu\partial\boldsymbol{u}/\partial z$) is non-zero only in a shallow layer — an Ekman layer — in the upper ocean. Below this layer we have, from (22.13a,b),

$$-fv = -\frac{\partial\phi}{\partial x} - ru, \qquad fu = -\frac{\partial\phi}{\partial y} - rv. \tag{22.17a,b}$$

and using (22.16) we obtain

$$-fv' = -\frac{\tau^x}{H} - ru', \qquad fu' = -\frac{\tau^y}{H} - rv'. \tag{22.18a,b}$$

where $u' \equiv u - U/H$ and $v' \equiv v - V/H$ is the deviation of the flow from the vertical average (i.e., the deviation from Sverdrup balance). That is to say, we may solve the equations assuming the Sverdrup flow is zero, and add it back in at the end of the day, noting also that the presence of a Sverdrup flow makes no difference to the vertical velocity. Given this, we'll drop the prime on u' and v' unless ambiguity would arise. Solving for u and v gives the expressions for the deep flow, namely

$$u = \frac{-\tau^x r - \tau^y f}{H(r^2 + f^2)}, \qquad v = \frac{\tau^x f - \tau^y r}{H(r^2 + f^2)}. \tag{22.19a,b}$$

The transport in the Ekman layer at the surface is in the opposite direction to the deep flow, in order to satisfy the integral constraints that $\int u\,dz = \int v\,dz = 0$. To complete the solution we use the mass continuity equation, (22.13c), to obtain w, giving

$$w = -\frac{(z+H)}{H}\,\frac{\beta(r^2 - \beta^2 y^2)\tau^x + 2r\beta^2 y\tau^y}{(r^2 + \beta^2 y^2)^2}. \tag{22.20}$$

To better understand these solutions it is useful to look at the nondimensional form and we obtain that by setting

$$(u, v) = (\widehat{u}, \widehat{v})\frac{\tau}{2\Omega H}, \quad y = \widehat{y}a, \quad (\tau^x, \tau^y) = (\widehat{\tau}^x, \widehat{\tau}^y)\tau, \quad \beta = \frac{2\Omega}{a} \tag{22.21}$$

where a hat denotes a nondimensional quantity and a is the radius of Earth. The nondimensional versions of (22.18) are then

$$-\widehat{y}\widehat{v} = -E_r\widehat{u} - \widehat{\tau}^x, \qquad \widehat{y}\widehat{u} = -E_r\widehat{v} - \widehat{\tau}^y, \tag{22.22}$$

where $E_r = r/(2\Omega)$ is a horizontal Ekman number and if, for example, the wind is zonal and westward then $\widehat{\tau}^x = -1$ and $\tau^y = 0$. The nondimensional versions of (22.19) are

$$\widehat{u} = \frac{-E_r\widehat{\tau}^x - \widehat{\tau}^y\widehat{y}}{E_r^2 + \widehat{y}^2}, \qquad \widehat{v} = \frac{\widehat{\tau}^x\widehat{y} - \widehat{\tau}^y E_r}{E_r^2 + \widehat{y}^2}. \tag{22.23a,b}$$

The overall strength of the undercurrent scales, unsurprisingly given the nature of the model, with the wind stress but the Ekman number determines the width and height of the profile. A typical solution is plotted in Fig. 22.6 (along with a solution using harmonic friction that we discuss later). The parameters are $\widehat{\tau}^x = -1, \widehat{\tau}^y = 0$ and $E_r = 8 \times 10^{-3}$, which corresponds to a purely westward wind and a frictional decay timescale of about 10 days. If we further suppose that the dimensional value of the stress is about $4 \times 10^{-2}\,\mathrm{N/m^2}$ and take $H = 100\,\mathrm{m}$ we obtain the dimensional values shown in the plot. The zonal flow as given by (22.19a) is then *eastward*, for the reason we have mentioned before, namely that, overall, the wind is balanced by an opposing pressure gradient and the deep ocean feels the pressure gradient but not the wind stress; thus, the deep zonal flow is in the opposite direction to the surface wind. The deep meridional flow is zero at the equator, where $f = 0$, but is toward the equator in both hemispheres and so induces equatorial upwelling. The shallow Ekman-layer flow is directed away from the equator, in order that the vertically integrated flow is zero. A consequence of this is that the vertical velocity is positive — that is, there is *upwelling* at the equator, as can be seen directly from (22.20) when $\tau^y = 0$, $y = 0$ and $\tau^x < 0$.

The zonal undercurrent falls with latitude with a width proportional to E_r. The peak value at the equator is proportional to E_r^{-1}, so that by reducing the drag we make the equatorial peak sharper. However, and as that scaling suggests, the overall transport is independent of E_r (at least for a constant, zonal stress). To see this we integrate (22.23) with $\widehat{\tau}^x = -1$ and $\widehat{\tau}^y = 0$:

$$\widehat{U}_T = \int_{-\infty}^{\infty} \widehat{u}\,\mathrm{d}\widehat{y} = \int_{-\infty}^{\infty} \frac{E_r}{E_r^2 + \widehat{y}^2}\,\mathrm{d}\widehat{y} = \left[\tan^{-1}\frac{\widehat{y}}{E_r}\right]_{-\infty}^{\infty} = \pi. \tag{22.24}$$

Dimensionally, this translates to

$$U_T = H\int_{-\infty}^{\infty} u\,\mathrm{d}y = \frac{-\pi a \tau^x}{2\Omega}. \tag{22.25}$$

It is pleasing that the total transport of the undercurrent does not depend on the rather poorly-constrained frictional coefficient, although the transport as given by (22.25) is somewhat smaller than observed. This can be guessed from Fig. 22.6 where the parameters are such that the width of the undercurrent is similar to that observed but its magnitude is too low (compare with Fig. 22.1). If we take $\tau^x = 4\times10^{-2}\,\mathrm{N/m^2}$ then using (22.25) we obtain a transport of about $5\times10^6\,\mathrm{m^3\,s^{-1}}$ or 5 Sv whereas the observed transport, with the vertical average (i.e, the Sverdrup flow) removed is 10–15 Sverdrups. Part of the discrepancy may come from the neglect of nonlinearity and stratification, and part of it from there being an inertial component to the equatorial undercurrent that is not a local response to the wind field, as we discuss in section 22.4.

The expressions are also useful when the wind is not purely zonal. In the somewhat less realistic situation in which the wind is northward ($\tau^y > 0, \tau^x = 0$), the deep flow is southward. If the wind blows toward the northwest the undercurrent flows down the pressure gradient to the southeast.

Vertical structure at the equator

Because there is no lateral friction the solution at the equator is independent of the solution elsewhere and an analytic form for the vertical profile may easily be obtained. The Coriolis parameter is zero and so, from (22.13), the equations of motion become

$$0 = -\frac{\partial\phi}{\partial x} + \nu\frac{\partial^2 u}{\partial z^2} - ru, \qquad 0 = -\frac{\partial\phi}{\partial y} + \nu\frac{\partial^2 v}{\partial z^2} - rv. \tag{22.26a,b}$$

If the meridional wind stress at the surface is zero (i.e., $\nu_z\partial v/\partial z = 0$ at $z = 0$) then $v = 0$ everywhere. The zonal pressure gradient is given by (22.16a) and the zonal flow is then given by the solution of

$$\nu_z\frac{\partial^2 u}{\partial z^2} - ru = \frac{\tau^x}{H}, \tag{22.27}$$

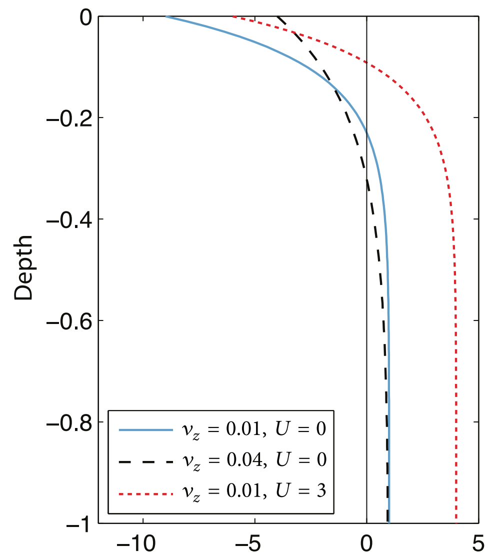

Fig. 22.5 Vertical profile of the zonal current at the equator, obtained using the analytic solutions (22.29) and (22.30), with $H = 1$, $r = 1$, $\tau^x = -1$ and the values of ν_z and U indicated in the legend.

with boundary conditions

$$\nu_z \frac{\partial u}{\partial z} = \begin{cases} \tau^x & \text{at } z = 0 \\ 0 & \text{at } z = -H. \end{cases} \tag{22.28}$$

The solution is easily found to be

$$u = A\mathrm{e}^{\alpha z} + B\mathrm{e}^{-\alpha z} - \frac{\tau^x}{Hr}, \tag{22.29}$$

where $\alpha = \sqrt{r/\nu_z}$ and A and B are obtained from the boundary conditions. We find

$$A = \frac{\tau^x}{\sqrt{\nu_z r}}\left(\frac{\mathrm{e}^{\alpha H}}{\mathrm{e}^{\alpha H} - \mathrm{e}^{-\alpha H}}\right), \qquad B = \frac{\tau^x}{\sqrt{\nu_z r}}\left(\frac{\mathrm{e}^{-\alpha H}}{\mathrm{e}^{\alpha H} - \mathrm{e}^{-\alpha H}}\right). \tag{22.30a,b}$$

A key parameter is the depth scale $d = \alpha^{-1} = \sqrt{\nu_z/r}$ that determines the depth to which the surface flow extends: if ν_z/r is small the flow in the direction of the wind is confined to a shallow layer near the surface with the undercurrent beneath. A few example solutions are illustrated in Fig. 22.5.

These solutions indicate one failing of this simple model: the undercurrent is too deep and in fact extends all the way to the bottom of the ocean; evidently the model fails to reproduce a coherent, focussed eastward flowing jet of finite vertical extent such as is seen in Fig. 22.1. The main ingredient that can overcome this limitation is stratification, with nonlinearity an important secondary effect, and we'll consider how this constrains the vertical extent in section 22.3.4.

A note on the undercurrent in the presence of a Sverdrup flow

The zonal winds in the tropics have a minimum in the westward flow, that is a local maximum in u, at the equator and produce an eastward vertically integrated (Sverdrup) flow, as sketched by the solid line in Fig. 22.4. It seems natural to associate this flow with the eastwards undercurrent, but in and of itself this be misleading. The Sverdrup flow is produced by the wind-stress curl whereas the undercurrent is a consequence of the wind itself, and the two are not necessarily in the same direction. If, for example, the meridional variation of the wind differed, and were more akin to the dashed line in Fig. 22.4, then the Sverdrup flow would be westward. Whether the undercurrent would be eastward or westward now depends on the relative strength of the Sverdrup flow as well as other parameters, as the following very simple calculation shows.

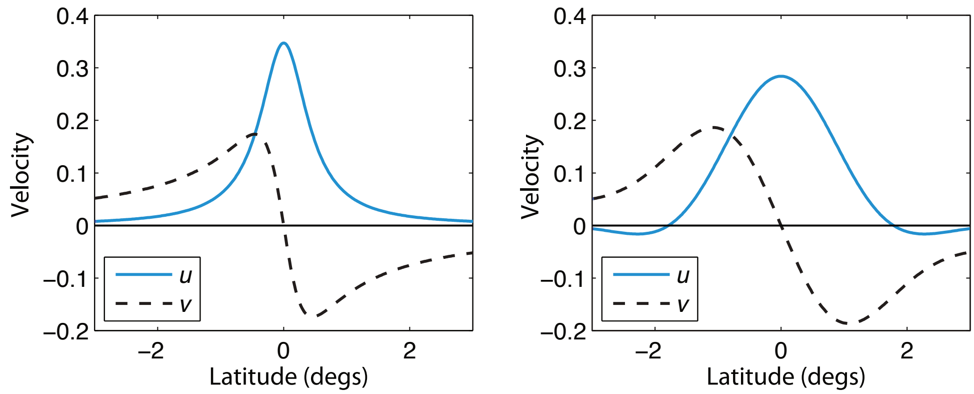

Fig. 22.6 Horizontal profiles of the undercurrent with friction represented by a linear drag [left, using (22.19)] and by a harmonic viscosity [right, using (22.43)], nominally in dimensional units (metres/second and degrees).

The deep flow is the superposition of the Sverdrup flow, U, and the vertically varying flow, so that at the equator and with $\tau^y = 0$ the deep flow is given by

$$u = \frac{-\tau^x + rU}{Hr}, \qquad v = V/H. \tag{22.31}$$

Plainly, if the magnitude of U is sufficiently large then the zonal undercurrent u will take the sign of U, rather than automatically opposing the direction of the wind stress. In this simple linear model, the deep flow is just the sum of two components, one proportional to and opposing the surface wind stress, and one in the direction of the Sverdrup flow. With the wind as it is today, the two effects reinforce each other and for that reason the undercurrent is significantly stronger than the surface flow, but this is not a general rule.

22.3.3 ✦ Effect of horizontal viscosity

We now explore to the effects of using a horizontal viscosity in place of a linear drag. Neither horizontal viscosity nor linear drag are wholly defensible representations of frictional effects, so that one purpose of this exercise is to see what aspects of the solution are robust to the choice made.

Formulating the problem

As we see from Fig. 22.2, meridional variations tend to occur on a smaller scale than zonal variations so we'll neglect the zonal derivatives in the lateral friction. Our equations of motion then become

$$-fv = -\frac{\partial \phi}{\partial x} + \nu_z \frac{\partial^2 u}{\partial z^2} + \nu_h \frac{\partial^2 u}{\partial y^2}, \tag{22.32a}$$

$$fu = -\frac{\partial \phi}{\partial y} + \nu_z \frac{\partial^2 v}{\partial z^2} + \nu_h \frac{\partial^2 v}{\partial y^2}, \tag{22.32b}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \tag{22.32c}$$

The Mean Equatorial Currents

The main observed features of tropical currents are as follows.

- Vertically integrated flow that is in approximate Sverdrup balance, but with non-negligible contributions from nonlinearity and friction. At the equator this flow is eastward, flanked by narrow westward and eastward moving strips, transitioning to broader westward flow polewards of about 10° that is part of the main subtropical gyres.
- A shallow westward flow at the equator, no more than a few tens of meters deep and a few degrees wide, with speeds of a few tens of centimetres per second.
- A strong eastward flowing undercurrent, typically from about 50 m to 200 m depth and a few degrees wide, with velocities up to a metre per second.

The leading order dynamics of this flow is roughly as follows.

- The zonal Sverdrup flow is proportional to the meridional derivative of the wind-stress curl, and so roughly to $-\partial^2 u_s/\partial y^2$ where u_s is the surface zonal wind. The vertically integrated eastward flow at the equator is thus a response to the *minimum* in the westward trade winds at the equator.
- The shallow surface westward flow, and the strong eastward undercurrent, are primarily a response to the westward winds themselves, rather than the curl of the winds, and so are very robust features. (See the shaded box on page 877 for more about the equatorial undercurrent.)
- If the surface zonal winds were uniformly westward in the tropics, or had a westward maximum at the equator, the vertically integrated flow would be quite different and might be westward because of the dependence of the zonal flow on the second derivative of the wind stress in Sverdrup theory. However, there might well still be an eastward equatorial undercurrent, depending on the strength of the Sverdrup flow.

where $f = \beta y$ and with boundary conditions given by (22.14), as before. The vertically integrated horizontal flow, (U, V), satisfies

$$-fV = -H\frac{\partial\phi}{\partial x} + \tau^x + \nu_h\frac{\partial^2 U}{\partial y^2}, \tag{22.33a}$$

$$fU = -H\frac{\partial\phi}{\partial y} + \tau^y + \nu_h\frac{\partial^2 V}{\partial y^2}, \tag{22.33b}$$

$$\frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} = 0, \tag{22.33c}$$

and cross-differentiating leads to an equation similar to (22.15), namely

$$\nu_z\nabla^2\frac{\partial^2\psi}{\partial y^2} + \beta\frac{\partial\psi}{\partial x} = \mathrm{curl}_z\boldsymbol{\tau}. \tag{22.34}$$

Once again, in the absence of a wind-stress curl, the vertically integrated flow is zero and the wind stress is balanced by a pressure gradient. The flow relative to the vertical average, (u', v'), is given by subtracting (22.33) (divided by H) from (22.32) giving

$$-fv' = \nu_z \frac{\partial^2 u'}{\partial z^2} + \nu_h \frac{\partial^2 u'}{\partial y^2} - \frac{\tau^x}{H}, \tag{22.35a}$$

$$fu' = \nu_z \frac{\partial^2 v'}{\partial z^2} + \nu_h \frac{\partial^2 v'}{\partial y^2} - \frac{\tau^y}{H}, \tag{22.35b}$$

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w}{\partial z} = 0. \tag{22.35c}$$

This is independent of the vertical average itself (as was the case with the linear drag) and henceforth, we'll take the vertical averaged flow to be zero and drop the prime on the velocity, with the understanding that it may be added back as needed. A full solution of (22.35) is both difficult to obtain and uninformative, so we will concentrate on various special cases, as follows.

Solution away from the equator

Away from the equator we neglect the horizontal friction terms and (22.35) becomes

$$-fv = \nu_z \frac{\partial^2 u}{\partial z^2} - \frac{\tau^x}{H}, \qquad fu = \nu_z \frac{\partial^2 v}{\partial z^2} - \frac{\tau^y}{H}, \tag{22.36a,b}$$

The particular solution to this is the depth independent flow,

$$v_p = \frac{\tau^x}{fH}, \qquad u_p = \frac{-\tau^y}{fH}. \tag{22.37a,b}$$

To this we must add the solution of the homogenous equation

$$\nu_z \frac{\partial^2 u}{\partial z^2} + fv = 0, \qquad \nu_z \frac{\partial^2 v}{\partial z^2} - fu = 0. \tag{22.38a,b}$$

These are the equations for an Ekman layer, as encountered in section 5.7. As there, the solution spirals down from the surface while decaying exponentially with an e-folding depth of $\sqrt{2\nu_z/f}$. The transport in the Ekman layer, $(\tau^y/f, -\tau^x/f$, is equal and opposite to the transport of the particular solution so that the total transport, relative to the vertical average, is indeed zero.

Solution below the Ekman layer

When f is small the lateral friction terms cannot be ignored and we are left with the full problem again. However, below the surface layer (which is the Ekman layer itself except very close to the equator) the vertical friction may be neglected and we can obtain a solution analogous to (22.19). The flow in this deep layer satisfies

$$-fv = \nu_h \frac{\partial^2 u}{\partial y^2} - \frac{\tau^x}{H}, \qquad fu = \nu_h \frac{\partial^2 v}{\partial y^2} - \frac{\tau^y}{H}, \tag{22.39}$$

where $f = \beta y$, and these equations are very similar to (22.18). We nondimensionalize by setting

$$(u, v) = (\hat{u}, \hat{v}) \frac{\tau}{2\Omega H}, \quad y = \hat{y} a, \quad (\tau^x, \tau^y) = (\hat{\tau}^x, \hat{\tau}^y)\tau, \quad \beta = \frac{2\Omega}{a} \tag{22.40}$$

where a hat denotes a nondimensional quantity and a is the radius of Earth. The nondimensional versions of (22.39) are then

$$-\hat{y}\hat{v} = E_h \frac{\partial^2 \hat{u}}{\partial \hat{y}^2} - \hat{\tau}^x, \qquad \hat{y}\hat{u} = E_h \frac{\partial^2 \hat{v}}{\partial \hat{y}^2} - \hat{\tau}^y, \tag{22.41}$$

where $E_h = \nu_h/(2\Omega a^2)$ is a horizontal Ekman number.

The easiest way to obtain a solution is to multiply the second equation by i (i.e., $\sqrt{-1}$) and add to the first, to give

$$E_h \frac{\partial^2 Z}{\partial \hat{y}^2} - \mathrm{i}\,\hat{y} Z = T \tag{22.42}$$

where $Z \equiv \hat{u} + \mathrm{i}\hat{v}$ and $T = \hat{\tau}^x + \mathrm{i}\hat{\tau}^y$, which we henceforth take to be equal to -1 (i.e., a purely westward stress). Equation (22.42) is a particular form of Airy's equation and its solution is given by[6]

$$Z(\hat{y}) = \int_0^\infty \exp\left[- E_h \alpha^3/3 - \mathrm{i}\, y\alpha \right] \mathrm{d}\alpha \tag{22.43}$$

This solution asymptotes to the geostrophic balance $Z = 1/(\mathrm{i}\,\hat{y})$ (i.e., $u = 0, v = \partial\phi/\partial x = \tau^x/fH$) for large $|\hat{y}|$. The solution, just like the one obtained using a linear drag, has total transport that is independent of the frictional coefficient; that is

$$\int_{-\infty}^{\infty} \hat{u}\,\mathrm{d}\hat{y} = \pi \qquad \text{or} \qquad H \int_{-\infty}^{\infty} u\,\mathrm{d}y = -\tau^x \frac{\pi a}{2\Omega}. \tag{22.44}$$

The mathematical derivation of this is left as a (tricky) exercise for the reader or a literature search. The integral is in fact exactly the same as that obtained using a linear drag, so that the quantitative underestimate of the magnitude of the undercurrent remains. The lack of dependence of the total transport on the viscosity arises because the width of the undercurrent increases with the (one third power of the) horizontal viscosity but the peak value diminishes with the (one third power of the) viscosity. The dependence on the one third power follows from a simple scaling of (22.42): at large $\hat{y}$ the flow is geostrophic and lateral friction unimportant, whereas at small $\hat{y}$ the lateral friction is required to remove the equatorial singularity. Thus, the nondimensional width of the undercurrent, $\hat{L}$ say, is determined by the requirement that the terms on the left-hand side of (22.44) are both important and so that

$$\frac{E_h}{\hat{L}^2} \sim \hat{L}. \tag{22.45}$$

Dimensionally, this translates to

$$L \sim E_h^{1/3} a = \left(\frac{\nu_h a}{2\Omega} \right)^{1/3} \sim 100\,\mathrm{km}, \tag{22.46}$$

if $E_h \approx 10^{-6}$ (which implies $\nu_h \approx 10^4\,\mathrm{m^2\,s^{-1}}$, but this value should not be seen as fundamental).

Horizontal profiles of u and v obtained from (22.43) are plotted in the right-hand panel of Fig. 22.6, and may be compared with the corresponding solutions obtained with a linear drag in the left-hand panel. The results shown are obtained with $E_h = 2 \times 10^{-6}$ but otherwise the same values as were used with a linear drag. Evidently, the results with the two frictional schemes display the same qualitative features, with a peak at the equator and a decay away, and a meridional velocity directed toward the equator in both hemispheres, which gives rise to equatorial upwelling. Thus, as with the Stommel (linear drag) and Munk (harmonic friction) solutions for an ocean-gyre, the similarity of solutions with two different forms of solution gives confidence in the insensitivity of the solution to the form of the frictional terms.

22.3.4 Effects of Stratification: A Layered Model of the Undercurrent

One unrealistic aspect of the models described above is that the undercurrent appears to extend all the way to the bottom of the ocean, whereas in reality it is confined to the upper few hundred meters of the ocean, with the deeper fluid being almost quiescent. A potential reason for this discrepancy is that we have neglected stratification, which tends to limit vertical communication

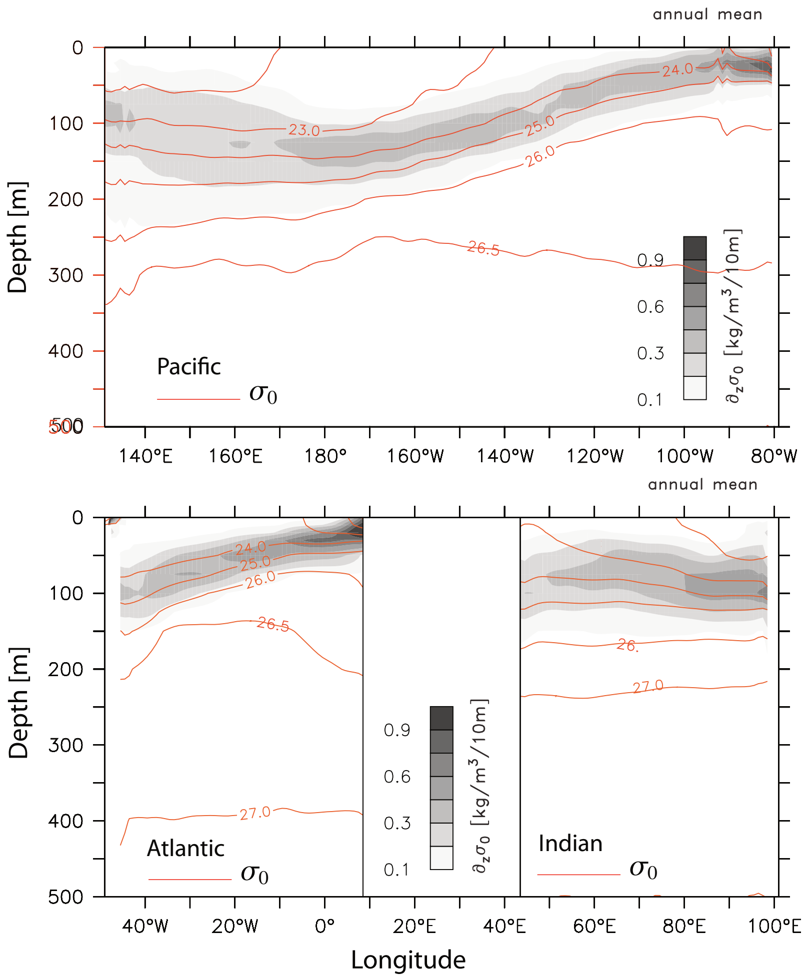

Fig. 22.7 Zonal sections of annual mean density at the equator in the Pacific (top) and Atlantic and Indian oceans. The contours are of potential density and the shading is the vertical derivative of potential density, with data from the World Ocean Atlas.

within an ocean column. Let's try to model this with a simple layered model, reverting to the use of linear drag.

Let us suppose that the ocean consists of two homogeneous layers. The continuous, homogeneous model described above describes the solution in the upper layer while the lower layer, of slightly greater density, represents the abyssal ocean and is assumed stationary. The pressure gradient must therefore be zero in the lower level, and we will see that this requires that the interface between the layers must slope, and indeed will usually slope upwards toward the east. The interface is, of course, a crude representation of the equatorial thermocline.

The zonal pressure gradient at the equator at the base of the upper layer is given by (22.16a), namely $\partial\phi/\partial x = (\tau^x - rU)/H$, and this is usually negative. If the upper layer has a density ρ_1 and the lower layer has a density ρ_2 then, in order for there to be no pressure gradient in the lower layer the interface must slope by an amount

$$s \equiv \frac{\partial z}{\partial x} = -\frac{1}{g'}\frac{\partial \phi}{\partial x} \tag{22.47}$$

where $g' \equiv g\Delta\rho/\rho_1 \equiv g(\rho_2 - \rho_1)/\rho_1$ is the reduced gravity and, as we recall, $\phi \equiv p/\rho_1$. Thus, an

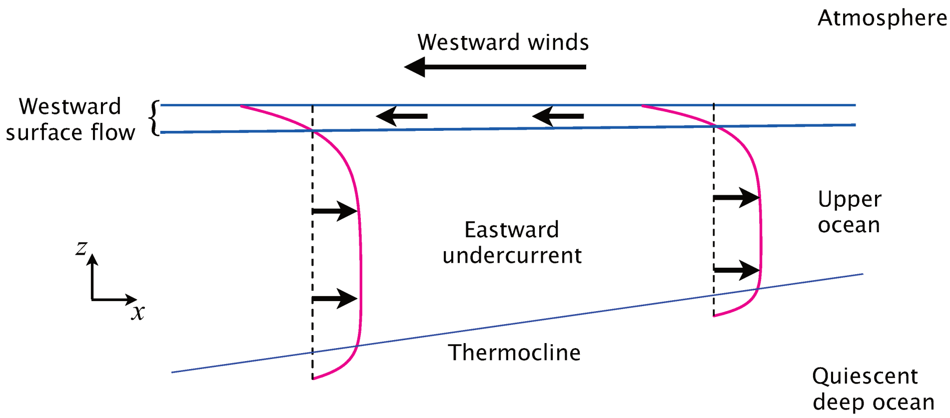

Fig. 22.8 A schematic zonal section of the local model of the undercurrent and thermocline.

estimate of the slope of the thermocline is

$$s \approx \frac{1}{g'H}(\tau^x - rU). \tag{22.48}$$

The quantitative effects of the Sverdrup flow are hard to gauge because of the rather ill-constrained frictional coefficient r. The mean wind stress at the equator is westward and about $0.04\,\mathrm{N\,m^{-2}}$, and with $\Delta\rho = \rho_2 - \rho_1 = 3\,\mathrm{kg\,m^{-3}}$ and $H = 200\,\mathrm{m}$ we find

$$s \approx \frac{\tau^{*x}}{g\Delta\rho H} \approx \frac{0.04}{10 \times 3 \times 200} = 6.7 \times 10^{-6}. \tag{22.49}$$

This suggests that over the 15,000 km extent of the equatorial Pacific we might expect the thermocline to shoal upwards toward the east by about 100 m. This slope is comparable to that observed (see Fig. 22.7), although considering the simplicity of the model the agreement is perhaps a little fortuitous. The thermocline slopes up toward the east in both the Atlantic and Pacific, where the prevailing winds are westward, but not in the Indian Ocean where the prevailing winds are seasonally variable because of the monsoons. The undercurrent itself is also a seasonal phenomenon in the Indian Ocean.

Except for the presence of the frictional coefficient, (22.48) is fairly insensitive to the details of the model; a virtually identical expression results if we model the ocean as two immiscible layers of fluid using the shallow water equations. The parameters determining the thermocline slope are just the thickness, H, of the upper layer and the density difference, $\Delta\rho$, between the upper and lower layers. A schematic of the flow is given in Fig. 22.8.

22.4 AN IDEAL FLUID MODEL OF THE EQUATORIAL UNDERCURRENT

The model of the equatorial undercurrent presented in the previous sections is physically appealing and transparent, but it has two potential shortcomings.

(i) The detailed results depend on the frictional parameters chosen. This is unsatisfactory because we have no well-founded basis on which to choose those parameters.

(ii) The model makes no connection to the extra-tropical circulation of the ocean; that is, all the dynamics are essentially local.

The Equatorial Undercurrent

What is it?

The equatorial undercurrent is the single most striking feature of the low latitude ocean circulation. It is an eastward flowing subsurface current, mostly confined to depths between about 50 m and 250 m and to latitudes within 2° of the equator, with speeds of up to $1\ \mathrm{m\,s^{-1}}$ (Fig. 22.1). It is sometimes connected to an eastward flowing current a few degrees north or south of the equator. The undercurrent is a permanent feature of the Atlantic and Pacific Oceans, but varies with season in the Indian Ocean because of the monsoon winds.

What are its dynamics?

Most models of the equatorial undercurrent lie between two idealized end members that we refer to as the *local theory* and the *inertial theory*.[5]

- The local theory regards the undercurrent as a direct response to the westward winds at the equator. The winds push water westward and create a balancing eastward pressure gradient force. Below a frictional surface layer the influence of the wind stress is small and the pressure gradient leads to an eastward undercurrent.
  - In the frictional surface layer the flow is away from the equator and there is upwelling at the equator. The circulation is closed in the equatorial region. Continuous stratification may be included in the theory, although if there is upwelling through stratified water the diapycnal diffusivity must be non-zero.
  - The dynamics of the simplest models of this ilk are linear, but their quantification relies on the use of somewhat poorly constrained frictional and mixing coefficients.

- In the inertial theory, the equatorial current system is connected to the extra-equatorial region. A subsurface current moves inertially from higher latitudes, conserving its potential vorticity (which includes, crucially, a relative vorticity component) and Bernoulli function into the equatorial region. A pressure head is created in the western equatorial basin, which then pushes the undercurrent along.
  - Even if there were no wind at the equator the theory, in its simplest form, would still predict the presence of an undercurrent.
  - The theory, which is unavoidably nonlinear, contains parameters that must be specified somewhat arbitrarily but to which results are not especially sensitive.

- In reality, the undercurrent contains aspects of both theories, and more. Neither theory can be entirely correct; the local theories do not properly take into account distant effects and, in contrast to the inertial theory, numerical experiments show that the undercurrent *does* depend on the wind at the equator.
  - Part of the undercurrent is closed within the equatorial region, and part connects to higher latitudes. A more complete model involves treating the equatorial undercurrent as one branch of a more complex tropical current system.
  - It would be hard, perhaps impossible, to construct a theory of the whole system that is elegant, complete and correct. Understanding arises via careful treatments of special cases along with numerical and conceptual models of the areas in between.

The second shortcoming is of no import in itself — the real ocean might be that way. However, observations suggest that at least some of the water in the equatorial undercurrent has its origins in the subtropical gyre: temperatures in the core of undercurrent are mostly in the range of 16° C to 22° C, rather lower than the surface temperatures in the equatorial region except at the eastern end of the ocean basins — that is, at the end of the undercurrent. Furthermore, as we see from the upper two panels Fig. 22.1, the current gains strength as it moves eastward, implying that it is drawing water from higher latitudes as it moves.

Modern observational analyses of the equatorial ocean indeed suggest that the equatorial current system is a three-dimensional beast, connecting smoothly with the subtropical current system described in earlier chapters.[7] As subsurface water approaches the equator it largely rises along isopycnal surfaces as it moves eastward, with the cross-isopycnal velocity being only a small fraction of the total vertical velocity. This is in some contrast to the more local picture imagined in section 22.3 in which there is overturning in the vertical-meridional plane, and hence (to the extent that the water is stratified) with cross-isopycnal upwelling at the equator.

The above discussion suggests that it would be useful to construct a model that both connects to the subtropics and does not depend in any essential way or dissipative processes. That is, we should try to construct an ideal fluid model of the equatorial ocean. We'll do this in an way that is analogous to our treatment of the ventilated thermocline in chapter 20. That is, we'll represent the vertical structure of the ocean with a small number (one or two) of immiscible layers, and we'll assume that the subsurface layer conserves its potential vorticity.[8]

22.4.1 A simple barotropic model

Suppose that a fluid parcel at some latitude moves toward the equator, preserving its potential vorticity in a shallow-water system. If the fluid parcel originates from a latitude y_0 where, we suppose, its relative vorticity is negligible then, as it moves its vorticity, ζ, is determined by

$$\frac{f+\zeta}{h} = \frac{f_0}{h_0} \tag{22.50}$$

where, on the equatorial beta plane, $f = \beta y$, $f_0 = f(y_0) = \beta y_0$ and h_0 is the depth of the fluid column at y_0. If, simplifying still further, the depth of the fluid column is assumed constant and meridional derivatives are much larger than zonal derivatives so that $\zeta = \partial v/\partial x - \partial u/\partial y \approx -\partial u/\partial y$, we have

$$\beta y - \frac{\partial u}{\partial y} = \beta y_0. \tag{22.51}$$

Integrating this expression, with $u = 0$ at $y = y_0$, gives

$$u = \frac{\beta}{2}(y - y_0)^2. \tag{22.52}$$

Interestingly, at $y = 0$, $u = \beta y_0^2/2$ which is positive. That is, conservation of absolute vorticity has, virtually by itself, produced an eastward flowing current at the equator (Fig. 22.9). The solution resembles the angular momentum conserving solution to the equinoctial Hadley Cell discussed in section 14.3, specifically equation (14.38) but with a different constant of integration: essentially, in the atmospheric case $y_0 = 0$, because the meridionally moving air in the upper branch of the Hadley Cell originates at the equator in the equinoctial case. However, in the oceanic case we do not expect angular momentum to be conserved because of the presence of a zonal pressure gradient, absent in the zonally averaged atmospheric case. Rather, it is absolute vorticity conservation, in its simplest form, that leads to (22.52).

However, from a quantitative standpoint the solution is not very satisfactory. It depends heavily on the value of y_0, and for y_0 greater than a few degrees the value of the zonal flow at the equator as

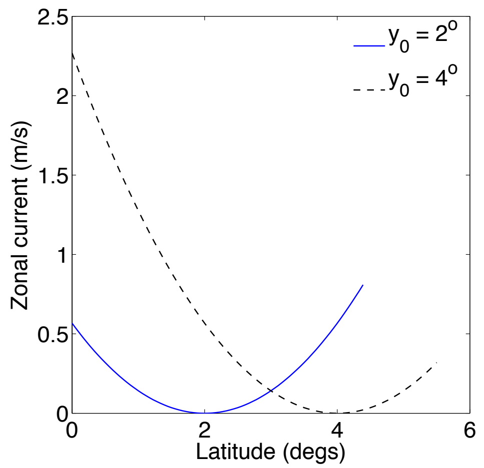

Fig. 22.9 Zonal current as produced by the absolute vorticity conserving model. Specifically, solutions are plotted of (22.52) with $y_0 = 2°$ and $y_0 = 4°$ $(\times 2\pi a/360)$ and $\beta = 2\Omega/a = 2.27 \times 10^{-11}\,\mathrm{m^{-1}\,s^{-1}}$.

predicted by the model is far too large, as can be inferred from Fig. 22.9. Also, the model eastward flow at the equator is not as jetlike as the undercurrent in the real ocean (Fig. 22.1). Nevertheless, the qualitative success suggests that it might be useful to proceed with a more complete model, in particular one in which the value of h does vary with latitude, perhaps accounting for a good fraction of the variation of the potential vorticity.

22.4.2 A two-layer model of the inertial undercurrent

We now extend the barotropic model to two moving layers, with the flow in the lower level presuming to be conserve potential vorticity, and with the height field h determined in a self-consistent fashion rather than being fixed. Thus, the features of the model are as follows.

1. The use of the ideal form (i.e., inviscid, no dissipation) of the two-layer shallow water equations, with the lower level shielded from the wind's influence and conserving potential vorticity and giving rise to the equatorial undercurrent.

2. At low latitudes the equations are solved in a boundary-layer approximation, with variations in y being of smaller scale than variations in x. Unlike our treatment of the ventilated thermocline in midlatitudes, no assumption is made that the flow satisfies the planetary geostrophic equations. It is the inertial terms that prevent the solution from becoming singular at the equator.

3. At higher latitudes the solutions are constructed to blend in with the solution of a midlatitude ventilated thermocline model, described in section 20.6. Put another way, the ventilated thermocline provides a high-latitude boundary condition for the model.

To get a sense of the overall flow the reader may look ahead to Fig. 22.10.

Equations of motion

Our primary concern will be the lower layer (layer 2) for which the momentum and mass continuity equations are, respectively,

$$\frac{\mathrm{D}\boldsymbol{u}_2}{\mathrm{D}t} + \boldsymbol{f} \times \boldsymbol{u}_2 = -\frac{1}{\rho_0}\nabla p_2 = -g_2'\nabla h \tag{22.53a}$$

$$\frac{\partial h_2}{\partial t} + \nabla\cdot(h_2\boldsymbol{u}_2) = 0, \tag{22.53b}$$

where h_2 is the thickness of the layer and $\boldsymbol{u}_2$ the horizontal velocity within it, and $g_2' = g(\rho_3-\rho_2)/\rho_0$. We remain on the equatorial beta plane so that $\boldsymbol{f} = f\,\mathbf{k} = \beta y\,\mathbf{k}$, where $\mathbf{k}$ is the unit vector in the vertical direction, and we will consider only the steady versions of these equations. We may also write the momentum equation in terms of the Bernoulli function,

$$\frac{\partial u_2}{\partial t} + (f + \zeta_2)v_2 = -\frac{\partial B_2}{\partial x}, \qquad \frac{\partial v_2}{\partial t} + (f + \zeta_2)u_2 = -\frac{\partial B_2}{\partial y} \tag{22.54}$$

where $B_2 = g_2'h + \boldsymbol{u}_2^2/2$.

The above equations conserve potential vorticity, $Q_2 = (f + \zeta)/h_2$ and, because the flow is presumed steady, the Bernoulli function. That is,

$$\boldsymbol{u}_2 \cdot \nabla Q_2 = 0, \qquad \boldsymbol{u}_2 \cdot \nabla B_2 = 0. \tag{22.55a,b}$$

Also, because of the form of the mass continuity equation in the steady state, namely $\nabla\cdot(h_2\boldsymbol{u}_2) = 0$, we can define a streamfunction, ψ, such that

$$h_2\boldsymbol{u}_2 = \mathbf{k}\times\nabla\psi \qquad \text{or} \qquad hu_2 = -\frac{\partial\psi}{\partial y}, \quad hv_2 = \frac{\partial\psi}{\partial x} \tag{22.56}$$

Using the streamfunction, conservation of potential vorticity and Bernoulli function may be written as

$$J(\psi, Q_2) = 0, \qquad J(\psi, B_2) = 0, \tag{22.57a,b}$$

where $J(a,b) = \partial_x a\,\partial_y b - \partial_y a\,\partial_x b$. Equations (22.57a) and (22.57b) imply, respectively that isolines of Q_2 and ψ, as well as isolines of B_2 and ψ, are everywhere parallel to each other. Thus, in general, Q_2 is a function of B; that is,

$$Q_2 = F(B_2), \tag{22.58}$$

where the function, F, is as yet unknown. It is also the case that Q_2 is a function of ψ; that is, $Q_2 = G(\psi)$ where G is some other function. However, it is not the case that, in general, Q_2 is a function of the height field h, because h is not proportional to the streamfunction for the flow. This is in contrast to the midlatitude case in which geostrophic balance may be written as $\boldsymbol{u}_2 = (g_2'/f)\mathbf{k}\times\nabla h$, and so the relation $\boldsymbol{u}_2 \cdot \nabla Q_2$ implied that Q_2 is a function of h itself. We do not assume geostrophic balance in the equatorial region.

The equations and the properties of the equations so far discussed are quite general (save for the restriction to the beta plane). Let us now consider the equatorial region, and then how it connects to the subtropics.

Equatorial dynamics

Let us first derive some elementary scaling relations between the variables. Consider motion within a narrow strip of distance no more than L_y from the equator where L_y is the characteristic meridional scale of the undercurrent, as yet undetermined. If L_x is the characteristic zonal scale, typically the scale of the ocean basin itself, then $L_y \ll L_x$. We expect that L_y will be the scale over which the relative vorticity becomes comparable to the planetary vorticity, or equivalently the scale such that the beta Rossby number is $\mathcal{O}(1)$. If the scale of the zonal velocity is U then this requirement is $U/(\beta L_y^2) = 1$ or

$$L_y = \left(\frac{U}{\beta}\right)^{1/2} \qquad \text{or} \qquad U = \beta L_y^2. \tag{22.59}$$

The disparity between zonal and meridional scales implies that there will also be a disparity between the zonal and meridional velocities, and in particular from the mass continuity equation we expect that

$$V = \frac{UL_y}{L_x}, \tag{22.60}$$

and so $V \ll U$, where V is the scale of the meridional velocity.

At a (non-zero) distance L_y from the equator the relevant Rossby number in the meridional momentum equation is given by $U/(\beta L_x^2)$, and this remains small. Thus, essentially because U is so much larger than V, even very close to the equator the *zonal* flow will be in near geostrophic balance with the meridional pressure gradient. The meridional momentum equation then becomes

$$\beta y u_2 = -g_2' \frac{\partial h}{\partial y}, \tag{22.61}$$

implying the scaling

$$H = \frac{\beta L_y^2 U}{g_2'} = \frac{\beta^2 L_y^4}{g_2'}, \tag{22.62}$$

using (22.59), where H is the scale of the variation of thickness in layer 2.

Now let's consider the equations themselves. As we noted the flow conserves potential vorticity, Q_2. Close to the equator $Q_2 \approx (f - \partial u_2/\partial y)/h_2$ so that, using (22.58),

$$\frac{\beta y - \partial u_2/\partial y}{h_2} = F(B_2), \tag{22.63}$$

where $B_2 = g_2' h + u_2^2/2$, noting that $|u_2| \gg |v_2|$ in the equatorial region. There is an obvious similarity between (22.63) and (22.51). Note also that (22.63) and (22.61) are ordinary differential equations, although of course u_2 and h do vary in x. If we knew the function $F(B_2)$, and we knew the upper layer thickness h_1, then the equations would be closed and we could find a solution. For this we turn to the dynamics in the subtropics.

Extra-equatorial dynamics

The role of the extra-equatorial region in out treatment is to provide a boundary condition for the equatorial dynamics, and to determine the functional relationship between potential vorticity and the Bernoulli function, $F(B_2)$. We will suppose that the fluid obeys the dynamics of the two-layer model of the ventilated thermocline discussed in section 20.6, in which the fluid obey the planetary geostrophic equations. The total depth of the moving fluid, H, is given by

$$h^2 = -z_2 = \frac{D_0^2}{\left[1 + (g_1'/g_2')(1 - f/f_2)^2\right]}, \tag{22.64}$$

where

$$D_0^2(x, y) = -\frac{2f^2}{\beta g_2'} \int_x^{x_e} w_E(x', y)\, \mathrm{d}x', \tag{22.65}$$

with $w_E = \mathrm{curl}_z(\boldsymbol{\tau}/f)$ being the vertical velocity at the base of the Ekman layer. Assuming the stress is zonal then, at low latitudes, $w_E \approx \beta\tau^x/f^2 = \tau^x/(\beta y^2)$. If the stress is also independent of longitude then we find

$$D_0^2 = \frac{-2(x_e - x)\tau^x}{g_2'} \tag{22.66}$$

so that

$$h^2 = \frac{-2(x_e - x)\tau^x}{g_2'\left[1 + (g_1'/g_2')(1 - f/f_2)^2\right]}. \tag{22.67}$$

The extra-equatorial solution is completed by noting the expressions of the depths of each layer as a function of the total depth, as in (20.100)

$$h_2 = z_1 - z_2 = \frac{f}{f_2} h \qquad \text{and} \qquad h_1 = -z_1 = \left(1 - \frac{f}{f_2}\right) h. \tag{22.68a,b}$$

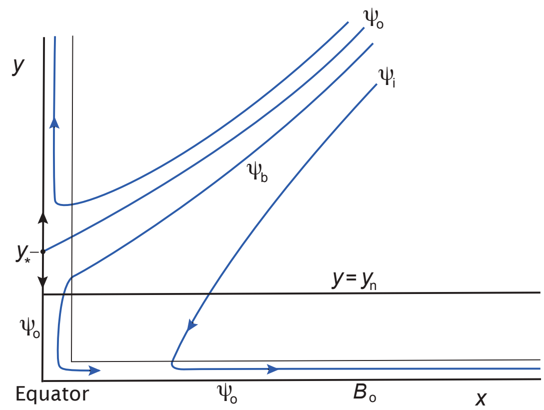

Fig. 22.10 Schematic of the flow streamlines leading to an equatorial undercurrent. Subsurface low from the subtropics heads westward and equatorward, before veering equatorward and back eastward to form the equatorial undercurrent. The critical streamline ψ_0 hits the stagnation point on the western boundary at y_*, and the value of the Bernoulli function at the equator is equal to the value on this streamline.

Connections

We now start to connect the extra-equatorial solution to the tropical one. We first note that (22.67) provides a scaling relation for h, namely

$$H = \left(\frac{L_x \tau}{g_2'}\right)^{1/2}. \tag{22.69}$$

Now, we are assuming that the equatorial dynamics transition smoothly to the extra-equatorial solution, so that (22.69) must be consistent with (22.62). Taken together they give us estimates for the meridional scale, the zonal velocity and the depth of the moving fluid purely in terms of external parameters, to wit:

$$\boxed{L_y = \left(\frac{L_x \tau g_2'}{\beta^4}\right)^{1/8}, \qquad H = \left(\frac{L_x \tau}{g_2'}\right)^{1/2}, \qquad U = (g_2' \tau L_x)^{1/4}.} \tag{22.70a,b,c}$$

These scalings are important results of the model, just as much as the precise form of the solution discussed below. Note that the scaling for zonal velocity is qualitatively different from that derived earlier using the frictional model — compare (22.70c) with (22.19a) or (22.25), for example. The dependance of U on the wind stress in (22.70c) is perhaps surprisingly weak, although both the layer thickness and the horizontal scale also increase with the wind so that the total transport increases almost linearly with wind stress. To the extent that the thickness of the upper layer stays constant, the transport of the lower layer scales as

$$HU = \left(\frac{L_x^3 \tau^3}{g_2'}\right)^{1/4} \qquad \text{and} \qquad HUL = \left(\frac{L_x^7 \tau^7}{g_2' \beta^4}\right)^{1/8}. \tag{22.71}$$

We now obtain the functional connection between Q_2 and B_2 necessary to close (22.63). In the extra-equatorial region, the horizontal shear becomes small compared to the Coriolis term so that (22.63) becomes

$$Q_2 = \frac{\beta y}{h_2} = F(B_2), \tag{22.72}$$

and the Bernoulli function itself, $B_2 = g_2' h + \boldsymbol{u}^2/2$, may be approximated by $B_2 = g_2' h$. Therefore, at the edge of the equatorial region,

$$Q_2 = \frac{f}{h_2} = \frac{f_2}{h}, \tag{22.73}$$

using (22.68a). This functional form holds throughout the equatorial region, and therefore $Q_2(h) = f_2/h$. More generally, $Q_2(\phi) = f_2/\phi$ for any variable ϕ and in particular,

$$Q_2(B_2) = \frac{f_2}{B_2} = \frac{f_2}{g_2'h + u_2^2/2} \tag{22.74}$$

Our quest for the solution is now all over bar the shouting, in the sense that we can write down the equations of motion and the boundary conditions. Using geostrophic balance, (22.74) and(22.63) we write down the equations determining the subsurface flow in the equatorial region, namely

$$\frac{\beta y - \partial u_2/\partial y}{h_2} = \frac{f_2}{g_2'h + u_2^2/2}, \tag{22.75}$$

$$\beta y u_2 = -g_2'\frac{\partial h}{\partial y} \tag{22.76}$$

$$h_2 = h - h_1 \tag{22.77}$$

In addition to specifying the value of the upper layer field, h_1, we need to specify the boundary conditions. At large values of y ($y/L_y \gg 1$) the value of h should be that given by (22.67). A second boundary condition may be applied at the equator, and if we suppose that the flow is hemi-spherically symmetric we have

$$v_2 = 0 \qquad \text{at } \ y = 0. \tag{22.78}$$

Taken with (22.54) this equation implies that, for steady flow, $\partial B_2/\partial x = 0$ and so that

$$B_2 = g_2'h + \frac{u_2^2}{2} = B_0 \qquad \text{at } \ y = 0, \tag{22.79}$$

where B_0 is a constant. That is to say, the equator is a streamline of the flow. (If there is flow across the equator the problem becomes more complicated, but we leave that for another day.) The value of B_0 is plausibly given by supposing it to be the value of B_2 at the western edge of the basin just outside the equatorial region but other choices might be made. Finally, we need to specify the field h_1, and there are a number of reasonable ways to proceed, although no obviously correct one. One choice would be to suppose that, just as in the extra-tropics, the total thickness of the moving layers is given by Sverdrup balance. If we were to do this we would essentially be extending the ventilated thermocline model all the way to the equator, with the addition of inertial terms. Although Sverdrup balance is qualitatively reasonable in equatorial regions (Fig. 22.2), quantitatively it is not particularly good and a simpler recipe is appropriate. One option is to choose h_1 to be a function of x only, such that the value of h_1 is equal to the value that it has at the high latitude edge of the equatorial region, at $y = y_n \gg L_y$. Using (22.68b) and (22.64) this gives

$$h_1^2 = \frac{D_0^2(1 - y_n/y_2)^2}{\left[1 + (g_1'/g_2')(1 - y_n/y_2)^2\right]} = \frac{-2(x_e - x)\tau^x(1 - y_n/y_2)^2}{g_2'\left[1 + (g_1'/g_2')(1 - y_n/y_2)^2\right]} \tag{22.80}$$

using (22.66). The choice is simple albeit a little special, but it turns out that the solution is not especially sensitive to it. That it is a reasonable choice can be seen by noting that for $y_n \ll y_2$, $h \to h_1$ so that

$$h \to h_1 = \left[\frac{-2(x_e - x)\tau^x}{g_2'\left[1 + (g_1'/g_2')\right]}\right]^{1/2}. \tag{22.81}$$

Re-arranging and differentiating this expression with respect to x we obtain

$$\frac{\partial}{\partial x}\left(1 + \frac{g_1'}{g_2'}\right)h_1 = \frac{\tau^x}{g_2'h_1}. \tag{22.82}$$

That is, there is a balance between the applied wind stress and the pressure gradient force in the upper layer.

Equations of Motion for the Inertial Undercurrent

The dimensional form of the equations of motion are

$$\frac{\partial u_2}{\partial y} - \beta y = \frac{f_2 h_2}{h + u_2^2/2}, \qquad (h_2 = h - h_1) \tag{U.1a}$$

$$\beta y u_2 = -g_2' \frac{\partial h}{\partial y}, \tag{U.1b}$$

where h_1 is a specified function of x. One plausible choice is given by choosing it to be the value of h_1 just outside the equatorial region, as given by (22.80).

Two boundary conditions are needed. The first is that in the extra-equatorial region h approaches the value given by the ventilated thermocline model, for example

$$h^2 = \frac{-2(x_e - x)\tau^x}{g_2' \left[1 + (g_1'/g_2')(1 - f/f_2)^2\right]}. \tag{U.2}$$

At the equator the boundary condition on h is obtained by setting $v_2 = 0$ and specifying the Bernoulli function, B_2 there. That is, we specify

$$B_2 = g_2' h + \frac{1}{2} u_2^2 = B_0 \qquad \text{at } y = 0. \tag{U.3}$$

Here, B_0 is a constant, chosen to be equal to the value of the Bernoulli function on the western edge of the basin just outside the equatorial region (that is, using (U.2) at $x = 0$ and $y = y_n$). There is then a pressure head at the western edge of the equatorial region, and the flow accelerates zonally along the equator preserving its Bernoulli function.

The nondimensional form of (U.1) is

$$\frac{\partial \hat{u}_2}{\partial \hat{y}} - \hat{y} = \frac{-y_2 \hat{h}_2}{\hat{h} + \hat{u}_2^2/2}, \qquad y \hat{u}_2 = -\frac{\partial \hat{h}}{\partial \hat{y}}. \tag{U.4}$$

where $y_2 = f_2/(\beta L_y)$.

The solution and its properties

The equations of motion and the boundary conditions for this model are summarized in the shaded box above. They are nonlinear and rather complex, and solutions must in general be obtained numerically by an iterative method, and an example solution is given in Fig. 22.11 and Fig. 22.12. However, some qualitative properties may be deduced from the form of the equations. For familiarity we use Northern Hemisphere terminology, but the ideas apply equally to the Southern Hemisphere.

The ventilated thermocline gives rise to fluid that, at low latitudes, flows southward and westward. As it flows equatorward, potential vorticity conservation must lead to, in the absence of changes in layer thickness, an increase in relative vorticity — an anticlockwise or cyclonic turning — and the flow veers more southward and then eastward, so giving rise to the equatorial undercurrent. This property is of course present in the barotropic model of absolute vorticity conservation discussed in section 22.4.1. However, the two-layer model differs from the barotropic model in

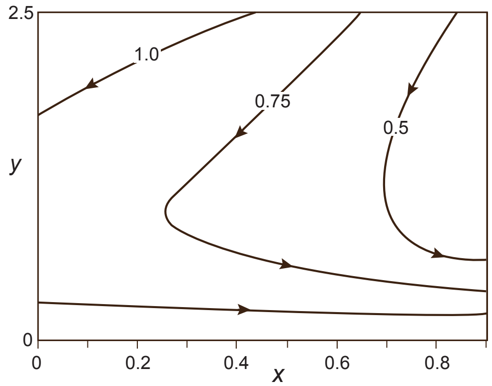

Fig. 22.11 The streamlines in a solution of the equations for an inertial equatorial undercurrent. The wind stress is constant, $g_1'/g_2' = 1$, $y_2 = 5$ and $B_0 = 1.26$.

two important regards. First, the layer thickness is allowed to change in a self-consistent fashion. Second, the flow is not particularly sensitive to the matching latitude at which we connect the equatorial equations of motion to the extra-equatorial region. This is because the ventilated thermocline model is itself based on conservation of potential vorticity, so that changing the matching latitude will have little effect on the potential vorticity entering the equatorial region. The two layer model does have some parameters that cannot be deduced a priori, in particular the thickness of the top layer and the value of the Bernoulli parameter, but the solutions are not especially sensitive to it. That is, and in common with some other models in geophysical fluid dynamics (for example, the Stommel model of western intensification in a gyre), the behaviour of the solutions is quite robust and transcends the detailed limitations of the model itself.

A numerically obtained solution is illustrated in Fig. 22.11. We see the streamlines sweeping westward and equatorward before taking a sharp equatorward and then eastward turn, with the flow being purely eastward at the equator. The solutions of u_2 and h are shown in Fig. 22.12, illustrating the formation of the undercurrent and its intensification as it moves eastward at the equator. Note also the latitudinal variation of the layer depth, h_2 ($h_2 = h - h_1$). In the extra-tropical ventilated thermocline, the thickness of layer 2 diminishes as we move equatorward, and if the limit of $y \to 0$ were taken the thickness of layer 2 would goes to zero (a consequence of f going to zero and f/h being preserved). However, in the equatorial boundary layer the layer thickness actually increases as the equator is approached, and so the relative vorticity must increase to compensate, because now $(f + \zeta)/h$ is preserved. That is, the cyclonic intensification of the flow is somewhat more pronounced than in the barotropic model. Note also that the layer depth diminishes eastward; that is, the thermocline slopes up toward the east.

22.4.3 Relation of Inertial and Frictional Undercurrents

In the above sections we have discussed two different conceptual models of the equatorial thermocline. The first one is frictional and local, the second one is inertial and remote. In the local model, the westward winds set up a compensating pressure gradient, and below the frictional layer near the surface the pressure gradient dominates leading to an eastward flow. The only cross-latitudinal effects come from a lateral friction, and if this is replaced with a linear drag the zonal flow at the equator is wholly independent of the dynamics at other latitudes. In contrast, in the inertial model the undercurrent arises as a consequence of potential vorticity conservation of the subsurface flow, with the value of the potential vorticity set in the extra-equatorial region. The undercurrent is fed by extra-equatorial waters at all longitudes and so builds up as it moves eastward (as is observed). There is a pressure head at the western edge of the equatorial basin, so that the flow accelerates eastward *without the need for any winds at all at the equator.* It is the link with the geostrophically balanced motion in the extra-equatorial region that determines the structure of the equatorial un-

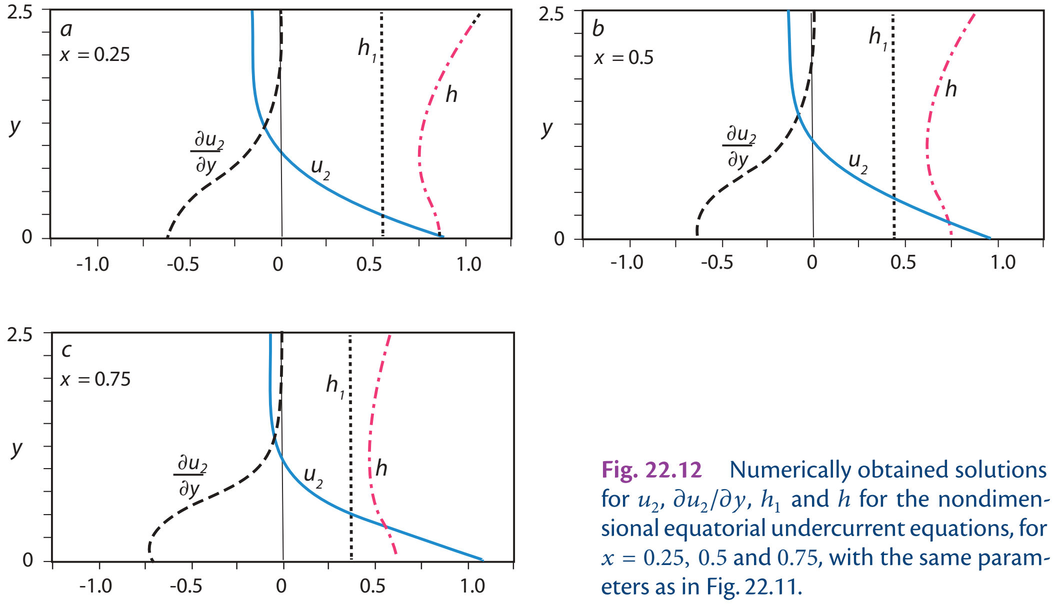

Fig. 22.12 Numerically obtained solutions for u_2, $\partial u_2/\partial y$, h_1 and h for the nondimensional equatorial undercurrent equations, for $x = 0.25$, 0.5 and 0.75, with the same parameters as in Fig. 22.11.

dercurrent, not the local winds.

Are these two views of the equatorial undercurrent in complete opposition, to the extent that only one can be true? In fact, the real ocean may have elements of both, and the two models are best through of as endmembers, or limiting cases, of the a model of the true system, as summarized in the shaded box on page 877 and discussed more in the literature here.[9]

22.5 AN INTRODUCTION TO EL NIÑO AND THE SOUTHERN OSCILLATION

El Niño! One of the most famous phenomenon in the climate sciences, and certainly one with an enormous impact on humankind. It is an anomalous warming of the surface waters in the eastern equatorial Pacific, and its appealing name (el niño, without capitalization, is Spanish for male infant, el Niño refers to the Christ Child and El Niño to the oceanic phenomenon[10]) belies its enormous power and global effects, bringing heavy rains to California and Northern Argentina and anomalously dry weather to South East Asia and Northern and Eastern Australia; it also raises the globally average surface temperature by about half a degree Celsius. Taken with the associated changes in the atmosphere, in which case the whole phenomenon is known as the El Niño–Southern Oscillation (ENSO), it is the largest and most important source of global climate variability on interannual timescales, and it is one of only two unequivocal 'oscillations' in the atmosphere-ocean system that is not directly forced by an external agent (the other one being the Quasi-Biennial Oscillation; however, neither phenomena are modes of a linear system). Our focus will be on the phenomenon itself, not its global effects.[11]

22.5.1 A descriptive overview

Every few years the temperature of the surface waters in the eastern tropical Pacific rises quite significantly. The strongest warming takes place between about 5° S to 5° N, and from the west coast of Peru (a longitude of about 80° W) almost to the dateline, at 180° W, as illustrated in Fig. 22.13. The warming is large, with a difference in temperature up to 6° C from an El Niño year to a non-El Niño year. The warmings occur rather irregularly, with typical intervals between warmings of 3 to 7 years, as seen in Fig. 22.14, and with major events in 1887–88, 1982–83, 1997–98.

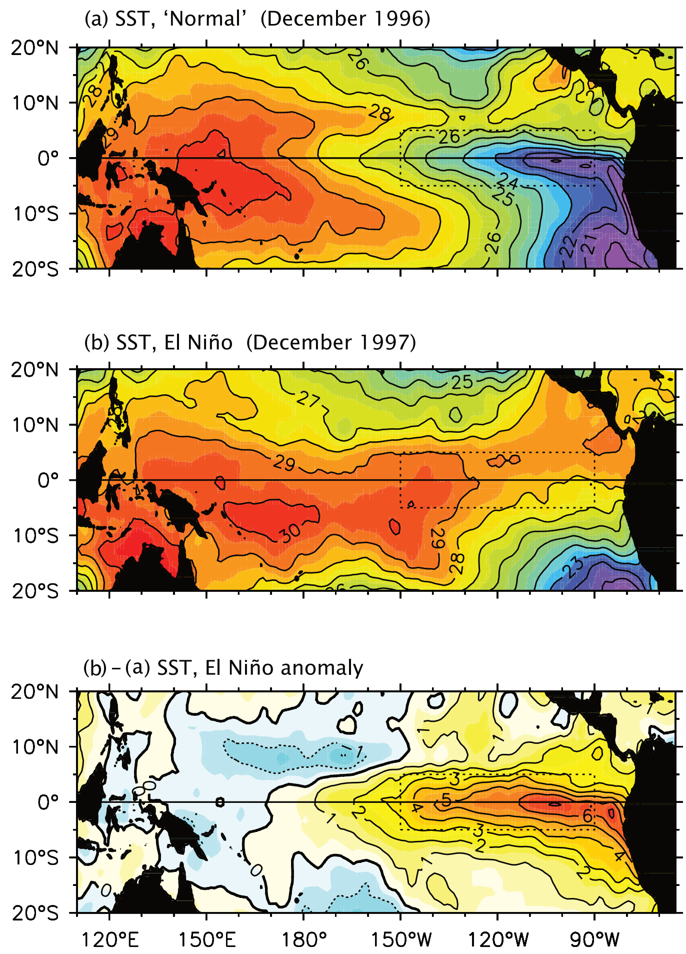

Fig. 22.13 The sea-surface temperature in December of a non El Niño year (December 1996, top panel), a strong El Niño year (December 1997, middle panel) and their difference (bottom panel). An El Niño year is typically characterized by an anomalously warm tongue of water in the eastern tropical Pacific.[12]

The warmings have become known as El Niño events or (abusing the Spanish language) El Niños. The warmings typically last for several months, occasionally up two years, and appear as an enhancement to the seasonal cycle with high temperatures appearing at a time when the waters are already warming. If definiteness is desired, an event may be said to occur when there is a warming of at least 0.5° C averaged over the eastern tropical Pacific lasting for six months or more.[14] Ocean temperatures tend to fluctuate between warm El Niño years and those years in which the equatorial ocean temperatures are colder in the east and warmer in the west, and a *La Niña* is said to occur when it is particularly warm in the west (la niña being Spanish for young girl). We have direct observational evidence — temperature measurements from ships and buoys — of El Niño events for over a century, but the events have almost certainly gone on for a much longer period of time, probably many millennia, judging from proxy records of tree rings and coral growth.[15]

The overlying atmospheric winds and the surface pressure in the equatorial Pacific tend to covary with the SST and during El Niño events the equatorial Pacific trade winds, become much weaker and may even reverse. A measure of this is the pressure difference between Darwin (12° S, 130° E) and Tahiti (17° S, 150° W), and the normalized record of this is known as the *Southern Oscillation.*[18]The Southern Oscillation and the SST record of El Niño are highly correlated (Fig. 22.14), and the combined El Niño–Southern Oscillation phenomenon is denoted *ENSO.*

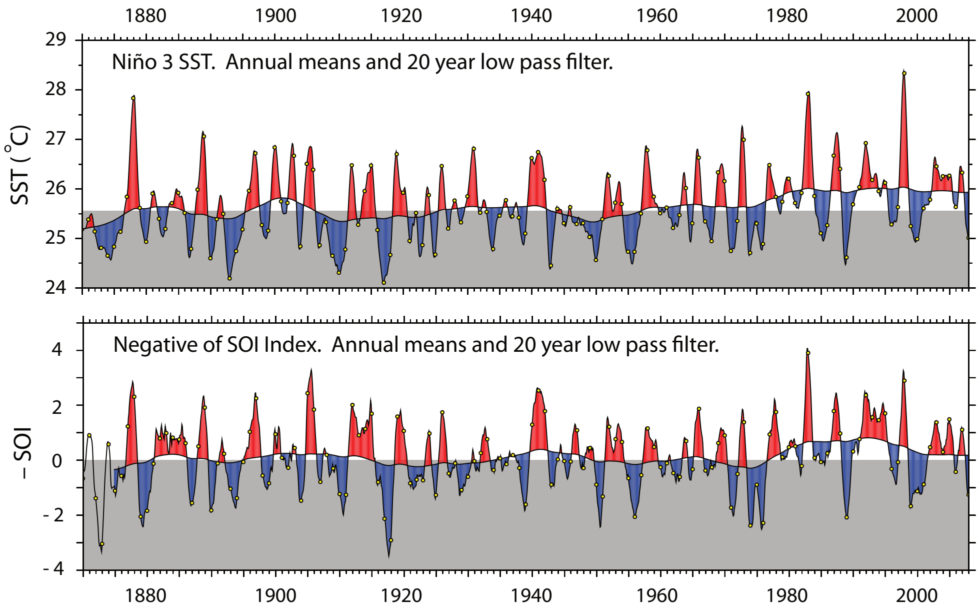

Fig. 22.14 Upper: Times series of the sea-surface temperature (SST) in the Eastern Equatorial Pacific (Niño 3) region. Lower: Negative of the Southern Oscillation Index (SOI), the anomalous pressure difference between Tahiti and Darwin. The pointy curves show the annual means, the dots are values in December, the smoother curves show the variable after application of a 20 year low-pass filter, and the top of the grey shading is the 1876–1975 mean.[13]

22.5.2 A qualitative view of the mechanism

ENSO is the only unambiguous example in the climate system of dynamical coupling between the atmosphere and ocean on timescales of months to years, and the essential mechanism is sketched in Fig. 22.16. First consider the mean state. The trade winds blow predominantly from higher latitudes toward the equator, and from the east to the west. This leads to a current system as described earlier in this chapter, with equatorial divergence of surface waters and upwelling — particularly in the east because here it must also replenish the surface waters moving westward away from the South American continent. The upwelling water here is cold, because much of it comes from below the thermocline, so that the SST of eastern equatorial Pacific is relatively low, and the surface waters warm as they move westward. Furthermore, the thermocline deepens further west and so the upwelling does not bring as much cold abyssal water to the surface. The result of all this is that the SST is high in the western Pacific, up to about 30° C, and low in the eastern Pacific, about 21° C (see Fig. 22.13, top panel).

The strong zonal gradient of SST affects the atmosphere. The warm western Pacific region becomes more convectively unstable with respect to convection, the eastern Pacific is correspondingly cool and, as sketched in Fig. 22.16, an east-west overturning circulation is set up in the atmosphere — the *Walker circulation*, which we discuss more in section 22.6. Evidently, the oceanic and the atmospheric states reinforce each other: the westward trade winds give rise to an east–west SST gradient in the ocean which generates the Walker Cell, so the surface winds are stronger than they would be if there were no ocean or if the ocean extended all the way round the globe.

By their nature positive feedbacks reinforce initial tendencies, whatever those tendencies may be, and such a feedback is at the core of El Niño. If the trade winds weaken then so will the zonal SST gradient, and the trades will weaken further and soon we have a full fledged El Niño event.

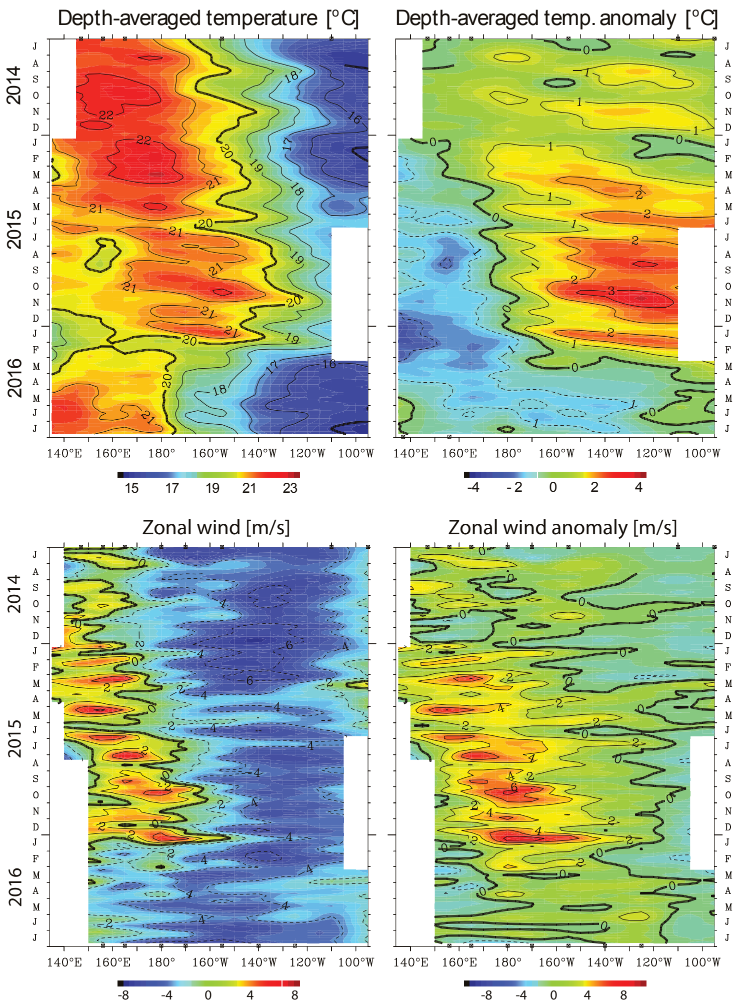

Fig. 22.15 The development of the 2015–2016 ENSO event. The top panels show the evolution of the depth-averaged temperature over the upper 300 m from mid 2014 to mid 2016, and the bottom panels show the surface zonal wind, both over a strip from 2° S to 2° N.

An eastward propagation in both fields can be seen over the second half of 2015, ceasing in early 2016 just after the temperature anomalies in the eastern tropical Pacific reach at their maximum in the East.[16]

The feedback cannot amplify without bound and, in fact, there is are natural damping mechanisms for El Niño, one of them being the warm pool in the east is able to leak out of the tropics along the coast of the Americas, both north and south, through the mechanism of coastal Kelvin waves. No such leakage can occur in the western boundary, and this may be one reason that there is an asymmetry between El Niño events and La Niña events (after subtracting out the basic state asymmetry between east and west because of the direction of the trade winds). The El Niño event then begins to decay and the whole feedback then occurs in the opposite sense and eventually the system finally reverts to its normal state. The mechanism that gives rise to the ENSO cycle is known as the *Bjerknes feedback*.[19] The Bjerknes feedback is generally regarded as being stronger in the eastern ocean because there the thermocline in shallower. Thermocline depth and SST are correlated, with a shallow thermocline corresponding to low SST because cold water can then upwell through the thermocline, with the correlation getting weaker as the thermocline thickens.

22.5.3 Why the tropics?

Why do these dynamics occur in the tropics, and not in midlatitudes? The tropics are different because there is a close and reasonably direct connection between SST and the winds, so enabling a feedback to occur that reinforces initial tendencies, and that arises because of a combination of

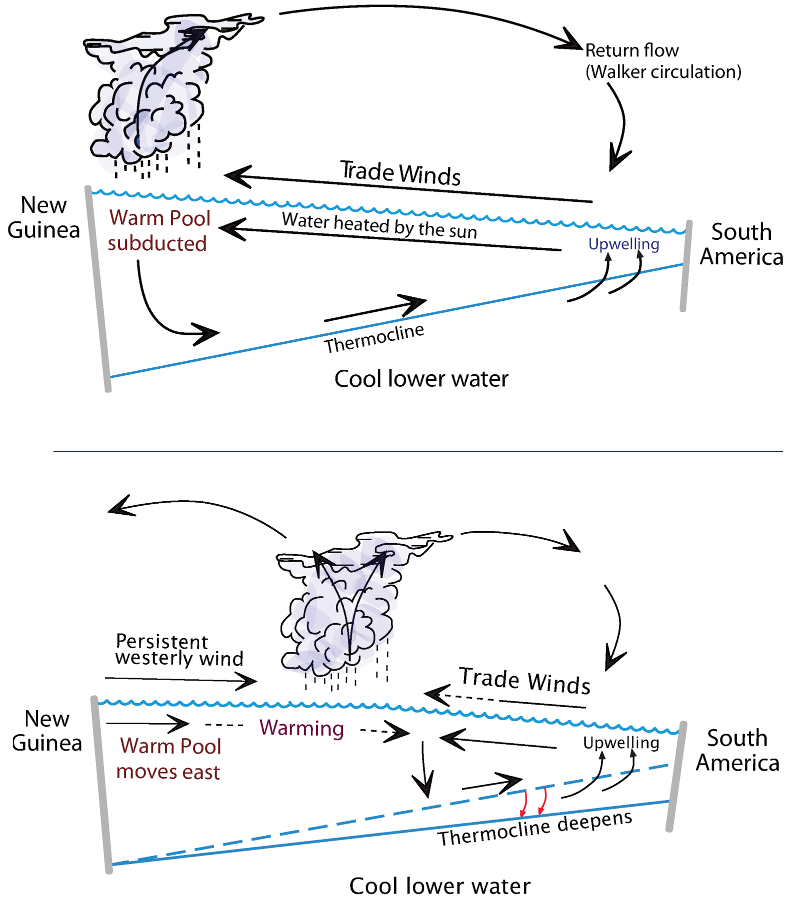

Fig. 22.16 Sketch of the end members of the ENSO cycle. The top panel shows a cross-section along the equator during non-El Niño, years, and the bottom panel at the peak of El Niño. Not to scale, and slopes are exaggerated.[17]

the following factors.

(i) Equatorial sea surface temperatures are generally high, around 27° C (Fig. 22.13), and convection is readily triggered if the temperature further increases (although there is nothing magical about 27° C). Ascending motion occurs over warm regions with associated low-level convergence, and the surface winds thus directly respond to a changing SST gradient, as in the Matsuno–Gill problem.

(ii) The equatorial thermocline is quite shallow, varying from 200 m in the west to 50 m in the east, and is thus sensitive to changing wind patterns. Furthermore, because the thermocline is shallow upwelling can occur *through* it (Fig. 22.16), leading to a close connection between thermocline depth and SST, especially in the east.

(iii) Equatorial Rossby and Kelvin waves are quite fast, certainly compared to advective motion, so allowing cross-basin communication to occur on short, but not immediate, timescales. The slight delay (of order months) may contribute to the ability of the system to oscillate.

In mid-latitudes none of the above are as effective. Furthermore, here the atmosphere is internally highly variable and its response to an SST anomaly has a much smaller signal to noise ratio and the induced winds may have little immediate local correspondence with the anomaly. Adding to this, the midlatitude thermocline is deep and anomalous winds do not necessarily not reinforce an existing anomaly. For all of these reasons midlatitude ocean-atmosphere coupling is much weaker and slower than its tropical counterpart.[20] None of this, of course, is to imply that midlatitude SST

El Niño and the Southern Oscillation

- *El Niño* is the name given to the aperiodic warming of the surface in the eastern equatorial Pacific Ocean. The interval between warm events is typically from two to seven years but is quite irregular (Fig. 22.14).
- El Niño events are associated with a weakening of the trade winds and an eastward shift of the region of convection, conveniently measured by a pressure difference between Tahiti and Darwin and known as the *Southern Oscillation*. The combined phenomenon is known as the El Niño–Southern Oscillation, or ENSO.
- El Niño, and its complement La Niña (an anomalous, weaker, warming in the western equatorial Pacific) are caused by the mutual interaction between the atmosphere and ocean in the equatorial Pacific. The key ingredients are:
  - *(i)* A close correlation between the thermocline thickness and surface temperature, especially in the east, with a shallow thermocline associated with a cool surface. A shallow thermocline allows water to upwell through it, bringing cold abyssal water to the surface (Fig. 22.16).
  - *(ii)* A positive feedback between the winds and the SST. Surface warming in the eastern Pacific leads to convergence of the winds (Fig. 22.17), a deepening of the thermocline, and a further warming of the surface.
  - *(iii)* The back-and-forth 'sloshing' of the thermocline depth anomalies, mediated by Kelvin and Rossby waves propagating quickly eastward and more slowly westward, respectively, with corresponding basin crossing times of about 70 and 200 days.
  - *(iv)* The interaction of these timescales with the annual cycle of the trade winds and thermodynamic forcing, and with the natural ('stochastic') variability of the atmosphere on shorter timescales.
  - *(v)* Damping by way of leakage in the west via coastal Kelvin waves and loss of heat to the atmosphere and deep ocean, and a delayed negative feedback because of the finite crossing time of the waves.
- The above factors combine to produced irregular oscillations, partially phase-locked to the annual cycle.
- It seems that the crossing times in the Atlantic are too short for such phase-locking and amplification, and there is no equivalent phenomenon there.

anomalies do not affect the atmosphere.

When described in these terms the El Niño phenomenon seems readily understandable, but it glosses over a host of issues. If the above feedback is robust why doesn't it occur in the tropical Atlantic for example? Why doesn't the system find a stable fixed point, or oscillate in a regular manner? What determines the interval between events, and the magnitude of the events? We can't answer all these questions, but to begin let us consider the dynamics in a little more detail. We first consider the atmosphere where the response is, essentially, to modify the Walker circulation.

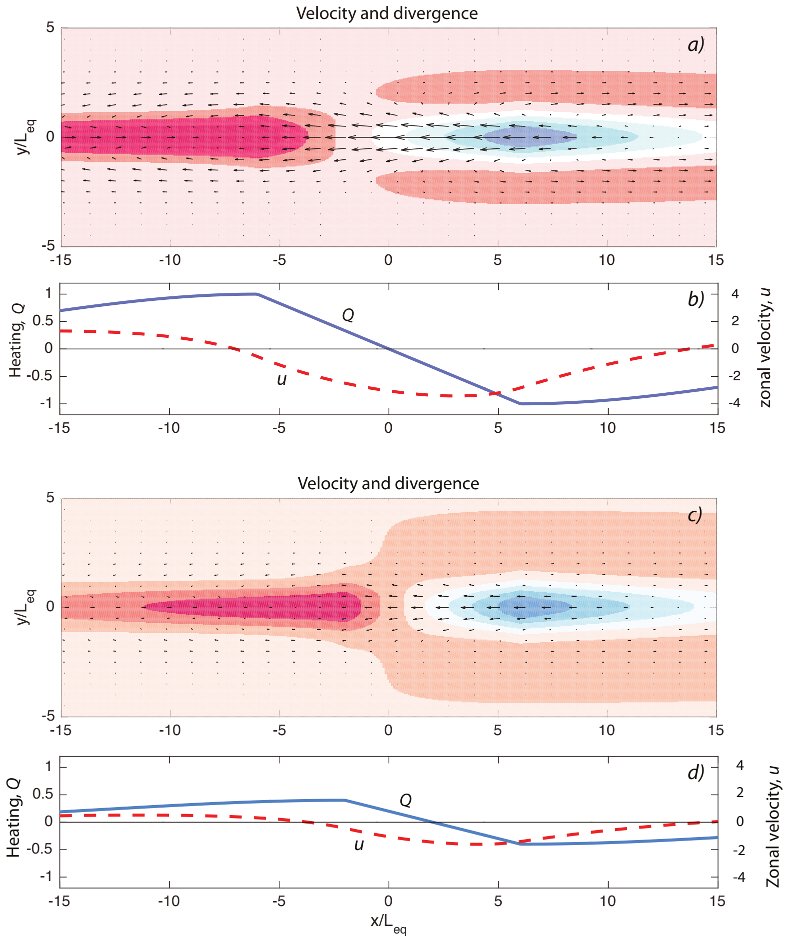

Fig. 22.17 Two solutions to the Matsuno–Gill model for the atmosphere, each with different distributions of heating, corresponding to a normal year (panels (a) and (b), top) and El Niño (panels (c) and (d)).

Panels (a) and (c) show the low-level velocity (arrows) and velocity divergence (shading) and (b) and (d) show the heating, Q, and zonal velocity, u, along $y = 0$. (a) and (b) have a maximum heating a $x = -6$ and a cooling at $x = 6$, and (c) and (d) have heating and cooling at $x = -2$ and $x = 6$ with a much reduced amplitude.

In both cases, there is descent (blue shading) in the cooling region, westward flow toward the heating where air ascends (red shading), but the westward winds are weaker and extend less in (c) and (d).

22.6 THE WALKER CIRCULATION

The *Walker circulation*[21] is an overturning cell in the tropical atmosphere predominantly in the zonal, or x–z plane, as schematically illustrated in the top panel of Fig. 22.16, and as a solution of the Matsuno–Gill problem in the top panel of Fig. 8.12. Convection is embedded within the cell, especially at the warm western end, as in Fig. 18.14 on page 691, but the Walker Cell is not 'driven' by convection anymore than the meridional overturning circulation in ocean is driven by convection in the North Atlantic. Rather, if there is a driver, it is the warm surface waters in the western equatorial Pacific that arise because of the westward surface flow and that are heated by the Sun as they move (see also the shaded box on page 872). These winds are part of the trade wind system and at leading order are independent of the presence of the ocean — they arise by the action of the Coriolis force on the equatorward surface branch of the Hadley Cell. There are other zonal overturning cells in the tropics but the one in the Pacific — the Walker Cell — is the strongest.

22.6.1 A Matsuno–Gill model

Perhaps the simplest dynamical description of the Walker circulation is the Matsuno–Gill model.[22] The model (described in section 8.5) finds the steady solutions of the linear shallow water equations on an equatorial beta-plane forced with a mass source. As discussed in section 18.7, the solution of these equations may be interpreted as being the first baroclinic mode of a continuously stratified convecting atmosphere in which the static stability is maintained by convection, with the mass source representing thermal forcing and coming in this instance from a warm ocean.

What does such a solution look like? Figure 22.17 shows the solution to the Matsuno–Gill problem in which the thermal forcing linearly increases with longitude, crudely representing a sea-surface temperature gradient increasing westward, falling of exponentially away from the equator with an e-folding scale of an equatorial deformation radius. The solution shows a westward surface velocity between the cooling in the east and heating in the west, with ascent over the heating and descent over the cooling. Ascent is largest over the heating region, and so the trades weaken and contract as the heating moves east, as in the schematic (Fig. 22.16).

We can make some rough estimates of the importance of the above effects. Suppose that an SST anomaly of ΔT produces a heat flux of $Q = \lambda \Delta T$, where λ has units $\mathrm{m^2\,s^{-1}\,K^{-1}}$. The nondimensional parameter, γ^* say, that determines the importance of the heating is, using the expression involving Q in (8.115),

$$\gamma^* = \frac{\lambda\,\Delta T\,L_{eq}}{c_a^3}, \tag{22.83}$$

where c_a is the gravity wave speed in the atmosphere. Using (8.136a) a rough estimate for the magnitude of the wind response is then

$$u \sim \frac{Q}{c\alpha} = \frac{\lambda \Delta T}{c_a r_a}, \tag{22.84}$$

where r_a is the linear drag on the wind (units of $\mathrm{s^{-1}}$) in the Matsuno–Gill problem. If $r_a \approx 1/10\,\mathrm{days}^{-1}$ and $c_a = 50\,\mathrm{m\,s^{-1}}$ then u increases by $3 \times 10^4 \lambda$ per degree Kelvin. Observations show that $\lambda \sim 10^{-4}\,\mathrm{m^2\,s^{-1}\,K^{-1}}$, suggesting that sea-surface temperature anomalies can have a measurable impact on surface winds in the tropics.

22.7 THE OCEANIC RESPONSE

The oceanic response during an ENSO cycle is mediated by Kelvin and Rossby waves propagating westward and eastward respectively, as discussed in chapter 8. These waves do not intrinsically preferentially cause upwelling or downwelling, nor do they cause a systematic change in the thermocline thickness or sea-surface temperature. Rather, the waves are the means by which the ocean transitions from one state to another, just as gravity waves effect an adjusment to geostrophic balance.

The linearized equations of motion for the equatorial thermocline

$$\frac{\partial u}{\partial t} - \beta y v = -g' \frac{\partial h}{\partial x} - ru + \widetilde{\tau}^x, \qquad \frac{\partial v}{\partial t} + \beta y u = -g' \frac{\partial h}{\partial y} - ru + \widetilde{\tau}^y, \tag{22.85a,b}$$

$$\frac{\partial h}{\partial t} + H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0 \tag{22.85c}$$

where $\widetilde{\tau} = \tau/\rho_0 H$ and H is the undisturbed mean thickness of the thermocline.

If the trade winds blow steadily westward, with no y-variation, then there is a stationary, steady state situation with

$$u = 0, \quad v = 0, \qquad g\frac{\partial h}{\partial x} = \frac{\widetilde{\tau}^x}{H}, \quad \frac{\partial h}{\partial y} = 0. \tag{22.86}$$

where h is the perturbation thickness of the upper layer (a positive number) and H is its mean thickness. For negative τ_x (westward winds) the thermocline thickness deepens going west. If the trade winds slacken (as during an El Niño) then the slope of the thermocline will, in its equilibrium

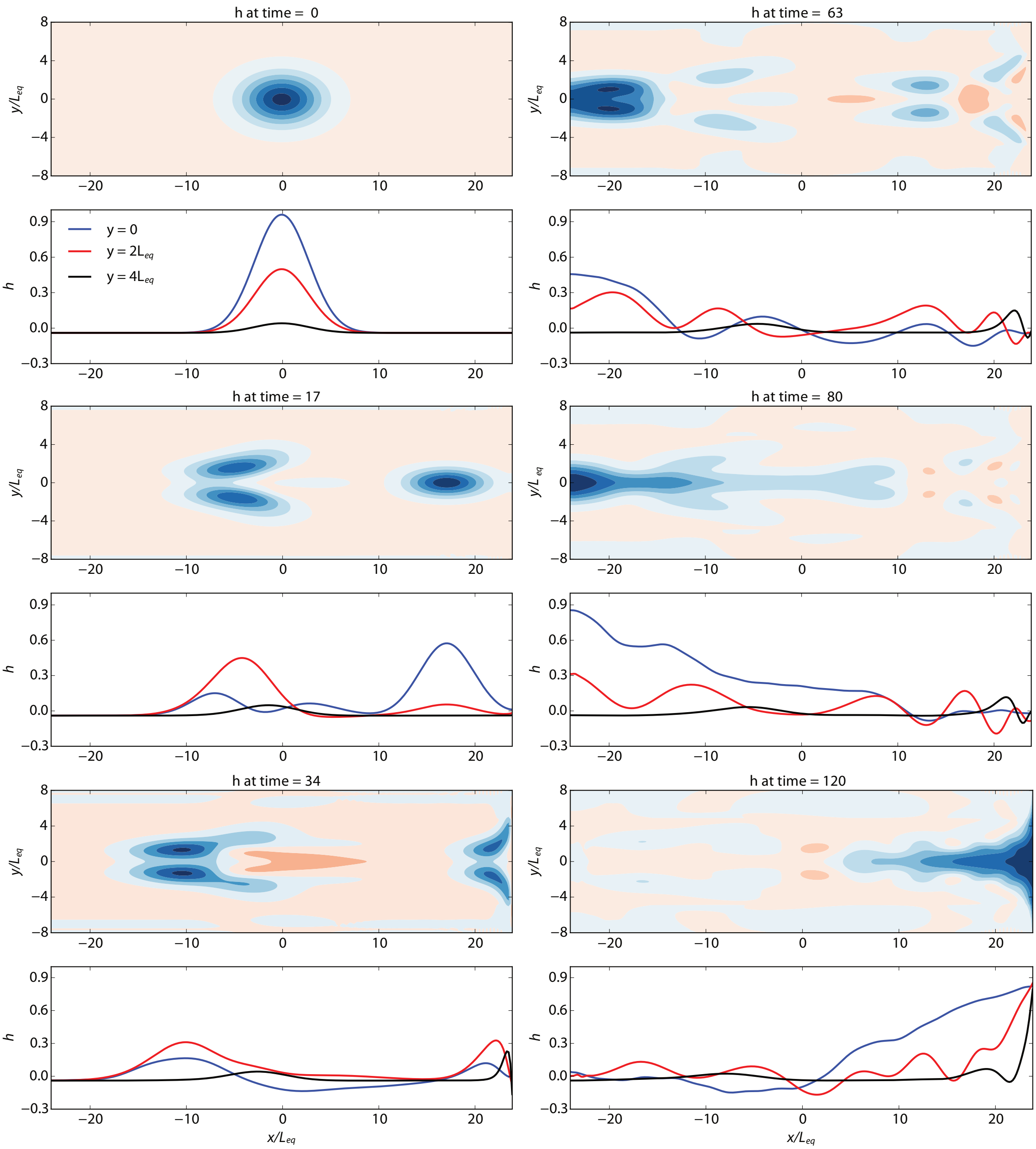

Fig. 22.18 Numerical evolution of the height field on an equatorial beta plane shown at the times indicated, starting from a Gaussian hump centred at the equator. The blue curve shows the height field along the equator and the red and black curves at two and four deformation radii poleward. All quantities are nondimensional, so the Kelvin wave (the hump propagating eastward) moves at a speed of one with off-equatorial Rossby waves moving more slowly westward. At time 34 we see the equatorial Kelvin wave partially reflected back as Rossby waves and partially propagating polewards as coastal Kelvin waves. At times 63 and 80 we see the Rossby waves reflected back as an equatorial Kelvin wave.

state, diminish (as in Fig. 22.16) but the new equilibrium is not achieved instantly; rather, it is mediated by Kelvin and Rossby waves, just as the passage to geostophic balance is mediated by gravity waves. This means that any feedback that the ocean may have on the atmosphere will be delayed by a time that is of same order as, and perhaps somewhat more than, the time it takes Rossby and Kelvin waves to traverse the Pacific.

To see these waves consider how the ocean responds to a perturbation on the equator in the middle of the ocean, and specifically we set up the initial value problem with a perturbation height field, and no velocity perturbation, as illustrated in Fig. 22.18. This perturbation will potentially project onto multiple modes and at the equator we expect there to be a significant Kelvin wave response, decaying away from the equator as $\exp(-y^2/L_{eq}^2)$ as in section 8.2.2. The Kevin wave moves east at a speed $(g'H)^{1/2}$, eventually colliding with the eastern boundary. Here it generates Kelvin coastal waves that move along the coast away from the boundary and are eventually dissipated, and a weak reflected Rossby wave.

Rossby waves will also generated by the perturbation, the gravest of which has an $m = 1$ structure in the v field, where m is the index of the Hermite polynomial (section 8.2). This mode contains both $m = 0$ and $m = 2$ components in the height field (see (8.161) on page 332) represented in physical space as lobes slanted away from the equator, initially rather like the lobes in the stationary problem illustrated in Fig. 8.11. These lobes propagate westward (second panel of Fig. 22.18), but at about a third the speed of the Kelvin wave, before they eventually collide with the western boundary. Now, in midlatitudes, when Rossby waves hit a western wall they can only reflect as short Rossby waves, in order to have a eastward directed group velocity (section 6.6) and in practice they often dissipate in the western boundary current. But here at the equator the wave energy can (and does) return eastward as an equatorial Kelvin wave, because the frequencies of Kelvin and Rossby waves can be the same — see Fig. 8.6 — as required for reflection. The whole sequence is illustrated numerically in Fig. 22.18 and schematically in Fig. 22.19. The initial conditions for the numerical simulation are single postive depth anomaly (top left panel), whereas the wording on the schema assumes that the wind anomaly produces a deepening of the thermocline on the equator and a shallowing off the equator (generated by the *curl* of the wind stress) which are propagated away by Kelvin and Rossby waves respectively.

22.8 COUPLED MODELS AND UNSTABLE INTERACTIONS

22.8.1 Equations of motion

The equatorial atmosphere and ocean are governed by very similar sets of shallow-water equations, although the scales of the two systems, and how the variables are to be interpreted, differ. In the ocean the equations represent the water above the equatorial thermocline and should be interpreted as reduced gravity equations, with the variable h being the thermocline depth and g' being a direct measure of the density difference between the upper and deep ocean. In the atmosphere the equations represent the first baroclinic mode of the atmosphere, and the deformation radius is about 1000 km, several times that of the ocean.

Ocean

In the longwave approximation (discussed in section 8.2) the zonal scales are much greater than meridional scales and (22.85) reduces to geosrophic balance and the above equations become

$$\frac{\partial u}{\partial t} - \beta y v = -\frac{\partial \phi}{\partial x} - ru + \tilde{\tau}^x, \qquad \beta y u = -\frac{\partial \phi}{\partial y}, \qquad \frac{\partial \phi}{\partial t} + c_o^2\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0, \qquad (22.87a,b,c)$$

where we have also written $\phi = g'h$ and $c_o^2 = g'H$; in the equatorial ocean c_o has a value of 1.5–2 m s^{-1}. The above equations are complete, but do not provide a prediction for the surface

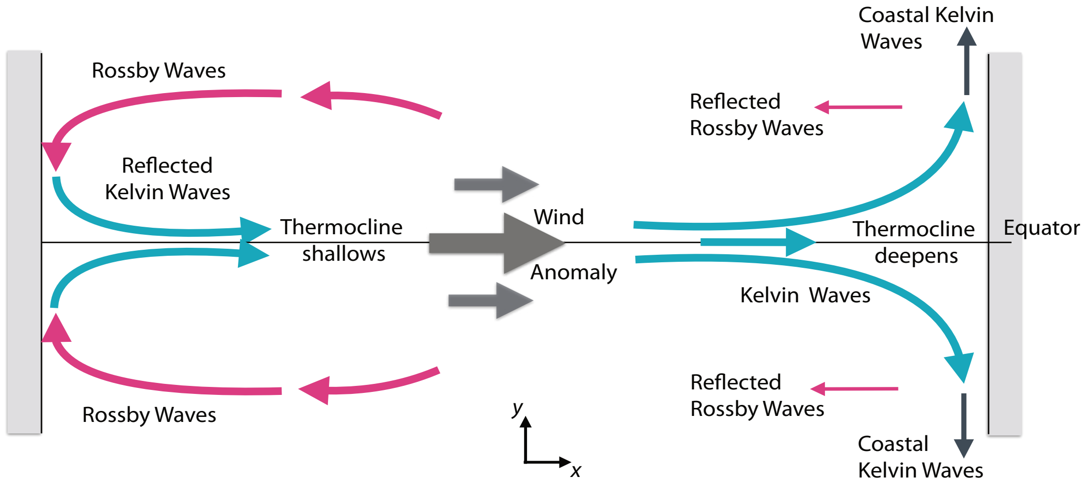

Fig. 22.19 Oceanic waves induced by a wind anomaly on the equator, leading to a local thickening of the thermocline and a propagating Kelvin wave that carries the anomaly westward, where it is partially propagated away as coastal Kelvin waves. The wind-stress curl may generate an off-equatorial shallow anomaly that is carried westward by Rossby waves and then reflected back as Kelvin waves.

temperature, which is the field that the atmosphere actually cares about. We may heuristically add a mixed-layer temperature variable by supposing that the temperature is passive and advected by the flow, and is heated or cooled from above. Then, linearizing around a mean temperature $\overline{T}$, we write

$$\frac{\partial T}{\partial t} + u\frac{\partial \overline{T}}{\partial x} + v\frac{\partial \overline{T}}{\partial y} + \gamma_1 \frac{\overline{w}}{H'}(T - T_d) = -\lambda_1 (T - T_a). \tag{22.88}$$

The term $\gamma_1 \overline{w}(T - T_d)/H'$ represents upwelling from some depth H' where the temperature is T_d, and the term on the right-hand side represents exchange with the atmosphere, and γ_1 and λ_1 are constants. The derivations of equations like this are all somewhat empirical,[23] so it is appropriate to maximally simplify. The most important processes are exchange with the atmosphere and upwelling, and the efficacy of the upwelling is largely dependent on the depth of the thermocline, h, since this determines the temperature of the upwelling water. Thus, simplifying further, we have

$$\frac{\partial T}{\partial t} = +\gamma_2 h - \lambda T \quad \text{or, with still more approximation,} \quad T = Ah. \tag{22.89a,b}$$

where γ_2 is a constant and $A = \gamma_2/\lambda$ and $h = \phi'/g'$. Equation 22.89b) assumes that equatorial upwelling is the dominant factor in determining surface temperature, and states that the anomalous temperature is proportional to the anomalous thickness of the upper layer.

Atmosphere

On the atmospheric side the equations are almost the same, and using capital letters for the variables and making the longwave approximation we have

$$\frac{\partial U}{\partial t} - \beta y V = -\frac{\partial \Phi}{\partial x} - r_a U, \qquad \beta y U = -\frac{\partial \Phi}{\partial y}, \qquad \frac{\partial \Phi}{\partial t} + c_a^2\left(\frac{\partial U}{\partial x} + \frac{\partial V}{\partial y}\right) = Q_a, \tag{22.90a,b,c}$$

where the constant r_a is a drag coefficient and c_a is the gravity wave speed of the first baroclinic mode, around $30\,\mathrm{m\,s^{-1}}$. The term on the right-hand side of (22.90) represents a thermodynamic forcing by SST anomalies, and the simplest parameterization of this is $Q_a = \alpha' T$ where α' is a constant. Then, using (22.89), we obtain $Q_a = \alpha h$, where $\alpha = A\alpha'$. Similarly, the stress on the ocean [represented by $\widetilde{\tau}^x$ in (22.87)] arises from the linear drag term $r_a U$ in (22.90), since stress is continuous across the atmosphere-ocean interface. Thus, we write $\widetilde{\tau}^x = -\gamma U$ where γ is a constant — the negative sign arising because (22.90) describes the first baroclinic mode and the surface wind is proportional to $-U$. Coupling between the atmosphere and ocean in (22.90) and (22.87) is thus, in the simplest case, effected by

$$\widetilde{\tau}^x = -\gamma U, \qquad\qquad Q_a = \alpha\phi. \tag{22.91a,b}$$

Equations (22.85), (22.90) and (22.91), are a complete set of equations for a coupled, equatorial, ocean-atmosphere system. Nearly all extant theories of ENSO are essentially theories of the behaviour of these or related sets of equations. The most ad hoc aspects of these equations are the relationships in (22.91), and the coefficients γ and α should be regarded as semi-empirical — a problem that would remain even if we were to use (22.89a). The model could be further extended by a more complete treatment of diabatic effects and the coupling terms — for example, one could add an evolution equation for the sea-surface temperature with advection and boundary layer processes, and decouple it from the thermocline depth. But even with a comprehensive numerical model (e.g., a coupled GCM) a degree of empiricism is unavoidable.

22.8.2 Unstable air-sea interactions

We can expose some of the essential dynamics of the coupled system if we severely approximate (22.85), (22.90) and (22.90) by omitting y-derivatives, Coriolis terms and meridional velocities. We also neglect time derivatives in the atmosphere, since it achieves a quasi-equilibrium with the state of the ocean on a shorter timescale than that of the ocean. The coupled set then reduces to[24]

$$\frac{\partial u}{\partial t} = -\frac{\partial \phi}{\partial x} - r_o u - \gamma U, \qquad \frac{\partial \phi}{\partial t} + c_o^2 \frac{\partial u}{\partial x} = 0, \tag{22.92a,b}$$

$$0 = -\frac{\partial \Phi}{\partial x} - r_a U, \qquad c_a^2 \frac{\partial U}{\partial x} = \alpha\phi, \tag{22.92c,d}$$

These equations combine into a single equation for the ϕ (a proxy for ocean surface temperature), namely

$$\frac{\partial^2 \phi}{\partial t^2} - c_o^2 \frac{\partial^2 \phi}{\partial x^2} + r_o \frac{\partial \phi}{\partial t} - \frac{c_o^2}{c_a^2}\gamma\alpha\phi = 0. \tag{22.93}$$

Seeking a harmonic solution of the form $\mathrm{e}^{\mathrm{i}(kx-\omega t)}$ yields the dispersion relation

$$\omega^2 = -\mathrm{i} r_o \omega + c_0^2 k^2 - \gamma\alpha \frac{c_o^2}{c_a^2}. \tag{22.94}$$

The terms on the right-hand side give rise to damping (the term involving r_o, the drag on the oceanic flow), waves (the term $c_o^2 k^2$), and, potentially, exponential growth (the term $\gamma\alpha$, which is the feedback between atmosphere and ocean). If this feedback is large enough the flow will be unstable, and this is a simple mathematical representation of the feedback described verbally in section 22.5.2. The solution of (22.94) is

$$\omega = -\frac{\mathrm{i} r_o}{2} \pm \sqrt{c_o^2 k^2 - \frac{r_o^2}{4} - \gamma\alpha \frac{c_o^2}{c_a^2}} \qquad \text{or} \qquad \sigma = -\frac{r_o}{2} \pm \sqrt{\gamma\alpha \frac{c_o^2}{c_a^2} + \frac{r_o^2}{4} - c_o^2 k^2}\,, \tag{22.95a,b}$$

where $\sigma \equiv -i\omega$ is the growth rate, and depending on the size of $\alpha\gamma$ the solution may be decaying or a growing, and if decaying it may be oscillatory. Note that the frequency of the oscillations is reduced (from its pure oceanic value, $c_o k$) by the interaction with the atmosphere. Also, (22.95) has both westward and eastward propagating gravity waves; however, since these are at the equator they really represent Kelvin waves that propagate only eastward; that is, the westward propagating solution in (22.95) is artifactual. To obtain steady self-sustained oscillations we need to delve further, and we will illustrate that by numerical integrations and toy models.

22.9 SIMPLE CONCEPTUAL AND NUMERICAL MODELS OF ENSO

We now discuss two conceptual, or toy, models of ENSO. Toy models are *ad hoc* models, meaning that their governing equations cannot be derived in a rigorous or even a systematic fashion from the fundamental physics or dynamics. Rather, the models often arise using verbal reasoning and/or severe approximation, and their purpose is to illustrate mechanisms and suggest developments rather than to be accurate predictive tools or proper reductions of the full equations. However, the models may be very useful and the behaviour they describe may be quite robust.[25]

We first construct and analyse a *delayed-oscillator* model, which provides a basis for understanding how the finite propagation time of equatorial waves in conjunction with the Bjerknes feedback can give rise to oscillatory behaviour. We then look at the perhaps more realistic *recharge-discharge* model and finally show some numerical simulations.

22.9.1 A delayed-oscillator model

The mechanisms involved in the delayed-oscillator model are illustrated in Fig. 22.19.[26] To begin, suppose there an eastward wind anomaly at the equator, and that this develops a positive (i.e., deeper) thermocline anomaly in mid-basin at the equtor. This anomaly propagates eastward as a Kelvin wave, thickening the thermocline, warming the sea surface and generating anomalous eastward winds, that further thicken the thermocline (the Bjerknes feedback). At the same time a negative thermocline anomaly is generated off the equator, which is propagated westward as Rossby waves before being reflected back as a Kelvin wave. When this reaches the eastern Pacific it lowers the sea-surface temperature, so providing a delayed negative feedback. The appendix (page 904) attempts to derive a simple equation governing such a process from the equations of motion and the result is

$$\frac{\partial T}{\partial t} = aT(t) - bT(t-\delta) - rT^3(t), \tag{22.96}$$

where a, b, c and δ are constants and T is the SST anomaly, nominally in the eastern equatorial Pacific. The three terms on the right-hand side of this equation may be interpreted as follows.

(i) The term $aT(t)$ represents a local wind-surface temperature positive feedback. A positive SST anomaly in the East Pacific producing an anomalous eastward wind stress and eastward propagating downwelling Kelvin wave, deepening the thermocline, further warming the sea surface.

(ii) The term $bT(t-\delta)$ represents a delayed negative feedback. A wind-stress curl anomaly leads to off-equatorial thermocline shallowing and westward propagating Rossby waves. These reflect back as a Kelvin wave that, after a time δ, cools the eastern equatorial ocean.

(iii) The cubic nonlinear damping term, rT^3, represents all other dissipative processes in the system.

In some ways the verbal reasoning above is as convincing as the appendix.

Analysis of the oscillator

Equation (22.96) simplifies if we write $\hat{t} = ta$ and $\widehat{T} = T\sqrt{r/a}$, giving the nondimensional equation

$$\frac{\mathrm{d}\widehat{T}}{\mathrm{d}\hat{t}} = \widehat{T} - \alpha\widehat{T}(\hat{t} - \widehat{\delta}) - \widehat{T}^3, \tag{22.97}$$

where $\widehat{\delta} = \delta a$ is the nondimensional delay and $\alpha = b/a$. Equations such as (22.97) are not 'simple'. Formally, the equation has an infinite number of degrees of freedom — note that we need to specify initial conditions not just at a single time but at all times from $t = -\delta$ to $t = 0$ — and corresponds to an infinite number of ordinary differential equations. Nevertheless, we can obtain useful information about the solution by way of a linear analysis. Dropping the hats from (22.97), equilibria occur at $T = T_0$ where

$$T_0 = 0 \quad \text{and} \quad T_0 = \pm\sqrt{1-\alpha}. \tag{22.98}$$

We set $T = T_0 + T'$ and linearize then (22.97) becomes

$$\frac{\mathrm{d}T'}{\mathrm{d}t} = T'(1 - 3T_0^2) - \alpha T'(t - \delta). \tag{22.99}$$

In the usual manner we now let $T' = \widetilde{T}\mathrm{e}^{\sigma t}$, where $\sigma = \sigma_r + \mathrm{i}\sigma_i$ and obtain

$$\sigma = 1 - 3T_0^2 - \alpha\mathrm{e}^{-\sigma\delta}, \qquad \sigma_r = 1 - 3T_0^2 - \alpha\mathrm{e}^{-\sigma_r\delta}\cos\sigma_i\delta, \quad \sigma_i = \alpha\mathrm{e}^{-\sigma_r\delta}\sin\sigma_i\delta. \tag{22.100}$$

These are transcendental equations, since σ appears on the right-hand sides in the sine and cosine terms. For $T_0 = 0$ we get

$$\sigma_r = 1 - \alpha\mathrm{e}^{-\sigma_r\delta}\cos\sigma_i\delta, \qquad \sigma_i = \alpha\mathrm{e}^{-\sigma_r\delta}\sin\sigma_i\delta. \tag{22.101a,b}$$

A solution of this satisfies $\sigma_i = 0$ and $\sigma_r = 1 - \alpha\mathrm{e}^{-\sigma_r\delta}$. A value for σ_r can easily be obtained graphically, but even without doing that we can see that σ_r will generally be positive, because the term $\alpha\mathrm{e}^{-\sigma_r\delta}$ is positive and less than unity. The solution is therefore unstable.

For the case in which $T_0 = \pm\sqrt{1-\alpha}$ the stability equation is

$$\sigma_r = 3\alpha - 2 - \alpha\mathrm{e}^{-\sigma_r\delta}\cos\sigma_i\delta, \qquad \sigma_i = \alpha\mathrm{e}^{-\sigma_r\delta}\sin\sigma_i\delta. \tag{22.102a,b}$$

The neutral curves, where $\sigma_r = 0$, are given by

$$\delta = \frac{1}{\sigma_i}\cos^{-1}[(3\alpha - 2)/\alpha], \qquad \sigma_i = [\alpha^2 - (3\alpha - 2)^2]^{1/2}. \tag{22.103a,b}$$

These are plotted in Fig. 22.20 and the lower curve corresponds to a boundary between stable and unstable regions: for the shaded regions the solution to (22.97) converges to a fixed point, whereas in the unshaded region the fixed points are unstable and the solutions are oscillatory. (Because the signal will diminish as the Rossby waves propagate westward and return as Kelvin waves, we might expect that the negative feedback in the delay term will be weaker than the positive feedback in the linear term and therefore $\alpha < 1$.)

Unstable modes arise more readily for *larger* values of the negative feedback parameter α and for *larger* values of the delay time δ. To help understand this consider the case with zero delay, for which the equation of motion is just

$$\frac{\mathrm{d}\widehat{T}}{\mathrm{d}\hat{t}} = (1 - \alpha)\widehat{T} - \widehat{T}^3, \tag{22.104}$$

This equation has three stationary states: an unstable state at $T = 0$ and stable states at $T = \pm\sqrt{1-\alpha}$, as in the case with finite delay, but no oscillatory modes are present. As α diminishes the effect of the delay also diminishes, and for $\alpha = 0$ stable states occur at $T = \pm 1$ and no oscillatory modes are present. The period of the oscillations is not easily obtained except on the neutral curves, and for that we turn to numerical solutions.

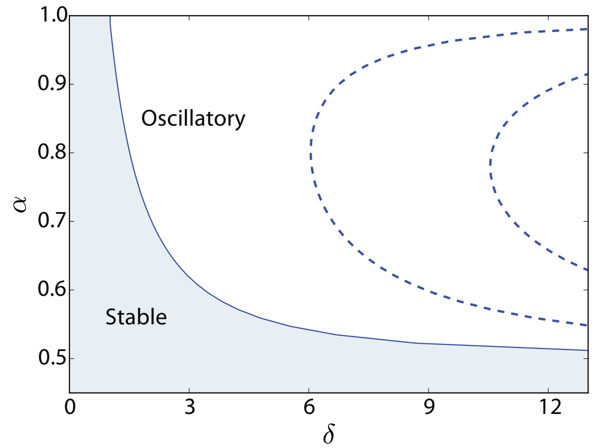

Fig. 22.20 Stability regime for a delayed-oscillator model. Plotted are values of α vs δ as given by (22.103).

Numerical solutions

Numerical solutions of (22.97) may be obtained easily and some typical solutions are shown in Fig. 22.21. The actual periods close the neutral line are in fact similar to those given by the linear analysis, (22.103b). The period gets longer for a longer delay, although the ratio of the period to the delay itself diminishes slightly as the delay increases. Typically, the period is two or three times the delay, although for small values of δ the period can be five times the delay. What is that delay time, dimensionally? Given the speed of Rossby waves and the size of the Pacific, a typical value is about eight months, with a maximum possible value of about 12 months. This gives a period of about 24 months, which is rather less than the typical observed interval between El Niño events. However, it is hard to be precise because we need to determine what nondimensional value of δ corresponds to a dimensional value of 8 months: we have scaled time by the value of the amplifying parameter a in (22.96), and this is uncertain to within a factor of a few. Unless the parameter regime of the Pacific ocean is such that a dimensional delay of 8 months corresponds to a nondimensional delay of about 2, which would give a period of four or five times 8 months, then the period delayed oscillator model is somewhat shorter than that of the real El Niño.

The delayed-oscillator model illustrates two properties that may be important in the real system, namely the positive feedback by the wind (the underlying Bjerknes hypothesis) and the importance of a delayed damping, so enabling oscillations to occur. Still, it has ad hoc aspects, especially its treatment of the negative feedback from the Rossby waves and its introduction of a cubic damping.

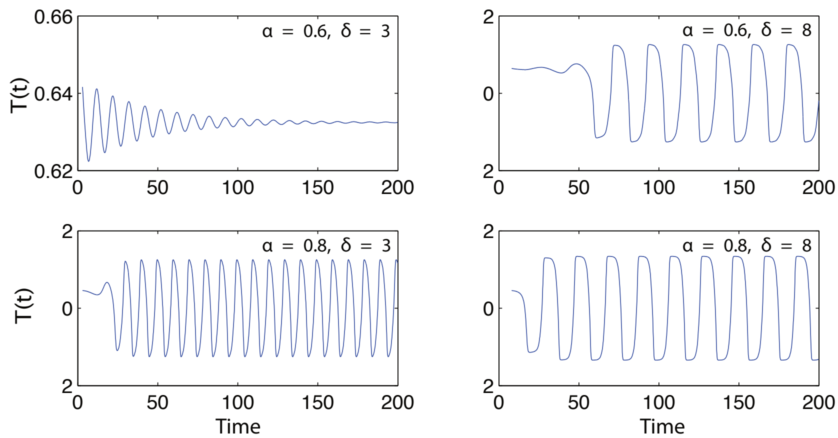

Fig. 22.21 Time series obtained by numerically integrating (22.97) for the parameters indicated. The initial conditions are the value at the fixed point, $T = \sqrt{1-\alpha}$, plus a small perturbation.

22.9.2 The recharge-discharge oscillator

A basin-wide mechanism that has become known as the *recharge-discharge* hypothesis seems to capture many of the salient features of the ENSO phenomenon in an appealingly simple way, roughly as follows.[27]

The prevalent westward trade winds push warm water westwards, depressing the thermocline in the west and building up a significant amount of warm water. At some point a fluctuation in the atmospheric circulation will occur that causes the trade winds to relax and give rise to an eastward Kelvin wave surge, warming the central and Eastern equatorial ocean, further weakening the trades and allowing El Niño to develop fully. Then, along with the wave reflection that occurs at the eastern boundary, some of that warm water drains away polewards along the coast in the form of coastal Kelvin waves, meaning that some of the warm pool is irretrievably lost to the equatorial ocean, and this discharge is the demise of the event. The trade winds then resume, slowly building up the warm pool in the west and priming the system for the onset of another El Niño. The timescale of the oscillation — that is, the time between events — is then determined by the time taken to build up the warm water, rather than being a multiple of the wave propagation time across the basin. The two are related, since Kelvin and Rossby waves are still the mechanism that determine the timescale of propagation of the signal across the ocean basin, and so determine the timescales of the El Niño genesis and recharge. In this model, the positive feedback is (still) the Bjerknes mechanism, the growth rate (or, perhaps more relevantly, the onset time) is determined by the passage of the equatorial ocean waves across the basin, the damping is the drainage of warm water into coastal Kelvin waves as well as heat loss to the atmosphere and deeper ocean, and the interval between events is determined by the time it takes to refill the basin with warm water, along with the interaction with the seasonal cycle.

In some ways the recharge-discharge mechanism is more straightforward and physically grounded than that of the delayed-oscillator, although the two are not wholly in opposition — waves propagate back and forth, leading to delayed negative feedbacks. Similar to the delayed-oscillator case, we can write down, or try to derive more systematically, simple ODE models mimicking the recharge-discharge behaviour, and readers are invited to roll their own.[28] The models themselves are transparent and simple but the derivations have unavoidably ad hoc aspects. Rather than proceeding down that path let illustrate the ENSO phenomenon with a simple numerical model, namely the shallow water equations of section 22.8 for the ocean, with a very simplified atmosphere.

22.10 NUMERICAL SOLUTIONS OF THE SHALLOW WATER EQUATIONS

We now show some numerical solutions of the time-dependent shallow water equations on an equatorial beta-plane, (22.85). The model is in a regime in which the linear equations are a very good approximation, and have a mean thermocline depth of 50 m and $g' = 0.05\,\mathrm{m\,s^{-2}}$ (so that $c = (g'H)^{1/2} \approx 2.2\,\mathrm{m\,s^{-1}}$) in a domain about 15,000 km across. The equatorial deformation radius, $L_{eq} = (c/\beta)^{1/2}$ is then about 300 km, the nondimensional time $T_{eq} = 1/(c\beta)^{1/2}$ is 1.6 days and the basin width is about 50 L_{eq}. The model also has a linear damping in all fields with a timescale of 16 months.

The simplest illustration of El Niño arises from the initial value problem illustrated in Fig. 22.22. The initial thermocline thickness is set to deepen linearly toward the west and falls off exponentially away from the equator over a few deformation radii, as if held by a westward wind. The flow is released (i.e., the nominal wind is 'turned off') and a Kelvin wave propagates eastward, colliding with the eastern wall some time later, generating poleward propagating coast Kelvin waves and, most obviously, a warm tongue that appears to spread from the east — this is the model analogue of an El Niño event. The disturbance is partially damped by those coastal waves but also propagates back across the basin as a Rossby wave, and then back again as a Kelvin wave, producing another El Niño event, and so forth, with the oscillations eventually dying out.

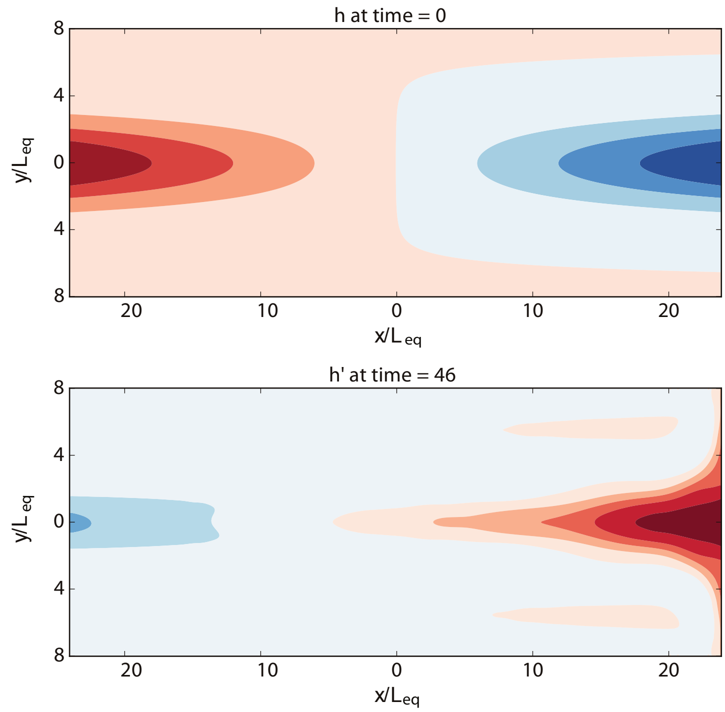

Fig. 22.22 A numerical solution of the shallow water equations showing the initial thermocline depth and the perturbation depth (i.e., final minus initial) at $t = 46/\sqrt{\beta c} \approx 70$ days, approximately the Kelvin wave crossing time for the basin. Red shading indicates deeper, warmer, water.

The initial field increases linearly toward the west, and the later state has a tongue of warm water in the east.

Self-sustained oscillations can be generated by adding a wind forcing that depends on the ocean state. Although it is straightforward to couple the ocean to a corresponding shallow water atmosphere, the mechanism is seen transparently if we reduce (22.90) and (22.91) to the simple equation

$$\widetilde{\tau}^x = \gamma'(\overline{h}_E - \overline{h}_W) - \widetilde{\tau}^x_0, \tag{22.105}$$

where the constant $\widetilde{\tau}^x_0$ represents the trade winds and $\overline{h}_E$ and $\overline{h}_W$ are the averaged thermocline thicknesses in the eastern and western parts of the domain, so representing the Bjerknes feedback in its most basic form. The results are quite sensitive to the value of the coupling coefficient γ' but much less so to the way $\overline{h}_E$ and $\overline{h}_W$ are calculated.

Various results are shown in Fig. 22.23, all with the initial condition of Fig. 22.22. The plots all show timeseries of the thermocline thickness at the eastern and western edges, h_E and h_W, and the wind stress, in rather arbitrary, order one, units. Panel (a) shows the continuation of the integration of Fig. 22.22, showing a damped oscillator (the wind stress has no effect) with a period of approximately the sum of Kelvin wave and Rossby wave crossing times, or about 190 time units, somewhat less than one year. The difference in the two crossing times is apparent in the saw-tooth nature of the thickness time series. If we now force the model with seasonal cycle in winds, but do not allow the thermocline thickness to feedback on the wind, then the oscillations become phase-locked to the seasonal cycle and eventually become quite steady [panel (b)]. If we allow a feedback using (22.105), but remove the seasonal cycle, the system develops self-sustained, nearly periodic, oscillations with a period that is significantly longer than the unforced period [panel (c)]. The combined effects of a seasonal cycle and a wind feedback produce more irregular oscillations, partially locked to the seasonal cycle but sometimes skipping a year. Even more irregular behaviour can be obtained if some stochasticity is added to the forcing. If the feedback is too strong (i.e., γ' too large in (22.105)) the model goes to a fixed point — the wind feedback is too strong to allow the model to escape from either a permanent El Niño or La Niña.

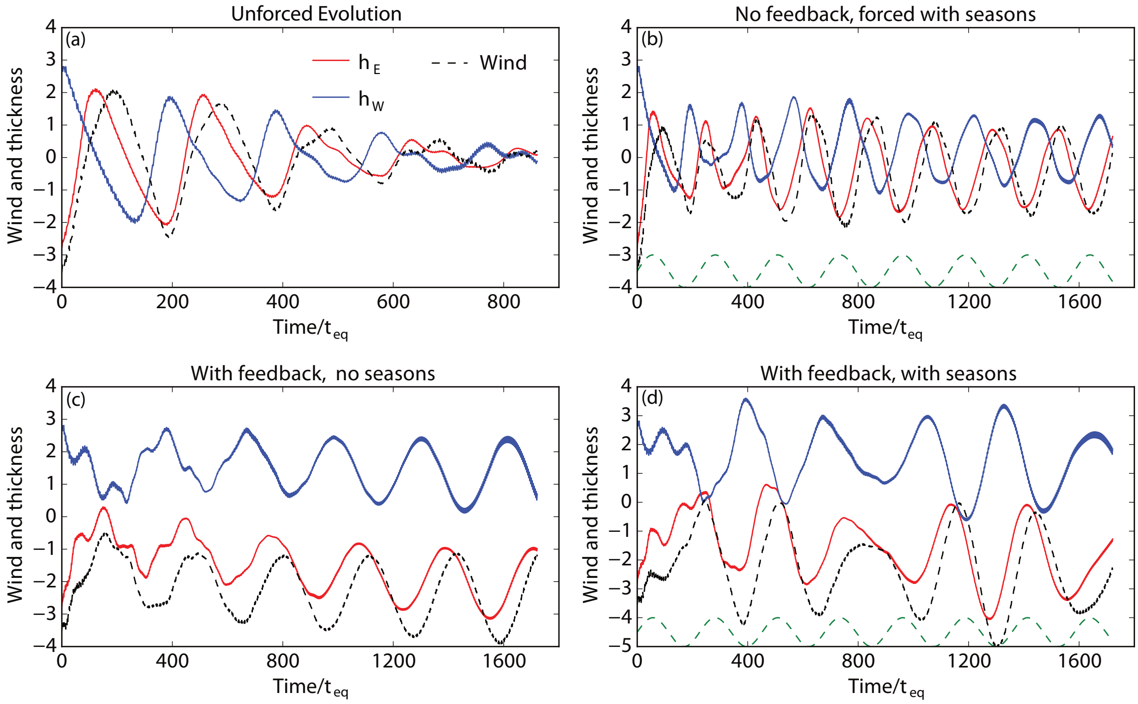

Fig. 22.23 Time series of thermocline thickness in the east and west, h_E and h_W, and the wind given by (22.105), which has no effect in the top two panels. Time is in units of $\sqrt{1/(c\beta)}$ or 1.6 days. The panels variously show the effects of a wind feedback and a seasonal cycle: (a) shows the unforced evolution, (b) includes an annual cycle of winds but no feedback on those winds, (c) includes a wind feedback as in (22.105), and (d) includes both a wind feedback and annual cycle. A year is 226 time units ($226/\sqrt{c\beta}$) and its cycle is shown in the dashed lines at bottom. A Kelvin wave crossing time is 48 time units and a Rossby wave crossing time is about three times that.

Although still some distance from reality, the model illustrates the most important and robust features of the ENSO cycle (see also the shaded box on page 891).

(i) The back-and-forth of Kelvin and Rossby waves, illustrated even in the unforced simulation of panel (a),

(ii) The sustaining effects of the feedback of the thermocline thickness on the wind, which also significantly lengthens the period of oscillation (compare the periods in panels (a) and (c)).

(iii) The damping that arises through leakage at the western boundary and the explicit damping in the model reflecting loss to the atmosphere and the deeper ocean, and the delay inherent through the finite propagation speed of the waves.

(iv) The irregularity arising from interactions with another timescale, in this case the seasonal cycle.

The quantitative importance of these various effects — for example the relative importance of western boundary leakage versus loss to the atmosphere — remains a matter of investigation, as is the the exact mechanism that produces the great irregularity in the observed El Niño events — the role of the seasonal cycle, chaos, stochasticity, and so forth.[29]

22.A ✦ APPENDIX: DERIVATION OF A DELAYED-OSCILLATOR MODEL

We begin with (22.87) and include a damping on the height field to give

$$\frac{\partial u}{\partial t} - \beta y v = -\frac{\partial \phi}{\partial x} - r_o u + \widetilde{\tau}^x, \qquad \beta y u = -\frac{\partial \phi}{\partial y}, \qquad \frac{\partial \phi}{\partial t} + c_o^2\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = -r_o\phi. \quad (22.106\text{a,b,c})$$

The properties of this equation were explored in section 8.4, but we proceed ab initio. We are interested in thermocline depth variations so we combine the above into a single equation for ϕ,

$$\beta y^2\left(\frac{\partial}{\partial t} + r_o\right)\phi + \frac{c_o^2}{\beta}\left[\frac{2}{y} - \frac{\partial^2}{\partial y^2}\right]\left(\frac{\partial}{\partial t} + r_o\right)\phi - c_o^2\frac{\partial \phi}{\partial x} + \left[\widetilde{\tau}^x - y\frac{\partial \widetilde{\tau}^x}{\partial y}\right] = 0. \qquad (22.107)$$

Solutions of these equations will have the meridional form discussed in sections 8.2 and 8.4, namely Hermite functions multiplied by a Gaussian, and we focus on the gravest of these modes.

Equatorial response

If the wind perturbation $\widetilde{\tau}^x$ is peaked at the equator then it will excite a Kelvin wave that decays as $\exp(-y^2/L_{eq}^2)$ away from the equator (as in (8.62) on page 311). Thus motivated, we seek a solution to (22.107) of the form

$$\phi(x, y, t) = \Phi(x, t)\exp(-y^2/2L_{eq}^2), \qquad (22.108)$$

where $L_{eq}^2 = c_o/\beta$. Substituting into (22.108) gives, after a few lines of algebra,

$$\left(\frac{\partial}{\partial t} + r_o\right)\Phi + c_o\frac{\partial \Phi}{\partial x} = \frac{1}{c_o}\left[\widetilde{\tau}^x - y\frac{\partial \widetilde{\tau}^x}{\partial y}\right]. \qquad (22.109)$$

At the equator this is the equation for a forced-damped Kelvin wave, propagating eastward at speed c_o, forced by the wind (and not its curl). We will also excite Rossby waves at the equator, but as seen in the previous section the Kelvin wave response will normally be dominant.

Off equatorial response

Because of the fast meridional decay of the Kelvin wave, off the equator we expect the response to wind perturbations to be largely in the form of Rossby waves. To see this explicitly, note that the relative sizes of the first and second terms in (22.107) is $\beta y^2 : (c_o^2/\beta y^2) = y^4 : L_{eq}^4$. Thus, if $y \gg L_{eq}$ then the second term will be small and if we neglect it we have

$$\left(\frac{\partial}{\partial t} + r_o - \frac{c_o^2}{\beta y^2}\frac{\partial}{\partial x}\right)\phi = \frac{1}{\beta}\frac{\partial}{\partial y}\left(\frac{\widetilde{\tau}^x}{y}\right). \qquad (22.110)$$

This is an equation for forced-dissipative equatorial Rossby waves travelling westward at speed $c_o^2/\beta y^2$ — it is in fact nothing other than forced-dissipative version of the planetary-geostrophic wave equation (8.16). It may seem odd that the speed decreases as beta increases, but in the mid-latitude case the speed is $c_o^2\beta/f_0^2$, which is the same if we replace f_0 by βy. Note that the wave is forced by the wind-stress curl (divided by f) rather than the wind itself, and typically the two have the opposite sign as the reader may verify with a few examples.

Wave characteristics and solution

By following the Rossby and Kelvin waves as they propagate we may see their remote effects on sea-surface height as time progresses. Suppose that a Rossby wave takes a time t_R to cross the basin, and a Kelvin wave takes t_K, and that we are interested in the depth of the thermocline at time t in the eastern equatorial Pacific (where the thermocline is shallowest at its variations have most effect on the SST and thence the wind). The Kelvin wave influence at the east, Φ_E, is given by the solution to (22.109), which may be obtained using the method of characteristics — we introduce a new variable $t' = t + c_o x$, whence (22.109) becomes an integrable ordinary differential equation from which we obtain the solution

$$\Phi_E(t) = \Phi_W(t - t_K)\mathrm{e}^{-r_o t_K} + \frac{1}{c_o}\widetilde{\tau}^x(t - t_K/2)\mathrm{e}^{-r_o t_K/2}. \tag{22.111}$$

Here, Φ_W is the thermocline thickness in the west, to be evaluated at time $t - t_K$, and the wind, assumed to be confined to the centre of the domain, is evaluated at $t - t_K/2$. Variations in Φ_W arise from Rossby waves propagating westward from the wind anomaly and being reflected back at the equator as a Kelvin wave, and are thus given by the solution of (22.110):

$$\Phi_W(t - t_K) = -\frac{1}{\beta}\frac{\partial}{\partial y}\left(\frac{\widetilde{\tau}^x(t - t_K - t_R/2)}{y}\right)\mathrm{e}^{-r_o t_R/2} \tag{22.112}$$

Combining these last two expressions gives

$$\Phi_E(t) = \frac{1}{c_o}\widetilde{\tau}^x(t - t_K/2)\mathrm{e}^{-r_o t_K/2} - \frac{1}{\beta}\frac{\partial}{\partial y}\left(\frac{\widetilde{\tau}^x(t - t_K - t_R/2)}{y}\right)\mathrm{e}^{-r_o(t_K + t_R/2)}. \tag{22.113}$$

There may be additional contributions from successive wave crossings but these will be weak, especially since reflection does not occur without loss. Although the derivation of (22.113) is far from rigorous, the essential aspect is that *it is the wind at some time in the past that affects the thermocline thickness in the present.* If we now use (22.89a) to relate the SST to the thermocline thickness we have

$$\frac{\partial T_t}{\partial t} = A\widetilde{\tau}^x(t - t_K/2) - B\widetilde{\tau}^x(t - t_K/2 - t_R/2) - \lambda T \tag{22.114}$$

where A and B are dimensional coefficients that incorporate the constants in (22.113) and account for the decay of the waves as they propagate. The delay in the first term on the right-hand side, due to Kelvin waves, is much smaller than that in the second term due to Rossby waves and Kelvin waves, and we will neglect it.

It remains to relate the wind stress to the SST. Using the Matsuno–Gill model of section 8.5 the atmosphere responds approximately to an SST anomaly with a structure of the form

$$\tau^x \sim \mu A(T_e, x, y)\mathrm{e}^{-y^2/2L_a^2} \tag{22.115}$$

where T_e is the SST in the east, L_a is the atmospheric equatorial deformation radius and A is a function that depends linearly on the SST and only slowly on y. (More generally the response is a Green function multiplied by $T_e(x, y)$ and integrated over the domain.) The wind-stress curl, $\partial(\tau^x/y)/\partial y$ is then proportional to the negative of τ^x, which means that in (22.113) the Rossby waves will carry a height anomaly of *opposite* sign to that of the Kelvin waves. Also, the atmospheric response is much faster than that of the ocean, and occurs over larger zonal scales and thus the wind-stress in the ocean centre can be approximately related to the SST in the East by

$$\widetilde{\tau}^x(t) = CT_e(t), \qquad \frac{\partial}{\partial y}\left(\frac{\widetilde{\tau}^x(t)}{y}\right) = -DT_e(t) \tag{22.116}$$

where C and D are positive constants. Using (22.116) in (22.114) finally gives

$$\frac{\partial T}{\partial t} = aT(t) - bT(t - \delta) - rT^3(t). \tag{22.117}$$

where we have neglected the delay in the Kelvin wave term, we introduced a cubic damping term for stability, and $a = AC + \lambda$, $b = BD$ and $\delta = t_K + t_R/2$.

Notes

1 *"I'm getting youngerly every day."* Paul Kushner, Gordon Conference, Santorini, 2007. Meteorologists tend to talk about westerly winds — the winds that come from the west — because it is where the winds come from that determines the weather. Oceanographers tend to talk about eastward currents, because this is where the currents will take things (or perhaps oceanographers are just a more forward looking crowd). We follow the lead of the oceanographers and talk about eastward and westward flow, for both currents and winds. We also use westward as both adjective and adverb, eschewing westwards, and similarly for other directions.

2 Figure kindly made by Neven Fučkar, using a state estimation from NCEP/GODAS (http://www.esrl.noaa.gov/psd/data/gridded/data.godas.html). Many of the observations themselves are made with acoustic Doppler current profilers (ADCP), which measure the currents by measuring the Doppler shift from a sonar.

3 Adapted from Kessler *et al.* (2003).

4 The undercurrent itself seems to have been first discovered in the Atlantic by J. Y. Buchanan in the 1880s. He measured a southeastward flowing current with speeds of more than 1 knot (about $0.5\,\mathrm{m\,s^{-1}}$) at depths around 30 fathoms (55 metres) at the equator and 13° W from the steamship *Buccaneer,* which was charted to do a survey prior to the laying of a telegraph cable (Buchanan 1886). The discovery of the undercurrent in the Pacific is often credited to Townsend Cromwell (1922–1958) in the early 1950s, and there the current is called the Cromwell Current. Cromwell also provided the first credible theoretical model of the undercurrent, as noted below. He tragically died in 1958 in plane crash while en route to an oceanography expedition.

5 The local theories began with a description by Cromwell (1953) of the currents produced by a westward wind at the equator and were extended and put into mathematical form by Stommel (1960) with thermal effects added by Veronis (1960) and with a later variation by Robinson (1966). This class of model, which at its core is essentially linear and unavoidably dissipative, was further developed and clarified by Gill (1971), McKee (1973) and Gill (1975) and we mostly follow their treatment. The effects of nonlinearities were looked at first by Charney (1960) and then by McKee (1973) and Cane (1979a,b). The linear model was significantly extended by McCreary (1981) to include the effects of continuous stratification.

The inertial theories of the undercurrent, described in section 22.4, have their seeds in Fofonoff & Montgomery (1955) and were developed by Pedlosky (1987b). This viewpoint was extended and, in part, reconciled with the local viewpoint by McCreary & Lu (1994) who considered the equatorial undercurrent as part of a larger and more complex subtropical current system, with both local and inertial aspects. A complete description of the integrated dynamics is perforce largely numerical.

6 The canonical Airy equation is $\partial^2 y/\partial x^2 - xy = 0$. The solution, the Airy function, is discussed in many books on ordinary differential equations and special functions (e.g., Jeffreys & Jeffreys 1946, Abramowitz & Stegun 1965) and, perhaps of more relevance to the modern reader, may be calculated using mathematical software such as Maple and Python. The form of solution we use was presented by McKee (1973).

7 One of the first observational analyses to unambiguously link the equatorial ocean to higher latitudes was Bryden & Brady (1985).

8 Much of our discussion follows Pedlosky (1987b).

9 Read McCreary & Lu (1994) and work backward and forward.

10 The name El Niño was originally used by fishermen along the coasts of Ecuador and Peru to refer to a warm ocean current that often appears in December (i.e., around Christmas) between Paita and Pascamayo, and lasts for several months. An early, perhaps the first, written account of El Niño is to be found in Carillo (1892), a report by a Navy captain on a warm ocean current know to the fishermen as *Corriente del Niño,* or Current of the Christ Child. These days, the name is applied only when the warming is particularly strong and to the warming over the whole Eastern tropical Pacific.

11 My thanks to Eli Tziperman for a number of conversations on the topic, and for kindly providing some unpublished lecture notes.

12 Figure kindly constructed by A. Wittenberg by passing satellite observations through an optimal interpolation analysis.

13 Data from http://www.cgd.ucar.edu/cas/catalog/climind/soi.html. See also Wittenberg (2009). The Niño 3 region is the rectangular region outlined by dotted lines in Fig. 22.13.

14 Trenberth (1997) provides more detail.

15 See, for example, Tudhope *et al.* (2001) and Wittenberg (2009).

16 Data from the Tropical Atmosphere Ocean (TAO) project, http://www.pmel.noaa.gov/tao/.

17 Adapted from drawings by Dr. Billy Kessler.

18 Any convenient measure of the subtropical high pressure zone of the South Pacific might be used instead of Tahiti, and Easter Island was used by Quinn (1974). For more methodology on calculating the SOI, see https://www.ncdc.noaa.gov/teleconnections/enso/indicators/soi/

19 Jacob Bjerknes (1897–1975), son of Vilhelm Bjerknes (see note 3 on page 169), was a leading player in the Bergen school of meteorology. He was responsible for the now-famous frontal model of cyclones (Bjerknes 1919), and was one of the first to seriously discuss the role of cyclones in the general circulation of the atmosphere. With Halvor Solberg and Tor Bergeron the frontal model led to a picture of the lifecycle of extratropical cyclones (see chapter 12), in which a wave grows initially on the polar front (akin to baroclinic instability with the meridional temperature gradient compressed to a front, but baroclinic instability theory was not then developed), develops into a mature cyclone, occludes and decays. In 1939 Bjerknes moved to the U.S. and, largely because of World War II, stayed, joining UCLA and heading its Department of Meteorology after its formation in 1945. He developed an interest in air-sea interactions, and notably proposed the feedback between sea-surface temperatures and the strength of the trade winds (Bjerknes 1969). See also Friedman (1989), Cressman (1996), Shapiro & Grønas (1999), and the memoir by Arnt Eliassen at http://www.nap.edu/readingroom/books/biomems/jbjerknes.html.

20 Thus, for example, Latif & Barnett (1996) used a coupled model to identify a plausible coupling between atmosphere and ocean in the midlatitudes that might give rise to decadal variability, but the mechanism is not robustly seen in other models.

21 Gilbert Walker (1868–1958) was a British meteorologist who described this eponymous circulation in the 1920s. He has also been credited with discovery of the Southern Oscillation and the North Atlantic Oscillation. Walker began his career as an applied mathematician but was afforded greater renown by his analysis of meteorological observations.

22 Studies of the Walker circulation using models that make quasi-equilibium assumptions and/or use variations of the Matsuno–Gill model include Bretherton & Sobel (2002), Stechmann & Ogrosky (2014) and, in the first instance, Gill (1980). Stechman and Ogorsky argue that the Matsuno–Gill model can in fact give quantitatively accurate results if the forcing and damping are properly chosen. An exploration of the Matsuno–Gill problem in the weak-temperature-gradient (WTG) limit is given by Bretherton & Sobel (2003).

23 See, for example, Hirst (1986) and Zebiak & Cane (1987).

24 A related system of equations was considered by Lau (1981).

25 Oxymoronically, so-called toy models have a rather distinguished role in geophysical fluid dynamics in general, and in ENSO dynamics in particular. Toy models are neither a theory (such as the theory of the western boundary current in chapter 19) nor a realistic model (such as a GCM might aim to be), and they thus occupy rather questionable territory. Still, the models can sometimes spawn great insight and be very useful. The Stommel box model of the deep overturning circulation, Energy Balance Models of the global climate, and the Lorenz model of chaotic convection are important toy models in related areas.

26 Such models began with Suarez & Schopf (1988) and Battisti (1988), and we also draw from Galanti & Tziperman (2000).

27 The recharge, or recharge-discharge, or refill, hypothesis was suggested by Wyrtki (1985) and Cane

& Zebiak (1985). Wyrtki had for some time been of the view that El Niño was a basinwide phenomenon (see Wyrtki 1975), resembling seiches in smaller enclosed seas in which winds cause water to pile up at one end of the basin before sloshing back if and when the winds relax (Wyrtki 1952). An interesting historical perspective is provided by McPhaden *et al.* (2015).

28 Versions of the mechanism have been put in the form of toy-like models by Jin (1997a,b) and Clarke *et al.* (2007).

29 Papers exploring stochastic, chaotic and seasonal effects include Vallis (1988a), Tziperman *et al.* (1994), Chang *et al.* (1996) and Samelson & Tziperman (2001). See also the book by Clarke (2008).

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